

TABLE 17.5. HUMIDIFICATION CHARACTERISTICS: RASCHIG RINGS AND BERL SADDLES*

$$K_{xa} = C_1 G^\gamma$$

Where gas loading, $G = \text{lb}/(\text{hr})(\text{ft}^2 \text{ ground area})$ Liquid loading, $L = \text{lb}/(\text{hr})(\text{ft}^2 \text{ ground area})$

$$K_{xa} = \text{lb}/(\text{hr})(\text{ft}^3)(\text{lb}/\text{lb}) \text{ potential}$$

Packing, in.†	Depth, in.†	L	G	γ	K_{xa}	C_1
1 Raschig.....	24	500	250	0.50	226	14.3
		1500	250	0.50	468	29.6
		3000	250	0.50	635	40.2
1½ Raschig...	20.6	500	250	0.43	208	19.4
		1500	250	0.55	370	17.9
		3000	250	0.60	445	16.4
2 Raschig.....	19.1	500	250	0.47	190	14.3
		1500	250	0.54	301	15.3
		3000	250	0.53	351	18.9
½ Berl.....	15.5	500	250	0.61	320	11.1
		1500	250	0.61	468	16.3
		3000	250	0.61	595	20.7
1 Berl.....	20.3	500	250	0.42	245	24.2
		1500	250	0.50	464	29.4
		3000	250	0.69	569	12.7
1½ Berl.....	22	500	250	0.52	200	11.4
		1500	250	0.52	305	17.4
		3000	250	0.52	383	21.8

* From M. D. Parekh Report, MIT, courtesy of Professors T. K. Sherwood and W. H. McAdams.

† ½-in. Raschig rings do not follow the equation.

usually distributed only at the top, the tendency for more and more liquid to reach and descend along the wall and by-pass the packing increases with the height of the tower. A tower packed to a height of 20 ft does not actually deliver twice as many diffusion units as a tower packed to a height of 10 ft with the same packing, or fill. Another difficulty in large packed towers is the increased opportunity for *channeling*. With a large cross section there is a tendency for a disproportionate quantity of the liquid to descend through half the cross section while a disproportionate quantity of air rises in the other half. Still another note of caution refers particularly to data obtained on spray towers. As explained earlier, most of the performance occurs in the upper few feet of the tower. If a tower is equipped with sprays only at its top, the height of the tower will not be an indication of the available surface.

Consideration must also be given to the fact that many of the available data have been reported on experiments in which gas and liquid loadings of from 200 to 5000 lb/(hr)(ft²) have been used. The twenty-five-fold range of variables provides considerable insight into the influence of these variables on performance. By the same token it is easier to obtain good data on a small packed height in which the approach to equilibrium is not very close and in which the precise experimental determination of the humidities is not too significant. None of these should be considered representative of the ranges employed in modern cooling towers. The purpose of a cooling tower is solely to produce cooling water, which, next to air itself, is the cheapest utility. The most important operating cost is that of the fan power for circulating the air. Accordingly a small allowable pressure drop of less than 2 in. of water is the standard practice. In all but extraordinary services the liquid loading on droplet-forming fills is from 1 to 4 gpm/ft² or 500 to 2000 lb/(hr)(ft²). Gas loadings are between 1300 and 1800 lb/(hr)(ft²), corresponding to gas velocities of roughly 300 to 400 fpm.

Another factor to be considered is that of flooding the fill in which the orderly counterflow of air and water is disrupted. Lobo, Friend, *et al.*¹ have made available the results of a study on the influence of liquid and gas loadings on the flooding points of packed towers. In slat-fill towers forming droplets there are two flooding points. One exceeds about 15 gpm/(ft²) above which the liquid loading is so great that the water falls in curtains thereby reducing the production of water droplets. Another, occurring at higher liquid loadings, is the true flooding point in which the distribution of air and water is impaired. The liquid loading at the flood points is not independent of the gas loading. At the other extreme of low liquid loading is incipient wetting, in which the liquid flow is so small that the film surface may not be entirely wetted. In the case of the slat-fill tower few droplets are produced under these conditions and the surface is principally film surface.

Only limited performance data are available in the literature on commercial cooling towers. This is natural, since such information is usually confidential with fabricators, and operators only occasionally make test data available. By comparison with experimental towers the values of K_xa or HDU obtained in large towers are relatively small. At gas and liquid loadings of about 1000 lb/(hr)(ft²) many laboratory-scale data have been reported with values of K_xa ranging from 200 to 600 lb/(hr)(ft³)(lb/lb). It is doubtful that any full-sized apparatus erected in the United States at present is capable of a K_xa exceeding 100 within the usual limits of pressure drop permitted in industry.

¹ Lobo, W. E., L. Friend, F. Hashmall, and F. Zenz, *Trans. AIChE*, **41**, 693 (1945).

The Calculation of Cooling-tower Performances. Usually operators buy cooling towers rather than erect them themselves. This is undoubtedly the wisest policy, since it makes available to the operator a great deal of "know-how" in a field in which it is of great value. The operator will specify the quantity of water and the temperature range he requires for his process. The fabricator will propose a tower which will fulfill the conditions furnished by the operator for the 5 per cent wet bulb in the locality of the plant and guarantee the fan power with which it will be accomplished. With the first cost, fan power, and the approximate height of the pumping head the operator can compute the cost of the cooling-tower water based on a period of depreciation of about twenty years.

Assume that a cooling tower has been erected and placed in operation on this basis. An acceptance run is made to determine whether or not the cooling tower is meeting the guarantee. This consists of a wet-bulb determination on the windward side of the tower and the determination of the air rate by velometer or other pitot arrangement. The air leaving the cooling tower is always assumed to be saturated at its outlet temperature. Tests have shown it to be anywhere from 95 to 99 per cent saturated.

In a heat exchanger the performance is satisfactory if the measured overall coefficient during initial operation at the process conditions equals or exceeds the stipulated clean coefficient. In cooling towers the basis for design is one which is very rarely present. A cooling tower designed for a given approach to the 5 per cent wet bulb and erected in the fall of the year will probably not encounter the design condition for another 8 or 9 months. On the basis of its performance in autumn will it operate at the design conditions when they eventually return? A series of calculations answering questions of this nature will be undertaken presently.

Each performance calculation may be divided into two parts: (1) How many diffusion units correspond to the process requirement, and (2) how many diffusion units is or can the tower actually perform? Obviously at all conditions (2) should exceed (1).

Example 17.2. Calculation of the Number of Diffusion Units. A plant is being laid out in a restricted-water locality. The total heat load to be removed from several processes by the cooling tower is 26,000,000 Btu/hr. The locality has a 5 per cent wet-bulb temperature of 75°F. The water will leave the tower with a 10° approach to the wet bulb, or 85°F. Being water of ordinary air and mineral content it will emerge from equipment at a maximum temperature of 120°F. The water equivalent to this range is 1500 gpm.

A tower 24 by 24 ft has been erected with a fan capacity of 187,000 cfm. How

many diffusion units must the tower be capable of performing to fulfill the process requirements?

Determine (a) by numerical integration and (b) by using the log mean enthalpy difference.

Solution. (a) On coordinates of enthalpy vs. water temperature such as Fig. 17.12 plot the saturation line from the data of Table 17.2.

Next determine the enthalpy of the inlet air which is one terminal of the operating line. In this problem it corresponds to a 75°F wet bulb or air saturated at 75°F. From Table 17.2 $H_1 = 39.1$ Btu/lb dry air at an outlet water temperature of 85°F.

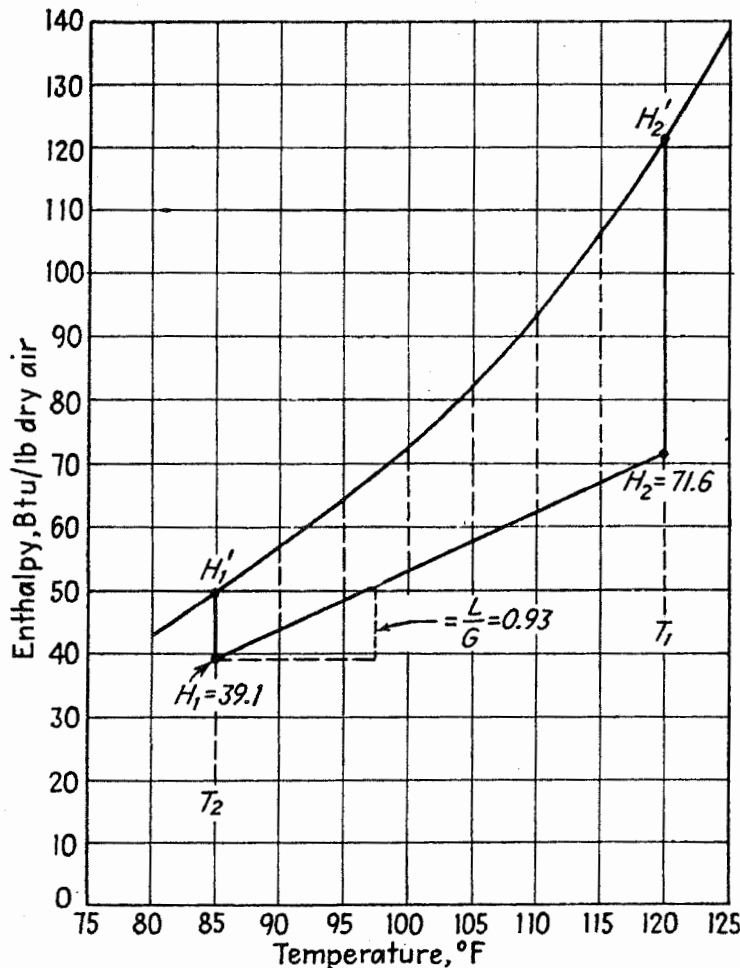


FIG. 17.12. Solution of Example 17.2.

The liquid and air loading will establish the slope of the operating line throughout the tower, starting at H_1 . The liquid loading is simply the total hourly water rate divided by the ground area. The air loading was given in the proposal as 187,000 cfm at 75°F wet bulb. The density of the dry air in a cubic foot of mixture is $1/13.88 = 0.072$ lb/ft³ from Table 17.2.

$$\text{Ground area} = 24 \times 24 = 576 \text{ ft}^2$$

$$L = 1500 \times \frac{50}{576} = 1302 \text{ lb/(hr)(ft}^2\text{)}$$

$$G = 187,000 \times 0.072 \times \frac{60}{576} = 1400 \text{ lb/(hr)(ft}^2\text{)}$$

$$\frac{L}{G} = \frac{1302}{1400} = 0.93$$

In Fig. 17.12 from $H_1 = 39.1$ draw a line of positive slope equal to 0.93 or calculate H_2 from Eq. (17.83) and draw a line between H_1 and H_2 .

$$H_2 = H_1 + \frac{L}{G}(T_2 - T_1) = 39.1 + 0.93(120 - 85) = 71.6 \text{ Btu}$$

The area between the saturation line and the operating line represents the potential for heat transfer. It can be determined either by counting squares, in which case a plot of $1/(H' - H)$ vs. T is more convenient, or by numerical integration. The latter is employed here:

T	H'	H	$H' - H$	$(H' - H)_{av}$	$\frac{dT(= 5.0^\circ\text{F})}{(H' - H)_{av}}$
85	50.0	39.1	10.9		
90	56.7	43.7	13.0	11.45	0.418
95	64.2	48.4	15.8	14.4	0.347
100	72.7	53.1	19.7	17.7	0.282
105	82.5	57.7	24.8	22.2	0.225
110	93.8	62.4	31.4	28.1	0.178
115	106.7	67.0	39.7	35.55	0.1405
120	121.5	71.6	49.9	44.8	0.1115
$n_d = K_x a \frac{V}{L} = \int \frac{dT}{H' - H} = 1.70$					

In order to fulfill process requirements the tower must be capable of performing 1.70 diffusion units. But to perform to that extent under process requirements it must also be capable of that much performance under any other conditions.

(b) Using the log mean enthalpy difference,

At the top of the tower $H'_2 - H_2 = 49.9 \text{ Btu/lb}$

At the bottom of the tower $H'_1 - H_1 = 10.9 \text{ Btu/lb}$

$$\text{Log mean } (H' - H) = \frac{49.9 - 10.9}{2.3 \log 49.9/10.9} = 25.8 \text{ Btu/lb}$$

$$n_d = \frac{K_x a V}{L} = \frac{dT}{H' - H} = \frac{120 - 85}{25.8} = 1.35 \text{ vs. } 1.70$$

The error is naturally larger the greater the range, and 35°F is nearly the extreme for a cooling tower. This method is acceptable only where the range is very small.

Example 17.3. Calculation of the Required Height of Fill. Suppose it is known that, for a particular fill at $L = 1302$ and $G = 1400$, $K_x a$ is $115 \text{ lb}/(\text{hr})(\text{ft}^3)(\text{lb}/\text{lb})$. What height of fill should be supplied, and what is the HDU when a fill has a $K_x a$ of $115 \text{ lb}/(\text{hr})(\text{ft}^3)(\text{lb}/\text{lb})$?

Solution. Height of fill:

Since the loading is based on 1 ft^2 of ground area,

$$n_d = \frac{K_x a(Z)}{L}$$

$$Z = \frac{n_d L}{K_x a} = 1.70 \times \frac{1302}{115} = 19.1 \text{ ft}$$

The height of a diffusion unit is Z/n_d .

$$HDU = \frac{19.1}{1.70} = 11.3 \text{ ft}$$

This last is contrasted to values of 3 ft obtained by London, Mason, and Boelter¹ on a streamline shape film tower of known surface or of 1½ ft obtained by Parekh (extrapolated) for Raschig rings.

Example 17.4. Determination of a Cooling-tower Guarantee. Under steady atmospheric conditions during an autumn test the 24- by 24-ft tower of Example 17.2 in actual operation showed a steady water temperature of 114.3°F on the tower and 79.3° off the tower. The wet bulb determined by psychrometer was 65°F. Air and water loadings were maintained at their design values.

Will this tower be capable of producing 1500 gpm of 120 to 85°F water when the wet bulb is 75°F?

Solution. This is a solution based on the observed data and is the same as before except that the operating line starting at H_1 corresponds to the 65°F wet bulb or 30.4 Btu/lb.

T	H'	H	$H' - H$	$(H' - H)_{av}$	$\frac{dT}{(H' - H)_{av}}$
79.3	43.4	30.4	13.0		
85	50.0	35.7	14.3	13.65	0.417
90	56.7	40.3	16.4	15.35	0.326
95	64.2	45.0	19.2	17.8	0.281
100	72.7	49.6	23.1	21.5	0.237
105	82.5	54.3	28.2	25.65	0.195
110	93.8	58.5	35.3	31.75	0.158
114.3	103.3	62.9	40.4	37.85	0.113
$n_d = 1.72$					

In order to fulfill the process requirements the cooling tower must be capable of providing 1.70 diffusion units when the wet bulb is 75°F. The same calculation applied to the *actual* test conditions shows it to be delivering 1.72 diffusion units, and it is therefore acceptable. The fan power can be checked independently from the electric circuit or from the steam rate for a turbine drive.

The Influence of Process Conditions upon Design. It is rewarding to study the effects of various process changes upon the height and cross section of the apparatus or the cost of its operation. Six of the considerations which affect the size of the tower are indicated in Fig. 17.13. They are analyzed best by means of the enthalpy-temperature diagram, since the area between the saturation and operating lines is a measure of the

¹ London, A. L., W. E. Mason, and L. M. K. Boelter, *Trans. ASME*, **62**, 41-50 (1940).

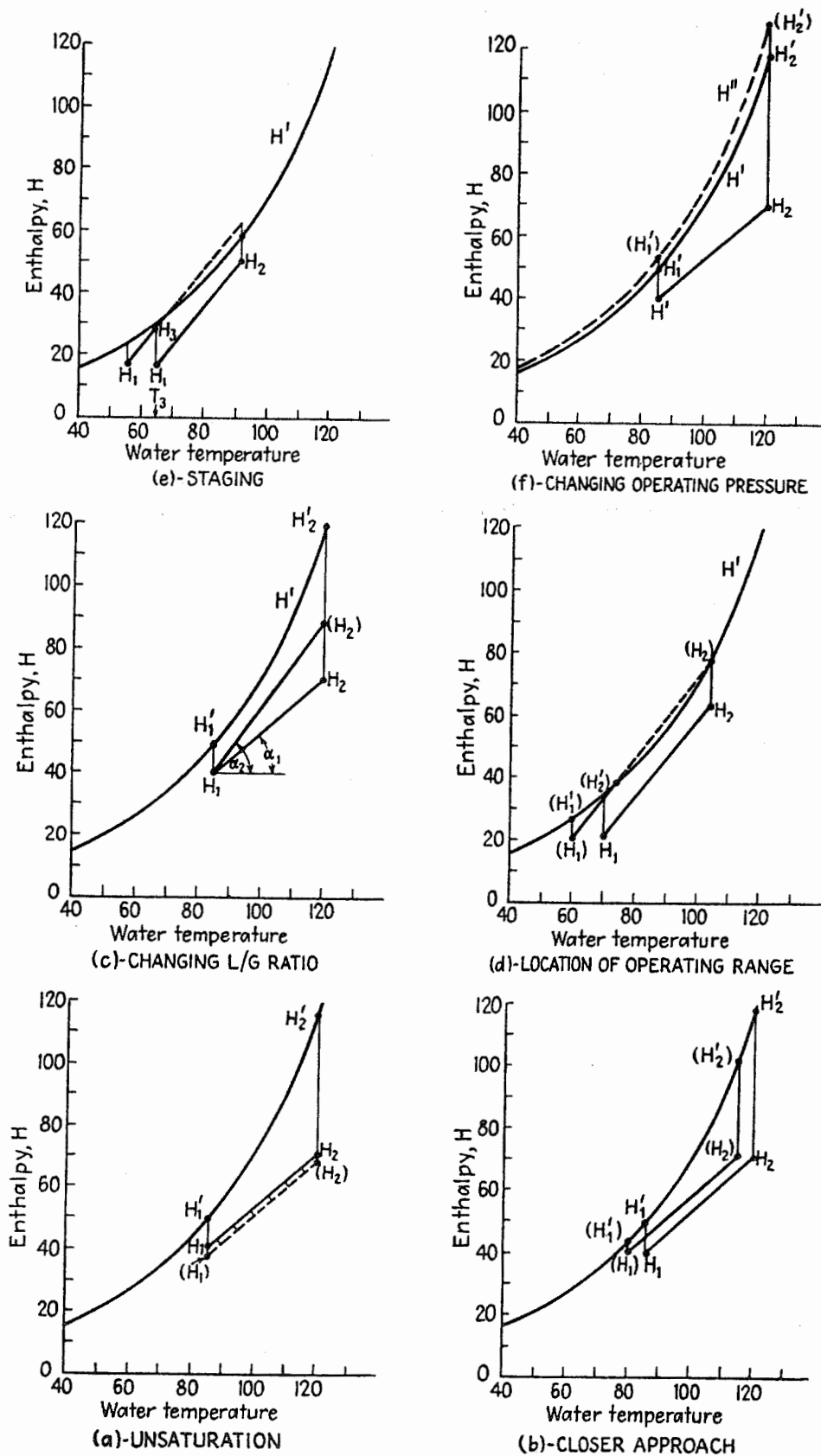


FIG. 17.13. Effect of design variables on the size of the potential.

total potential. The smaller the area the greater the height of tower required for fulfillment of the process conditions.

a. Unsaturation of the Inlet Air. Heretofore reference has been made only to the wet-bulb temperature of the inlet air and not to its dry bulb. In each case it has been assumed identical with the wet bulb, *i.e.*, adiabatically saturated. In Example 17.2 the enthalpy of the inlet air at the 75°F wet bulb was 39.1 Btu/lb. Suppose the air is at a dry bulb of 85°F when the wet bulb is 75°F. The air will be unsaturated, and its enthalpy will be 41.7 Btu/lb instead of 39.1. In Fig. 17.13a this will drop the operating line insignificantly from $H_1 - H_2$ to $(H_1) - (H_2)$, the cross-hatch area representing the increase in potential. Failure to correct the enthalpy for the dry bulb gives results which are slightly on the safe side, and for this reason it is customary to specify only the wet bulb.

b. Close Approach. Both operating lines in Fig. 17.13b have the same L/G ratio (same slopes) and equal ranges of 35°F for the removal of the same process heat load. The operating line $(H_1) - (H_2)$ attempts to do the same cooling as $H_1 - H_2$ and with the same inlet air but between the temperatures of 115 and 80°F instead of 120 and 85°F. The area between the saturation curve and the operating line is greatly decreased by $(H_1) - (H_2)$. Similarly it might be desired to obtain water from 120 to 85°F with 80° wet-bulb instead of 75° wet-bulb air. This will raise the operating line $H_1 - H_2$ vertically and also decrease the potential.

c. Changing the L/G Ratio. If the ground area is too limited, such as when a cooling tower is erected atop a building, it may be necessary to employ a relatively high liquid loading without increasing the air quantity, since 400 fpm is about the maximum economical air velocity. This will decrease the cross section of the tower but increase the slope of the operating line $H_1 - H_2$ to $H_1 - (H_2)$ as in Fig. 17.13c, resulting in a decreased potential and a higher tower. This is the simple observation that the less air circulated per pound of water the less the extent of cooling.

d. Location of the Operating Range. The saturation line has a varying curvature. In Fig. 17.13d an operating line $H_1 - H_2$ is shown with a range from 105 to 70°F. Suppose it is desired to design a tower using the same inlet air but to cool water from 95 to 60°F. This would be impossible with the same L/G ratio, since the operating line $(H_1) - (H_2)$ would intersect the saturation line. Obviously heat transfer would stop at the intersection (H'_2) , since the potential would be zero at that point. A considerably lower L/G ratio would be required, which in turn means more air must be circulated for the removal of the same number of Btu.

e. Staging. One of the means of overcoming the small L/G ratio in the preceding paragraph can be accomplished by the use of two towers. This is *staging*. The water at the top of the first tower is hot and contacts air of enthalpy H_2 along the operating line $H_1 - H_2$ as shown in Fig. 17.13e. The water is removed from the basin at temperature T_3 and is pumped over the second tower, which also uses atmospheric air of enthalpy H_1 . The second tower operates between H_1 and H_3 . In this way both operating lines may have large slopes without intersecting the saturation line. The fixed charges and operating cost of two towers increase the cost of water considerably, but water produced in this manner should be regarded as chilled water, and its cost and range compared with that of refrigerated water.

f. Elevation. Some plants are located at high elevations. Should this be mentioned in the process conditions? At reduced atmospheric pressure, as seen in Fig. 17.13f, the saturation line is higher, which in turn increases the potential and reduces the required size of tower if all other conditions are the same. This occurs because the partial pressure of the water is fixed whereas the total pressure has been decreased. The humidity of the saturated air at higher elevation is also greater.

The Influence of Operating Variables. Since a cooling tower uses the atmosphere as a cooling medium, it is also subject to the variations in the atmosphere. When operating at the design wet bulb, the tower should produce water of the range and at the temperature specified in the guarantee. When the wet bulb falls, however, it is the same as increasing the potential in the tower. If the heat load on the tower and the values of L and G circulated through the tower are all to be kept constant, the water still undergoes the same number of degrees of cooling in it but the inlet and outlet temperatures will be colder than guaranteed. The cooling tower is only able to remove the same heat load from the water by automatically reducing the potential difference, the water temperatures declining accordingly with the wet bulb. From the standpoint of operating the coolers and condensers employed in a plant, this arrangement requires the simplest instrumentation. When a cooler receives water at a constant rate which is colder than that for which it was rated, the hot fluid is cooled below the desired outlet temperature. To prevent this from occurring in a cooler, the flow rate of the colder water through the cooler is reduced by by-passing, so that less water is used but with a constant design outlet temperature. The constant-temperature outlet water is then recombined with the by-pass water before returning to the cooling tower. In this way all coolers or condensers designed for water, say, from 85 to 120°F have outlet-water temperatures of 120°F

the year around. This was, in fact, the basis for the temperatures in Example 17.4.

If the water and air loadings on a tower are changed in any way, the number of diffusion units the tower is capable of providing are also altered. Ordinarily the loadings on a single cell cannot be changed very greatly. In fact a variation of 20 per cent from the mean of the design water loading is the maximum which may be anticipated, since the droplets are produced by nozzles which, in turn, have been sized for a given flow rate at a given head. The maximum discharge capacity will be about 120 per cent of design and when less than 80 per cent of the design water rate is used, the dispersion of the droplets is reduced along with the total quantity of water. The air loading can be regulated by varying the pitch of the fan blades, which can usually be rotated to a plus or minus 3° position from the mean. In summer the blades will be in a $+3^\circ$ position and in winter at -3° . In winter the -3° position of the fan delivers about 80 per cent of the $+3^\circ$ air quantity, but the power saving is 40 per cent. If requested, the cooling-tower fabricator will specify the temperature range assumed by the cooling-tower water at 80 and 120 per cent of its design loading when the design air quantity is at the guarantee wet bulb. Sometimes, when the operating wet bulb is lower than the design wet bulb, it is desirable to use the increased potential to produce more cooling water of the original design temperature range. When the wet bulb varies, how much more water can be produced? This can be calculated from the 80, 100, and 120 per cent values, using the assumption that the number of diffusion units performed by a tower is dependent only upon the L/G ratio and not upon either L or G separately. This is a useful approximation within the range of cooling-tower operation and from 80 to 120 per cent of design capacity.

Example 17.5. The Recalculation of Cooling-tower Performance. In the guarantee of the 24- by 24-ft tower of Example 17.2 for cooling 1500 gpm from 120 to 85°F at a 75°F wet bulb the following data were given by the fabricator for overload and underload

<i>Liquid loading, %</i>	<i>Temperature range, $^\circ\text{F}$</i>
120	122.2–87.2
100	120.0–85.0
80	117.5–82.5

When the wet bulb is 70°F , how much water from 120 to 85°F can the tower provide?

Solution. First reconstruct the portion of the actual performance curve from which the tower was designed. On the assumption that the performance curve is only a function of L/G and not L or G separately and, having already integrated the design condition, the same must be done for the overload and underload.

For 1500 gpm, $L/G = 0.93$

For 120 per cent of design, $L/G = 1.20 \times 0.93 = 1.115$

At 87.2°F, $H_2 = 39.1$ Btu

At 122.2°F, $H_2 = 39.1 + 1.115 \times 35 = 78.1$ Btu

T	H'	H	$H' - H$	$(H' - H)_{av}$	$\frac{dT}{(H' - H)_{av}}$
87.2	53.1	39.1	14.0		
90	56.7	42.0	14.7	14.35	0.195
95	64.2	47.6	16.6	15.65	0.320
100	72.7	53.3	19.4	18.0	0.278
105	82.5	58.8	23.7	21.55	0.232
110	93.8	64.3	29.5	26.6	0.188
115	106.7	70.0	36.7	33.25	0.150
120	121.5	75.6	45.9	41.3	0.121
122.2	128.2	78.1	50.1	48.0	0.0458
$K_{xa} \frac{V}{L} = 1.53$					

For 80 per cent of design, $L/G = 0.80 \times 0.93 = 0.74$

At 82.5°F, $H_1 = 39.1$

At 117.5°F, $H_2 = 39.1 + 0.74 \times 35 = 65.0$ Btu

T	H'	H	$H' - H$	$(H' - H)_{av}$	$\frac{dT}{(H' - H)_{av}}$
82.5	47.2	39.1	8.1		
85	50.0	40.8	9.2	8.7	0.286
90	56.7	44.6	12.1	10.7	0.465
95	64.2	48.3	15.9	14.0	0.357
100	72.7	52.0	20.7	18.3	0.273
105	82.5	55.6	26.9	23.8	0.210
110	93.8	59.3	34.5	30.7	0.163
115	106.7	63.0	43.7	39.1	0.128
117.5	121.5	65.0	56.5	50.1	0.050
$K_{xa} \frac{V}{L} = 1.92$					

The values of $K_{xa}V/L$ vs. L/G are plotted in Fig. 17.14.

How much water from 120 to 85°F can be circulated over the tower when the wet bulb falls to 70°F? In Fig. 17.14 both the ordinate $K_{xa}V/L$ and the abscissa L/G contain L , which in turn determines the desired water quantity. A value of L can be assumed, and if the value obtained for both $K_{xa}V/L$ and L/G falls on the curve of Fig. 17.14, the requirements of performance will be satisfied. The solution is consequently by trial and error.

Trial 1. Assume $L/G = 1.10$

$$H_1 = 34.5$$

$$H_2 = 34.5 + 1.10 \times 35 = 73.0$$

T	H'	H	$H' - H$	$(H' - H)_{av}$	$\frac{dT}{(H' - H)_{av}}$
85	50.0	34.5	15.5		
90	56.7	40.0	16.7	16.1	0.309
95	64.2	45.5	18.7	17.7	0.282
100	72.7	51.0	21.7	20.2	0.247
105	82.5	56.5	26.0	23.85	0.210
110	93.8	62.0	31.8	28.9	0.173
115	106.7	68.5	38.2	35.0	0.143
120	121.5	73.0	48.5	43.35	0.115
					1.48

From Fig. 17.14, for $K_X aV/L = 1.48$, $L/G = 1.19$

Assumed, $L/G = 1.10$ No check

Trial 2: Assume $L/G = 1.20$

$$H_1 = 34.5$$

$$H_2 = 34.5 + 1.20 \times 35 = 76.5$$

T	H'	H	$H' - H$	$(H' - H)_{av}$	$\frac{dT}{(H' - H)_{av}}$
85	50.0	34.5	15.5		
90	56.7	40.5	16.2	15.9	0.313
95	64.2	46.5	17.7	16.95	0.295
100	72.7	52.5	20.2	18.95	0.264
105	82.5	58.5	24.0	22.1	0.226
110	93.8	64.5	29.3	26.65	0.188
115	106.7	70.5	36.2	32.75	0.153
120	121.5	76.5	45.0	40.6	0.123
					1.56

From Fig. 17.14, for $K_X aV/L = 1.56$, $L/G = 1.08$

Assumed, $L/G = 1.20$ No check

By interpolation the conditions are satisfied when the value of $L/G = 1.14$. The total water which can be produced with an 85°F temperature will be $1500 \times \frac{1.14}{0.93} = 1840$ gpm.

Dehumidifiers. The dehumidification of hot humid air by washing with cold water is a common practice in air conditioning. Although it is generally carried out in spray washers, it can be effectively achieved through the use of a cooling tower or similar contacting device. Con-

sider humid air saturated and at a higher temperature than the cold-water spray. The humidity or partial pressure of the water vapor in the hot air body is greater than the saturation humidity in the air film, which, in the absence of a liquid film resistance, is presumed to be saturated at the

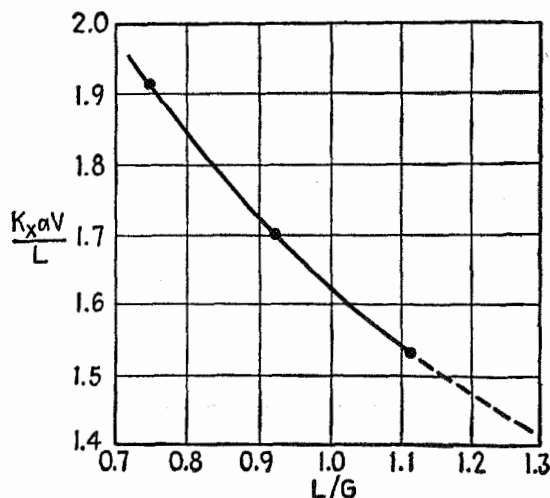


FIG. 17.14. Solution of Example 17.5.

liquid is a mechanism apart from condensation, although the heat of the water removed by mass transfer is equal to the latent heat of vaporization. In true diffusion the mass transfer from the gas body to the liquid is treated as a molecular phenomenon, in which case no dew is assumed to form. Humidification and dehumidification may be considered identical except in the direction of the mass transfer. Studies indicate that the rate of dehumidification equals the rate of humidification, the same $K_x a$ data being applicable to either. Thus the number of diffusion units for dehumidification can likewise be determined by integrating the area between the upper operating line and the saturation line. The basic equation for the case of dehumidification where the Lewis number is unity becomes

$$n_d = K_x a \frac{V}{L} = \int \frac{dT}{H - H'} \quad (17.86)$$

which differs from Eq. (17.81) only in the position of H and H' .

Heat Transfer from Gases. All the preceding illustrations have been concerned with cooling warm water. Suppose it is desired to cool a gas at a moderate pressure as in Chap. 9. If the gas is passed through a heat exchanger with a reasonable allowable pressure drop, a low heat-

water temperature. The potential difference for mass transfer is then in the direction from the gas body to the water body, the reverse of the direction of humidification. The enthalpy at every point in the air body is greater than the corresponding value of the air film. This is shown graphically in Fig. 17.15. The operating line again has the slope of L/G , but it is above the saturation line.

The mass transfer of the water vapor from the air body into the

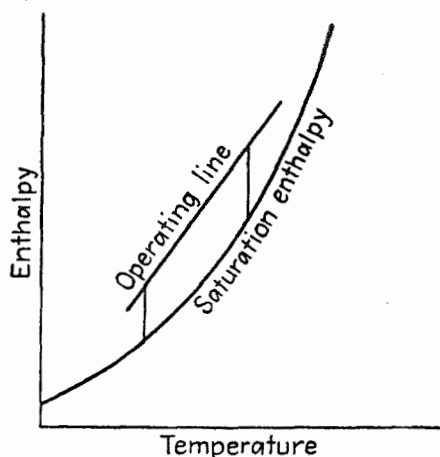


FIG. 17.15. Dehumidification potentials.

transfer coefficient results. This is particularly true where the static pressure on the gas is developed by a blower or fan. The same hot gas may be brought into contact with water in a packed or filled tower with droplet surface, and the heat transfer will be accomplished very cheaply. In many instances a cylindrical vessel containing a few feet of dumped packing material may replace a large tubular exchanger. The application of direct-contact heat transfer to the solution of gas-cooling problems has not yet received the widespread acceptance it appears to merit.

There is an unfounded prejudice in some quarters against the direct contact of water with high-temperature gases. It is sometimes felt that unpredictable results will be obtained such as frothing and boiling. Consider the typical cases shown in Fig. 17.16. In Fig. 17.16a a hot gas (air) enters a tower at 300°F with a dew point of 120°F and leaves at 200°F. The water has a range from 85 to 120°F. What will happen at

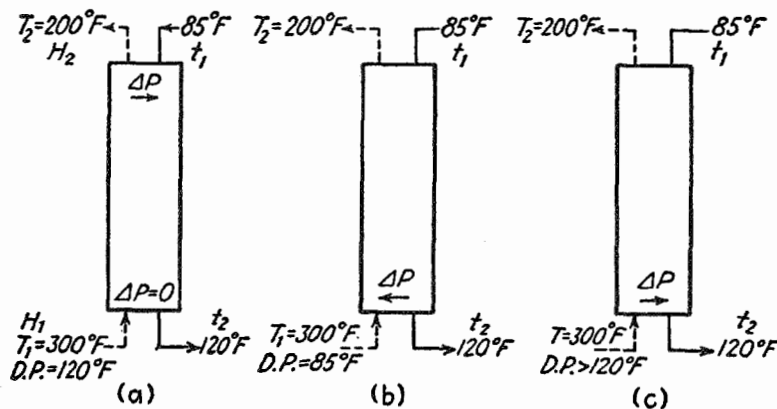


FIG. 17.16. Influence of the inlet dew point on the direction of mass transfer.

the bottom of the tower? The gas, by virtue of its dew point, has a vapor pressure of 1.69 psia. The water leaving the tower is at 120°F and has the same vapor pressure. There will be no potential for the mass transfer of water from the water body into the gas, so that there can be no vaporization, whether the temperature of the inlet gas be 200 or 2000°F. The mass transfer is established solely by the difference between the dew point of the inlet gas and the partial pressure corresponding to the outlet water temperature. At a differential height upward in the tower the relationship between the potentials changes: The water temperature is lower than the gas dew point, and water vapor starts to diffuse from the gas to the water. The diffusion process continues up to the gas outlet at the top.

In Fig. 17.16b the process conditions have been altered so that the dew point of the inlet gas is 85 instead of 120°F. Here the partial pressure of the outlet cold water is greater than that of the inlet gas, but the pressure exerted by the outlet water is still 1.69 psia, no matter how high the

gas temperature may be. This case is somewhat different from that of a cooling tower, since the gas will be cooled and humidified whereas in the cooling tower it is heated and humidified. In Fig. 17.16c a gas enters with a higher dew point than that of the water and the movement of the vapor is from the gas to the water body throughout. Other interesting arrangements are possible. Analysis shows that sensible-heat transfer by direct contact must be accompanied by some mass transfer. When a gas is to be cooled over a great range, measured in hundreds of degrees, and the dew point corresponds to that of flue gases such as from 110 to 130°F, the amount of mass transfer may be very small.

Calculations without a Simplified Lewis Number. In the derivation of equations for the performance of cooling towers or other air-water humidifiers and dehumidifiers, a simple expression was obtained. This was owing very largely to the assumption that the Lewis number for air-water diffusion is nearly unity. It thus becomes possible to eliminate the last term of Eq. (17.75) from further evaluation in cooling-tower calculations. When $Le \neq 1.0$, Eq. (17.75), instead of reducing to Eq. (17.81), reduces to

$$n_a = K_{xa} \frac{V}{L} = \int \frac{dT}{(H' - H) + c(T - t)(Le - 1)} \quad (17.87)$$

Equation (17.87) cannot be evaluated in a simple, straightforward manner because of the last term of the denominator. In Fig. 17.4 the Lewis number for air-water has a value of 1.0 at 600°F. For the diffusion of water into other gases such as H_2 and CO_2 the Lewis number is about 2.0 and 0.6, respectively.

In the case of gas cooling if no simplifying assumption can be made for the Lewis number, it is necessary to return to the three basic equations for humidification and dehumidification. These are

$$dq_c = h_a dV(T - t) = Gc dT \quad (17.62)$$

$$dq_d = K_{xa} dV(X - X') = \lambda dL \quad (17.67)$$

$$L dt = G dH \quad (17.78)$$

where T now refers to the hot gas and t to the cold water. These equations are, respectively, the convection, diffusion, and overall heat balances. The first two can be related by the fact that $h = (Le)K_{xc}$. The overall heat balance given by Eq. (17.78) obviously is the sum of Eqs (17.62) and (17.67), respectively.

The problem is to determine the number of diffusion units corresponding to process requirements by the simultaneous integration of all three equations. This calls for a typical trial-and-error solution. Suppose a hot gas as in Fig. 17.16a contacts cold water in a counterflow tower. The process conditions leave two unknowns: the outlet air humidity and the quantity of water flowing over the tower. Although its temperature is

known, the outlet-air enthalpy cannot be calculated until its humidity is known. Without the outlet humidity the total heat to be removed $G dH = L dt$ cannot be calculated, and without it L cannot be determined. The number of diffusion units is dependent upon the L/G ratio, which cannot be set without knowing L . The solution of a gas-cooling problem can be seen to hinge entirely upon the value of the outlet-air humidity. However, by assuming the outlet humidity for a required gas-outlet temperature the enthalpy can be determined, the heat balance closed, and the L/G ratio is found.

Neglecting mechanical problems of distribution, the required height of the tower is proportional to the number of diffusion units it must contain. Having obtained L by assuming the gas-outlet humidity, the quantity of water diffused in the tower and the quantity of heat transferred can be determined by $K_x a V/L$ or $h a V/L$, both of which are related by the Lewis number. Starting at the bottom of the tower at the gas inlet, an increment $K_x a \Delta V/L$ can be assumed and the amount of mass and heat transfer occurring over the increment calculated, since L is now known and $K_x a \Delta V$ is the actual number of pounds of mass transfer for the interval. Working up the column by increments, Eqs. (17.62) and (17.67) can be integrated. Since Eq. (17.78) is the sum of Eqs. (17.62) and (17.67), a height should be reached in the tower at which all the heat transferred corresponds to Eq. (17.78). If the assumed enthalpy of the outlet-gas and the inlet-water temperature does not occur at the same height, the outlet enthalpy has been assumed incorrectly and a new assumption must be made.

Sherwood and Reed¹ have given the solution of the three differential equations by the method of W. E. Milne, which is straightforward but fairly lengthy. If an estimate of the outlet humidity can be made, the method outlined above may be simpler.

Example 17.6. Calculation of a Direct-contact Gas Cooler. A cooling operation consists of passing 50,000 lb/hr of bone-dry gas of 29 molecular weight (essentially nitrogen) over trays of hot material from whose surface oxygen must be excluded. The gas is heated in passing over the trays from 200 to 300°F and emerges with a dew point of 120°F. It then goes to a direct-contact cooler where it is cooled back to 200°F and dehumidified with water which is heated from 85 to 120°F. A pressure drop of about 2 in. of water is permissible.

(a) How many diffusion units are required to accomplish this process?

(b) Using some standard low-pressure-drop data in the literature calculate the dimensions of a direct-contact tower. In the solution of this problem data will be used for a simple fill described by Simpson and Sherwood.

Solution. Basis: 1 ft² of ground area.

¹ Sherwood, T. K., and C. E. Reed, "Applied Mathematics in Chemical Engineering," 134, McGraw-Hill Book Company, Inc., New York, 1939.

(a) Tower fill of suspended fiber board will be used. Such towers can easily operate with gas velocities of 450 fpm with reasonable pressure drops and drift elimination because they are film type towers. If the gas loading is assumed to be 1500 lb/(hr)(ft²), it will correspond to a gas velocity at the average temperature of the gas of 450 fpm. A higher gas rate cannot be justified, and a lower one might provide a tower of unnecessarily large cross section. This last can be checked only by a pressure-drop calculation once the height of the tower has been determined.

To close the heat balance and to determine the total heat transfer and liquid loading it is necessary to assume the outlet-gas humidity. Assumption: 20 per cent of the initial vapor content of the gas enters the water body.

$$\text{At the gas inlet } X_1 = \frac{1.69}{14.7 - 1.69} \times \frac{18}{29} = 0.0807 \text{ lb/lb}$$

$$G = 1500 \text{ lb/hr}$$

$$\text{Total water in inlet gas} = 1500 \times 0.0807 = 121.05 \text{ lb/hr}$$

The inlet gas is at 300°F and a 120°F dew point. Use 0.25 Btu/(lb)(°F) for the specific heat of nitrogen

$$H_1 = 0.0807 \times 120 + 0.0807 \times 1025.8 + 0.45 \times 0.0807(300 - 120) + 0.25 \times 300 = 174.0 \text{ Btu/lb dry air (17.55)}$$

20 per cent of the vapor has been assumed to diffuse into the water body.

$$\text{Outlet gas humidity, } X_2 = \frac{121.05(1.0 - 0.20)}{1500} = 0.06456 \text{ lb/lb}$$

$$\text{Dew point of outlet gas, } \frac{p_w}{14.7 - p_w} \times \frac{18}{29} = 0.06456$$

$$p_w = 1.388 \text{ psia} \approx 112.9^\circ\text{F dew point (Table 7)}$$

The outlet gas has a temperature of 200°F and a 112.9°F dew point.

$$H_2 = 0.06456 \times 112.9 + 0.06456 \times 1029.8 + 0.06456 \times 0.45(200 - 112.9) + 0.25 \times 200 = 126.4 \text{ Btu/lb dry air (17.55)}$$

$$\text{Total heat load, } q = G(H_1 - H_2) = 1500(174.0 - 126.4) = 71,500 \text{ Btu/hr}$$

$$\text{Water loading, } L = \frac{71,500}{120 - 85} = 2040 \text{ lb/hr}$$

This water loading corresponds to 4.0 gpm/(ft²), which is reasonable for this type of apparatus as explained earlier.

Interval 1:

$$\frac{K_{xa} \Delta V}{L}: 0 \text{ to } 0.05$$

From Fig. 17.4 at 300°F, $Le = 0.93$

$$C = 0.25 + 0.45 \times 0.0807 = 0.283 \text{ Btu/(lb)(°F)}$$

$$haV = K_{xa} \frac{V}{L} L(Le)C = 0.05 \times 2040 \times 0.93 \times 0.283 = 26.9 \text{ Btu/(hr)(°F)}$$

$$q_c = haV(T - t) = 26.9(300 - 120) = 4850 \text{ Btu/hr}$$

$$\Delta T = \frac{4,850}{0.283 \times 1500} = 11.4^\circ\text{F}$$

$$T_{0.05} = 300 - 11.4 = 288.6^\circ\text{F}$$

Since the dew point of the gas and the water outlet are the same in this problem, there is no diffusion in the first interval. In any other problem diffusion may occur in the first interval, and it can be treated the same as the second interval below:

$$\Delta t = \frac{4850}{2040} = 2.38^\circ\text{F}$$

$$t_{0.05} = 120 - 2.38 = 117.6^\circ\text{F}$$

The temperatures for the first interval are shown in Fig. 17.17.

Interval 2: $\frac{K_{xa} \Delta V}{L}$: 0.05 to 0.15

$$haV = 0.10 \times 2040 \times 0.93 \times 0.283 = 53.8 \text{ Btu}/(\text{hr})(^\circ\text{F})$$

For the interval:

$$q_c = 53.8(288.6 - 117.6) = 9200 \text{ Btu/hr}$$

$$\Delta T = \frac{9200}{0.283 \times 1500} = 21.7^\circ\text{F}$$

$$T_{0.15} = 288.6 - 21.7 = 266.9^\circ\text{F}$$

$$X'_{117.6^\circ\text{F}} = 0.0748 \text{ lb/lb}$$

Lb water diffused during interval = $K_{xa}V(X - X')$

$$K_{xa}V = K_{xa} \frac{V}{L} \times L = 0.10 \times 2040 = 204.0 \text{ lb}/(\text{hr})(\text{lb/lb})$$

$$K_{xa}V(X - X') = 204.0(0.0807 - 0.0748) = 1.203 \text{ lb/hr}$$

$$\text{Lb water remaining} = 121.05 - 1.203 = 119.85 \text{ lb/hr}$$

$$\lambda_{117.6^\circ\text{F}} = 1027 \text{ Btu/lb}$$

$$q_d = 1.203 \times 1027 = 1235 \text{ Btu/hr}$$

$$q = 9200 + 1235 = 10,435 \text{ Btu/hr}$$

$$\Delta t = \frac{10,435}{2040} = 5.12^\circ\text{F}$$

$$t_{0.15} = (117.6 - 5.1) = 112.5^\circ\text{F}$$

$$X'_{112.5^\circ\text{F}} = 0.0640 \text{ lb/lb}$$

$$X_{112.5^\circ\text{F}} = \frac{119.85}{1500} = 0.0798 \text{ lb/lb}$$

Interval 3: $\frac{K_{xa} \Delta V}{L}$: 0.15 to 0.25

$$haV = 53.8 \text{ Btu}/(\text{hr})(^\circ\text{F})$$

For the interval:

$$q_c = 53.8(266.9 - 112.5) = 8300 \text{ Btu/hr}$$

$$\Delta T = \frac{8300}{0.283 \times 1500} = 19.5^\circ\text{F}$$

$$T_{0.25} = 266.9 - 19.5 = 247.4^\circ\text{F}$$

$$\text{Lb water diffused during interval} = 204.0(0.0798 - 0.0640) = 3.22 \text{ lb/hr}$$

$$\text{Lb water remaining} = 119.85 - 3.22 = 116.63 \text{ lb/hr}$$

$$\lambda_{112.5^\circ\text{F}} = 1030 \text{ Btu/lb}$$

$$q_d = 3.22 \times 1030 = 3320 \text{ Btu/hr}$$

$$q = 8300 + 3320 = 11,620 \text{ Btu/hr}$$

$$\Delta t = \frac{11,620}{2040} = 5.70^\circ\text{F}$$

$$t_{0.25} = 112.5 - 5.7 = 106.8^\circ\text{F}$$

$$X'_{106.8^\circ\text{F}} = 0.0533 \text{ lb/lb}$$

$$X_{106.8^\circ\text{F}} = \frac{116.63}{1500} = 0.0775 \text{ lb/lb}$$

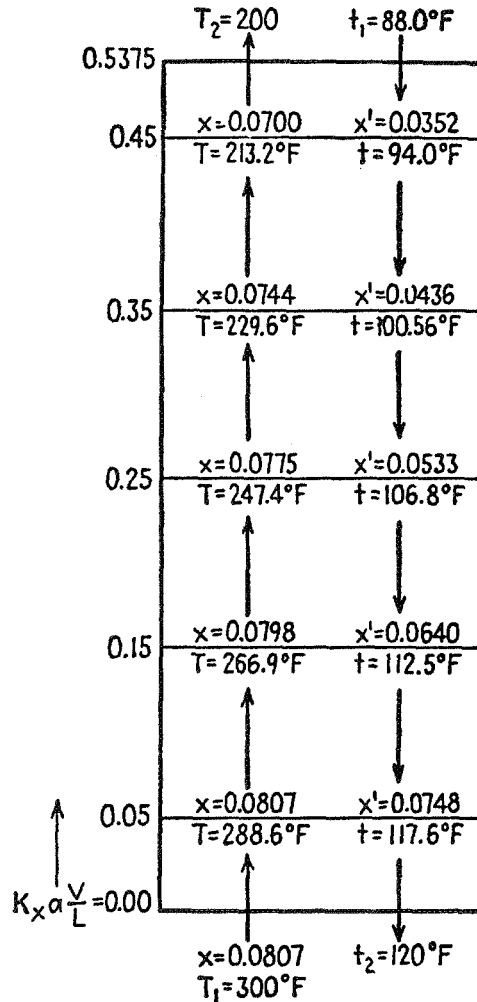


FIG. 17.17. Solution of Example 17.6.

The calculations of the remaining intervals until a gas temperature of 200°F is reached are shown in Fig. 17.17 from the summary table below.

Interval	$K_{xa} \frac{V}{L}$	T	t	H ₂ O diffused, lb	q_c	q_d
	0	300	120			
1	0.05	288.6	117.6	0	4,850	0
2	0.15	266.9	112.5	1.20	9,200	1,235
3	0.25	247.4	106.8	3.22	8,300	3,320
4	0.35	229.6	100.6	4.99	7,560	5,150
5	0.45	213.2	94.0	6.29	6,950	6,520
6	0.538	200.0	88.0	6.22	5,640	6,460
	$n_d = 0.538$			21.92	42,500	22,685
$q = 65,185 \text{ Btu/hr}$						

Assumed diffusion = 20 per cent

Calculated diffusion = $21.92 \times \frac{100}{121.05} = 18.0$ per cent

It is seen that the above summary is somewhat off, since the water temperature ended at 88°F whereas the inlet temperature was specified to be at 85°F. If an answer of greater accuracy is desired, the temperature and humidity are found to check when the diffusion is assumed to be 18½ per cent and 0.55 diffusion units are obtained. Plots can be made to show the variation of T and t with height.

(b) The following smoothed data are given for an experimental tower with vertical sheets of composition board:

Tower cross section: 41⅝ by 23⅞ in.

Filled height: 41⅝ in.

Sheet thickness: ⅛ in.

Horizontal spacing: ⅝-in. centers

Number of sprays: 18

L , lb water/(hr)(ft ²)	G , lb air/(hr)(ft ²)	K_{xa}	ΔP , in. H ₂ O
882	700	190	0.040
	1100	258	0.083
	1500	312	0.136
1178	700	200	0.049
	1100	290	0.095
	1500	373	0.150
1473	700	206	0.060
	1100	315	0.106
	1500	420	0.162

For $G = 1500$, extrapolate to $L = 2040$ on logarithmic coordinates. $K_{xa} = 510$.

$$\text{Tower height, } Z = \frac{n_d L}{K_{xa}} = 0.54 \times \frac{2040}{510} = 2.16 \text{ ft}$$

$$\text{Cross section} = \frac{50,000}{1500} = 33.3 \text{ ft}^2$$

Note the small height. Even with an adequate factor for safety little height will be required to cool the gas.

The extrapolated pressure drop is 0.175 in. H_2O per 41 $\frac{3}{8}$ in. of height.

SENSIBLE-HEAT TRANSFER

Heating and Cooling without Mass Transfer. When a hot gas contacts a nonvolatile cooling medium, the mass transfer is very small. In addition, with the exception of water, methanol, and ammonia, the latent heat of vaporization is also small, from 100 to 200 Btu/lb, so that the diffusion heat load is insignificant. In such case it is possible to neglect the mass transfer altogether and simplify the method of calculation for gas coolers.

Most performance data on packed towers have been obtained for mass transfer, but convection alone can be computed from mass-transfer data such as K_{xa} or HDU by the simple relationship $haV/L = (Le)C(K_{xa}V/L)$. The enthalpy of the gas where there is no mass transfer is proportional to its temperature, and the potential for convection is the difference between the temperature of the gas and the temperature of the water at any cross section in the tower. Accordingly, $haV/L = \int dt/(T - t)$, and since only sensible-heat changes occur,

$$\frac{haV}{L} = \int \frac{dt}{\text{LMTD}} \quad (17.88)$$

For a given system in a given tower if HDU is known, Eq. (17.88) can in turn be converted to the number of diffusion units by

$$n_d = (Le) \int \frac{dt}{\text{LMTD}} = \frac{haV}{L(Le)C} \quad (17.89)$$

There is a dearth of information on the performance of different packings and fills during the saturation of gases with oils and other organics which are suitable as nonvolatile heat-transfer mediums for hot gases. This is also true for the Lewis numbers for such systems. Theoretical approximations can be obtained, but these may lead to considerable error.

In a cooling tower the convection-heat transfer represents only about 20 per cent of the total heat transfer. In the gas cooler of Example 17.6 convection accounted for 65 per cent of the total heat load. If the gas had had an inlet temperature of 1000°F and a dew point of 120°F, convection would have accounted for 95 per cent of the heat load. If a gas

with a dew point of 120 to 130°F or less is to be cooled by water without a close approach, diffusion represents a small portion of the heat load and the entire problem can be treated as one of sensible-heat transfer. The dew points referred to above are within the range of flue gases. Some allowance must be made in the heat load and liquid loading, although the height of the tower will not be affected, since both diffusion and convection will occur in the same height. The use of this short method is demonstrated below.

Example 17.7. Approximate Calculation of a Gas Cooler. 50,000 lb/hr of the gas in Example 17.6 (dew point = 120°F) is to be cooled from an initial temperature of 500 to 200°F. How many diffusion units of any type of tower are required?

Solution. Assume that C for the mixture is 0.28 Btu/(lb)(°F).

$$\text{Sensible-heat load} = 50,000 \times 0.28(500 - 200) = 4,200,000$$

$$\text{From Example 17.6 approximate diffusion} = \frac{50,000}{1,500} \times 22,685 = 758,000$$

4,958,000 Btu/hr

Actually, an allowance as high as 30 per cent of the sensible load can be made and the excess water compensated for by throttling when the tower is in operation.

$$\text{Total water quantity} = \frac{4,958,000}{120 - 85} = 142,000 \text{ lb/hr}$$

If the maximum liquid loading is taken as 2040 lb/(hr)(ft²), the required tower cross section will be

$$\frac{142,000}{2040} = 69.7 \text{ ft}^2$$

The new gas rate will be

$$\frac{50,000}{69.7} = 718 \text{ lb/(hr)(ft}^2\text{)}$$

Calculation of the number of diffusion units:

$$\frac{haV}{L} = \int \frac{dt}{\text{LMTD}} \quad (17.88)$$

The two terminal temperature differences are (200 - 85) and (500 - 120).

$$\text{LMTD} = \frac{(500 - 120) - (200 - 85)}{\ln (500 - 120)/(200 - 85)} = 222^\circ\text{F}$$

$$dt = 120 - 85 = 35^\circ\text{F}$$

$$\frac{haV}{L} = \frac{35}{222} = 0.16$$

$$n_d = \frac{haV}{L(Le)C}$$

$$= \frac{0.16}{0.93 \times 0.28} = 0.62 \text{ diffusion units}$$

For a tower employing fill as in Example 17.6:

By extrapolation for $G = 718$ and $L = 2040$, $K_{xa} = 215$

$$n_a = K_{xa} \frac{V}{L}$$

$$0.62 = 215 \times \frac{V}{2040} = 215 \times \frac{1Z}{2040}$$

$$Z = 6.8 \text{ ft high}$$

$$\text{Ground dimensions} = (69.7)^{1/2} = 8\frac{1}{3} \times 8\frac{1}{3} \text{ ft}$$

As before, the pressure drop will be negligible.

PROBLEMS

17.1. A cooling tower 30 by 30 ft (Example 15.2) has been designed to cool 1800 gpm of water from 110 to 85°F when the 5 per cent wet bulb is 75°F. The fans are capable of delivering 275,000 cfm of air. How many diffusion units are required? If the 5 per cent wet bulb was 80°F, how many diffusion units would be required?

17.2. A cooling tower 30 by 24 ft was designed to deliver 1200 gpm of water from 105 to 85°F when the 5 per cent wet bulb was 80°F. Fans are capable of delivering 230,000 cfm of air. (a) In an actual test at full loadings when the wet bulb was 70°F the water range was from 77.0 to 97.0°F. Was the tower fulfilling guarantee conditions? (b) If the range was 78.0 to 98.0°F, would the tower fulfill guarantee conditions?

17.3. In a Texas lube plant it is desired to install a water-cooled heat exchanger for a liquid with a waxy residue. A small portion of the cooling-tower water from a single cell is available, but there is some question that in winter the entering water will chill out wax and disrupt operation.

The tower operates in summer at an 80°F wet bulb with water on at 120°F and off at 85°F. Because of the main equipment on the water line the air and water quantities to the tower are fixed throughout the year at an $L/G = 0.86$. Similarly the load is also fixed.

As a winter extreme it is desired to know the temperature to the exchanger and on the tower when the wet bulb is 60°F.

17.4. A 6'0" ID tower 30 ft high packed with 10'0" of 3-in. coke was used to provide 40 gpm of water cooled from 120 to 80°F when contacted with 3600 cfm of air having a dry bulb of 85°F and a 75°F wet bulb.

To approximately what temperature could the same tower cool 6800 cfm of flue gas from an economizer if packed to a height of 20'0" with coke? In a previous test at a gas load of 1020 lb/(hr)(ft²) a diffusion coefficient of 555 lb/(hr)(ft³)(lb/lb) was obtained.

17.5. 100,000 lb/hr of nitrogen enters a direct-contact cooler at 350°F and a dew point of 130°F and is cooled to 200°F by water from 85 to 120°F. Assume that a maximum gas loading of 1400 lb/(hr)(ft²) and a maximum liquid loading of 2000 lb/(hr)(ft²) are permissible. What diameter of round tower is required? How many diffusion units are required?

NOMENCLATURE FOR CHAPTER 17

A	Heat-transfer surface, ft ²
a	Surface of packing or fill, ft ² /ft ³
C_H	Henry's law constant, (atm)(ft ³)/mol
C	Specific heat of hot liquid or humid heat of hot gas, Btu/(lb)(°F)
C_1	A constant

c	Specific heat of cold liquid or humid heat of cold gas, Btu/(lb)(°F)
$[c]$	Concentration in liquid, lb-mols/ft ³
D	Inside diameter of tube, ft.
g	Acceleration of gravity, ft/hr ²
G_m	Gas rate, lb-mols/(hr)(ft ² of ground area)
G	Gas rate, lb/(hr)(ft ² of ground area)
H, H'	Enthalpy of the gas, saturation enthalpy of the gas, Btu/lb dry air
HTU	Height of a transfer unit, ft
HDU	Height of a diffusion unit, ft
h	Overall coefficient of heat transfer h_G or h_L in a system in which one film actually controls, Btu/(hr)(ft ²)(°F)
h_G, h_L	Coefficient of heat transfer from the gas to the gas-liquid interface, coefficient from the liquid to the gas-liquid interface, Btu/(hr)(ft ²)(°F)
h_i, h_o	Coefficient of heat transfer referred to the inside diameter and the outside diameter of the tube, respectively, Btu/(hr)(ft ²)(°F)
j_d, j_h	Factor for diffusion, factor for heat transfer, respectively, dimensionless
K_G	Overall coefficient of mass transfer, lb-mols/(hr)(ft ²)(atm)
K_L	Overall coefficient of mass transfer, lb-mols/(hr)(ft ²)(unit concentration)
K_X	Overall coefficient of mass transfer, lb/(hr)(ft ²)(lb/lb)
k	Thermal conductivity, Btu/(hr)(ft ²)(°F/ft)
k_d	Diffusivity, ft ² /hr
k_G	Gas-film coefficient, lb-mol/(hr)(ft ²)(atm)
k_L	Liquid-film coefficient, lb-mol/(hr)(ft ²)(unit concentration)
k_X	Gas-film coefficient, lb/(hr)(ft ²)(lb/lb)
L	Liquid loading, lb/(hr)(ft ² of ground area)
$LMTD$	Logarithmic mean temperature difference, °F
L_o	Makeup water, lb/(hr)(ft ² of ground area)
Le	Lewis number, h_G/k_{Xc} or h/K_{Xc} , dimensionless
l	Film thickness, ft
M	Molecular weight, dimensionless
M_w, M_a	Molecular weight of water and air, respectively, dimensionless
N_A	Gas rate of diffusion, lb-mol/hr
n	Number of mols
n_d	Number of diffusion units, dimensionless
n_t	Number of transfer units, dimensionless
p	Pressure of diffusing component in the gas body, atm
p'	Equilibrium pressure of the diffusing component, atm
p_{Bm}	Log mean pressure of inert gas, atm
p_t	Total pressure, atm or psi
p_w	Partial pressure of water, consistent with units of p_t
Q	Total heat transfer, Btu/hr
q	Heat transfer based on 1 ft ² of ground area, Btu/(hr)(ft ² ground area)
q_c, q_d	Heat transfer by convection and diffusion, respectively, based on 1 ft ² of ground area, Btu/(hr)(ft ² of ground area)
R	Gas constant, 1544 ft-lb/Btu
S	Entropy, Btu/°F
T	Temperature of the hot fluid, °F
T_o	Makeup water, temperature °F
t	Temperature of the cold fluid, °F

t_{DB}, t_{WB}, t_{DP}	Dry-bulb, wet-bulb, and dew-point temperatures, respectively, °F
U_C	Clean overall coefficient of heat transfer, Btu/(hr)(ft ²)(°F)
u	Velocity, ft/hr
V	Tower volume, ft ³
v	Specific volume, ft ³ /lb
X, X'	Humidity of the gas, saturation humidity of the gas, lb/lb
x	Mol fraction of diffusing component in liquid phase
y	Mol fraction of diffusing component in gas phase
Z	Height of tower, ft
z	Height of free fall, ft
α_{AB}	Proportionality constant, hr/ft ²
γ	Exponent in Eq. (17.85)
δ	Molar density, lb-mols/ft ³
θ	Time, hr
λ	Average latent heat of vaporization, Btu/lb
λ_{DP}	Latent heat of vaporization at the dew point, Btu/lb
ρ	Density, lb/ft ³

Subscripts (except as noted above)

A	Diffusing component
B	Inert gas
i	Gas-liquid interface
1	Inlet
2	Outlet

CHAPTER 18

BATCH AND UNSTEADY-STATE PROCESSES

Introduction. The relationships of preceding chapters have appned only to the steady state in which the heat flow and source-temperature were constant with time. *Unsteady-state* processes are those in which the heat flow, the temperature, or both vary with time at a fixed point. Batch heat-transfer processes are typical unsteady-state processes in which discontinuous heat changes occur with specific quantities of material as when heating a given quantity of liquid in a tank or when a cold furnace is started up. Still other common problems involve the rates at which heat is conducted through a material while the temperature of the heat source varies. The daily periodic variations of the heat of the sun on various objects or the quenching of steel in an oil bath are examples of the latter. Other apparatuses based on the characteristics of the unsteady state are the regenerative furnaces used in the steel industry, the pebble heater, and equipment in processes employing fixed and moving-bed catalysts.

In batch processes for the heating of liquids the time requirement for heat transfer can usually be modified by increasing the circulation of the batch fluid, the heat transfer medium, or both. The reasons for using a batch rather than a continuous heat-transfer operation are dictated by numerous factors. Some of the common reasons are (1) the liquid being processed is not continuously available, (2) the heating or cooling medium is not continuously available, (3) the requirements of reaction time or treating time necessitates holdup, (4) the economics of intermittently processing a large batch justifies the accumulation of a small continuous stream, (5) cleaning or regeneration is a significant part of the total operating period, and (6) the simplified operation of most batch processes is advantageous.

In order to treat the commonest applications of batch and unsteady-state heat transfer systematically, it is preferable to divide processes between liquid (fluid) heating or cooling and solid heating or cooling. The commonest examples are outlined below.

1. Heating and cooling liquids
 - a. Liquid batches
 - b. Batch distillation

2. Heating and cooling solids
 - a. Constant medium temperature
 - b. Periodically varying temperature
 - c. Regenerators
 - d. Granular material in beds

HEATING AND COOLING LIQUIDS

1a. LIQUID BATCHES

Introduction. Bowman, Mueller, and Nagle¹ have derived an expression for the time required to heat an agitated batch by the immersion of a heating coil when the temperature difference is the LMTD for counterflow. Fisher² has extended the batch calculation to include an external counterflow exchanger. Chaddock and Sanders³ have treated agitated batches heated by external counterflow exchangers with the continuous addition of liquid to the tank and have also taken into account the heat of solution. Some of the derivations which follow apply to coils in tanks and jacketed vessels, although the method of obtaining overall coefficients of heat transfer for these elements is deferred until Chap. 20.

It is not always possible to distinguish between the presence or absence of agitation in a liquid batch, although the two premises lead to different requirements for the accomplishment of a batch temperature change in a given period of time. When a mechanical agitator is installed in a tank or vessel as in Fig. 18.1, there is no need to question that the tank fluid is agitated. When there is no mechanical agitator but the liquid is continuously recirculated, the conclusion that the batch is agitated is one of discretion. Where the heating element is an external exchanger, it is safer to assume agitation.

In the derivation of batch equations given below, T refers to the hot batch liquid or heating medium and t refers to the cold batch liquid or cooling medium. The following cases are treated here:

Heating and cooling agitated batches, counterflow

Coil-in-tank or jacketed vessel, isothermal medium
Coil-in-tank or jacketed vessel, nonisothermal medium

External exchanger, isothermal medium
External exchanger, nonisothermal medium

External exchanger, liquid continuously added to tank, isothermal medium
External exchanger, liquid continuously added to tank, nonisothermal medium

¹ Bowman, R. A., A. C. Mueller, and W. M. Nagle, *Trans. ASME*, **62**, 283-294 (1940).

² Fisher, R. C., *Ind. Eng. Chem.*, **36**, 939-942 (1944).

³ Chaddock, R. E., and M. T. Sanders, *Trans. AIChE*, **40**, 203-210 (1944).

Heating and cooling agitated batches, parallel flow-counterflow

External 1-2 exchanger

External 1-2 exchanger, liquid continuously added to tank

External 2-4 exchanger

External 2-4 exchanger, liquid continuously added to tank

Heating and cooling batches without agitation

External counterflow exchanger, isothermal medium

External counterflow exchanger, nonisothermal medium

External 1-2 exchanger

External 2-4 exchanger

Heating and Cooling Agitated Batches. There are several ways of considering batch heat-transfer processes. If it is desired to accomplish a certain operation in a given time, the surface requirement is usually unknown. If the heat-transfer surface is known as in an existing installation, the time required to accomplish the operation is usually unknown. A third possibility arises when the time and surface are known but the temperature at the conclusion of the time is unknown. The following assumptions are involved in the derivations

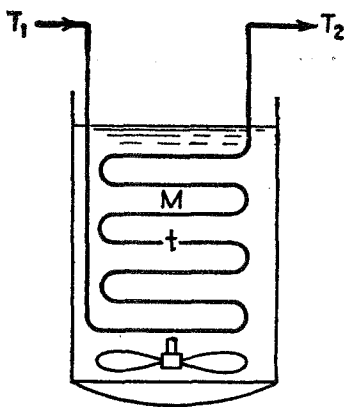


FIG. 18.1. Agitated batch. of Eqs. (18.1) through (18.23):

1. U is constant for the process and over the entire surface.
2. Liquid flow rates are constant.
3. Specific heats are constant for the process.
4. The heating or cooling medium has a constant inlet temperature.
5. Agitation produces a uniform batch fluid temperature.
6. No partial phase changes occur.
7. Heat losses are negligible.

Heating and Cooling Agitated Batches: Counterflow

COIL-IN-TANK OR JACKETED VESSEL, ISOTHERMAL HEATING MEDIUM. Consider the arrangement shown in Fig. 18.1 consisting of an agitated vessel containing M lb of liquid with specific heat c and initial temperature t_1 heated by a condensing medium of temperature T_1 . The batch temperature t_2 at any time θ is given by the differential heat balance. If Q' is the total number of Btu transferred, then per unit of time

$$dQ = \frac{dQ'}{d\theta} = Mc \frac{dt}{d\theta} = UA \Delta t \quad (18.1)$$

$$\Delta t = (T_1 - t) \quad (18.2)$$

$$\frac{dt}{d\theta} = \frac{UA}{Mc} \Delta t \quad (18.3)$$

$$\int \frac{dt}{T_1 - t} = \frac{UA}{Mc} \int d\theta \quad (18.4)$$

Integrating from t_1 to t_2 while the time passes from 0 to θ ,

$$\ln \frac{T_1 - t_1}{T_1 - t_2} = \frac{UA\theta}{Mc} \quad (18.5)$$

The use of an equation such as Eq. (18.5) requires the independent calculation of U for the coil or jacketed vessel as given in Chap. 20. With Q and A fixed by process conditions, the required heating time can be calculated.

COIL-IN-TANK OR JACKETED VESSEL, ISOTHERMAL COOLING MEDIUM. Problems of this type usually arise in low-temperature processes in which the cooling medium is a refrigerant which is fed to the cooling element at its isothermal boiling temperature. Consider the same arrangement as shown in Fig. 18.1 containing M lb of liquid with specific heat C and initial temperature T_1 cooled by a vaporizing medium of temperature t_1 . If T is the batch temperature at any time θ ,

$$\frac{dQ'}{d\theta} = -MC \frac{dT}{d\theta} = UA \Delta t \quad (18.6)$$

$$\Delta t = (T - t_1)$$

$$\ln \frac{T_1 - t_1}{T_2 - t_1} = \frac{UA\theta}{MC} \quad (18.7)$$

COIL-IN-TANK OR JACKETED VESSEL, NONISOTHERMAL HEATING MEDIUM. The nonisothermal heating medium has a constant flow rate W and inlet temperature T_1 but a variable outlet temperature

$$\begin{array}{ccc} (a) & (b) & (c) \\ \frac{dQ'}{d\theta} = MC \frac{dt}{d\theta} & = WC(T_1 - T_2) & = UA \Delta t \end{array} \quad (18.8)$$

$$\Delta t = \text{LMTD} = \frac{T_1 - T_2}{\ln (T_1 - t)/(T_2 - t)}$$

$$T_2 = t + \frac{T_1 - t}{e^{UA/WC}}$$

Let $K_1 = e^{UA/WC}$ and equate (a) and (b) of Eq. (18.8).

$$\begin{array}{l} Mc \frac{dt}{d\theta} = WC \left(\frac{K_1 - 1}{K_1} \right) (T_1 - t) \\ \ln \frac{T_1 - t_1}{T_1 - t_2} = \frac{WC}{Mc} \left(\frac{K_1 - 1}{K_1} \right) \theta \end{array} \quad (18.9)$$

COIL-IN-TANK, NONISOTHERMAL COOLING MEDIUM:

$$\frac{dQ'}{d\theta} = -MC \frac{dT}{d\theta} = wc(t_2 - t_1) = UA \Delta t \quad (18.10)$$

$$K_2 = e^{UA/wc}$$

$$\ln \frac{T_1 - t_1}{T_2 - t_1} = \frac{wc}{MC} \left(\frac{K_2 - 1}{K_2} \right) \theta \quad (18.11)$$

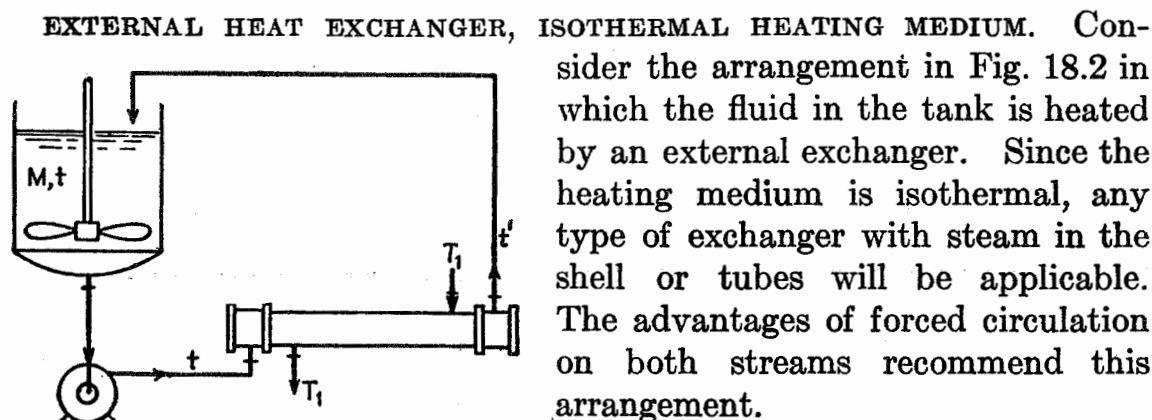


FIG. 18.2. Agitated batch with external 1-1 exchanger.

The variable temperature out of the exchanger t' will differ from the variable tank temperature t , and the differential heat balance is given by

$$\begin{matrix} (a) & (b) & (c) \\ \frac{dQ'}{d\theta} = Mc \frac{dt}{d\theta} = wc(t' - t) = UA \Delta t \end{matrix} \quad (18.12)$$

$$t' = T_1 - \frac{T_1 - t}{e^{UA/wc}}$$

Let

$$K_2 = e^{UA/wc}$$

$$\ln \frac{T_1 - t_1}{T_1 - t_2} = \frac{wc}{Mc} \left(\frac{K_2 - 1}{K_2} \right) \theta \quad (18.13)$$

EXTERNAL EXCHANGER, ISOTHERMAL COOLING MEDIUM:

$$\ln \frac{T_1 - t_1}{T_2 - t_1} = \frac{WC}{MC} \left(\frac{K_1 - 1}{K_1} \right) \theta \quad (18.14)$$

EXTERNAL EXCHANGER, NONISOTHERMAL HEATING MEDIUM. The differential heat balance is given by

$$\begin{matrix} (a) & (b) & (c) & (d) \\ \frac{dQ'}{d\theta} = Mc \frac{dt}{d\theta} = wc(t' - t) = WC(T_1 - T_2) = UA \Delta t \end{matrix} \quad (18.15)$$

There are two variable temperatures, t' and T_2 , which appear in the LMTD and which must first be eliminated. Equating Eq. (18.15), (a) and (b),

$$t' = t + \frac{Mc}{wc} \frac{dt}{d\theta}$$

Equating Eq. (18.15), (a) and (c),

$$T_2 = T_1 - \frac{Mc}{WC} \frac{dt}{d\theta}$$

Let

$$K_3 = e^{UA \left(\frac{1}{wc} - \frac{1}{WC} \right)}$$

and

$$\ln \frac{T_1 - t_1}{T_1 - t_2} = \frac{K_3 - 1}{M} \frac{wWC}{(K_3wc - WC)} \theta \quad (18.16)$$

EXTERNAL EXCHANGER, NONISOTHERMAL COOLING MEDIUM:

$$K_4 = e^{UA \left(\frac{1}{WC} - \frac{1}{wc} \right)}$$

$$\ln \frac{T_1 - t_1}{T_2 - t_1} = \frac{(K_4 - 1)}{M} \frac{Wwc}{(K_3WC - wc)} \theta \quad (18.17)$$

EXTERNAL EXCHANGER, LIQUID CONTINUOUSLY ADDED TO TANK, ISOTHERMAL HEATING MEDIUM. The elements of the process are shown in Fig. 18.3. Liquid is continuously added to the tank at the rate of L_0 lb/hr and at the constant temperature t_0 . It is assumed that no chemical heat effects attend the addition of the make-up liquid. Since M is the pounds of liquid originally in the batch and L_0 is in pounds per hour, the total liquid at any time is $M + L_0\theta$. The differential heat balance is given by

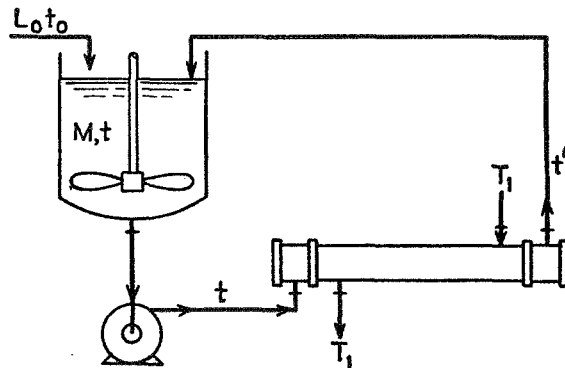


FIG. 18.3. Agitated batch with external 1-1 exchanger, liquid continuously added.

$$(M + L_0\theta) \frac{dt}{d\theta} + L_0c(t - t_0) = wc(t' - t) \quad (18.18)$$

and

$$wc(t' - t) = UA \Delta t$$

where

$$\Delta t = \text{LMTD} = \frac{t' - t}{\ln (T_1 - t)/(T_1 - t')}$$

Solving for t' ,

$$t' = \left(\frac{K_2 - 1}{K_2} \right) T_1 + \frac{t}{K_2}$$

Substituting in Eq. (18.18),

$$\begin{aligned} \int \frac{dt}{w \left(\frac{K_2 - 1}{K_2} \right) (T_1 - t) - L_0(t - t_0)} &= \int \frac{d\theta}{M + L_0\theta} \\ \ln \frac{t_1 - t_0 - \frac{w}{L_0} \left(\frac{K_2 - 1}{K_2} \right) (T_1 - t_1)}{t_2 - t_0 - \frac{w}{L_0} \left(\frac{K_2 - 1}{K_2} \right) (T_1 - t_2)} &= \left[\frac{w}{L_0} \left(\frac{K_2 - 1}{K_2} \right) + 1 \right] \ln \frac{M + L_0\theta}{M} \quad (18.19) \end{aligned}$$

If the addition of liquid to the tank causes an average endothermic or exothermic heat of solution, $\pm q_s$ Btu/lb of make-up, it may be included by adding $\pm q_s/c_0$ to both the numerator and denominator of the left side. The subscript 0 refers to the make-up.

EXTERNAL EXCHANGER, LIQUID CONTINUOUSLY ADDED TO TANK, ISOTHERMAL COOLING MEDIUM:

$$(M + L_0\theta)c \frac{dT}{d\theta} + WC(T - T') = L_0C(T_0 - T) \quad (18.20)$$

$$\begin{aligned} \ln \frac{T_0 - T_1 - \frac{W}{L_0} \left(\frac{K_1 - 1}{K_1} \right) (T_1 - t_1)}{T_0 - T_2 - \frac{W}{L_0} \left(\frac{K_1 - 1}{K_1} \right) (T_2 - t_1)} &= \left[1 - \frac{W}{L_0} \left(\frac{K_1 - 1}{K_1} \right) \right] \ln \frac{M + L_0\theta}{M} \quad (18.21) \end{aligned}$$

The heat-of-solution effects can be included by adding $\pm q_s/C_0$ to both the numerator and denominator of the left side.

EXTERNAL EXCHANGER, LIQUID CONTINUOUSLY ADDED TO TANK, NON-ISOTHERMAL HEATING MEDIUM. The heat balance is identical with Eq. (18.18) for heating except that Δt is written for the inlet and outlet temperatures of the heating medium.

$$\begin{aligned} \Delta t = \text{LMTD} &= \frac{(T_2 - t) - (T_1 - t')}{\ln (T_2 - t)/(T_1 - t')} = \frac{(t' - t) - (T_1 - T_2)}{\ln (T_2 - t)/(T_1 - t')} \\ (T_1 - T_2)WC &= (t' - t)wc = UA \Delta t \end{aligned}$$

Let

$$K_5 = e^{\frac{UA}{wc} \left(1 - \frac{wc}{WC}\right)}$$

$$\ln \frac{t_0 - t_1 + \frac{wWC(K_5 - 1)(T_1 - t_1)}{L_0(K_5WC - wc)}}{t_0 - t_2 + \frac{wWC(K_5 - 1)(T_1 - t_2)}{L_0(K_5WC - wc)}} = \left[\frac{wWC(K_5 - 1)}{L_0(K_5WC - wc)} + 1 \right] \ln \frac{M + L_0\theta}{M} \quad (18.22)$$

The heat-of-solution effects can be included by adding $\pm q_s/c_0$ to both the numerator and denominator of the left side.

EXTERNAL EXCHANGER, LIQUID CONTINUOUSLY ADDED TO TANK, NON-ISOTHERMAL COOLING MEDIUM:

$$K_6 = e^{\frac{UA}{WC} \left(1 - \frac{WC}{wc}\right)}$$

$$\ln \frac{T_0 - T_1 - \frac{Wwc(K_6 - 1)(T_1 - t_1)}{L_0(K_6wc - WC)}}{T_0 - T_2 - \frac{Wwc(K_6 - 1)(T_2 - t_1)}{L_0(K_6wc - WC)}} = \left[\frac{Wwc(K_6 - 1)}{L_0(K_6wc - WC)} + 1 \right] \ln \frac{M + L_0\theta}{M} \quad (18.23)$$

The heat-of-solution effects can be included by adding $\pm q_s/C_0$ to both the numerator and denominator of the left side.

Heating and Cooling Agitated Batches. 1-2 Parallel flow-Counterflow. The derivations for the preceding cases included assumption 7, which required that all external exchangers operate in counterflow. With non-isothermal cooling and heating mediums this will not always be advantageous, since it sacrifices the construction and performance advantages of multipass apparatus such as the 1-2 exchanger. The external 1-2 exchanger can be included by using the temperature difference as defined by Eq. (7.37).

$$\frac{UA}{wc} = \frac{1}{\sqrt{R^2 + 1}} \ln \frac{2 - S(R + 1 - \sqrt{R^2 + 1})}{2 - S(R + 1 + \sqrt{R^2 + 1})} \quad (7.37)$$

where

$$R = \frac{T_1 - T_2}{t' - t} = \frac{wc}{WC} \quad \text{and} \quad S = \frac{t' - t}{T_1 - t}$$

$$\frac{2 - S(R + 1 - \sqrt{R^2 + 1})}{2 - S(R + 1 + \sqrt{R^2 + 1})} = e^{(UA/wc)\sqrt{R^2 + 1}} = K_7$$

$$S = \frac{2(K_7 - 1)}{K_7(R + 1 + \sqrt{R^2 + 1}) - (R + 1 - \sqrt{R^2 + 1})} \quad (18.24)$$

and so S as well as R is a constant which is independent of the exchanger outlet temperatures.

EXTERNAL 1-2 EXCHANGER, HEATING. Using the same heat balance as defined by Eq. (18.15),

$$t' = t + \frac{M}{w} \frac{dt}{d\theta}$$

$$S = \frac{t' - t}{T_1 - t} = \frac{(M/w)(dt/d\theta)}{T_1 - t}$$

Rearranging,

$$\int \frac{dt}{T_1 - t} = \frac{Sw}{M} \int d\theta$$

$$\ln \frac{T_1 - t_1}{T_1 - t_2} = \frac{Sw}{M} \theta \quad (18.25)$$

where S is defined by Eq. (18.24).

EXTERNAL 1-2 EXCHANGER, COOLING:

$$\ln \frac{T_1 - t_1}{T_2 - t_1} = S \frac{wc}{MC} \theta \quad (18.26)$$

where S is defined by Eq. (18.24).

EXTERNAL 1-2 EXCHANGER, LIQUID CONTINUOUSLY ADDED TO TANK. HEATING:

$$(M + L_0\theta)c \frac{dt}{d\theta} = wc(t' - t) - L_0c(t - t_0) \quad (18.27)$$

$$S = \frac{t' - t}{T_1 - t} = \frac{\left(\frac{M}{w} + \frac{L_0\theta}{w}\right) \frac{dt}{d\theta} + \frac{L_0}{w} (t - t_0)}{T_1 - t}$$

Simplifying

$$\frac{dt}{ST_1 + \frac{L_0}{w} t_0 - \left(S + \frac{L_0}{w}\right) t} = \frac{d\theta}{\frac{M}{w} + \frac{L_0\theta}{w}}$$

$$\ln \frac{ST_1 + \frac{L_0}{w} t_0 - \left(S + \frac{L_0}{w}\right) t_1}{ST_1 + \frac{L_0}{w} t_0 - \left(S - \frac{L_0}{w}\right) t_2} = \frac{Sw + L_0}{L_0} \ln \frac{(M + L_0\theta)}{M} \quad (18.28)$$

where S is defined by Eq. (18.24). The heat-of-solution effects can be included by adding $\pm q_s/c_0$ to the numerator, and denominator of the left side.

EXTERNAL 1-2 EXCHANGER, LIQUID CONTINUOUSLY ADDED TO TANK, COOLING:

$$(M + L_0\theta)C \frac{dT}{d\theta} = L_0C(T_0 - T) - WC(T - T') \quad (18.29)$$

$$\begin{aligned} \frac{1}{L_0 + WSR} \ln \frac{L_0T_0 + WSRt_1 - (L_0 + WSR)T_1}{L_0T_0 + WSRt_1 - (L_0 + WSR)T_2} \\ = \frac{1}{L_0} \ln \left(\frac{M + L_0\theta}{M} \right) \end{aligned} \quad (18.30)$$

where S is defined by Eq. (18.24). The heat-of-solution effects can be included by adding $\pm q_s/C_0$ to the numerator and denominator of the left side.

HEATING AND COOLING AGITATED BATCHES, 2-4 PARALLEL FLOW-COUNTERFLOW. Equation (8.5) gives the true temperature relations for the 2-4 exchanger. This can be rearranged in terms of S to give

$$S = \frac{2(K_s - 1)[1 + \sqrt{(1 - S)(1 - RS)}]}{(K_s - 1)(R + 1) + (K_s + 1)\sqrt{R^2 + 1}} \quad (18.31)$$

and

$$K_s = e^{(UA/2wc)\sqrt{R^2 + 1}} \quad (18.32)$$

Since S cannot be expressed simply, Eq. (18.31) can be solved by trial and error, assuming different values of S until an equality is reached. The equations for heating and cooling are the same as those developed for the 1-2 exchanger except that the value of S from Eq. (18.31) replaces the value of S from Eq. (18.24). The heat-of-solution effects can be accounted for in the same manner as for the 1-2 exchanger.

Heating and Cooling without Agitation. It will be seen in Chap. 20 that agitation increases the film coefficient and thereby decreases the time requirement when heating or cooling liquids by means of a coil in tank. With external exchangers the presence of agitation, either intended or unavoidable, much to the contrary increases the time required to heat or cool a batch. This can be appreciated by simple analysis. Referring to Fig. 18.4, the batch with initial temperature t passes through an external exchanger and is returned to the tank where it builds up as a layer of temperature t_1 . Such might be the case if the liquid was relatively viscous or the vessel tall and narrow. All the liquid enters the exchanger at the tank temperature t during the initial circulation and

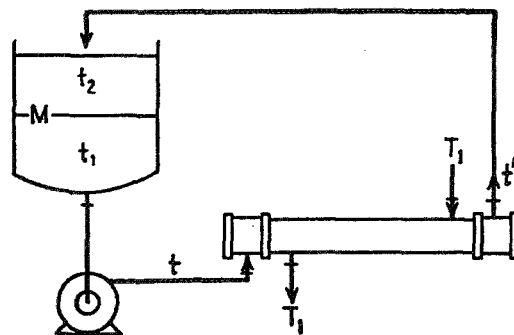


FIG. 18.4. Batch without agitation.

emerges with a temperature t_1 which is the feed temperature to the exchanger on the next recirculation. With agitation, however, the first liquid emerging from the exchanger mixes with the batch liquid and immediately raises its temperature above the initial temperature t . This in turn reduces the temperature difference in the exchanger and increases the time required for a given heat transfer.

Suppose the initial quantity of batch liquid is M lb and that it is circulated through the exchanger at the rate of w lb/hr. Since there is a discrete temperature change with each recirculation, the process is not described by a differential change. If the number of circulations required for the attainment of a final batch temperature is N , the time is given by $\theta = NM/w$.

EXTERNAL COUNTERFLOW EXCHANGER, ISOTHERMAL HEATING MEDIUM:

$$wc(t_1 - t) = UA \frac{(t_1 - t)}{\ln (T_1 - t)/(T_1 - t_1)}$$

For the initial circulation

$$t_1 = T_1 - \frac{1}{K_2} (T_1 - t)$$

For the first recirculation

$$t_2 = T_1 - \frac{1}{K_2} (T_1 - t_1)$$

In terms of T_1 and t

$$t_2 = T_1 - \frac{1}{K_2^2} (T_1 - t)$$

or

$$t_N = T_1 - \frac{1}{K_2^N} (T_1 - t) \quad (18.33)$$

When solved for N circulations,

$$\theta = \frac{NM}{w} \quad (18.34)$$

Provision can be made for the continuous addition of liquid by computing the mix temperature to the exchanger after each circulation. In such an event the size of the batch itself must be treated as increasing with each circulation, so that Eq. (18.34) does not apply unless M is increased by $L_0\theta$ pounds during the recirculation. The total time will be the sum of the individual calculations as above.

EXTERNAL COUNTERFLOW EXCHANGER, ISOTHERMAL COOLING MEDIUM:

$$T_N = t_1 - \frac{1}{K_1^N} (t_1 - T) \quad (18.35)$$

EXTERNAL COUNTERFLOW EXCHANGER, NONISOTHERMAL HEATING MEDIUM. The outlet temperature of the batch and the heating medium after each recirculation is not known. This case does not factor so simply as the preceding with an isothermal medium. Although the answer can be expressed in a series form, it is tedious to evaluate and greater speed can be attained by calculating the temperature changes after each recirculation. The temperature relations after each recirculation can be defined when

$$S = \frac{K_9 - 1}{K_9 R - 1} \quad (18.36)$$

$$K_9 = e^{(UA/wc)(R-1)}$$

Initial circulation

$$t_1 = t + S(T_1 - t) \quad (18.37)$$

Recirculation

$$t_2 = t_1 + S(T_1 - t_1)$$

Solve for each circulation by introducing the temperature from the preceding circulation.

EXTERNAL COUNTERFLOW EXCHANGER, NONISOTHERMAL COOLING MEDIUM. After each recirculation,

$$T_1 = T - S(T - t_1) \quad (18.38)$$

EXTERNAL 1-2 EXCHANGER, HEATING AND COOLING. This case can be calculated in the same manner as the preceding but using S as defined by Eq. (18.24). An even greater simplification can be obtained by using the Ten Broeck chart in Fig. 7.25 and calculating each step separately. The continuous addition of liquid can be accounted for in each step along with its heat of solution.

EXTERNAL 2-4 EXCHANGER, HEATING AND COOLING. This is the same as the preceding except that S is defined by Eq. (18.31) or Fig. 8.7.

Example 18.1. Calculation of Batch Heating. 7500 gal of liquid benzene under pressure at 300°F is required for a batch extraction process. The storage temperature of the benzene is 100°F. Available as a heating medium is a 10,000 lb/hr 28°API oil stream at a temperature of 400°F. A pump connected to the tank is capable of circulating 40,000 lb/hr of benzene. Available for the service is 400 ft² of clean double pipe exchanger surface which, in counterflow streams, yields a U_C of 50 calculated for the flow rates above.

(a) How long will it take to heat the agitated batch using the double pipe counterflow assembly?

(b) How long will it take using a 1-2 exchanger with the same surface and coefficient?

(c) How long will it take using a 2-4 exchanger with the same surface and coefficient?

(d) How long will it take in (a) if the batch vessel is very tall and the batch is not considered to be agitated?

Solution:

(a) This corresponds to Eq. (18.16).

Specific gravity of benzene = 0.88

Specific heat of benzene = 0.48 Btu/(lb)(°F)

$M = 7500 \times 8.33 \times 0.88 = 55,000$ lb

$wc = 40,000 \times 0.48 = 19,200$ Btu/(hr)(°F)

$WC = 10,000 \times 0.60 = 6000$ Btu/(hr)(°F)

$$K_3 = e^{UA \left(\frac{1}{wc} - \frac{1}{WC} \right)}$$

$$= e^{50 \times 400 \left(\frac{1}{19,200} - \frac{1}{6000} \right)} = 0.101$$

Substitute in Eq. (18.16).

$$\ln \frac{400 - 100}{400 - 300} = \frac{(0.101 - 1)}{55,000} \times \frac{40,000 \times 6000}{(0.101 \times 6000 - 19,200)} \theta$$

$$\theta = 5.18 \text{ hr}$$

(b) This case corresponds to Eq. (18.25) in which S is defined by Eq. (18.24) and θ by Eq. (18.25).

$$R = \frac{wc}{WC} = \frac{19,200}{6000} = 3.20$$

$$K_7 = e^{UA/wc} \sqrt{R^2 + 1}$$

$$= e^{\frac{50 \times 400}{19,200} \sqrt{3.20^2 + 1}} = 33.1$$

$$S = \frac{2(33.1 - 1)}{33.1(1 + 3.20 + \sqrt{3.20^2 + 1}) - (1 + 3.20 - \sqrt{3.20^2 + 1})} = 0.266$$

$$\ln \frac{400 - 100}{400 - 300} = \frac{0.266 \times 40,000}{55,000} \theta$$

$$\theta = 5.63 \text{ hr}$$

(c) Use S from Eq. (18.31).

$$K_8 = e^{UA/2wc} \sqrt{R^2 + 1}$$

$$= e^{\frac{100 \times 200}{2 \times 19,200} \sqrt{3.20^2 + 1}} = 5.75$$

Solve Eq. (18.31) by trial and error:

$$S = \frac{2(5.75 - 1)[1 + \sqrt{(1 - S)(1 - 3.20S)}]}{(5.75 - 1)(3.2 + 1) + (5.75 + 1) \sqrt{3.20^2 + 1}}$$

$$= 0.282$$

$$\ln \frac{400 - 100}{400 - 300} = \frac{0.282 \times 40,000}{55,000} \times \theta$$

$$\theta = 5.33 \text{ hr}$$

(d) Use Eq. (18.37) and S from Eq. (18.36).

$$K_9 = e^{UA/wc(R-1)}$$

$$= e^{\frac{50 \times 400}{19,200} (3.20 - 1)} = 9.87$$

$$S = \frac{K_9 - 1}{K_9 R - 1} = \frac{9.87 - 1}{9.87 \times 3.20 - 1} = 0.290$$

$$\begin{aligned}
 t_1 &= t + S(T_1 - t) = 100 + 0.29(400 - 100) = 187^\circ\text{F} \\
 t_2 &= t_1 + S(T_1 - t_1) = 187 + 0.29(400 - 187) = 249^\circ\text{F} \\
 t_3 &= t_2 + S(T_1 - t_2) = 249 + 0.29(400 - 249) = 293^\circ\text{F} \\
 t_4 &= t_3 + S(T_1 - t_3) = 293 + 0.29(400 - 293) = 324^\circ\text{F}
 \end{aligned}$$

Actually, a fractional number of circulations are required. If the problem may be treated from the standpoint of total heat input to the batch, let

x = fractional circulation

$$293(1 - x) + 324 = 300$$

$$x = 0.23$$

$$\text{Total circulations} = 3 + 0.23 = 3.23$$

$$\theta = 3.23 \times \frac{55,000}{40,000} = 4.44 \text{ hr}$$

This value compares with 5.18 hr for the agitated batch.

1b. BATCH DISTILLATION

Introduction. Typical arrangements for batch distillation are shown in Figs. 18.5 and 18.6. The still pot is charged with a batch of liquid,

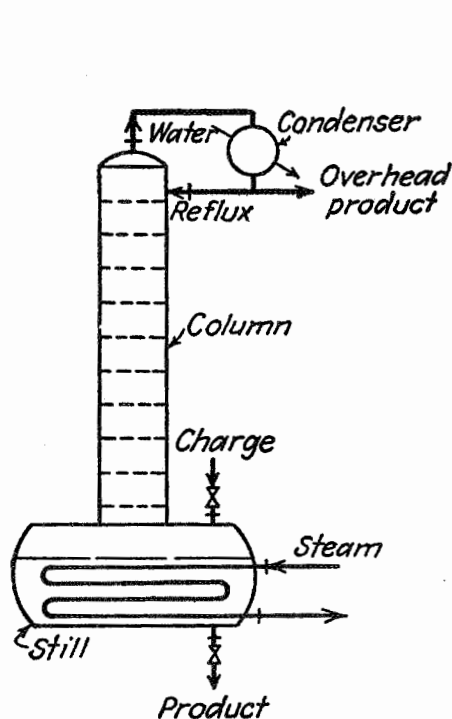


FIG. 18.5. Coil-heated batch still.

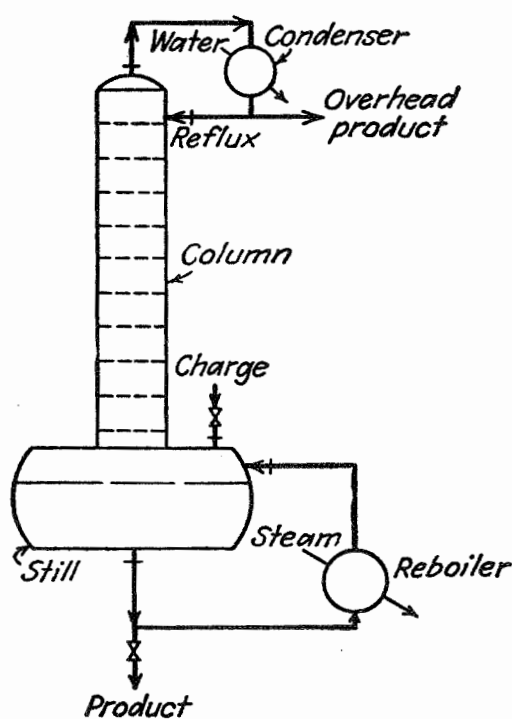


FIG. 18.6. Batch still with external reboiler.

and heat is supplied by a coil or a natural- or forced-circulation reboiler. In some high-temperature installations the still pot may be directly fired. Batch distillation is usually employed when there is an insufficiency of charge stock to warrant continuous operation and the assemblies are often relatively small. In batch distillation the composition and temperature of the residual liquid in the still constantly vary, and usually the same

applies to the condensate except when the still has been charged with or forms a constant-boiling mixture. In batch distillation it is possible to obtain an initial overhead fraction which is purer than that obtainable with the same reflux by continuous distillation. This is particularly true when the overhead product is sold in different grades with a premium for purity. It is also possible by constantly varying the reflux ratio to obtain a nearly-uniform overhead composition although its quantity constantly decreases. The latter is usually too costly to be general.

The composition change during the batch distillation of a binary mixture is given by the Rayleigh equation.

$$\ln \frac{L_1}{L_2} = \int_{x_2}^{x_1} \frac{dx}{y - x} \quad (18.39)$$

where L_1 = mols liquid charge to still

L_2 = mols residue after distillation

x_1 = mol fraction of volatile component in charge liquid

x_2 = mol fraction of volatile component in residue

y = mol fraction of vapor in equilibrium with x

The temperature must be obtained from a boiling-point curve if the mixture is not ideal and does not follow Raoult's and Henry's laws.

The Rayleigh equation does not contain any term with a unit of time. The time allotted to the distillation is therefore independent of any feed quantity. If a batch represents the intermittent accumulation of several hours' flow of charge stock, the rate of distillation must be such that the still pot will be vacated and ready to receive the next charge. If the distillation occurs infrequently, the rate of distillation can be determined economically from the optimum relationship between fixed and operating charges. In batch distillation the cost of labor is particularly large and favors a rapid distillation. The cost of the equipment, on the other hand, favors slower rates of distillation.

The Reboiler and Condenser. The design conditions for both the reboiler and the condenser are usually based upon the limiting conditions of operation. Very often batch distillation is controlled automatically by a program or elapsed-time controller (see Fig. 21.28) such that the heating medium is supplied at a rate producing a steady increase in the boiling temperature. If a heating medium such as steam were supplied to a batch reboiler at a fixed rate, much of it would not be condensed in the reboiler following the initial period of rapid vaporization whence the residue is heated sensibly. The charge stock is a mixture having volatile components which leave the residue at a decreasing rate as the distillation progresses. The boiling temperature of the residue consequently rises as the volatile components are depleted. As the sensible

load in the still pot increases, the weighted heat transfer coefficient to the residue decreases. Suppose steam at 300°F is used to vaporize a charge having an initial boiling point of 200°F and the distillation is to be cut off at a residue composition corresponding to a boiling point of 250°F. A distillation curve can be prepared using the methods of Chaps. 13 and 15. Instantaneous coefficients can be computed for both the start and the finish from adjacent intervals of heat input. At the start when the film coefficient and heat load are high, the temperature difference is

$$300 - 200 = 100^{\circ}\text{F}$$

but at the cutoff temperature, where the film coefficient and heat load are low, the temperature difference is only $300 - 250 = 50^{\circ}\text{F}$. Both conditions must be tested for U and Δt to determine which one requires the greater surface.

The factors affecting the condenser are different. The water rate is usually kept constant. At the start of distillation the temperature of the overhead vapor may be near 200°F and the water, with a range from 85 to 120°F, provides an LMTD of 96.2°F. At the cutoff the overhead may be near 250°F, corresponding to a temperature difference of 150°F or greater, and the heat load and water range will be smaller. The condensing coefficient will change little over the entire run. The limiting condition for the condenser therefore is at the start of distillation, where the maximum heat load and minimum temperature difference usually occur together.

A common practice for obtaining the heat load for the reboiler and condenser without resorting to a distillation curve is to take the total heat load and divide it by the time allotted to the distillation. This will give a fictitious average hourly heat load which is greater than that at the cutoff but usually less than the initial heat load. The value of Q so obtained is combined with the value of U and Δt occurring at both the start and the cutoff temperatures, and the larger calculated surface plus some allowance for the error is used. If the heating element is also used to preheat the charge, it is entirely likely that the preheat rate may be limiting. The time for preheating can be obtained from one of the batch heating equations in the preceding section.

HEATING AND COOLING SOLIDS

2a. CONSTANT MEDIUM TEMPERATURE

Introduction. Since the appearance of Fourier's¹ early work the conduction of heat in solids has attracted the interest and attention of many

¹ Fourier, J. B., "Théorie analytique de la chaleur," 1822.

leading mathematicians and physicists. It is possible to present only some of the simplest and most representative cases here and to suggest the overall nature of the study. The reader is referred to the excellent books on the subject listed below.¹ They treat the subject in greater detail and provide the solutions for a number of specific problems as well as many with more complex geometry.

In the treatment of unsteady-state conduction the simplest types of problems are those in which the surface of the solid suddenly attains a new temperature which is maintained constant. This can happen only when the film coefficient from the surface to some isothermal heat-transfer medium is infinite, and although there are not many practical applications of this type, it is an important steppingstone to the solution of numerous problems. Ordinarily, heating or cooling involves a finite film coefficient or else a contact resistance develops between the medium and the surface so that the surface never attains the temperature of the medium. Moreover, the temperature of the surface changes continuously as the solid is heated even though the temperature of the medium remains constant. It is also possible that the temperature of the medium itself varies, but this class of problem will be treated separately in the next section. The cases treated in this section include those with finite film coefficients or contact resistances as well as those with infinite coefficients. The following are considered:

Sudden change of the surface temperature (infinite coefficient)

Wall of infinite thickness heated on one side

Wall of finite thickness heated on one side

Wall of finite thickness heated on both sides (slab)

Square bar, cube, cylinder of infinite length, cylinder with length equal to its diameter, sphere

¹ Boelter, L. M. K., V. H. Cherry, H. A. Johnson, and R. C. Martinelli, "Heat Transfer Notes," University of California Press, Berkeley, 1946. Carslaw, H. S., and J. C. Jaeger, "Conduction of Heat in Solids," Oxford University Press, New York, 1947. Grober, H., "Einführung in die Lehre von der Wärmeübertragung," Verlag Julius Springer, Berlin, 1926. Ingersoll, L. R., O. J. Zobel, and A. C. Ingersoll, "Heat Conduction with Engineering and Geological Applications," McGraw-Hill Book Company Inc., New York, 1948. McAdams, W. H., "Heat Transmission," 2d ed., McGraw-Hill Book Company, Inc., New York, 1942. Schack, A., "Der industrielle Wärmeübergang," Verlag Atahleisen, Dusseldorf, 1929; English translation by H. Goldschmidt and E. P. Partridge, "Industrial Heat Transfer," John Wiley & Sons, Inc., New York, 1933. Sherwood, T. K., and C. E. Reed, "Applied Mathematics in Chemical Engineering," McGraw-Hill Book Company, Inc., New York, 1939. For a review of more recent methods see Dusenberre, G. M., "Numerical Analysis of Heat Flow," McGraw-Hill Book Company, Inc., New York, 1949, and Jakob, M., "Heat Transfer," Vol. 1, John Wiley & Sons, Inc., New York, 1949.

Changes due to media having contact resistances

Wall of finite thickness
 Cylinder of infinite length, sphere, semiinfinite solid
 Newman's method for common and composite shapes
 Graphical determination of the time-temperature distribution

Wall of Infinite Thickness Heated on One Side. A wall of infinite thickness and at a uniform original temperature is subjected to surroundings with constant temperature T_s . It is assumed that there is no contact resistance between the medium and the surface it contacts, so that the face temperature of the wall is also T_s . This differs from ordinary quenching in which there is a very definite contact resistance. The general equation for conduction has been given by Eq. (2.13). For a wall of infinite thickness it reduces to unidirectional heat flow as given by Eq. (2.12). The group $k/c\rho$ is the thermal diffusivity consisting only of the properties of the conducting material. Calling this group α , the conduction may be represented by

$$\frac{\partial t}{\partial \theta} = \alpha \frac{\partial^2 t}{\partial x^2} \quad (2.12)$$

Fourier has indicated that the time-temperature-distance relationship for a body of uniform temperature subjected to a sudden source of heat will be represented by the exponential $e^{-p\theta}e^{qx}$, where p and q are constants, θ is the time, and x the distance from the surface. With this as the starting point it is possible to set up a number of equations which describe the variation of the temperature with time and distance throughout a solid suddenly subjected to a heat source at one face. It is requisite, however, that the equation containing the exponential also fulfill all the boundary conditions imposed on the system. The most general equation of this type is given by

$$t = C_1 + C_2x + C_3e^{-p\theta}e^{qx} \quad (18.40)$$

in which C_1 , C_2 , and C_3 are constants. A modification of Eq. (18.40) which describes the case in question and fulfills numerous boundary conditions is given by Schack as

$$t = C_1 + C_2x + C_3 \frac{2}{\sqrt{\pi}} \int_{z=0}^{z=x/2\sqrt{\alpha\theta}} e^{-z^2} dz \quad (18.41)$$

where $\frac{2}{\sqrt{\pi}} \int_0^z e^{-z^2} dz$ is readily recognized as the probability integral or Gauss's error integral with values from 0 to 1.0. The boundary conditions for an infinite wall heated on one face are that, when $x = x$ and $\theta = 0$, $t = t_0$ and, when $x = 0$ and $\theta = 0$, $t = T_s$, where t_0 is the initial

uniform temperature of the solid. When $x = 0$ and $\theta = 0$, $t = T_s = C_1$ in which T_s is the temperature of the face of the slab directly upon being brought in contact with surroundings at T_s . When $x = x$ and $\theta = 0$, the temperature of the slab naturally possesses its original temperature t_0 or

$$t_{\theta=0} = C_1 + C_2x + C_3 = t_0 \quad (18.42)$$

This can be valid, however, only if $C_2 = 0$, for otherwise t_0 would have to vary with x whereas it was assumed to be uniform.

Thus

$$t_0 = C_1 + C_3$$

or

$$C_3 = t_0 - T_s$$

Substituting for the constants in Eq. (18.41),

$$t = T_s + (t_0 - T_s) \frac{2}{\sqrt{\pi}} \int_0^{x/2\sqrt{\alpha\theta}} e^{-z^2} dz$$

or in abbreviated form

$$t = T_s + (t_0 - T_s) f_1 \left(\frac{x}{2\sqrt{\alpha\theta}} \right) \quad (18.43)$$

where $f_1(x/2\sqrt{\alpha\theta})$ denotes the value of the error integral in terms of the

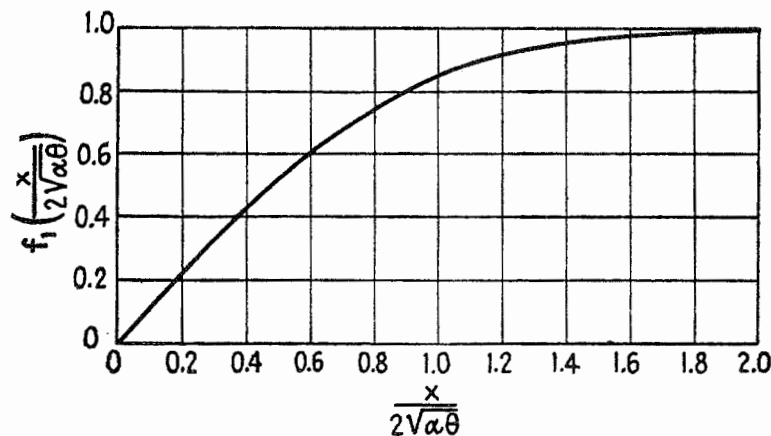


FIG. 18.7. Probability integral.

dimensionless group $x/2\sqrt{\alpha\theta}$. The values of the integral are plotted in Fig. 18.7. Equation (18.43) can be conveniently written

$$Y = \frac{T_s - t}{T_s - t_0} = f_1 \left(\frac{x}{2\sqrt{\alpha\theta}} \right) \quad (18.44)$$

Equation (18.44) is obviously the means by which the temperature t at any distance x and after time θ can be determined. A similar result can

be obtained by dimensional analysis. For unit area the heat flow can be taken from Eq. (2.5).

$$\frac{dQ}{dA} = k \frac{dt}{dx} \quad (18.45)$$

where Q is the heat flow Btu/hr. To obtain dt/dx from the expression for t in Eq. (18.44) the derivative of the error integral becomes

$$\frac{\partial}{\partial x} \int_0^{x/2\sqrt{\alpha\theta}} e^{-z^2} dz = \frac{e^{-x^2/4\alpha\theta}}{2\sqrt{\alpha\theta}} \quad (18.46)$$

and Eq. (18.45) reduces to

$$\frac{Q}{A} = \frac{k(T_s - t_0)}{\sqrt{\pi\alpha\theta}} e^{-x^2/4\alpha\theta} \quad (18.47)$$

Values of the exponential are plotted in Fig. 18.8.

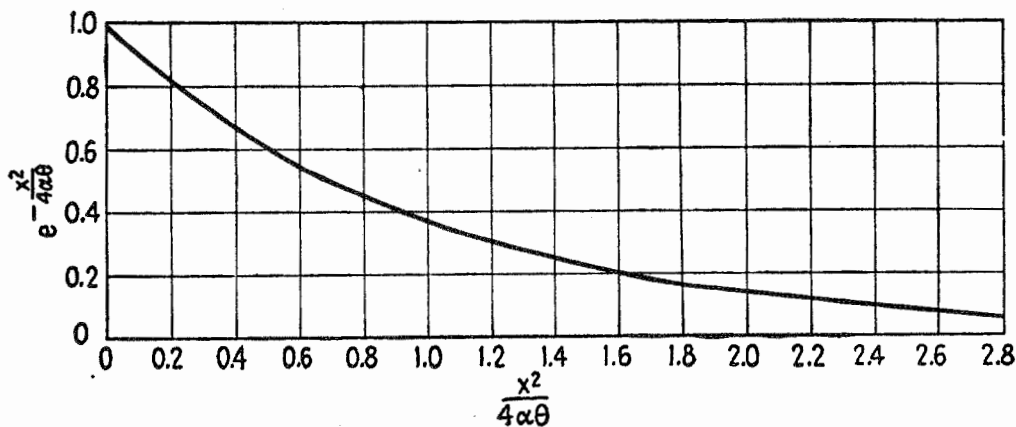


FIG. 18.8. Values of the exponential.

The heat flow through the surface is obtained when $x = 0$ or

$$\frac{Q}{A} = \frac{k(T_s - t_0)}{\sqrt{\pi\alpha\theta}} \quad (18.48)$$

and after θ hr the total heat which will have entered or left the wall will be

$$\frac{Q'}{A} = 2k(T_s - t_0) \sqrt{\frac{\theta}{\pi\alpha}} \quad (18.49)$$

where Q' is the number of Btu.

Example 18.2. Heat Flow through a Wall. It is desired to know the temperature of a thick steel wall 4 in. below its hot surface and 4 hr after its uniform temperature of 100°F was changed by suddenly applying a temperature of 1000°F to one face. How much heat will be passing into the wall at that time, and how much will already have passed into the wall?

Solution. Using Eq. (18.44):

Properties of steel: Assume

$$t_{av} = 500^{\circ}\text{F}$$

$$c = 0.12 \text{ Btu/(lb)}(^{\circ}\text{F})$$

$$k = 24.0 \text{ Btu/(hr)}(\text{ft}^2)(^{\circ}\text{F}/\text{ft})$$

$$\rho = 488 \text{ lb/ft}^3$$

$$\alpha = \frac{k}{c\rho} = \frac{24.0}{0.12 \times 488} = 0.41 \text{ ft}^2/\text{hr}$$

$$x = \frac{1}{2} = 0.333 \text{ ft}$$

$$\theta = 4$$

$$Y = \frac{T_s - t}{T_s - t_0} = f_1\left(\frac{x}{2\sqrt{\alpha\theta}}\right) = f_1\left(\frac{0.333}{2\sqrt{0.41 \times 4}}\right) = f_1(0.130) \quad (18.44)$$

From Fig. 18.7, for $x/2\sqrt{\alpha\theta} = 0.130$, $f_1(x/2\sqrt{\alpha\theta}) = 0.142$.

$$t = T_s + (t_0 - T_s)f_1\left(\frac{x}{2\sqrt{\alpha\theta}}\right) \quad (18.43)$$

$$t = 1000 + (100 - 1000)0.142 = 872^{\circ}\text{F}$$

The heat flow crossing a plane 4 in. from the surface at 4 hr after applying the heat is given by Eq. (18.47).

$$\frac{Q}{A} = \frac{k(T_s - t_0)}{\sqrt{\pi\alpha\theta}} \quad (18.47)$$

$$\frac{Q}{A} = \frac{24(1000 - 100)}{\sqrt{\pi \times 0.41 \times 4}} = 9525 \text{ Btu/(hr)}(\text{ft}^2)$$

The total heat which flowed through a square foot of wall in the 4 hr is given by

$$\begin{aligned} \frac{Q'}{A} &= 2k(T_s - t_0)\sqrt{\frac{\theta}{\pi\alpha}} \\ &= 2 \times 24(1000 - 100)\sqrt{\frac{4}{3.14 \times 0.41}} = 76,000 \text{ Btu/ft}^2 \end{aligned} \quad (18.49)$$

Wall of Finite Thickness Heated on One Side. The equations developed in the preceding section for the infinitely thick wall can be applied also with limitations to walls of finite thickness. If the finite wall is relatively thick, the temperature-distance distribution for a short period after the heat has been applied will be nearly the same as for the infinite wall. As the period is lengthened, however, the penetration of heat through the wall to its cold face increases. Schack¹ has analyzed this problem by noting that a finite wall can be made to duplicate an infinite wall. This will occur if the same heat flow is removed from the remote face of the finite wall as would ordinarily flow through a plane

¹ Schack, *op. cit.*

in an infinite wall at the same distance from the hot surface. If t_0 is the temperature at the remote face of a finite wall l ft thick, the heat flow per square foot from the remote face will be

$$Q = h(t - t_0) \quad (18.50a)$$

And this is equal to the heat flow in an infinite wall at the distance $x = l$ from the hot face.

$$Q = k(T_s - t_0) \frac{e^{-x^2/4\alpha\theta}}{\sqrt{\pi\alpha\theta}} \quad (18.50b)$$

By equating the two the rate of heat removal required to reproduce the distribution of an infinite wall will be

$$h = \frac{k(T_s - t_0)e^{-x^2/4\alpha\theta}}{(t - t_0)\sqrt{\pi\alpha\theta}} \quad (18.51)$$

If the value of $l/2\sqrt{\alpha\theta}$ is nearly 1.0, the temperature increase in the remote surface will be very small. If the value of $l/2\sqrt{\alpha\theta}$ exceeds 0.6, it will be possible for most industrial applications to apply Eq. (18.44) to the finite wall directly.

Wall of Finite Thickness Heated on Both Sides. In a study of the time-temperature distribution during the annealing of optical glass, Williamson and Adams¹ developed equations for obtaining the center, center-line, or center-plane temperature of a number of shapes whose surfaces had suddenly been exposed to a heat source with infinite film coefficient. Included among these are the infinitely wide slab, the square bar, the cube, the cylinder with infinite length, the cylinder with length equal to diameter, and the sphere. Since only in the slab is the flow of heat along a single axis, it may be expected that equations for the other forms will be more complex. Where the flow of heat is symmetrical, it is considerably more convenient to use the center line or center plane as the distance reference. The surface conditions will then correspond to $\pm l/2$, and the center line or center plane to $l/2 = 0$. Williamson and Adams obtained equations in terms of a Fourier series. For the infinite slab the equation is

$$Y = \frac{T_s - t}{T_s - t_0} = \frac{4}{\pi} \left[e^{-\left(\frac{\pi}{2}\right)^2 \frac{4\alpha\theta}{l^2}} \sin \pi \frac{x}{l} + \frac{1}{3} e^{-9\left(\frac{\pi}{2}\right)^2 \frac{4\alpha\theta}{l^2}} \sin \frac{3\pi x}{l} + \frac{1}{5} e^{-25\left(\frac{\pi}{2}\right)^2 \frac{4\alpha\theta}{l^2}} \sin \frac{5\pi x}{l} + \dots \right]$$

As $4\alpha\theta/l^2$ increases, the series converges even more rapidly until at $4\alpha\theta/l^2 = 0.6$ only the first term is important. The solutions for all the

¹ Williamson, E. D., and L. H. Adams, *Phys. Rev.*, **14**, 99-114 (1919).

above shapes can be expressed in terms of a series. Williamson and Adams have presented their calculations in the simplified form

$$Y = \frac{T_s - t}{T_s - t_0} = f_2 \left(\frac{4\alpha\theta}{l^2} \right) \quad (18.52)$$

or

$$t = T_s + (t_0 - T_s) f_2 \left(\frac{4\alpha\theta}{l^2} \right) \quad (18.52a)$$

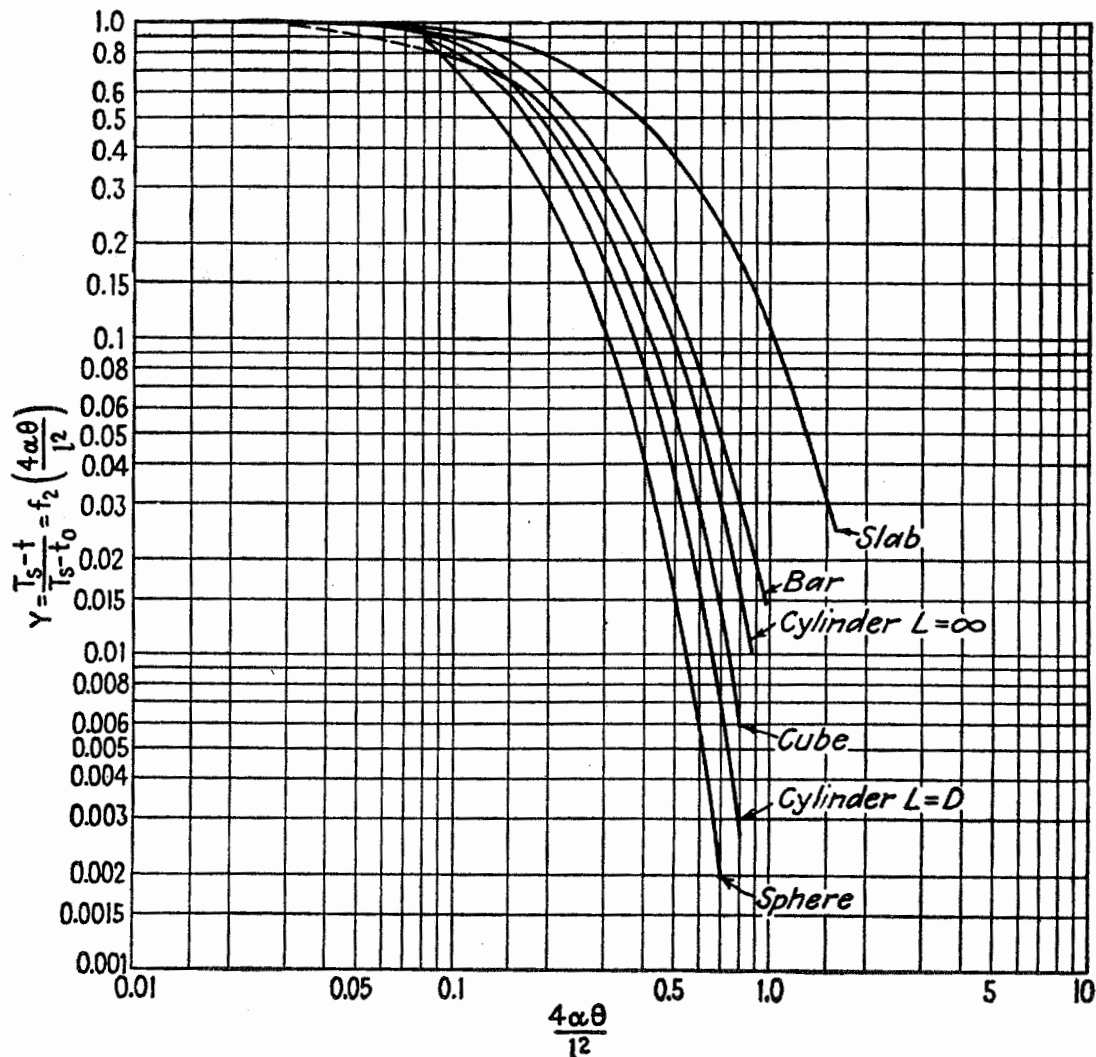


FIG. 18.9. Plot of $f_2(4\alpha\theta/l^2)$ for various shapes.

The series expression is presented by $f_2(4\alpha\theta/l^2)$, in which l is the principal depth or diameter and t the temperature at the center, center line, or center plane. For the various shapes and their respective final equations, the values of $f_2(4\alpha\theta/l^2)$ were tabulated by Williamson and Adams and have been plotted in Fig. 18.9.

Example 18.3. Center-line Temperature of a Shaft. Calculate the temperature at the center line of a long steel shaft 12 in. in diameter, initially at 100°F, 15 min

after its surface temperature has been suddenly changed to 1000°F. As before, α may be taken as 0.41 ft²/hr.

$$\frac{4\alpha\theta}{l^2} = \frac{4 \times 0.41 \times 15/60}{(1\frac{1}{2}/12)^2} = 0.41$$

From Fig. 18.9 when $L = \infty$

$$f_2\left(\frac{4\alpha\theta}{l^2}\right) = 0.155$$

From Eq. (18.52)

$$t = 1000 + (100 - 1000) \times 0.155 = 860^\circ\text{F}$$

Wall of Finite Thickness Heated by a Fluid with Contact Resistance. This practical condition has been treated by Grober¹ along with the evaluation of the functions contained in the final equations. This corresponds to quenching in that it considers a contact resistance between the heating or cooling medium and the wall face or both faces in the case of an infinitely wide slab of finite thickness. The reciprocal of the contact resistance is the coefficient of heat transfer between the liquid and solid and, as mentioned earlier, causes the surface temperature to vary even though the temperature of the heating medium remains constant. The method of evaluating the film coefficient may be selected approximately from among the methods of earlier chapters or those in Chap. 20. In many cases it is difficult to obtain an analogous mechanism for computing heat transfer during a quenching operation. In a large vessel at nearly constant temperature, such as might be employed for the quenching of steel plates, the limiting coefficient between the oil and metal is that of free convection, and it varies continuously with time as the temperature difference between the metal and oil decreases. Consider a plate with an initial temperature t_0 suddenly plunged into a gas or liquid of constant surrounding temperature T_s . The temperature of the surface will be given by

$$t_{x=0} = T_s + (t_0 - T_s)f_3\left(\frac{4\alpha\theta}{l^2}, \frac{hl}{2k}\right) \quad (18.53)$$

$$Y = \frac{T_s - t}{T_s - t_0} = f_3\left(\frac{4\alpha\theta}{l^2}, \frac{hl}{2k}\right) \quad (18.53a)$$

The temperature of the center plane will be

$$t_{l/2} = T_s + (t_0 - T_s)f_4\left(\frac{4\alpha\theta}{l^2}, \frac{hl}{2k}\right) \quad (18.54)$$

or

$$Y = \frac{T_s - t_{l/2}}{T_s - t_0} = f_4\left(\frac{4\alpha\theta}{l^2}, \frac{hl}{2k}\right) \quad (18.54a)$$

¹ Grober, *op. cit.*

Grober has graphically evaluated the functions

$$f_3\left(\frac{4\alpha\theta}{l^2}, \frac{hl}{2k}\right) \quad \text{and} \quad f_4\left(\frac{4\alpha\theta}{l^2}, \frac{hl}{2k}\right)$$

which have been plotted by Schack as shown in Fig. 18.10 and 18.11 for the surface and centers of rectangular shapes. When presented in this form the plots are referred to as Schack charts.

Example 18.4. Quenching: The Schack Chart. Iron castings in the form of large slabs 10 in. thick are held at red heat (1100°F) before being hung vertically in 70°F air to cool.

It is desired to start further operations on them after 4 hr have elapsed.

(a) What will the surface temperature be after 4 hr?

(b) What will the center-plane temperature be after 4 hr?

Solution. In order to obtain the average coefficient from the slab to air by radiation and convection an assumption must be made of the surface temperature after 4 hr.

(a) Assume the temperature is 500°F after 4 hr. The coefficient from plate to air is the sum of the radiation and convection coefficients. From Eq. (4.32) and (4.33)

$$h_r = \frac{\epsilon\sigma(T_1^4 - T_2^4)}{T_1 - T_2}$$

where the temperatures are in °R.

For the initial coefficient at 1100°F,

$$h_r = \frac{0.70 \times 0.173 \times 10^{-8}(1560^4 - 530^4)}{1100 - 70} = 6.9 \text{ Btu/(hr)(ft}^2\text{)(°F)}$$

From Eq. (10.10)

$$\begin{aligned} h_c &= 0.3(\Delta t)^{1/4} \\ &= 0.3(1100 - 70)^{1/4} = 1.7 \text{ Btu/(hr)(ft}^2\text{)(°F)} \end{aligned}$$

Total initial coefficient = 6.9 + 1.7 = 8.6 Btu/(hr)(ft²)(°F)

For the 4-hr coefficient at 500°F

$$\begin{aligned} h_r &= 2.2 \\ h_c &= 1.35 \end{aligned}$$

Total coefficient after 4 hr = 2.2 + 1.35 = 3.55 Btu/(hr)(ft²)(°F)

The average coefficient is somewhat lower than the mean of the initial and final coefficients, since the radiation coefficient decreases rapidly as the temperature falls. The mean coefficient gives a surface temperature somewhat larger than the actual.

Mean coefficient, $h = (8.6 + 3.55)/2 = 6.1 \text{ Btu/(hr)(ft}^2\text{)(°F)}$

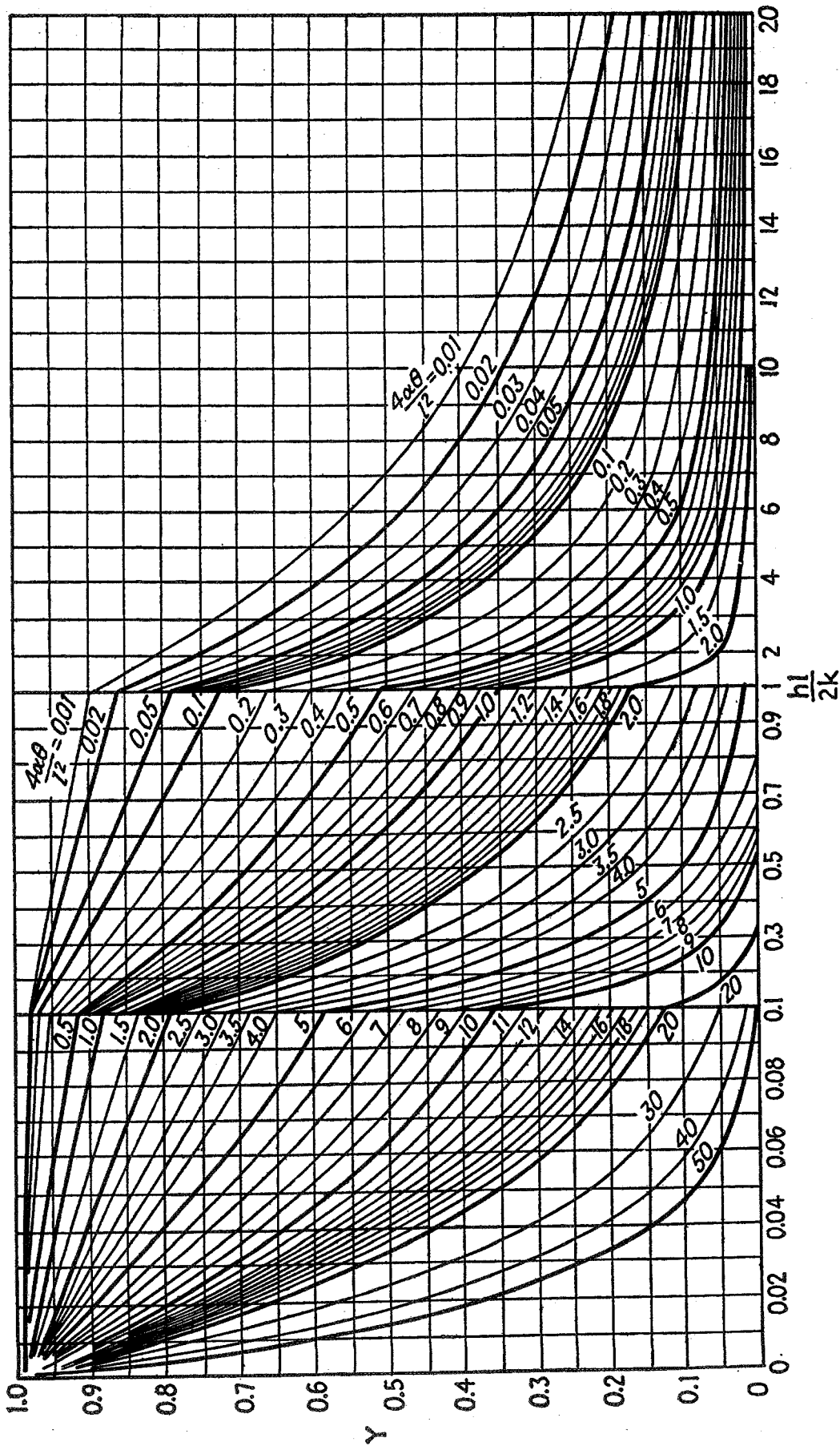


Fig. 18.10. Chart for determining the temperature history of points on the surfaces of rectangular shapes. (Newman, *Industrial Engineering Chemistry*, after Schack.)

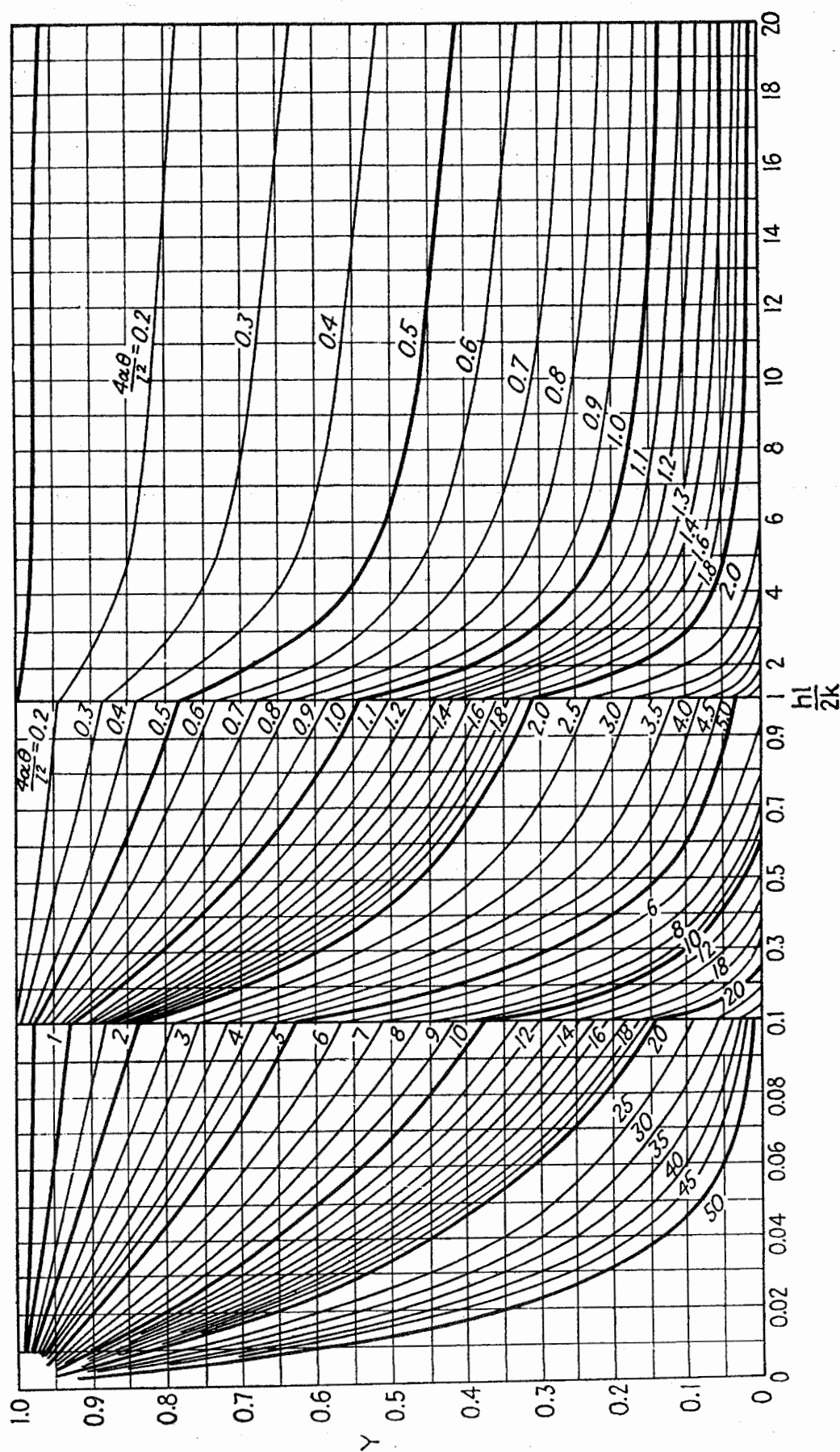


Fig. 18.11. Chart for determining the temperature history of points at the centers of rectangular shapes. (Newman, *Industrial Engineering Chemistry*, after Schack.)

From the Appendix, $k = 27$, $c = 0.14$, and $\rho = 490$ (approximately)

$$\begin{aligned}\alpha &= \frac{k}{c\rho} = \frac{27}{0.14 \times 490} = 0.394 \\ \frac{4\alpha\theta}{l^2} &= \frac{4 \times 0.394 \times 4}{(10/12)^2} = 9.1 \\ \frac{hl}{2k} &= \frac{6.1 \times (10/12)}{2 \times 27} = 0.094 \\ t_{x=0} &= T_s + (t_0 - T_s)f_3\left(\frac{4\alpha\theta}{l^2}, \frac{hl}{2k}\right) \\ &= 70 + (1100 - 70)f_3(9.1, 0.094)\end{aligned}\quad (18.53)$$

From Fig. 18.10, $f_3(9.1, 0.094) = 0.42$

$$t_{x=0} = 70 + (1100 - 70)0.42 = 502^\circ\text{F}$$

(b) The temperature of the center plane is given by

$$\begin{aligned}t_{l/2} &= T_s + (t_0 - T_s)f_4\left(\frac{4\alpha\theta}{l^2}, \frac{hl}{2k}\right) \\ &= 70 + (1100 - 70)f_4(9.1, 0.094)\end{aligned}\quad (18.54)$$

From Fig. 18.11, $f_4(9.1, 0.094) = 0.43$

$$t_{x=0} = 70 + (1100 - 70)0.43 = 512^\circ\text{F}$$

Finite and Semiinfinite Shapes Heated by a Fluid with a Contact Resistance. Referring to Eq. (18.52a) or (18.53a) Gurney and Lurie¹ noted that the relationships for heating various shapes with fluids having finite or infinite film coefficients could be represented by the four dimensionless groups $4\alpha\theta/l^2$, $(T_s - t)/(T_s - t_0)$, $2k/hl$, and $2x/l$. With these dimensionless parameters they made plots for the finite wall (slab), cylinder of infinite length, sphere, and semiinfinite solid. These are reproduced in Figs. 18.12 through 18.15. Gurney-Lurie charts are particularly useful, since they permit the calculation of the temperature not only at the surface and center of the object but at intermediate points as well. The ordinate Y is identical with the Schack chart for the same shape. Gurney-Lurie charts, however, are more difficult to interpolate than are Schack charts. The case of the semiinfinite solid is identical with the slab for the short initial period before the heat flow has reached the center plane of the slab. The use of these charts is demonstrated by the solution of an illustrative problem.

Example 18.5. The Gurney-Lurie Chart. Steel circular rods 8 in. in diameter and 12 ft long initially at 400°F are cooled in oil before being machined. The oil is maintained at a constant temperature of 200°F by means of a quench oil cooler. The average free convection coefficient from an 8-in. pipe to an oil of the properties of the quench oil is $50 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$.

What is the temperature 2 in. below the surface after 15 min?

¹ Gurney, H. P., and J. Lurie, *Ind. Eng. Chem.*, **15**, 1170 (1923).

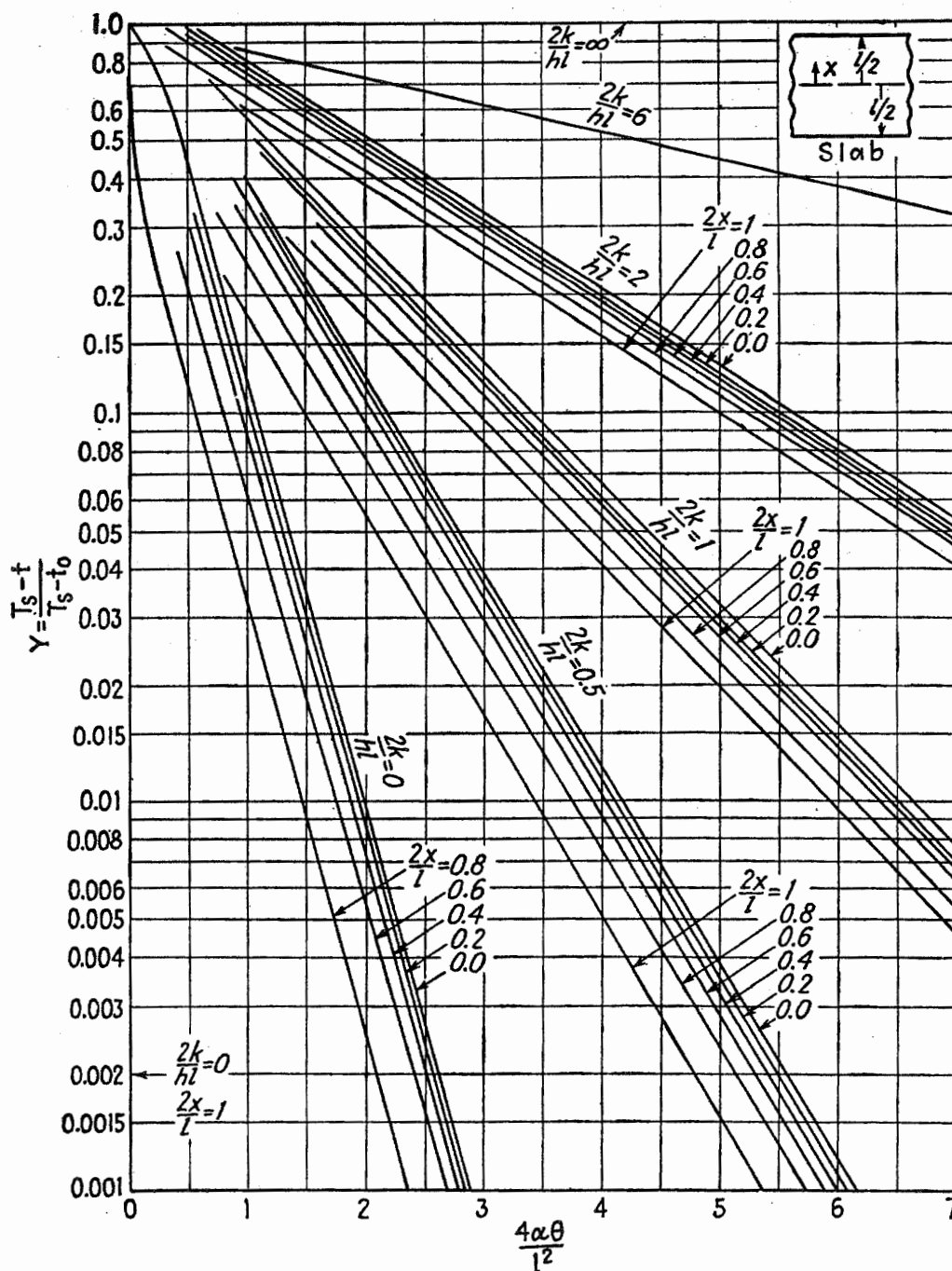


FIG. 18.12. Gurney-Lurie chart for slab.

Solution. This corresponds to an infinitely long cylinder, and Fig. 18.13 will be used. α will be evaluated at the mean temperature of 300°F

$$\begin{aligned}
 k &= 25 & c &= 0.12 & \rho &= 490 \\
 \alpha &= \frac{k}{c\rho} = \frac{25}{0.12 \times 490} = 0.425 \\
 \frac{4\alpha\theta}{l^2} &= \frac{4 \times 0.45 \times \frac{15}{60}}{(\frac{8}{12})^2} = 1.01 \\
 \frac{2x}{l} &= \frac{4}{8} = 0.5 \\
 \frac{1}{Nu} &= \frac{2k}{hl} = \frac{2 \times 25}{50 \times \frac{8}{12}} = 1.5
 \end{aligned}$$

From Fig. 18.13

$$Y = 0.31 = \frac{T_s - t}{T_s - t_0}$$

$$t_{x=2 \text{ in.}} = 400 + 0.31(200 - 400) = 338^\circ\text{F}$$

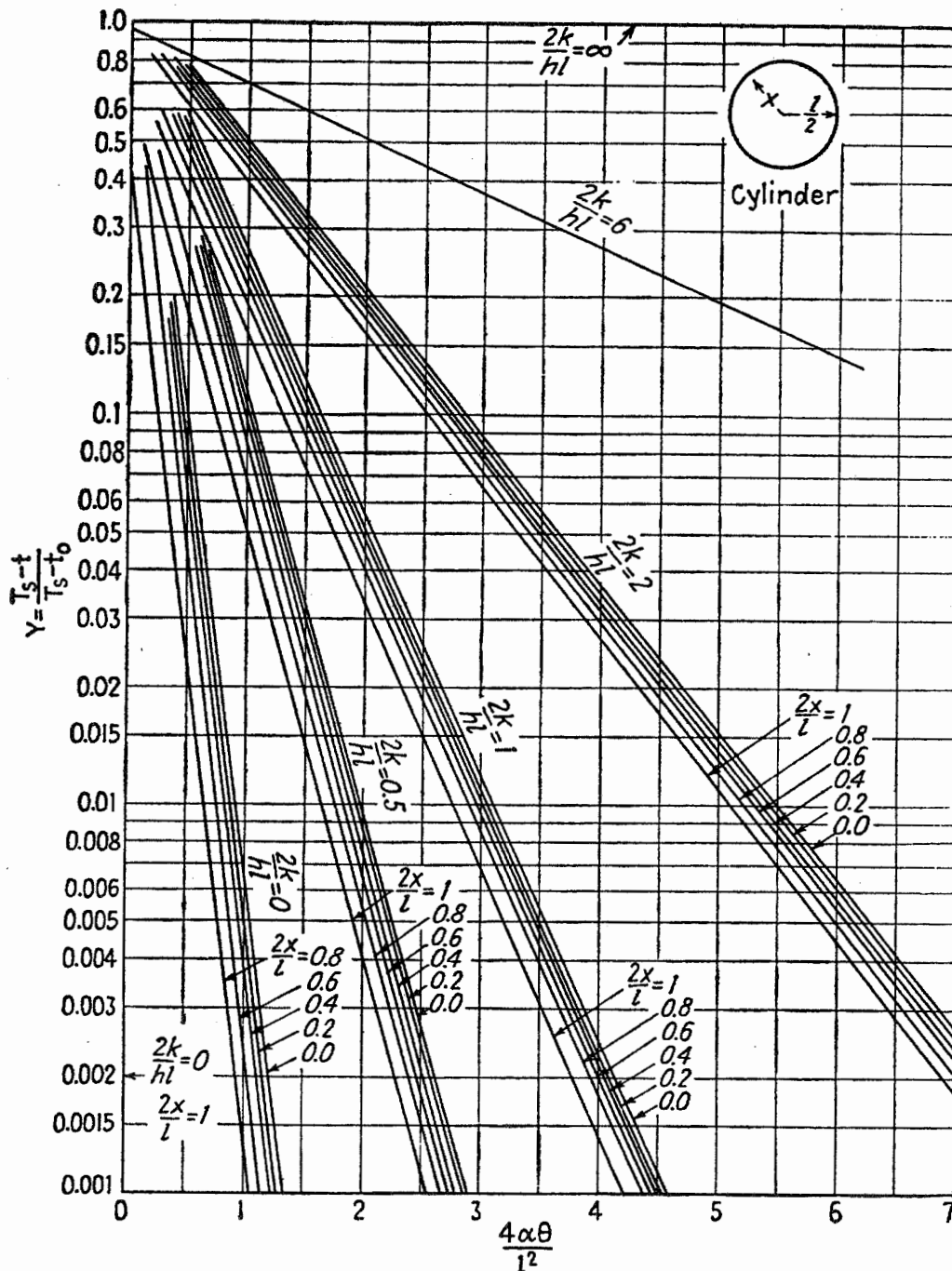


FIG. 18.13. Gurney-Lurie chart for cylinders.

Newman's Method for Common and Composite Shapes. Many of the objects which are regularly subjected to quenching do not conform with the simple shapes dealt with so far. Among such common shapes are the rectangular bar, the brick (rectangular parallelepiped), and the short cylinder of finite length. For heat flow along the x axis along the tem-

perature of the object is defined by Y or $Y_x = (T_s - t)/(T_s - t_0)$, where the subscript x indicates the direction of the heat flow. The center of the object is taken as the reference point. In a long rectangular bar

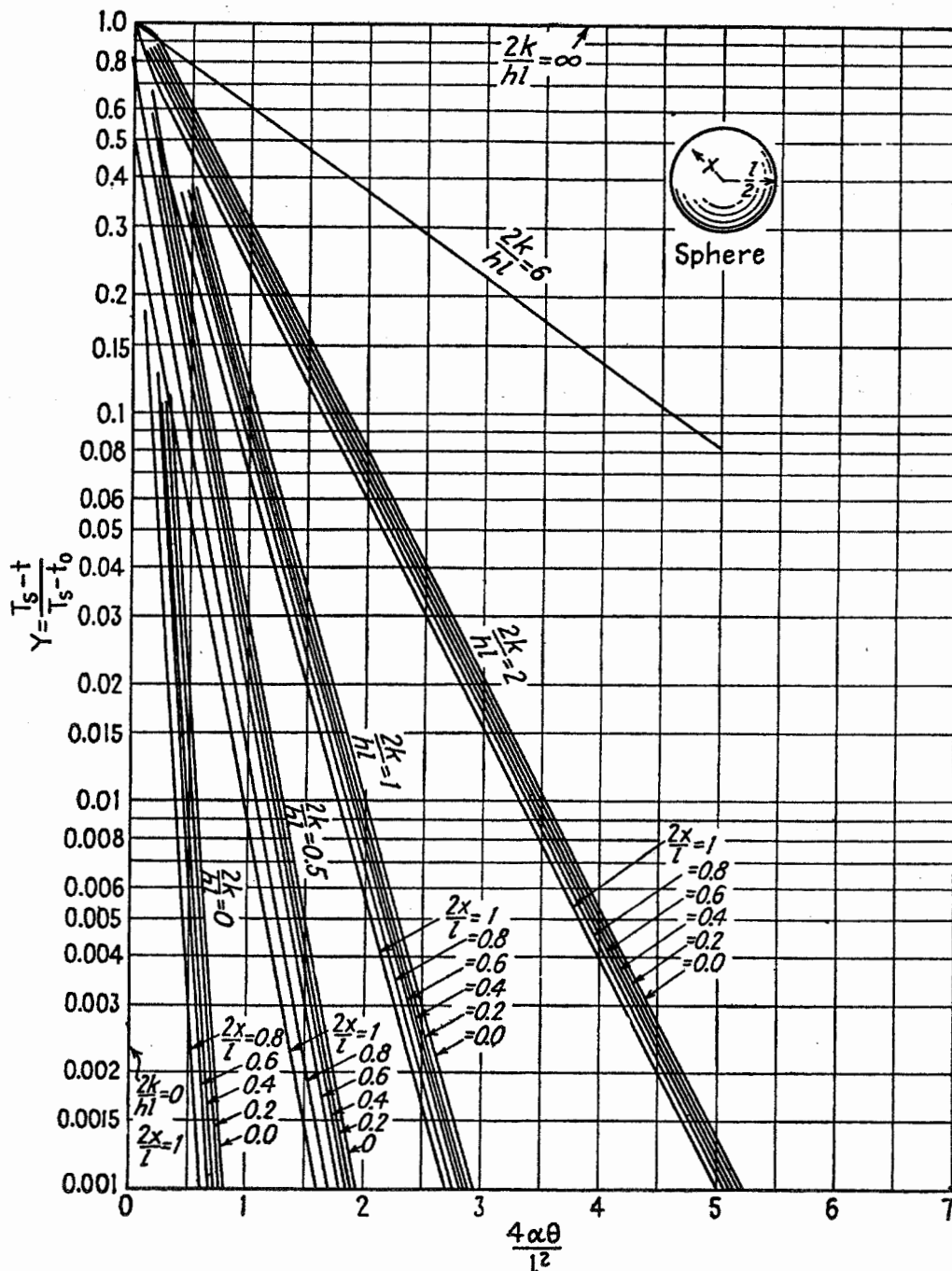


FIG. 18.14. Gurney-Lurie chart for spheres.

whose cross section is reckoned in the x and y directions, Newman¹ has proved that the temperature at any point in the cross section is given by $(T_s - t)/(T_s - t_0) = Y_x Y_y$, where both Y_x and Y_y have been calculated for the flow of heat through a finite slab. Similarly for a brick whose

¹ Newman, A. B., *Trans. AICHE*, 27, 203 (1931).

cross section is reckoned by x and y the temperature at any point can be defined by $(T_s - t)/(T_s - t_0) = Y_x Y_y Y_z$, where the value of Y_z has been determined for the long dimension of the brick as a slab with heat flow

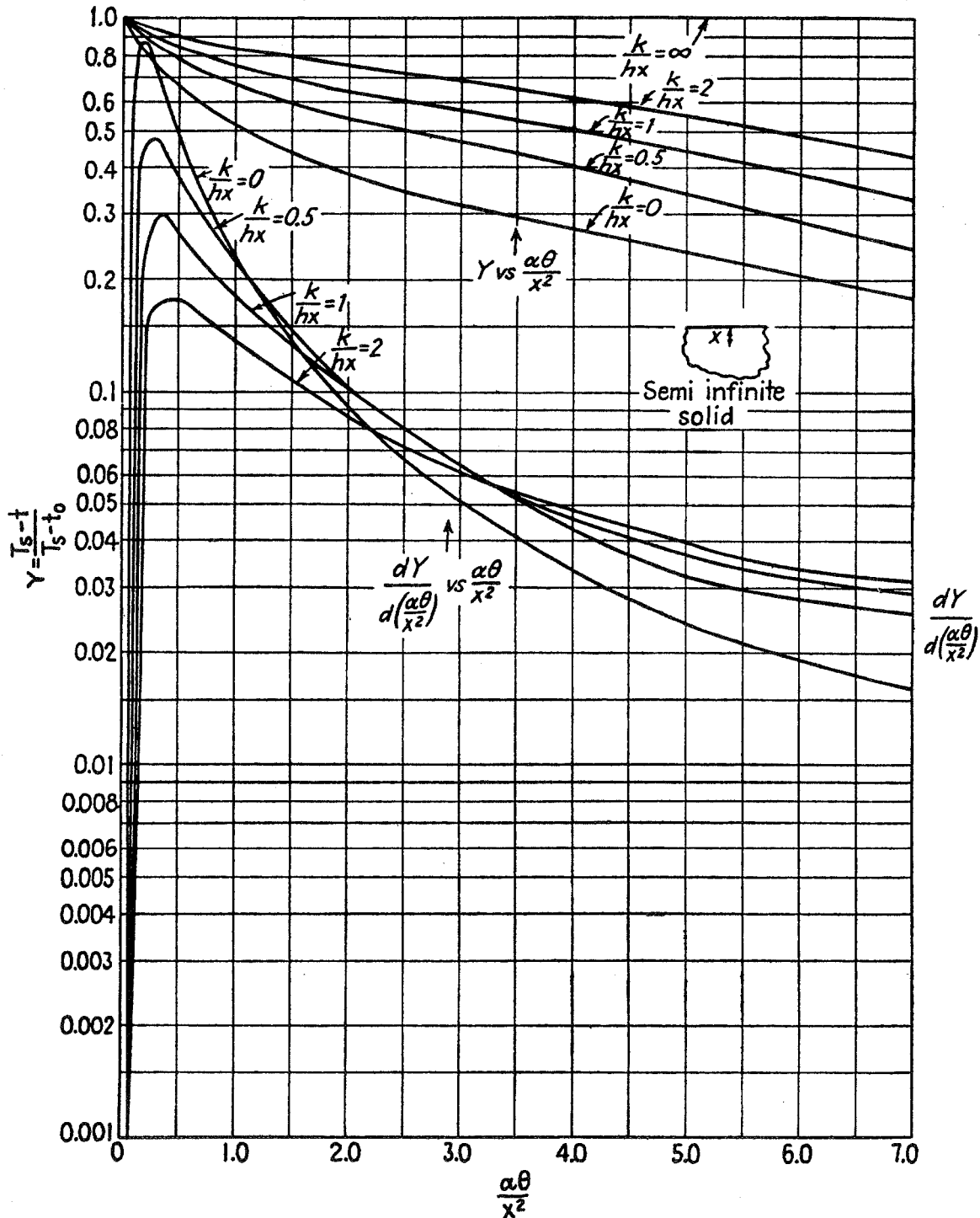


FIG. 18.15. Gurney-Lurie chart for semiinfinite solid.

in the z direction. For the cylinder of finite length if the length is l_z and the diameter l_z , the temperature can be defined by

$$\frac{T_s - t}{T_s - t_0} = Y_x Y_z.$$

In objects of finite length if one of the plane faces is insulated against the flow of heat, the calculation is treated as above except that the zero axis or reference point is moved to the insulated wall instead of the center and the distance perpendicular to the insulated face is doubled (l instead of $l/2$). If a pair of parallel faces is insulated so that there is no net heat flow in the x , y , or z direction, the entire direction is omitted. Similarly, it is possible that the film coefficient to the different faces of an object may not be identical on all sides, since the flow of fluid may be mostly parallel to some surfaces and perpendicular to others. This will not interfere with the solution of a problem for objects having parallel faces as long as the coefficient on any two parallel faces is the same and the heat-flow symmetry about the center is preserved. The values of Y for these shapes can be obtained from Schack or Gurney-Lurie charts for the finite slab or infinite cylinder and from which complex shapes can be built up accordingly. A Schack chart for the infinite cylinder has been prepared by Newman.¹

To illustrate this method, consider a brick which has the dimensions l_x , l_y , and l_z in the x , y , and z directions, respectively. The extreme distances of the faces of the brick reckoned from its center are $l_x/2$, $l_y/2$, and $l_z/2$. At its center it is apparent that $l_x/2 = 0$, $l_y/2 = 0$, and $l_z/2 = 0$. For the cylinder, at the circumference the dimension is $l_x/2$ and at the parallel end faces $l_z/2$.

Example 18.6. The Application of Newman's Method to Heating a Brick. This problem uses the data of an example by Newman with numerical values obtained from Gurney-Lurie charts.

A firebrick 9 by 4.5 by 2.5 in. initially at 70°F is suspended in a flue through which furnace gases at 300°F are traveling at such a rate that the resulting film coefficient on all faces is 4.1 Btu/(hr)(ft²)(°F). Estimate the temperatures at the following points at the end of 1 hr: (a) the center of the brick, (b) any corner of the brick, (c) the center of the 9- by 4.5-in. faces, (d) the center of the 9- by 2.5-in. faces, (e) the center of the 4.5- by 2.5-in. faces, (f) the middle of the long edges.

The following data are supplied:

$$k = 0.3 \text{ Btu/(hr)(ft)}^2 \text{ (°F/ft)}$$

$$\rho = 103 \text{ lb/ft}^3$$

$$c = 0.25 \text{ Btu/(lb)(°F)}$$

Solution:

$$\alpha = \frac{k}{c\rho} = \frac{0.3}{0.25} \times 10.3 = 0.01164 \text{ ft}^2/\text{hr}$$

$$\frac{4\alpha\theta}{l_x^2} = \frac{4 \times 0.01164 \times 1.0}{(9/12)^2} = 0.0828 \quad \frac{2k}{hl_x} = \frac{2 \times 0.3}{4.1 \times 9/12} = 0.195$$

$$\frac{4\alpha\theta}{l_y^2} = \frac{4 \times 0.01164 \times 1.0}{(4.5/12)^2} = 0.3313 \quad \frac{2k}{hl_y} = \frac{2 \times 0.3}{4.1 \times 4.5/12} = 0.390$$

$$\frac{4\alpha\theta}{l_z^2} = \frac{4 \times 0.01164 \times 1.0}{(2.5/12)^2} = 1.073 \quad \frac{2k}{hl_z} = \frac{2 \times 0.3}{4.1 \times 2.5/12} = 0.702$$

¹ Newman, A. B., *Ind. Eng. Chem.*, **28**, 545-548 (1936).

From Fig. 18.12

At the center	At the surface
$\frac{2x}{l} = 0$	$\frac{2x}{l} = 1.0$
$Y_x = 0.98^*$	$Y_x = 0.325^*$
$Y_y = 0.75$	$Y_y = 0.29$
$Y_z = 0.43$	$Y_z = 0.245$

For all cases,

$$Y = \frac{T_s - t}{T_s - t_0} = \frac{300 - t}{300 - 70} = \frac{300 - t}{230}$$

(a) Center of brick ($l_x = 0$; $l_y = 0$; $l_z = 0$):

$$\frac{300 - t}{230} = 0.98 \times 0.75 \times 0.43 = 0.316$$

$$t = 227.4^\circ\text{F}$$

(b) Corner of brick ($l_x/2$, $l_y/2$, $l_z/2$):

$$\frac{300 - t}{230} = 0.325 \times 0.29 \times 0.245 = 0.023$$

$$t = 294.7^\circ\text{F}$$

(c) Center of 9- by 4.5-in. face ($l_x/2 = 0$, $l_y/2 = 0$, $l_z/2$):

$$\frac{300 - t}{230} = 0.98 \times 0.75 \times 0.245 = 0.18$$

$$t = 258.5^\circ\text{F}$$

(d) Center of 9- by 2.5-in. face ($l_x/2 = 0$, $l_y/2$, $l_z/2 = 0$):

$$\frac{300 - t}{230} = 0.98 \times 0.29 \times 0.43 = 0.122$$

$$t = 272^\circ\text{F}$$

(e) Center of 4.5- by 2.5-in. face ($l_x/2$, $l_y/2 = 0$, $l_z/2 = 0$):

$$\frac{300 - t}{230} = 0.325 \times 0.75 \times 0.43 = 0.105$$

$$t = 275.8^\circ\text{F}$$

(f) Middle of long edge ($l_x/2 = 0$, $l_y/2$, $l_z/2$):

$$\frac{300 - t}{230} = 0.98 \times 0.29 \times 0.245 = 0.0695$$

$$t = 284^\circ\text{F}$$

The Graphical Determination of the Time-temperature Distribution.

The time-temperature distribution for many practical problems has not appeared in the literature because their solutions are lengthy or their mathematics is extremely complex. A useful and concise method of treating such cases graphically has been developed by E. Schmidt.¹ Consider a slab of infinite width and finite thickness in which the heat flows only through the thickness. As before, the time-temperature relationship can be obtained by solution of the basic conduction equation

$$\frac{dt}{d\theta} = \alpha \frac{d^2t}{dx^2} \quad (2.12)$$

* These values are read more easily from Schack charts.

¹ E. Schmidt, "Foppls Festschrift," pp. 179-189, Verlag Julius Springer, Berlin, 1924. See particularly, Sherwood and Reed, *op. cit.*, pp. 241-255.

The temperature in the slab at any point is a function of the time and distance. Divide the slab into a number of distance increments of Δx ft each, and consider an increment of time $\Delta \theta$. At a constant distance x from one face of the slab the incremental temperature change during a finite time change $\Delta \theta$ can be indicated by Δt_θ . For a constant value of θ the variation of the temperature with x can be indicated by Δt_x . Equation (2.12) can then be written

$$\frac{\Delta t_\theta}{\Delta \theta} = \alpha \frac{\Delta^2 t_x}{\Delta x^2} \quad (18.55)$$

or rearranging,

$$\Delta t_\theta = \alpha \frac{\Delta \theta}{\Delta x^2} \Delta^2 t_x \quad (18.56)$$

Referring to Fig. 18.16 in which the slab is divided into distance increments Δx , let $t_{n,m}$ represent the temperature at n distance increments from the origin $n \Delta x$ after m time increments, $m \Delta \theta$. When x is constant, the change of temperature with time in the layer $m \Delta x$ from the surface is

$$\Delta t_\theta = t_{n,m+1} - t_{n,m} \quad (18.57)$$

When θ is a constant, the variation of the temperature with distance is

$$\Delta t_x = t_{n+1,m} - t_{n,m} \quad (18.58)$$

and for the second change or difference between two differences

$$\Delta^2 t_x = \Delta(\Delta t_x) = (t_{n+1,m} - t_{n,m}) - (t_{n,m} - t_{n-1,m}) \quad (18.59)$$

Substituting these in Eq. (18.56),

$$t_{n,m+1} - t_{n,m} = \alpha \frac{\Delta \theta}{\Delta x^2} [(t_{n+1,m} - t_{n,m}) - (t_{n,m} - t_{n-1,m})] \quad (18.60)$$

If the increments of distance and time are chosen so that

$$\alpha \frac{\Delta \theta}{\Delta x^2} = \frac{1}{2} \quad (18.61)$$

Equation (18.60) reduces to

$$t_{n,m+1} = \frac{1}{2}(t_{n+1,m} + t_{n-1,m}) \quad (18.62)$$

Equation (18.62) is the basis of the graphic method. It implies that the temperature at any point at any time is the arithmetic mean of the two

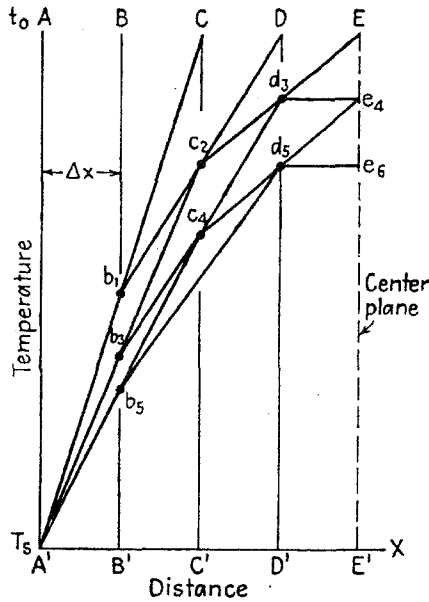


FIG. 18.16. Schmidt graphical method.

temperatures at $+$ and $- \Delta x$ during the preceding time increment $\Delta\theta$. A straight line drawn through the values of the temperatures at $(n - 1) \Delta x$ and $(n + 1) \Delta x$ intersects the vertical median at the arithmetic mean where the temperatures at $(n - 1) \Delta x$ and $(n + 1) \Delta x$ were the two previous temperatures.

The complete method is illustrated by Fig. 18.16. Consider a symmetrical object such as an infinite slab at initial temperature t_0 suddenly subjected at both faces to a cooling medium with zero contact resistance and temperature T_s . The heat flow is along the x axis. Since the temperature distribution about the center plane is symmetrical, only one-half of the slab need be considered and the half of the slab under consideration is broken up into the distance increments shown by the vertical lines. If the initial temperature upon contact is T_s at 0 and the temperature is t_0 at C , then at the time increment $\Delta\theta$ later the temperature in the plane $B-B'$ will be the arithmetic mean of t_0 and T_s , b_1 . Points C , D , and E remain unchanged during the first increment. During the second time increment the temperature at C in plane $C-C'$ falls to c_2 , D and E remaining unchanged. During the third time increment the temperature at b_1 falls to b_3 and the temperature at D falls to d_3 . During this increment the temperature at the center plane does not change, since it is the mean of the values at $\pm \Delta x$ from the center plane, which are both still at t_0 . During the fourth increment the temperature at C falls from c_2 to c_4 , but the temperature at the center plane e_4 is the mean of the two identical values of d_3 at $\pm \Delta x$ from the center plane and therefore lies on a horizontal line. The procedure can be continued indefinitely with each horizontal line across the center plane representing two time increments.

Example 18.7. The Graphical Determination of the Time-Temperature Distribution. A steel slab 20 in. thick and at an initial temperature of 500°F is suddenly subjected to a temperature of 100°F on both faces. What is the temperature distribution after 20 min?

Solution. For simplicity take $\alpha = 0.40 \text{ ft}^2/\text{hr}$. Take distance increments Δx 2 in. apart.

$$\Delta x = \frac{2}{12} = 0.167 \text{ ft}$$

From the conditions of Eq. (18.61) take time increments such that $\alpha(\Delta\theta/\Delta x^2) = \frac{1}{2}$.

$$\Delta\theta = \frac{0.167^2}{2 \times 0.40} = 0.035 \text{ hr or } 2.1 \text{ min}$$

Number of steps required = $20/2.1 = 9.5$

In Fig. 18.17 half of the slab is divided into distance increments of 2 in. and the time intervals are drawn accordingly. After 9 time increments corresponding to $9 \times 2.1 = 18.9 \text{ min}$, the temperature at the center plane is 413°F. After 11 increments or 23.1 min it is 383°F. The value at 20 min can be obtained by plotting a time-temperature curve for the center plane alone from which a temperature of 406°F is obtained.

Time-Temperature Distribution with Contact Resistance. The instances in which the heat source is applied with negligible contact resistance are exceedingly rare. A graphical method was developed by Schmidt for a slab when the film coefficient to it was finite. Sherwood and Reed¹ have given an excellent presentation of the method. The presence of a finite film coefficient or contact resistance indicates that a temperature drop occurs between the bulk temperature of the surround-

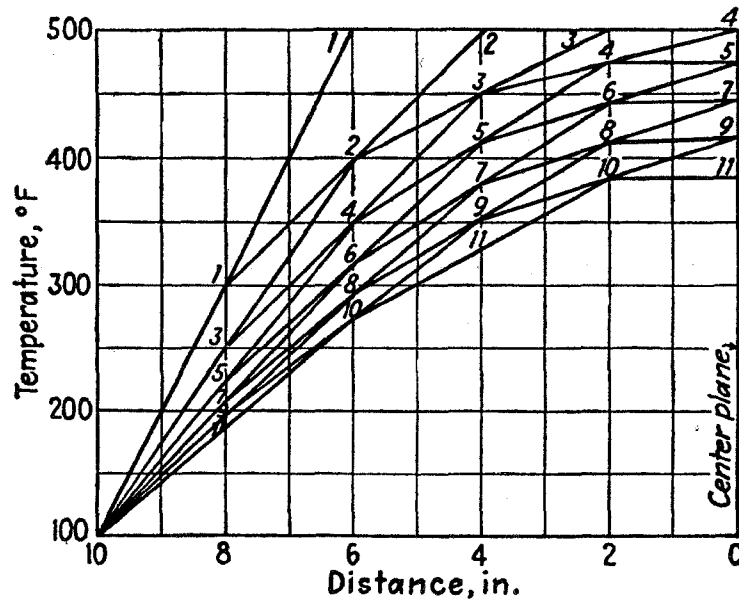


FIG. 18.17. Solution of Example 18.7 by the Schmidt method.

ing medium at T_s and the surface of the slab t_f . The heat balance across the surface is given by

$$k \left(\frac{dt}{dx} \right)_{x=0} = h(T_s - t_f) \quad (18.63)$$

where T_s is again the temperature of the surroundings and t_f is the surface temperature. Accordingly, the temperature gradient across the surface is given by

$$\left(\frac{dt}{dx} \right)_{x=0} = \frac{T_s - t_f}{k/h} \quad (18.64)$$

Any line cutting the surface of the slab on t vs. x coordinates such as in Fig. 18.16 must consequently have a numerical slope of $(T_s - t_f)/(k/h)$. Referring to Fig. 18.18, a slab is divided into distance increments Δx , except that they are marked off such that the surface of the slab corresponds to the middle of a distance increment. The reason will be apparent as the construction progresses. Next, the origin O is set off at a distance k/h ft to the left of the surface and at the temperature T_s .

¹ Sherwood and Reed, *op. cit.*

The slope of a line drawn from the origin O through the surface plane then has the slope $(T_s - t_f)/(k/h)$. By constructing the vertical at $\Delta x/2$ to the left of the surface, Schmidt's rule can be applied and each alternate time increment is represented by a line crossing the surface with the appropriate slope. The surface is thus the locus of values of t_f . The construction is carried out by the preceding Schmidt method until the center plane has been passed by a line drawn to $\Delta x/2$ at the right of the center plane. Since the distance increments are displaced by the distance $\Delta x/2$, the plane at $\Delta x/2$ to the right of the center plane is the

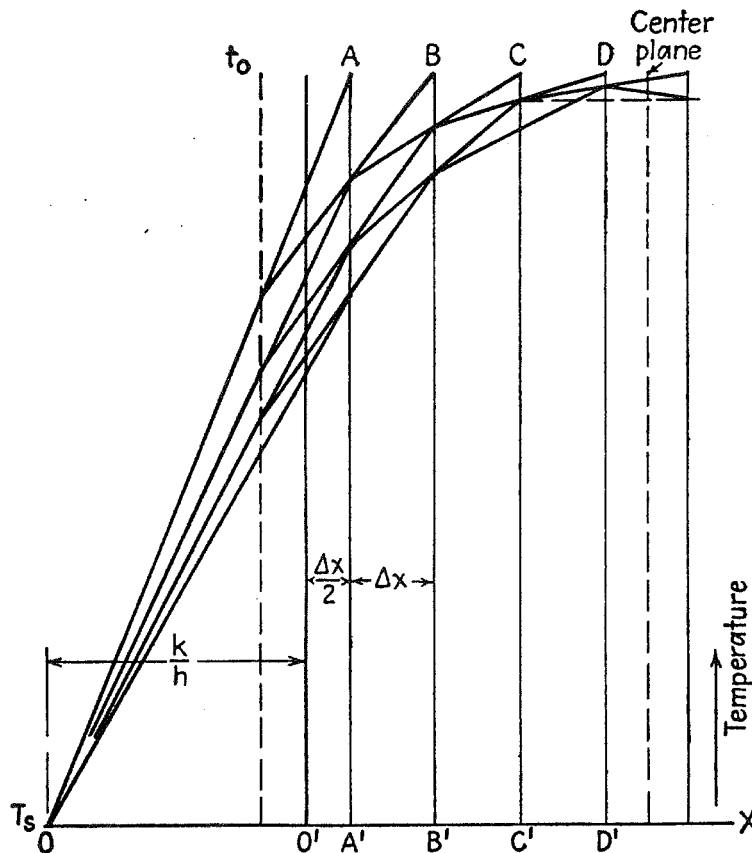


FIG. 18.18. Schmidt method with contact resistance.

mirror image of that at $\Delta x/2$ to the left with the temperatures drawn to corresponding points on the plane to the right. Thus for alternate increments of time the line slopes up and down.

Nonsymmetrical cases of heat flow can be studied by this approximate method when mathematical derivations lead to involved expressions. Such a condition arises when one face of a slab is exposed to a finite film coefficient while the other is exposed to a negligible resistance. Another common condition is that in which the faces are at two different temperatures or in which the temperatures undergo cyclical variations. For this last a number of repeated solutions are required, but they provide results which otherwise might be unobtainable. Schmidt has also provided

graphical solutions for a number of complex problems such as the heat flow through layers of dissimilar materials.

2b. PERIODICALLY VARYING TEMPERATURE

Periodic Variation of the Surface Temperature. There are a number of instances in which the temperature of the heating medium is not constant. In some, the medium and the surface temperatures vary harmonically. Typical examples of the latter are the temperature of the earth's surface or within an internal combustion cylinder, although there are not many practical applications. Figure 18.19 shows the variation at a distance x in a wall whose surface temperature varies as a sine function with time.

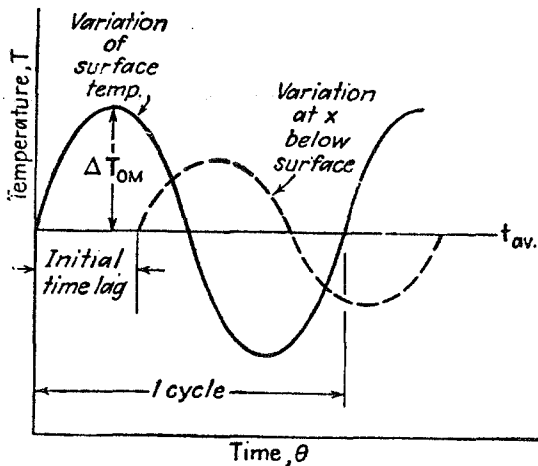


FIG. 18.19. Sine temperature variation at surface and at point x ft below.

Inasmuch as the heat must travel from the surface into the bulk of the material and the thermal diffusivity is finite, there is a time lag before the temperature-time variations at the surface are reproduced at the distance x . Naturally, the maximum amplitude at x is less than that at the surface, and it is still less at greater distances. It should be noted that Fig. 18.19 is a plot of time vs. temperature and not distance vs. temperature, so that the decline in the amplitude with

distance is not indicated. If f is the number of complete periodic changes per hour, the time required for a point within the body to respond to a variation at the surface is evidently given by

$$\theta = \frac{x}{2} \sqrt{\frac{1}{\alpha \pi f}} \quad (18.65)$$

where x is the distance in feet from the surface to the point. If the temperature variation is single harmonic, the surface temperature can be defined in terms of the maximum surface temperature by

$$\Delta t_{x=0} = \Delta t_{OM} \sin 2\pi f \theta \quad (18.66)$$

where t_{OM} is the maximum temperature at the surface where $x = 0$. At the distance x the temperature has the same period but lags by $(x/2) \sqrt{1/\alpha \pi f}$ hr, the time initially required to travel from the surface to x . The maximum temperature at x is given by

$$\Delta t_{x=x, \max} = \Delta t_{OM} e^{-x \sqrt{\pi f / \alpha}} \quad (18.67)$$

The equation for the variation of the temperature at any distance in typical problems requires the solution of a Fourier series which is beyond the limitations of space here. If the variation at the distance x at any time θ can be expressed as a sine function, it is given by

$$\Delta t_x = \Delta t_{OM} e^{-x\sqrt{\pi f/\alpha}} \sin \left(2\pi f\theta - x\sqrt{\frac{\pi f}{\alpha}} \right) \quad (18.68)$$

If the variation can be expressed as a cosine function instead of a sine function, the temperature at any point x is given by

$$\Delta t_x = \Delta t_{OM} e^{-x\sqrt{\pi f/\alpha}} \cos \left(2\pi f\theta - x\sqrt{\frac{\pi f}{\alpha}} \right) \quad (18.69)$$

The flow of heat through a flat wall for a half period can be determined from the basic conduction equation

$$dQ' = kA \left(\frac{dt}{dx} \right)_{x=0} d\theta$$

by differentiating Eq. (18.68) or (18.69) with respect to x , from which the following is obtained for Q for a half period:

$$Q = kA \Delta t_{OM} \sqrt{\frac{2}{\alpha \pi f}} \quad (18.70)$$

Example 18.8. Calculations for a Wall with Periodic Temperature Variation. The surface of a thick brick wall is subjected to a daily temperature variation from 60 to 120°F. The cosine variation may be assumed. The cement is assumed to have the same properties of the bricks, which are $k = 0.3$ Btu/(hr)(ft²)(°F/ft), $\rho = 103$ lb/ft³, $c = 0.25$ Btu/(lb)(°F), and $\alpha = k/c\rho = 0.01164$ ft²/hr.

- What is the temperature lag 6 in. below the surface?
- What is the amplitude at this depth?
- What is the temperature deviation from the mean after 2 hr?
- What is the heat flow during the half period?

Solution:

- Temperature lag 6 in. below the surface:

$$\begin{aligned} \theta &= \frac{x}{2} \sqrt{\frac{1}{\alpha \pi f}} \\ &= \frac{6/12}{2} \sqrt{\frac{1}{0.01164 \pi \frac{1}{24}}} = 6.45 \text{ hr} \end{aligned} \quad (18.65)$$

- Amplitude:

$$\begin{aligned} \Delta t_{OM} &= \frac{120 - 60}{2} = 30^\circ\text{F} \\ \Delta t_{x-x} &= \Delta t_{OM} e^{-x\sqrt{\pi f/\alpha}} \\ &= 30.e^{-0.5\sqrt{\pi/24 \times 0.01164}} = 5.8^\circ\text{F} \end{aligned}$$

(c) Temperature deviation after 2 hr:

$$\begin{aligned}\Delta t_x &= \Delta t_{OM} e^{-x\sqrt{\pi f/\alpha}} \cos \left(2\pi f\theta - x\sqrt{\frac{\pi f}{\alpha}} \right) \\ &= 5.8 \cos (2 \times \pi \times \frac{1}{24} \times 2 - 1.635) \\ &= 5.8 \cos 1.10 = 2.6^\circ\text{F}\end{aligned}\quad (18.69)$$

(d) Heat flow during the half period:

$$\begin{aligned}\frac{Q}{A} &= k \Delta t_{OM} \sqrt{\frac{2}{\alpha \pi f}} \\ &= 0.3 \times 30 \sqrt{\frac{2}{0.01164 \pi \frac{1}{24}}} = 326 \text{ Btu/hr(ft}^2\text{)}\end{aligned}\quad (18.70)$$

c. REGENERATORS

Introduction. A regenerator is an apparatus in which heat is alternately stored and removed. The broadest applications of regeneration have been in the blast and open-hearth furnaces of the steel industry. More recently regeneration applications have been extended to the "reversing exchanger" for large-scale air-separating plants in the Fischer-Tropsch process and other synthetic processes. A type of reversing exchanger which was popular in Germany employed Fränkl regenerator packing, which consists of metallic ribbons having corrugations at an angle of 45° . Two such ribbons are wound together in spirals which are then packed one above the other. Temperature histories and coefficients have been reported by Lund and Dodge.¹ Several of the proposed designs for the reversing exchanger also included the storage of heat by means of extended surfaces. Schack² presents an excellent survey of the literature on regenerators as well as his own work, which is recommended to the reader who is undertaking a comprehensive study of regenerator design. As treated here the regenerator is cited as another aspect of the unsteady state as well as to indicate the general methods of heat storage in processes. Actual regenerators require additional refinements in assumptions and practices.

Temperature Variations in Regenerators. The open-hearth furnace is charged with molten pig iron and scrap to supply the oxygen requirements for the oxidation of impurities. A mixture of hot fuel gas and air is burned above the pool of molten metal, thereby maintaining the temperature and removing the impurities arising from the pool. The products of combustion and oxidation are *waste gases* containing considerable heat which might profitably be recovered by preheating the fuel-gas-air mixture. When the waste gases leave the furnace, they are passed through

¹ Lund, G., and B. F. Dodge, *Ind. Eng. Chem.*, **40**, 1019-1032 (1948).

² Schack, *op. cit.*

a cold brick checkerwork to which they give up a large portion of their heat. In a matter of a few minutes the bricks become very hot, and at some optimum surface temperature the flow of waste gases is replaced by the passage of cold fuel-air mixture on its way to the furnace. This mixture is heated while cooling the brickwork back to a low temperature. In the meantime the flow of the waste gases continues over another brickwork only to be interrupted by the flow of more fuel-gas-air mixture. By employing a sufficient number of brick checkerworks it is possible to operate on a continuous basis with fuel-gas-air mixtures alternating with the waste gases in the transfer of heat.

It can be seen that the inlet temperature of either the waste gases or the fuel-gas-air mixture to the checkerwork is substantially constant but the outlet temperatures vary. This differs from preceding time-temperature relationships. The cyclic operation of a regenerator consists of two separate periods. During the heating period the brickwork absorbs heat, and during the cooling period the heat is removed, although the periods are usually of unequal duration. To simplify regenerator terminology the conventions of Schack and others will be followed. In these, gas refers to the hot medium (waste gas) and air to the cold medium. Heilgenstadt¹ has shown that the heat transfer in a generator can be represented by a modified form of the Fourier equation for 1 ft² of surface by

$$\frac{dq'}{dA} = H(T - t) \quad \text{Btu/}(\text{period})(\text{ft}^2) \quad (18.71)$$

where q' is the heat transferred during the period from the gas to air whose mean temperatures are T and t , respectively. H is the overall coefficient of heat transfer for the *period*, Btu/(period)(ft²)(°F), and differs from U in the unit of time. Integrating, since $A = 1$, Eq. (18.71) becomes

$$q' = H(T - t) \quad \text{Btu/}(\text{period})(\text{ft}^2) \quad (18.72)$$

Using θ_H and θ_C to indicate the duration of the gas and air half periods in hours, respectively, the heat transferred during the half periods is

$$q' = h_H(T - t_s)\theta_H = h_C(t_s - t)\theta_C \quad (18.73)$$

where t_s is the mean temperature of the brickwork, which is assumed to be identical for both half-periods in an ideal regenerator and h_H and h_C are the gas and air film coefficients, Btu/(hr)(ft²)(°F). Solving for an ideal regenerator,

$$t_s = \frac{h_H T \theta_H + h_C t \theta_C}{h_H \theta_H + h_C \theta_C} \quad (18.74)$$

¹ Heilgenstadt, W., *Mitt. Warmestelle Ver. deut. Eisenh.*, **73**, (1925).

From Eqs. (18.72) and (18.73)

$$H(T - t) = h_H T \theta_H - h_c t \theta_c \quad (18.75)$$

The value of H in Eq. (18.72) is thus the value for the ideal regenerator or H_{id} . Substituting in Eq. (18.73),

$$H_{id} = \frac{1}{1/h_H \theta_H + 1/h_c \theta_c} \quad (18.76)$$

In an actual regenerator the temperature of the gas decreases several times more rapidly than the air is heated. If T_s is the mean temperature of the surface during cooling only and t_s the mean during heating, the solution of Eq. (18.73) gives

$$H = H_{id} \left(1 - \frac{t_s - T_s}{T - t} \right) \quad (18.77)$$

Equation (18.71) can also be expressed in terms of the mean temperature of the brickwork.

$$dq' = h_H(T - t_s) dA d\theta \quad \text{Btu/(\text{period})(ft}^2\text{)} \quad (18.78)$$

Considering the brick as a slab of finite thickness whose temperature varies periodically on both sides, Grober has defined a heat storage factor γ which is the ratio of the heat actually stored in the wall to that which could have been stored if the conductivity of the wall were infinite or

$$q' = q'_{\max} \gamma$$

where γ is $f_5(4\alpha\theta/l^2)$. Values of $f_5(4\alpha\theta/l^2)$ are plotted in Fig. 18.20. In terms of the brick material the weight of the brickwork for 1 ft² of heat-transfer surface is M/A , where the depth is one-half the thickness of the brick. If c_s is the specific heat of the solid, the heat absorbed by the brickwork is then

$$dq' = Mc_s \gamma dt_s dA \quad \text{Btu/(\text{period})(ft}^2\text{)} \quad (18.79)$$

Equating (18.78) and (18.79),

$$Mc_s \gamma dt_s = h_H(T - t_s) d\theta \quad (18.80)$$

or rearranging,

$$\frac{dt_s}{d\theta} = \frac{h_H}{Mc_s \gamma} (T - t_s)$$

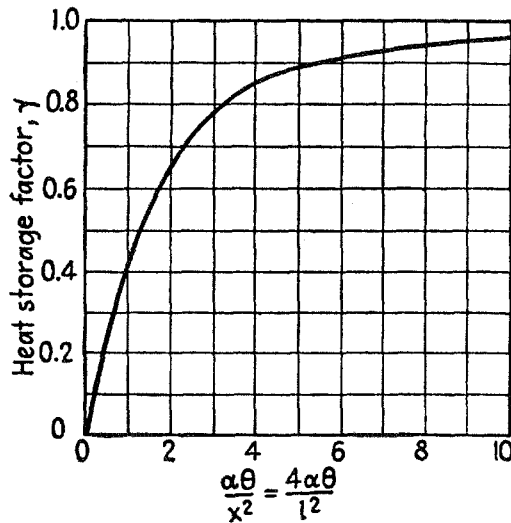


FIG. 18.20. Heat storage factor.

Differentiating with respect to θ ,

$$\frac{d^2 t_s}{d\theta^2} = \frac{h_H}{Mc_s \gamma} \frac{dT}{d\theta} - \frac{h_H}{Mc_s \gamma} \frac{dt_s}{d\theta} \quad (18.81)$$

Solving for $dT/d\theta$,

$$\frac{dT}{d\theta} = \frac{dt_s}{d\theta} + \frac{Mc_s \gamma}{h_H} \left(\frac{d^2 t_s}{d\theta^2} \right)$$

from which it will be seen that the temperature of the gas is not only dependent upon the temperature of the wall but also upon its rate of change with time. From Eq. (18.73) and based on 1 ft² of surface,

$$q = h_H(T - t_s) \quad \text{Btu/hr}$$

in which q is the quantity of heat absorbed per hour under the potential difference $T - t_s$. Differentiating with respect to θ and substituting in Eq. (18.81),

$$\frac{dT}{d\theta} = \frac{h_H}{Mc_s \gamma} (T - t_s) + \frac{d(T - t_s)}{d\theta} \quad (18.82)$$

Integrating for a finite heating period θ ,

$$\Delta T = \frac{q_{av} \theta}{Mc_s \gamma} + \frac{1}{h_H} (q_\theta - q_0) \quad (18.83)$$

where q_0 is the number of Btu transferred per hour at the start of the period and q_θ represents the transfer after time θ . If $T_{2,\theta=0}$ is the waste-gas exit temperature at the start of the period (the time for travel from inlet to outlet being negligible) and $T_{2,\theta=\theta}$ is the gas exit temperature at a finite time after the period has commenced, at the start

$$q_0 = \frac{WC}{A} (T_1 - T_{2,\theta=0}) \quad \text{Btu/(hr)(ft}^2\text{)} \quad (18.84)$$

and after θ hr

$$q_\theta = \frac{WC}{A} (T_1 - T_{2,\theta=\theta}) \quad (18.85)$$

where the inlet gas temperature T_1 is constant.

$$q_\theta - q_0 = \frac{WC}{A} (T_{2,\theta=0} - T_{2,\theta=\theta}) \quad \text{Btu/(hr)(ft}^2\text{)} \quad (18.86)$$

Substituting in Eq. (18.83),

$$\Delta T = \frac{q_{av} \theta}{Mc_s \gamma \left(1 - \frac{WC}{h_H A} \right)} \quad (18.87)$$

Although data are available on the heat-transfer coefficients to actual regenerators, they can be found roughly from Fig. 24 for smooth surfaces by using the equivalent diameter defined by Eq. (6.3) in place of D .

2d. HEAT TRANSFER TO GRANULAR MATERIALS IN BEDS

This case is of particular interest in processes employing fixed- and moving-bed catalysts. It can also be used to obtain coefficients for combination with regenerator equations when using granular materials. An excellent review of the literature on this subject has been given by Lof and Hawley.¹ Perhaps the most ambitious original undertaking in this field was by Schumann,² who formulated and solved the equations for the case of an incompressible fluid flowing through a bed of solids with infinite thermal conductivity. Schumann assumed that (1) any given particle could be considered to attain a uniform temperature throughout at any given instant, (2) the resistance to conduction in the solid itself was negligible, (3) the rate of heat transfer from fluid to solid at any point was proportional to the temperature difference between the two at the point, (4) the change in the volumes of fluid and solid with the temperature variation was negligible, and (5) thermal properties were independent of the temperature during a temperature-change cycle. Schumann calculated temperature-time curves in terms of the groups

$$\frac{T - t_0}{T_1 - t_0} \quad \frac{h'x}{C'v} = \frac{hx}{C'G} \quad \frac{h'\theta}{c'_s} (1 - f')$$

where

T = gas temperature at any time, °F

T_1 = inlet gas temperature, °F

t = solid temperature at any time, °F

t_0 = initial temperature of the solid bed, °F

C' = specific heat of unit volume of gas at constant pressure, Btu/(ft³) (°F)

c'_s = specific heat of unit volume of solids, Btu/(ft³) (°F)

f' = fraction of voids in the solid bed

G = mass velocity of fluid, lb/(hr)(ft²)

h' = volumetric heat-transfer coefficient, Btu/(hr)(ft³) (°F)

v = average volumetric fluid rate through bed, ft³/(hr)(ft² of bed section)

x = length of bed, ft

¹ Lof, G. O., and R. W. Hawley, *Ind. Eng. Chem.*, **40**, 1061-1070 (1948).

² Schumann, T. E. W., *J. Franklin Inst.*, **208**, 405 (1929).

Schumann's curves were later extended by Furnas,¹ and the data of both are given in Fig. 18.21. With a temperature-time curve of this form Furnas experimentally developed the equation for the volumetric heat-transfer coefficient.

$$h' = \frac{A' G^{0.7} T'^{40.31} 10^{0.68-3.56/f'^2}}{D_e'^{0.9}} \quad (18.88)$$

where T' = average of the inlet air and initial bed temperature, °F

A' = constant

D_e' = equivalent spherical diameter defined by Eq. (18.89), ft

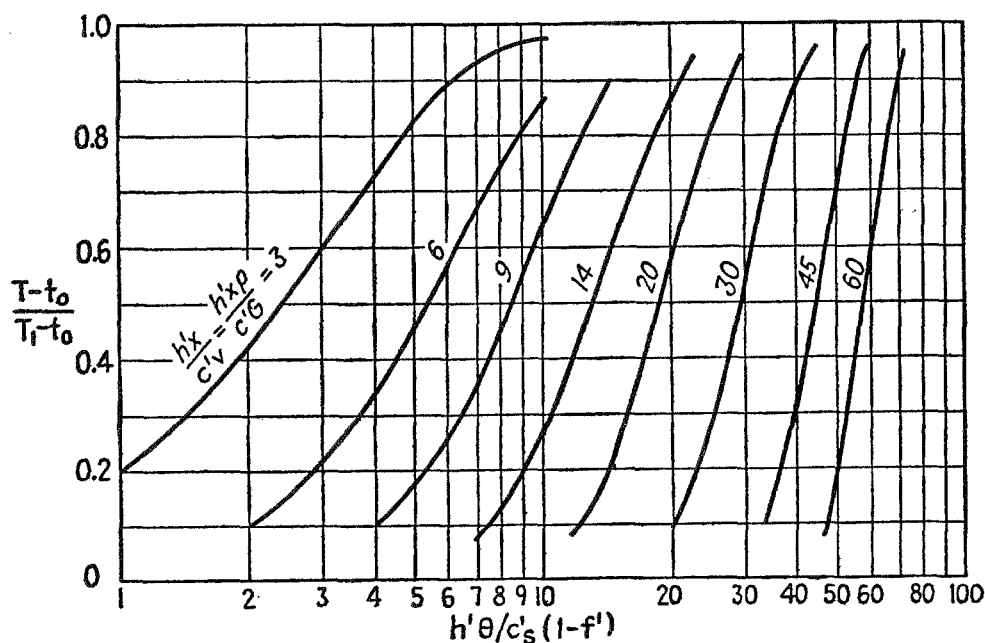


FIG. 18.21. Schumann curves. (Lof and Hawley, *Industrial Engineering Chemistry*)

$$D_e' = \left(\frac{6 \times \text{net volume of particles}}{\pi \times \text{number of particles}} \right)^{1/3} \quad (18.89)$$

Lof and Hawley carried out experiments on granitic gravel ranging in size from 4 mesh to 1.5 in. and air rates from 12 to 66 standard cubic feet of air per minute per square foot of gross cross-sectional area. The thickness of the bed was 36 in. They obtained the following dimensional equation

$$h' = 0.79 \left(\frac{G}{D_e'} \right)^{0.7} \quad (18.90)$$

Because the volumetric heat-transfer coefficient has not been introduced before, Table 18.1, computed by Lof and Hawley, is included for 0.394 in. particle diameters and an air rate of 50 standard cubic feet per minute per square foot, an inlet temperature of 200°F, and normal voids. The

¹ Furnas, C. C., *Trans. AIChE*, **24**, 142 (1930); *U.S. Bur. Mines Bull.* 361 (1932).

TABLE 18.1. SOLID BED COEFFICIENTS

Material	h' , Btu/(hr)(ft ²)(°F)
Iron ores.....	406-1200
Limestone.....	460
Coke.....	347
Blast-furnace charge.....	260
Coal.....	400
Firebrick.....	534
Iron balls.....	668
Gravel.....	399

authors suggest that Eq. (18.90) may be freely extrapolated to higher temperatures where the coefficient is increased by radiation.

The following problem has been adapted from Lof and Hawley.

Example 18.9. Calculation of the Length of a Bed. A bed of granitic-type gravel consisting of 1.0 in. particles has 45 per cent voids and is initially at a uniform temperature of 50°F. 60 lb/(hr)(ft²) cross-sectional area of hot air at 200°F is available for periods of 6 hr. It is considered wasteful from a heat-recovery standpoint to let the exit air leave the bed at a temperature above 90°F. How long a bed can be heated by the air stream?

Data:

Specific heat of gravel, 0.25 Btu/(lb)(°F) or 41.3 Btu/(ft³)(°F)

Actual density of gravel 165 lb/ft³

Specific heat of air 0.0191 Btu/(ft³)(°F)

Solution:

$$h' = 0.79 \left(\frac{G}{D_e} \right)^{0.7} = 0.79 \left(\frac{60}{1/12} \right)^{0.7} = 79.4$$

$$\frac{h'\theta}{c'_s(1-f')} = \frac{79.4 \times 6}{41.3 \times 0.55} = 21.0$$

$$\frac{T - t_0}{T_1 - t_0} = \frac{90 - 50}{200 - 50} = 0.267$$

From Fig. 18.21,

$$\frac{hx\rho}{c'G} = 24.5 \quad \rho = 0.0807 \text{ lb/ft}^3 \text{ air}$$

$$x = 24.5 \times \frac{0.0191 \times 60}{79.4 \times 0.0807} = 4.4 \text{ ft}$$

PROBLEMS

18.1. 6000 gal of liquid toluene is to be cooled from 300 to 150°F using water from 85 to 120°F. A pump in the tank is capable of circulating 30,000 lb/hr of toluene through an external exchanger. Available for the service is 400 ft² of heat-exchanger surface in various types, all affording approximately the same overall coefficient of 75 Btu/(hr)(ft²)(°F/ft).

(a) How long will it take to cool an agitated batch using, respectively, a double pipe exchanger, 1-2 exchanger, and 2-4 exchanger?

(b) How long will it take to cool a nonagitated batch using the same exchangers as in (a)?

(c) If the batch is cooled to 120°F, how long will it take in (a) and (b)?

(d) If the pumping rate is increased to 40,000 lb/hr and the coefficient increased to 90, how long will it take in (a) and (b)?

18.2. 6000 gal of liquid aniline is to be cooled from 300 to 150°F using water from 85 to 120°F. A pump in the tank is capable of circulating 30,000 lb/hr of aniline through an external exchanger. The overall clean coefficient is 40 Btu/(hr)(ft²)(°F).

(a) How much surface is required to accomplish the operation in 2 hr using double pipe, 1-2 and 2-4 exchangers with both agitated and nonagitated batches?

(b) How much surface is required for (a) if the time is reduced to 1 hr?

18.3. Cylindrical steel objects 4 in. long by 6 in. in diameter and at 1000°F are to be quenched by being dropped into a tank filled with recirculating water at a constant temperature of 100°F. The average overall coefficient on all faces will be about 75.

(a) How long will it take for the center of the cylinder to reach a temperature of 125°F?

(b) How long will it take for a depth 1 in. below the surface on the long axis to attain a temperature of 500°F?

18.4. A water main at 50°F passes through a mass of concrete at the same temperature, the whole being exposed to the atmosphere. The main is located 24 in. below the surface of the concrete. If the temperature were to fall to 0°F, how long would it take for the pipe to be in danger of freezing? Use $k = 0.65$ Btu/(hr)(ft²)(°F), $c = 0.20$, and $\rho = 140$ lb/ft³.

18.5. A steel slab 30 in. thick and at an initial temperature of 100°F is suddenly subjected to an air stream with a temperature of 300°F on both sides and a coefficient to steel of 5.0. What is the temperature distribution after $\frac{1}{2}$ hr?

18.6. The local daily range of temperatures on a concrete roof is between 80 and 40°F. Assume that the roof is 12 in. thick and that the data in Example 18.4 apply.

(a) What variations occur 6 in. below the surface?

(b) What is the time lag?

(c) What is the temperature 4 hr after the maximum has been reached?

NOMENCLATURE FOR CHAPTER 18

A	Heat-transfer surface, ft ²
C_0	Specific heat of hot fluid added to batch, Btu/(lb)(°F)
C	Specific heat of hot fluid, Btu/(lb)(°F)
C_1, C_2, C_3	Constants
c	Specific heat of cold fluid, Btu/(lb)(°F)
c_0	Specific heat of cold fluid added to batch, Btu/(lb)(°F)
c_s	Specific heat of solid, Btu/(lb)(°F)
C'	Specific heat of unit volume of gas, Btu/(ft ³)(°F)
c'_s	Specific heat of unit volume of solids, Btu/(ft ³)(°F)
f	Number of complete periodic temperature changes per hour, 1/hr
f'	Fraction of voids in solids, dimensionless
f_1, f_2 , etc.	Abbreviations for various functions by which Y is expressed
H	Overall coefficient or heat transfer per period, Btu/(period)(ft ²)(°F)
h	Heat-transfer coefficient, Btu/(hr)(ft ²)(°F)
h'	Volumetric heat-transfer coefficient, Btu/(hr)(ft ³)(°F)
h_H, h_C	Heat-transfer coefficient during heating and cooling half periods, respectively, Btu/(hr)(ft ²)(°F)

h_c, h_r	Heat-transfer coefficient by convection and radiation, respectively, Btu/(hr)(ft ²)(°F)
K_1, K_2 , etc.	Constants in batch heating and cooling equations, dimensionless
k	Thermal conductivity, Btu/(hr)(ft ²)(°F/ft)
L_0	Rate of liquid addition to a batch, lb/hr
L_1, L_2	Charge and residual quantities of liquid in batch distillation, mol
l	Length or width of object or diameter of cylinder, ft
M	Weight of batch liquid, lb; weight of solid, lb
m	Number of time increments, $\Delta\theta$
N	Number of circulations
n	Number of distance increments, Δx
Nu	Nusselt number, $hl/2k$
Q	Heat flow, Btu/hr
Q'	Heat, Btu
q	Heat flow during a heating or cooling period, Btu/(hr)(ft ² of regenerator surface)
q'	Heat flow during a heating or cooling period, Btu/(period)(ft ² of regenerator surface)
R	Temperature group $w_c/WC = (T_1 - T_2)/(t_2 - t_1)$, dimensionless
S	Temperature group $(t_2 - t_1)/(T_1 - t_1)$, dimensionless
T	Temperature of hot fluid or batch at any temperature or time, °F
T_0	Temperature of hot fluid continuously added to batch, °F
T_s	Temperature of surrounding medium, °F
T_1, T_2	Initial and final temperatures of hot fluid, °F
t	Temperature of cold fluid, batch or solid, °F
t_f	Temperature of solid surface in the Schmidt method, °F
t_s	Temperature of brickwork in a regenerator, °F
t_n, t_m	Temperature of solid after n distance and m time increments, °F
t_0	Original temperature of solid, °F
t_{OM}	Maximum surface temperature, °F
Δt	True temperature difference, °F
$\Delta t_\theta, \Delta t_x$	Temperature difference after time θ and distance x , respectively, °F
$\Delta t_{n,m}$	Temperature difference after n distance and m time increments, °F
U	Overall coefficients of heat transfer, Btu/(hr)(ft ²)(°F)
v	Average volumetric fluid rate through bed, ft ³ /(hr)(ft ² of bed cross section)
W	Weight flow of hot fluid, lb/hr
w	Weight flow of cold fluid, lb/hr
x	Mol fraction of volatile component in liquid, dimensionless; distance, ft
x_1, x_2	Initial and final mol fractions of volatile component in liquid, dimensionless
Y	Temperature group $(T_s - t)/(T_s - t_0)$, dimensionless
Y_x, Y_y, Y_z	Temperature group Y in the directions x, y , and z , respectively, dimensionless
y	Mol fraction of volatile component in vapor, dimensionless; distance, ft
z	Function in the probability integral; distance, ft
α	Thermal diffusivity, $k/c\rho$, ft ² /hr
γ	Heat-storage factor, dimensionless
ϵ	Emissivity, dimensionless
θ	Time, hr

ρ	Density, lb/ft ³
σ	Radiation constant, 0.173×10^{-8} Btu/(hr)(ft ²)(°R ⁴)

Superscripts and Subscripts (except as noted above)

0	Time or distance at start
p, q	Constants
x, y, z	Direction
θ	At time θ

CHAPTER 19

FURNACE CALCULATIONS

BY JOHN B. DWYER¹

Introduction. The most important commercial applications of radiant-heat-transfer calculations are encountered in the design of steam-generating boilers and petroleum-refinery furnaces. Since the art of the construction of these units developed before the theory, empirical methods were evolved for the calculation of radiant-heat transfer in such furnaces. Various contributions² to the literature on general and specific radiant-heat transfer problems, especially those of H. C. Hottel, have made possible a more fundamental approach to furnace design. Several semi-theoretical methods for furnace radiant-section heat-transfer calculations are now available. Often these methods can be adapted to the rapid solution of problems encountered in kilns, ovens, heat-treating and chemical furnaces, and miscellaneous equipment in which radiant-heat transfer is of importance.

It is the purpose of this chapter to present some of the empirical and semitheoretical methods of furnace radiant-section calculation, data for their use, and examples of their application. The limitations of these methods are pointed out, and their adaptability to miscellaneous heat-transfer problems indicated. Included is a brief description of several types of boilers and oil heaters currently in use. A discussion of the theoretical aspects of radiation from nonluminous gases is presented to illustrate the general scope of the problem, and the simplifications and assumptions used in reducing theories to practice are also pointed out.

While it is necessary to calculate the radiant-heat-transfer flux in order to design a furnace, many other factors often influence furnace arrangement, such as the permissible flux under various conditions and the extent and nature of boiler tube slagging on the efficiency of the surface. The precautions which must be taken to avoid the deposition

¹ The M. W. Kellogg Company. Acknowledgment is made of critical suggestions by W. E. Lobo of the same company.

² Perry, J. H., "Chemical Engineers' Handbook," 3d ed., pp. 483-498, McGraw-Hill Book Company, Inc., New York, 1950. McAdams, W. H., "Heat Transmission," Chap. 3 by H. C. Hottel, McGraw-Hill Book Company, Inc., New York, 1942. For additional references see McAdams, *op. cit.*, p. 430.

of coke in oil-heating or vaporizing furnaces have a pronounced effect on the actual design of refinery furnaces. The art of furnace design, in fact, often far exceeds the importance of the calculations.

Steam-generating Boilers. There are two general types of steam-generating boilers, the *fire-tube* boiler and the *water-tube* boiler. The former consists of a cylindrical vessel having tubes passing through it which are rolled into the heads at each end of the vessel. The tube bundle is generally horizontal, and the upper section of the vessel is not tubed. Combustion gases pass through the tubes, and a water level is carried in the vessel to immerse the tubes completely but at the same time allowing disengaging space between the water level and top portion of the vessel. In occasional vertical-tube boilers of this type, the tubes must be immersed in water for that portion of their length required to reduce the temperature of the gases sufficiently to avoid overheating the uncooled upper portion of the tubes. Some of the water-cooled parts such as shell or tube sheets can be subjected to radiation from the combustion gases, since these parts may form a portion of the enclosure of the combustion chamber. The major mechanism of heat transfer from the gases to the tubes is convection. Fire-tube boilers seldom exceed 8 ft in diameter, and steam pressure is generally limited to 100 to 150 psig. Fire-tube boilers are used for low-capacity services up to 15,000 to 20,000 lb/hr of steam production for domestic, industrial, and process heating and for small-scale power generation as in locomotives, etc. Fuels employed may be coal, oil, or gas, and in some cases, local combustibles such as wood, sludge, etc., are used.

Water-tube boilers, as their name implies, have water within their tubes. Combustion of stoker or pulverized coal and coke or of gas or oil fuels provides radiation to the boiler tubes, and further heat transfer is accomplished by arranging the flow of hot gases over the tubes to provide convection-heat transfer. There are three important classifications of water-tube boilers: longitudinal drum, cross-drum straight tube, and cross-drum bent tube. The last is the most important of the three and will be discussed here briefly. Information on the other types can be found in Gaffert.¹

Figure 19.1 shows a typical low-pressure boiler designed to generate 200,000 lb/hr of steam at 235 psig and 500°F. Since the saturation temperature at this pressure is only 401°F, 99°F of superheat is required. Only a small superheater is necessary, since the superheat duty is about 5 per cent of the total boiler duty. The radiant boiler tubes cover the entire wall and roof surface, forming a "water wall" by means of which

¹ Gaffert, G. A., "Steam Power Stations," 3d ed., McGraw-Hill Book Company, Inc., New York, 1946.

the temperature of the refractory walls is kept down, thus decreasing their maintenance. Often the water tubes are partially embedded in the walls. The radiant-section furnace walls sometimes are protected from overheating by circulating cooling air outside them. In the furnace shown, water is fed by gravity from the upper drums to headers at the bottom end of the water wall tubes on all four radiant walls. Circula-

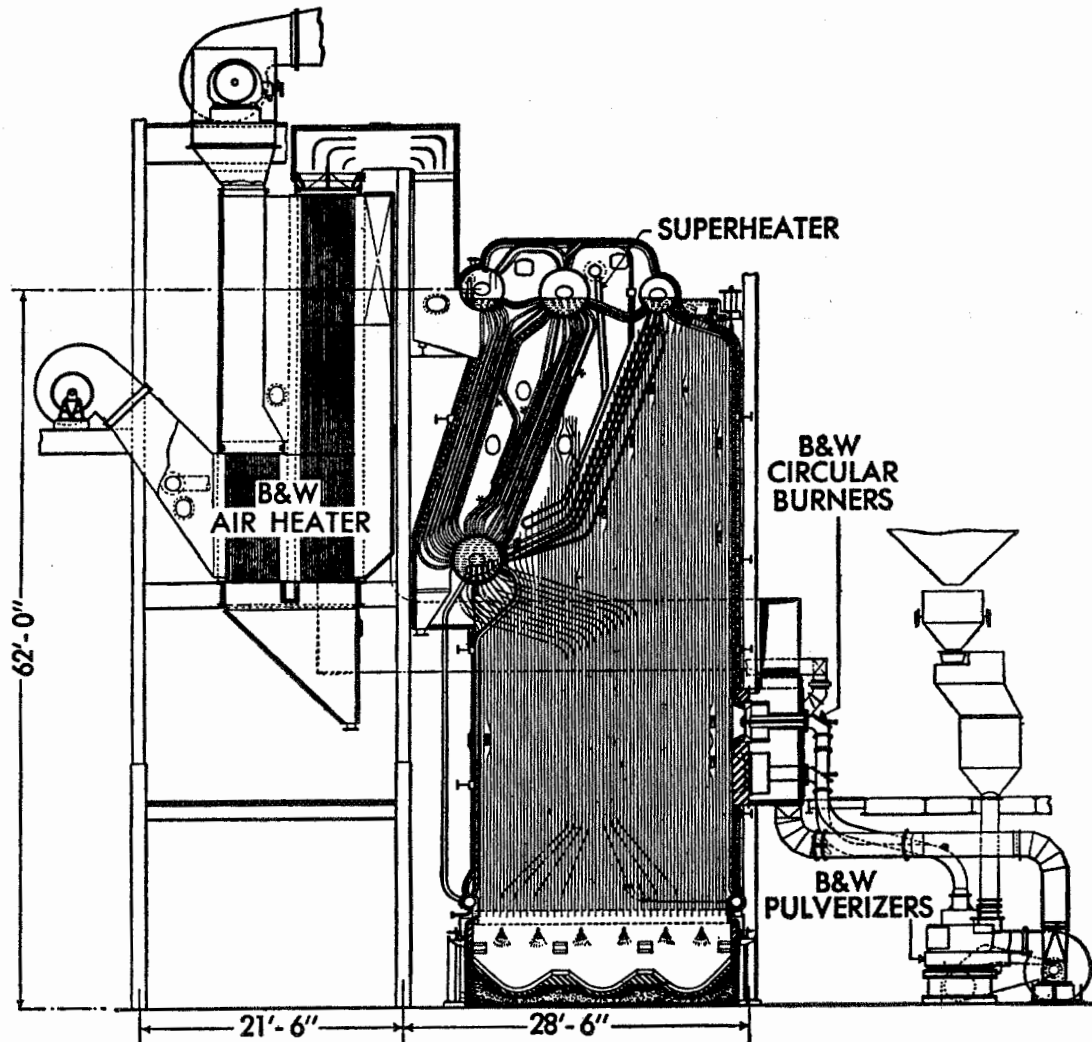


FIG. 19.1. Pulverized-coal-fired low-pressure boiler. (Babcock & Wilcox Company.)

tion is upward through these tubes, and the steam is disengaged from water in the upper drums, then passes through a steam separator before being superheated. In a low-pressure boiler, the convection tubes reduce the flue-gas temperature sufficiently that they proceed directly to the air preheater, obviating the need of an economizer (feed-water preheater). These convection tubes are the bent tubes running from the upper drums to the lower drum. Circulation in these tubes is, in general, downward in the left (cooler) bank and upward through the hotter bank.

A typical power-generating steam boiler is shown in Fig. 19.2. It has a capacity of 450,000 lb/hr of 900 psig steam delivered at 875°F. Since

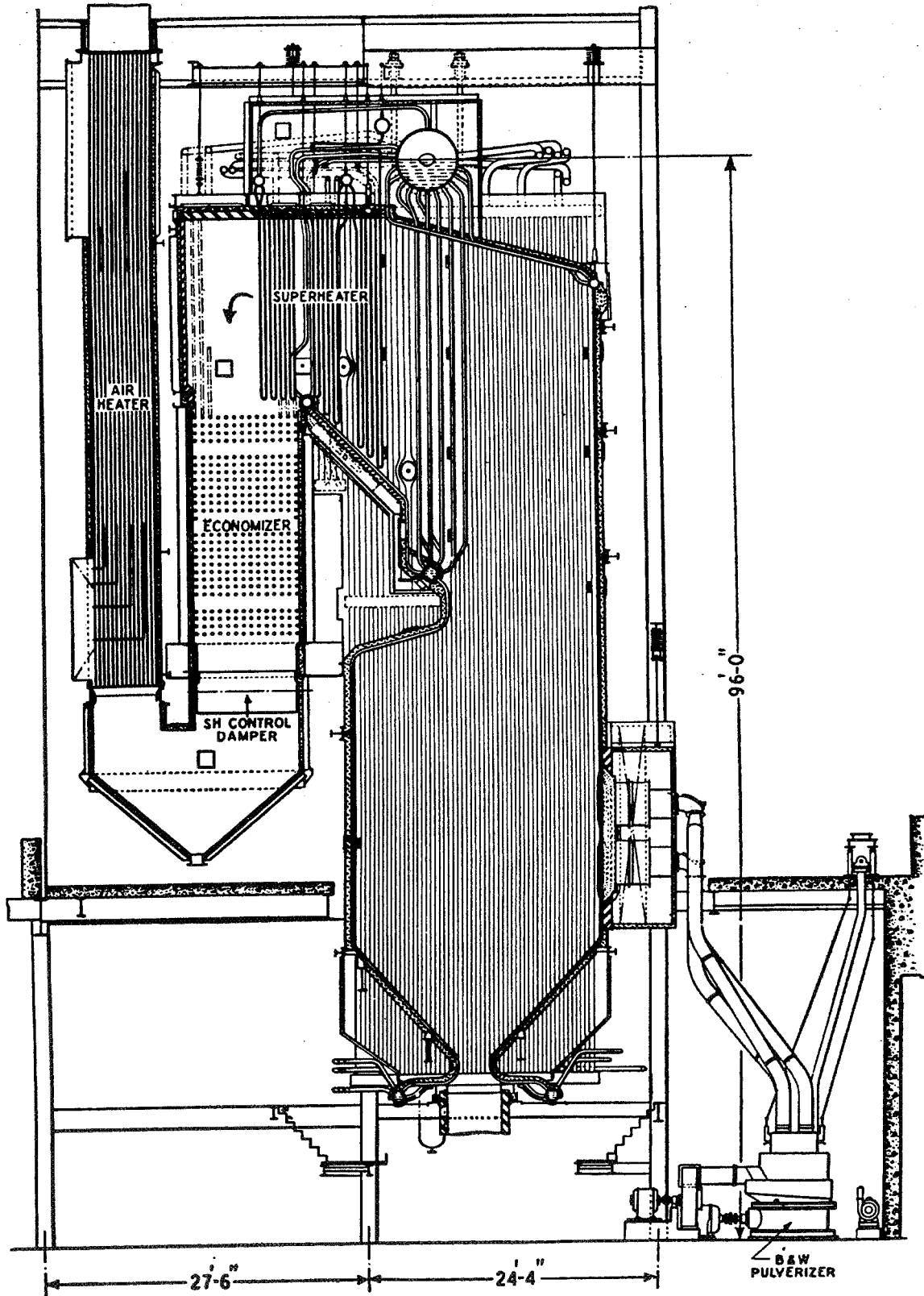


FIG. 19.2. Pulverized-coal-fired power-generation radiant boiler. (Babcock & Wilcox Company.)

the saturation temperature at 900 psig is 532°F, considerable superheat duty is required. Very little boiler convection surface can be placed between the radiant boiler and the superheater, since high-temperature combustion gases must be used to attain the required superheat temperature level with a reasonable amount of superheater tube surface. Since feed water must be brought essentially to the saturation temperature before it is admitted to the boiler drum, considerable heat is absorbed in the economizer section wherein the feed water is preheated, and the thermal efficiency of the unit is further increased by preheating the combustion air with the flue gases before they are sent to the stack.

Petroleum-refinery Furnaces. In atmospheric and vacuum crude distillation, thermal cracking, and modern high-temperature gas processing, the direct-fired tubular furnace is the primary factor in the refinery unit. Furnaces also are widely employed in various heating, treating, and vaporizing services. Refinery furnaces of various types are required for handling fluids at temperatures as high as 1500°F and at a combination temperature and pressure as severe as 1100°F and 1600 psig.

Oil or gas fuels are used exclusively in these furnaces, although in the near future the need may develop for firing with by-product petroleum coke. In general the thermal efficiency of refinery furnaces is considerably less than that of large boilers, since in many cases fuel has little worth in the refinery. With the trend toward utilization of a greater percentage of the crude oil produced, fuel is becoming scarcer and more valuable and refiners are recognizing the need for higher thermal efficiencies. It is expected that the range of thermal efficiencies will rise from 65 to 70 per cent employed in the past to 75 to 80 per cent in the future.

As in boilers, refinery furnaces usually contain both radiant- and convection-heat-transfer surface. Occasionally only radiant surface is employed for very low-capacity furnaces with duties up to 5,000,000 Btu/hr. Air preheaters have been used to a very limited extent because of the relative unimportance of fuel efficiency; however, even at moderate fuel prices their use generally can be shown to be economical.

In Fig. 19.3 is shown a box-type furnace fired from the end walls of the radiant section. Furnaces of this type might have capacities ranging from 25,000,000 to 100,000,000 Btu/hr heat input to the oil. Radiant tubes cover the side wall, roof, and bridge-wall (partition between radiant and convection section) surfaces. Oil is preheated in the bottom and top rows of the convection bank, then passes through the radiant tubes. After reaching an elevated temperature (900 to 1000°F) it is passed through a large number of convection-section tubes wherein it is main-

tained at a high temperature for sufficient time to accomplish the desired degree of cracking. These convection tubes are called the soaking section. The particular furnace shown employs flue-gas recirculation which serves to increase the convection-section duty and decrease the radiant-section duty. The amount of flue gas recirculated is controlled by two factors, namely, (1) limiting the radiant-section flux to prevent overheating the tubes and causing coke deposition within them and (2) controlling

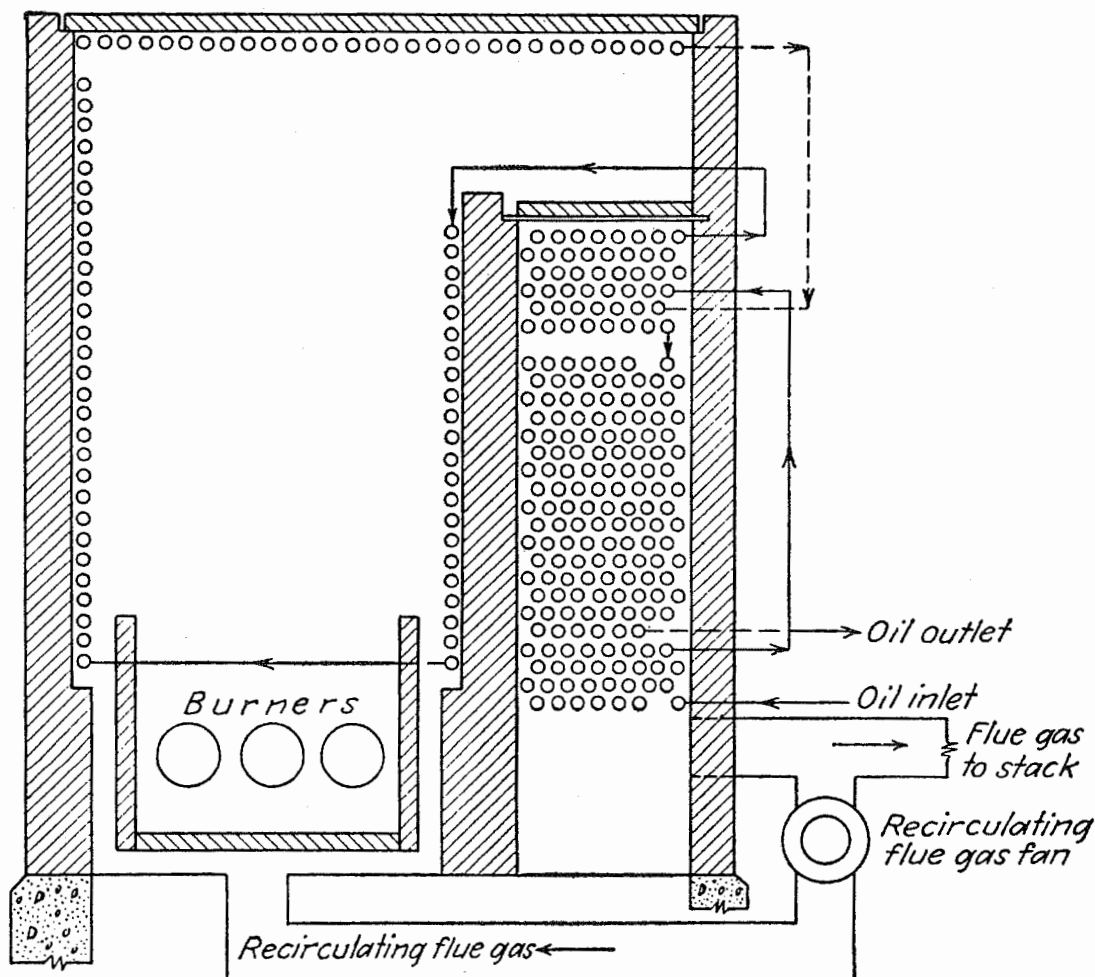


FIG. 19.3. Box-type furnace. (Lobo and Evans, *Transactions AIChE*.)

the temperature gradient through the convection soaker coil. The more nearly constant the oil temperature is held for a given outlet temperature the greater will be the "soaking factor" and extent of cracking. This assumes, of course, that the outlet temperature is the highest oil temperature. The endothermic reaction heat of cracking may result in a condition in which the oil temperature drops from soaker inlet to outlet. Such a temperature drop is inadvisable, particularly in vapor-phase cracking, since polymers formed in the vapor may condense on the tube walls and undergo further cracking to produce coke.

Figure 19.4 shows a De Florez type furnace, which is circular in cross section and employs vertical tubes. All the radiant tubes are equidistant from the burners, ensuring good circumferential heat distribution, but the flux may vary considerably from the bottom of the tubes to the top.

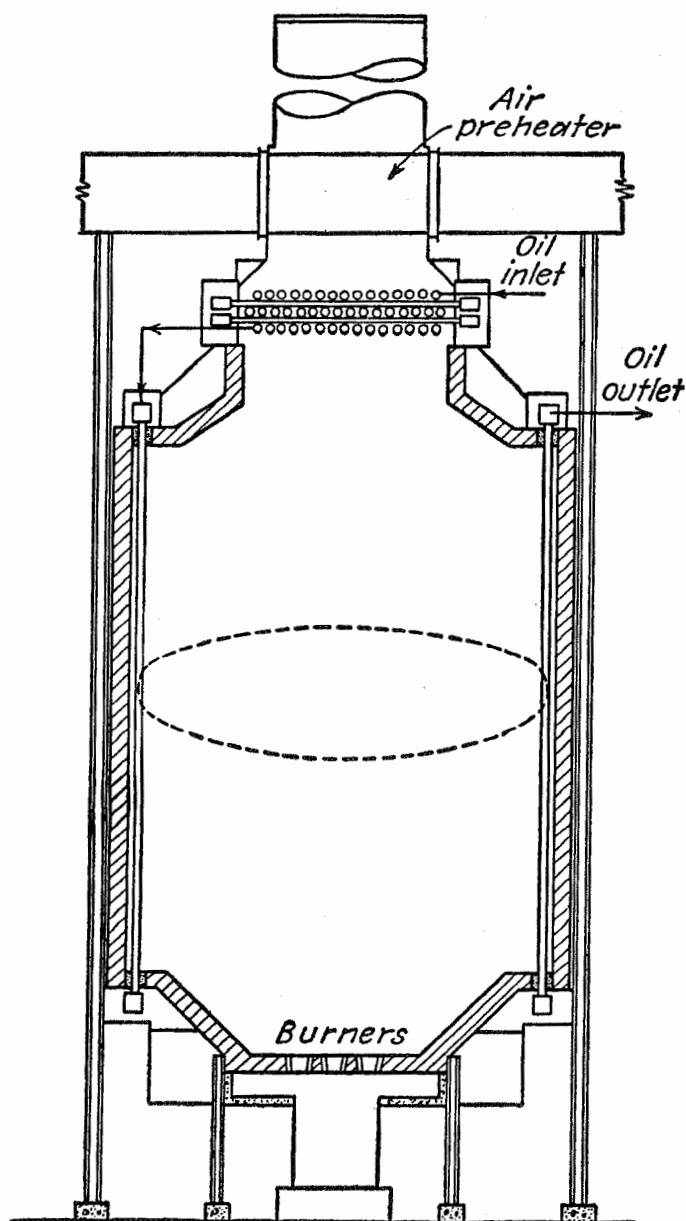


FIG. 19.4. De Florez circular furnace. (Lobo and Evans, *Transactions AIChE*.)

This furnace is fired from the bottom and has so little convection surface that an air preheater is employed to provide good thermal efficiency.

Figure 19.5 shows a double radiant-section box-type furnace. The convection-section tubes and those in one radiant section are employed for one service, while the other radiant section is independently controlled to perform another service. In Fig. 19.6 is an overhead convection-bank box-type furnace with the stack located on top of the con-

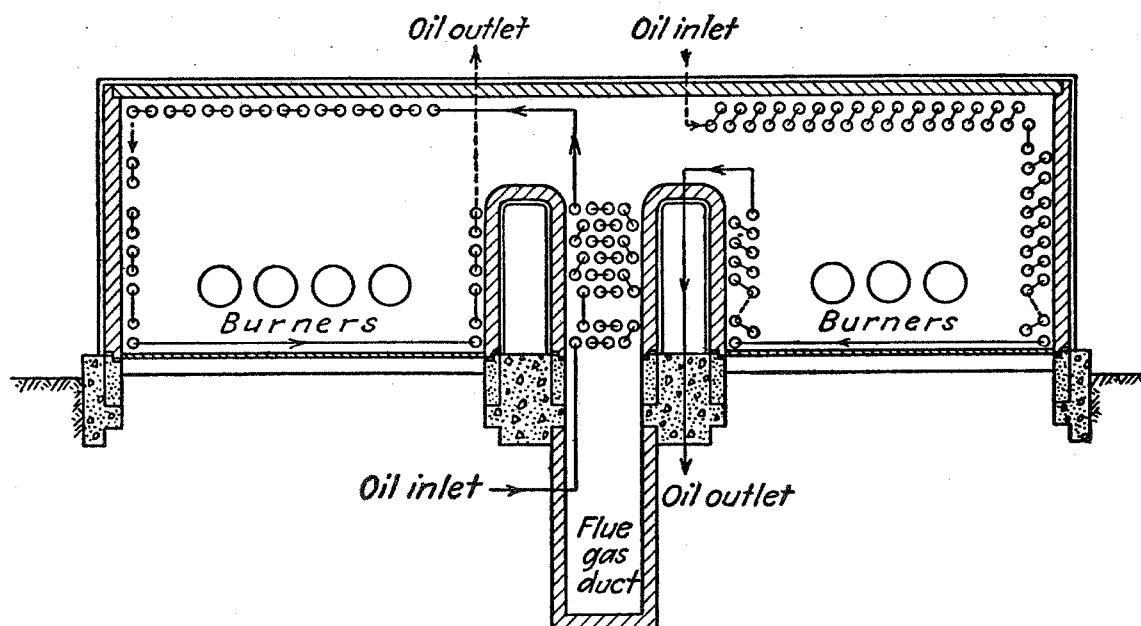


FIG. 19.5. Double-radiant-section box-type furnace. (Lobo and Evans, *Transactions AICHE*.)

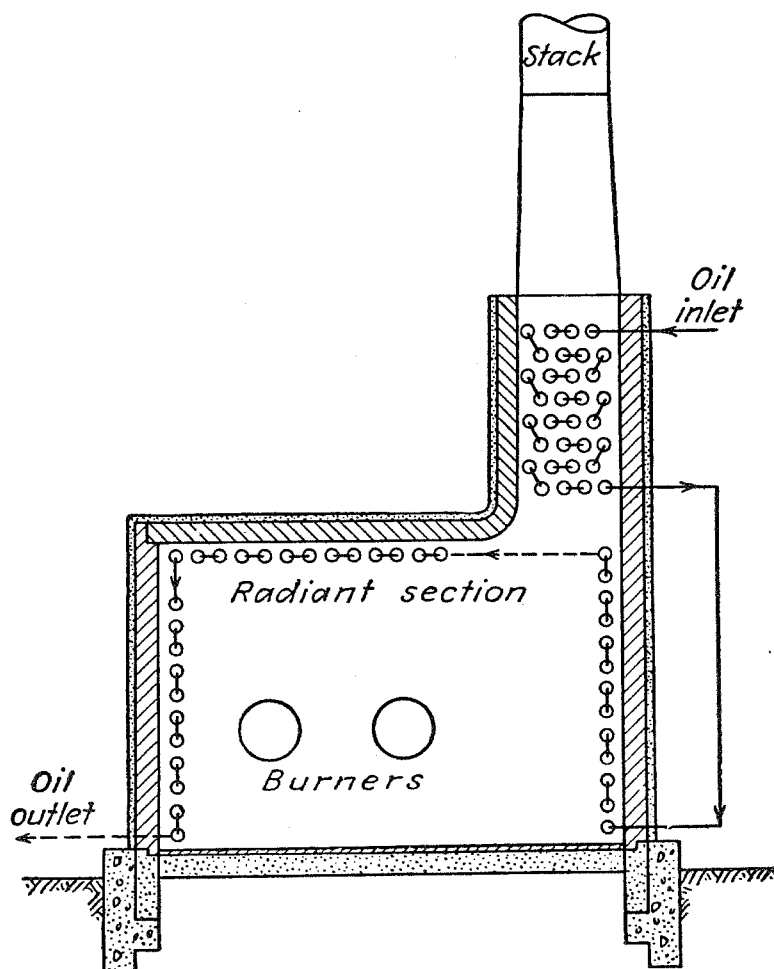


FIG. 19.6. Simple furnace with overhead convection bank. (M. W. Kellogg Co.)

vection bank. Such an arrangement saves on duct work and stack required as compared with a "down-draft" convection-bank arrangement as in Figs. 19.3 and 19.5. Figure 19.7 shows a somewhat similar furnace employing "A" frame type construction, utilizing an inherently stiff structural-steel arrangement to reduce steel costs.

Figure 19.8 presents a modern multiple radiant-section furnace. The convection bank is used for heating two separate oil streams. Each of

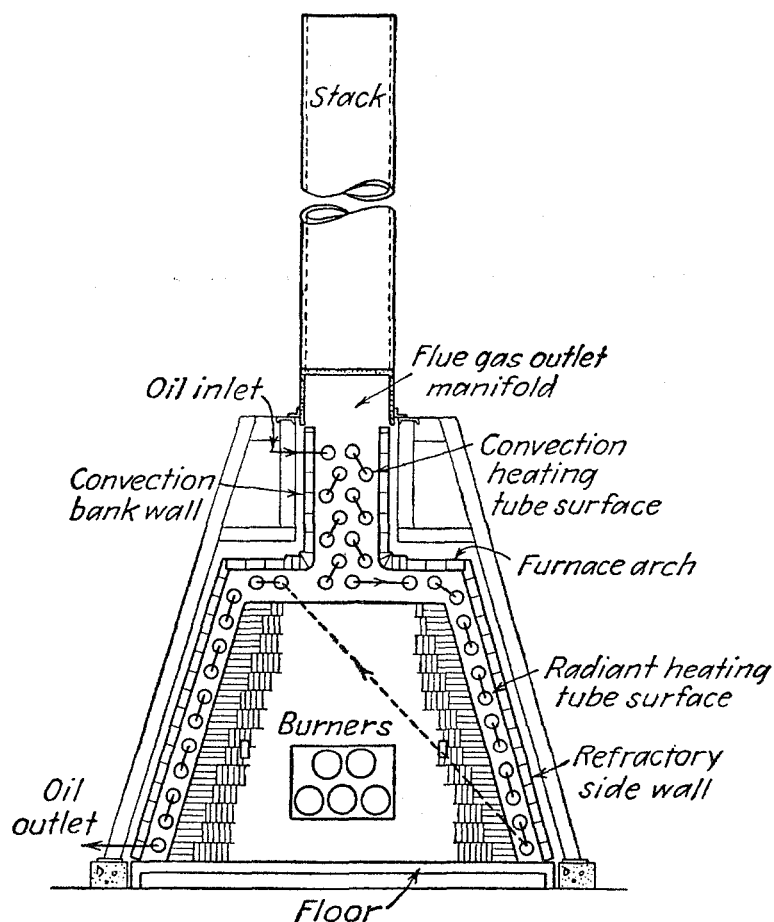


FIG. 19.7. Oil-heating "A" frame furnace. (M. W. Kellogg Co.)

these streams is identically heated in one of the outside radiant "heater" sections, and allowed to soak in one of the middle radiant "soaker" sections. Radiant soakers are preferred to convection soakers because they can be controlled better in so far as heat input is concerned. Furthermore, since the tubes can be seen during operation, any deformation of the tubes can be noted and tube failures with resultant fires avoided. Floor firing of the furnace permits the use of a large number of small burners distributed along the length of the tubes, ensuring even flux distribution. The small burners can be located close to the side-wall or bridge-wall tubes without danger of direct impingement of the burner flames on the tubes. As a result, the cross-sectional dimensions of the

furnace can be reduced and tubes made longer than in a furnace fired from the end walls with large burners. A saving in the number of expensive return bends or "headers" employed can thus be realized.

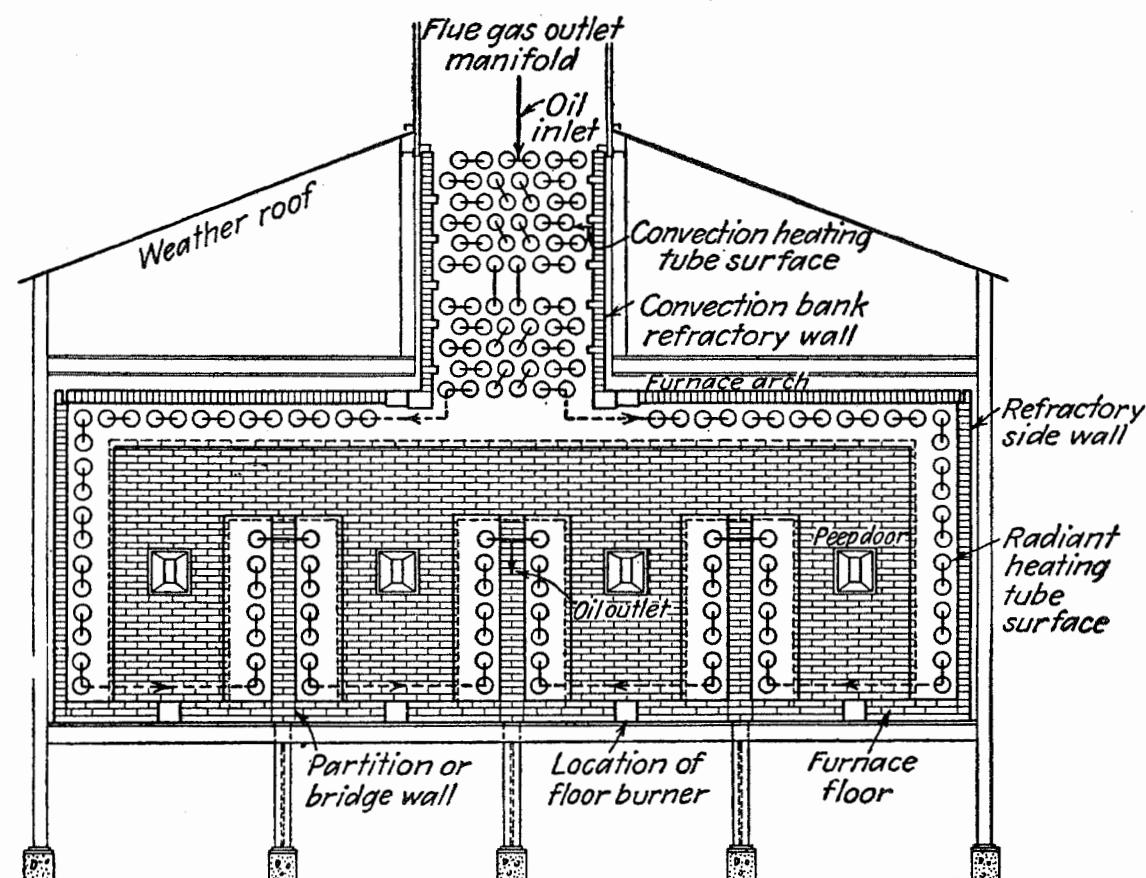


FIG. 19.8. Overhead convection bank oil-heating furnace provided with separately fired radiant sections. (M. W. Kellogg Co.)

Radiant-heat-transfer Factors. As pointed out at the conclusion of Chap. 4, the general equation for radiant-heat transfer can be represented by

$$Q = F_A F_e A \sigma (T_1^4 - T_2^4) \quad (4.42)$$

$$= \mathfrak{F} \sigma A' (T_1^4 - T_2^4) \quad (19.1)$$

where Q = heat flow by radiation alone to A' , Btu/hr

T_1 = temperature of source, °R

T_2 = temperature of sink, °R

$\mathfrak{F}$ = factor to allow for both the geometry of the system and the nonblack emissivities of the hot and cold bodies, dimensionless

A' = effective heat-transfer surface of sink or cold body, ft²

σ = Stefan-Boltzmann constant, 0.173×10^{-8} Btu/(hr)(ft²)(°R⁴)

It is obvious that the application of such an equation to practical engineering problems must incorporate simplifications and assumptions.

It will be helpful to develop the bases for those simplifications and indicate the assumptions.

In general, the furnace consists of a heat receiver or *sink*, a heat *source*, and *enclosing surfaces* (the last being made up in some part by the sink and/or source). While there is complex interaction among these three essential parts, they can be evaluated best in the order given.

Heat Sink. The usual heat receiver for industrial furnaces is composed of a multiplicity of tubes disposed over the walls, roof, or floor of the

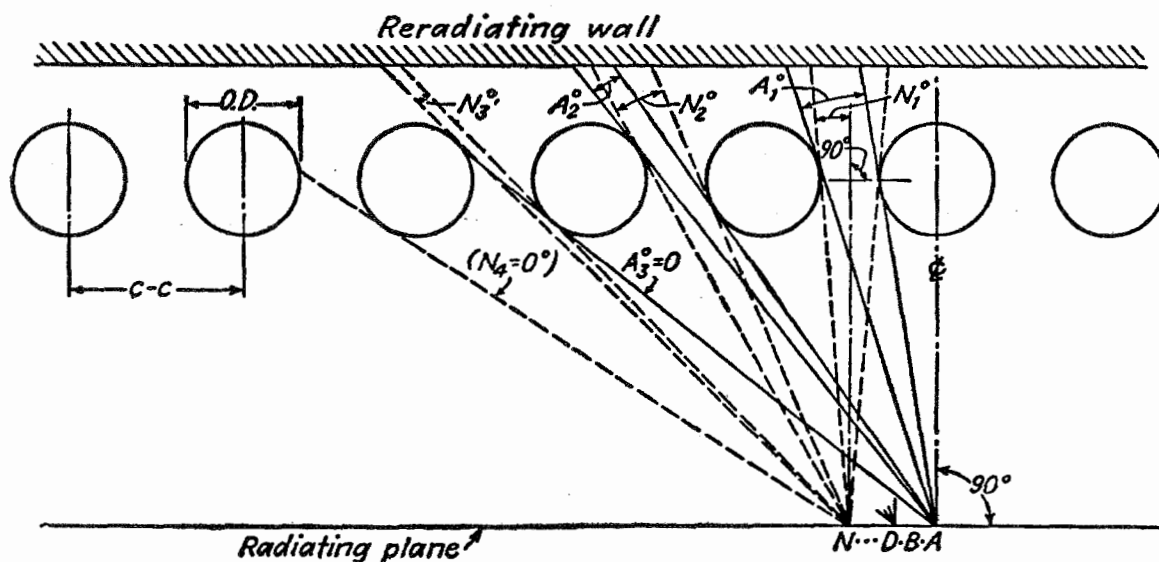


FIG. 19.9. Evaluation of the effectiveness factor α .

$$\alpha_{A, \text{ direct}} = \frac{90^\circ - A_1^\circ - A_2^\circ - A_3^\circ}{90^\circ}$$

$$\alpha_{N, \text{ direct}} = \frac{90^\circ - N_1^\circ - N_2^\circ - N_3^\circ - N_4^\circ}{90^\circ}$$

$$\alpha_{\text{direct}} = \frac{\alpha_{A, \text{ direct}} + \alpha_{B, \text{ direct}} + \dots + \alpha_{N, \text{ direct}}}{N}$$

$$\alpha_{\text{total}} = \alpha_{\text{direct}} + \alpha_{\text{direct}} (1.00 - \alpha_{\text{direct}})$$

Note that $\alpha_{A, \text{ direct}}$, $\alpha_{B, \text{ direct}}$, etc., approach equality as distance between radiating plane and tube row increases.

furnace or located more centrally in the firebox. The most common case is the one in which bare tubes are arranged in a single row in front of a refractory wall. Although there are many rather arbitrary means of evaluating the *effective* heat-transfer surface of such arrangements, a rational development has been proposed by Hottel,¹ which now is used almost exclusively. It already has been stated that the elements of the furnace are best handled individually, and in evaluating the effective or "seen" surface of the tube rows, the assumption is made that the heat source is a radiating plane parallel to the tube row. End effects as described in Chap. 4 are eliminated by assuming both the plane of the

¹ Hottel, H. C., *Trans. ASME*, **53**, 265 (1931).

tubes and the radiating plane to be infinite in extent. All surfaces are assumed to be black.

In Fig. 19.9 is shown a method of evaluating the *effectiveness factor* α by which the surface of a plane replacing the tube row with assumed emissivity of 1.0 must be multiplied to obtain an equivalent (cold) plane surface. The plane replacing the tubes A_{cp} is equal to the number of tubes times length times center-to-center spacing. The first method shown is one suggested by Hottel,¹ which is simple and rapid. It is assumed that any heat loss through the refractory is equal to the heat transferred to the refractory by convection; hence all radiation impinging on the wall is reradiated. It can be seen that, as the tubes are spread farther apart, the fraction of the radiation originating at a point at the source which will be intercepted by the tubes will diminish, hence α will decrease. At the same time A_{cp} per tube is increased. The net effect is an increase in effective surface per tube but a decrease in effective surface per unit surface of furnace wall. The effectiveness of the tube is increased because a greater portion of its circumferential area is irradiated. The radiation not intercepted by the tubes reaches the refractory, from which it is reradiated. If it is assumed that the refractory is at a uniform temperature (not quite true), the radiation leaving the refractory will be intercepted to the same extent as the radiation from the source. Hence the total radiation absorbed by the tube row will be the fraction $[\alpha_{\text{direct}} + \alpha_{\text{direct}}(1 - \alpha_{\text{direct}})]$ of the radiation from the source. The illustration shows that several points along the radiating plane must be investigated (covering only one-half of the tube center-to-center distance because of the symmetry of the system) to obtain the average fraction of interception. Also by symmetry, only the angle between 90 and 180 need be investigated at each point.

A better understanding of the distribution of the radiant rate on the tube circumference is obtained from Fig. 19.10 in which the values are developed from the standpoint of the tube surface. The point on a tube which is located on the diameter perpendicular to the radiating plane and on the side of the tube facing the plane receives radiation through an angle of 180°, and hence the α value for this point is 1.00. Other points on the circumference can "see" through a smaller and smaller angle as one proceeds to the rear of the tube, until a point is reached at which no direct radiation from the plane is received. The effectiveness of each incremental area of the entire *circumferential* tube surface A is evaluated, and the sum of these effective areas must be divided by A_{cp} to obtain α . Again by symmetry, only half the tube circumference need be investi-

¹ Hottel, H. C., Personal communication.

gated, and it is rather apparent that the front of the tube absorbs considerably more heat than at other points on its circumference.

The interception of reradiation can be evaluated in the same manner, recalling that the intensity of the reradiation is $(1 - \alpha_{\text{direct}})$ times the intensity of the source. The total of direct radiation plus reradiation can be indicated in the polar coordinate plot to show the actual heat distribution. It is important to note the poor ratio between the average and the maximum value of α for the various points on the tube circumference. For a normal tube spacing of about 1.8 times outside diameter

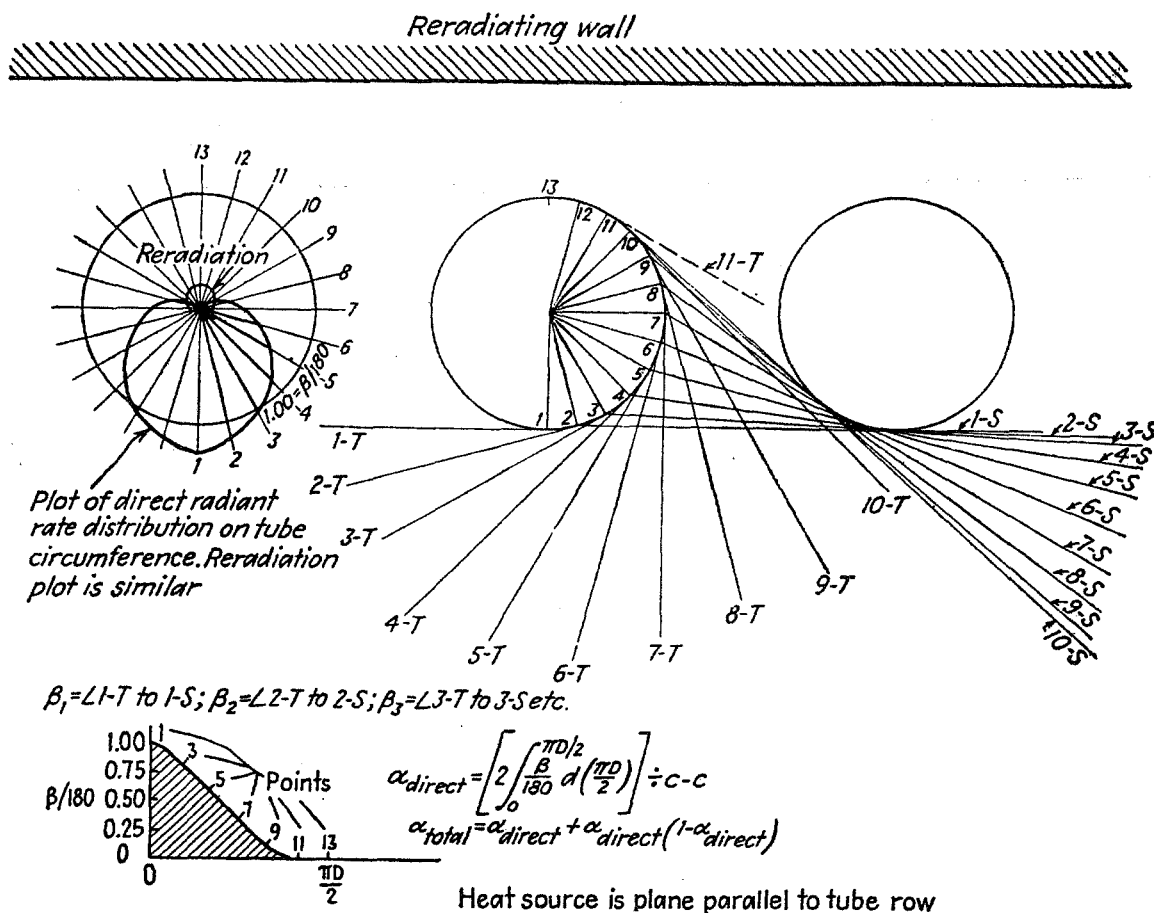


FIG. 19.10. Evaluation of α showing flux distribution on tube circumference.

in oil-refinery furnaces, the ratio is about 2.0, indicating that, at an average flux of 10,000 Btu/(hr)(ft²) on the total tube surface A , the maximum point flux (disregarding factors affecting heat distribution other than α) will be 20,000 Btu/(hr)(ft²) on the front face of the tube. The greater the ratio of center-to-center distance to outside diameter the lower the ratio of maximum rate to average rate, and since α_{max} in all cases is 1.00, it can be seen readily from this development of α that the effective surface of the tubes increases. Except in special cases, however, the cheapest furnace results when tubes are spaced as close together as the physical and mechanical limitations of return bends will permit.

In the case of double rows of tubes, the back row receives about one-quarter of the total heat transfer. It is again also important to note that the ratio of maximum flux to the average flux (circumferential) for both rows of tubes becomes much worse than for a single row. The ratio can be obtained by dividing the total circumferential surface for the two rows by the total product of αA_{cp} for the two rows. Actually the tube emissivity is not 1.00 as assumed, and there will actually be some reflection from one tube to another. The net effect is to increase the effective emissivity of the row 2 or 3 per cent which is neglected in design practice.

Figure 19.11 presents values of α , direct and total, for single and double rows of tubes with refractory behind them. From the values of α_{direct} for the first and second row, it can be seen that a nest of tubes more than two rows deep can be assumed to have a value of α_{total} equal to 1.0. For convection banks whose tubes are radiated directly by the furnace A_{cp} is merely the width times the length of the opening.

In boiler furnaces tubes are sometimes half imbedded in the refractory, are sometimes finned, and occasionally are equipped with metal- or refractory-faced blocks. For a more detailed development of α values for such arrangements reference should be made to Hottel.¹ Still another serious complication arises in boiler furnaces from the slagging of the tubes.

A method of evaluating the effective radiant-heat-transfer surface for various boiler tubes has been presented by Mullikin.² For slagged tubes of any type, the effective surface is

$$(\alpha A_{cp})_s = A_{cp} \alpha F_c F_s F_\epsilon \quad (19.2)$$

where A_{cp} and α are as before with the subscript indicating a slagged condition and

where F_c = conductivity factor, dimensionless

F_s = slag factor, dimensionless

F_ϵ = emissivity factor, dimensionless

Note that the tube emissivity is introduced into the evaluation of the effective surface in this case while previously the emissivity was considered in the exchange factor $\mathcal{F}$. Actually no confusion will result, since in the only method discussed here in which $(\alpha A_{cp})_s$ is employed F_ϵ is assumed to be unity. Practical values of F_c are 1.00 for bare and finned tubes, 0.70 for bare-faced metal blocks shrunk onto the tubes, and 0.33 for refractory-faced metal blocks. The slag factor will vary from a low of 0.6 to 0.9 or 1.0 for well-operated boilers. When the tubes are clean, F_s is 1.0, of course. From a review of published boiler tests the slag

¹ Hottel, *loc. cit.*

² Mullikin, H. F., *Trans. ASME*, **57**, 518 (1935).

factor appears to be ordinarily about 0.8 to 0.9. It is impossible to make generalizations about the slagging, which may occur with different coals (and some oil fuels) and with varying furnace temperatures and the slag

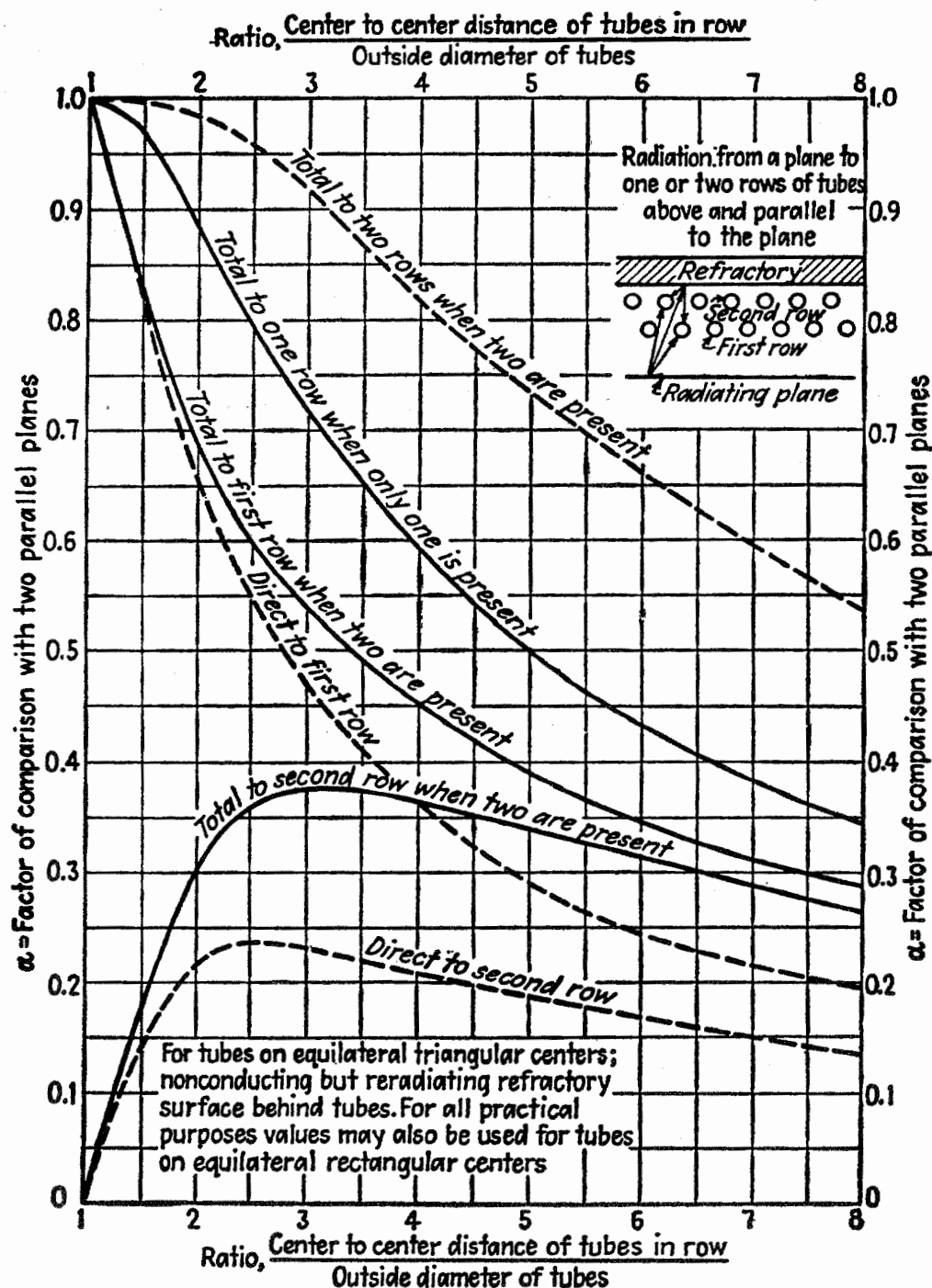


FIG. 19.11. Radiation between a plane and one or more tube rows parallel to the plane. (Courtesy of Hottel.)

factor in such cases is estimated from comparable information. A boiler in intermittent service tends to clean itself, while one in continuous service reaches an equilibrium state (steady or cyclic) at which the effects of the slag do not vary appreciably. Apparently there is consider-

able difference between the reduction in heat absorption due to dry ash on the tubes and that resulting when the slag is molten or running. A change in operating conditions may change the slag condition and cause variations in the accuracy of any equation which considered heat-transfer principles without the practical factors.

Heat Source. The heat to a furnace is provided primarily in the combustion reaction and in the sensible heat of the combustion air if it has been preheated. Gas fuels generally provide nonluminous flames. Oil fuels can be fired to provide flames of varying degrees of luminosity, depending upon burner design, extent of atomization, and percentage of excess air used. Pulverized coal burners produce a flame containing incandescent solid particles and of a greater degree of luminosity than the minimum obtainable with oil burners. Stoker firing provides an incandescent fuel bed.

The differences in the characteristics of the flames or heat patterns produced in the conventional firing of the various fuels have resulted in the development of methods of calculating radiant-heat transmission which apply on the one hand to oil- or gas-fired refinery furnaces and on the other hand to coal- (either stoker or pulverized) fired units. There is no simple, universally applicable method of calculating heat absorption in all types of furnaces. At the outset, then, the calculations must differentiate between furnaces fired with gas or oil and furnaces fired with solid fuels. It is justifiable to say that heat-absorption-calculation methods are further advanced in refinery-furnace design than in boiler work. Undoubtedly this is due at least in part to the greater complexity of the heat sources and sinks of boiler furnaces. While methods of calculation for coal-fired furnaces will be presented, the discussion of the heat source will be limited mainly to furnaces whose fireboxes are filled largely with nonluminous gases and whose bounding surfaces are not part of the heat source—as opposed to the case of a stoker-fired furnace. Luminous-flame calculations require information obtained either from experiment or from experience, and some data have been presented by Hottel.¹

Consider for the moment a furnace in which gaseous fuel is fired by burners producing nonluminous flames. Further consider that the path of the combustion products through the furnace is very short compared with the dimensions of the plane perpendicular to the path. In such a furnace, equipped with many small burners to give good distribution of firing, one would expect little or no variation in the temperature of the gases from point to point in the furnace. No luminous flames are present to radiate to the tubes or refractory. The only primary sources of radiation are the combustion products themselves, and these are assumed to be

¹ Perry, J. H., "Chemical Engineers' Handbook," 3d ed., pp. 493-495, McGraw-Hill Book Company, Inc., New York, 1950.

uniform in temperature. There is a great difference in the emissivity of various gases at the same temperature. The diatomic gases such as O_2 , N_2 , and H_2 have very poor emissivities, so low that they may be considered zero in furnace-design work. On the other hand, H_2O , CO_2 , and SO_2 have good emissivities (though much poorer than most solid materials), and CO has a fair emissivity. The sources of radiation may be referred to specifically as the radiating constituents in the combustion products. Ordinarily H_2O and CO_2 are the only radiating constituents which need be considered, since the small amount of sulfur in most fuels is negligible and furnaces are generally operated with sufficient excess air to eliminate CO .

The total radiation from a gas mass containing CO_2 and H_2O will depend upon the temperature of the gas and the number of radiating molecules present. The volume of the gas and the concentration of radiating molecules per unit of volume are therefore a measure of the radiation to be anticipated at a given temperature. In actuality the geometric shape of the gas mass must also be considered because of the angle factor involved in radiation.

The *mean beam length*, L ft, is the average depth of the blanket of flue gas in all directions for each of the points on the bounding surface of the furnace and is used instead of a cubical measure of the volume. The concentration of the radiating molecules is measured by their partial pressures. The emissivity of the gas mass in a furnace is a function of the product pL atm-ft, where p is the partial pressure of the radiating constituent. If more than one radiating constituent is present, the emissivities are additive, although a small correction must be made for the interference of one type of molecule with the radiation from the other. In calculating the emissivity of the gas mass, allowance must be made for the temperatures of both the source and the sink. For heat transfer to a black body, one would use the equation

$$Q_b = 0.173F_{bA} \left[\epsilon_g \left(\frac{T_g}{100} \right)^4 - a_g \left(\frac{T_b}{100} \right)^4 \right] A'_b \quad (19.3)$$

where Q_b = duty or heat transfer to the black body by radiation from gas,
Btu/hr

A'_b = effective heat-transfer surface of black body, ft^2

a_g = absorptivity of gas at T_b , dimensionless

F_{bA} = factor to allow for the geometry of a system with a black-body sink, dimensionless

T_g = temperature of gas, $^{\circ}R$

T_b = temperature of black body, $^{\circ}R$

ϵ_g = emissivity of gas at T_g , dimensionless

Practically, a_g can be replaced by ϵ_g evaluated at T_b . When two radiating constituents are present, H_2O and CO_2 , the equation can be rewritten (neglecting the correction factor for interference between the dissimilar molecules):

$$Q_b = 0.173\mathfrak{F}_b \left[(\epsilon_c + \epsilon_w)_{T_g} \left(\frac{T_g}{100} \right)^4 - (\epsilon_c + \epsilon_w)_{T_b} \left(\frac{T_b}{100} \right)^4 \right] A'_b \quad (19.4)$$

where $(\epsilon_c + \epsilon_w)_{T_g}$ = emissivity of gas at T_g

ϵ_c = emissivity of CO_2 at $p_{CO_2} \cdot L$ and T_g

ϵ_w = emissivity of H_2O at $p_{H_2O} \cdot L$ and T_g

$(\epsilon_c + \epsilon_w)_{T_b}$ = emissivity (substituted for absorptivity)

ϵ_c = emissivity of CO_2 at $p_{CO_2} \cdot L$ and T_b

ϵ_w = emissivity of H_2O at $p_{H_2O} \cdot L$ and T_b

It should be noted that in addition to the correction factor which should be introduced to allow for interference, the emissivity of water vapor has been found by Egbert¹ to be a function of its partial pressure. Values of L for furnaces of various geometric shapes have been determined by Hottel, and Table 19.1 presents a useful digest of these values for furnace

TABLE 19.1. MEAN LENGTH OF RADIANT BEAMS IN VARIOUS GAS SHAPES

<i>Dimensional ratio*</i>	<i>Mean length L, ft</i>
<i>Rectangular furnaces:</i>	
1. 1-1-1 to 1-1-3	$\frac{2}{3} \sqrt[3]{\text{Furnace volume, ft}^3}$
1-2-1 to 1-2-4	
2. 1-1-4 to 1-1- ∞	$1.0 \times \text{smallest dimension}$
3. 1-2-5 to 1-2-8	$1.3 \times \text{smallest dimension}$
4. 1-3-3 to 1- ∞ - ∞	$1.8 \times \text{smallest dimension}$
<i>Cylindrical Furnaces:</i>	
5. $d \times d$	$\frac{2}{3}$ diameter
6. $d \times 2d$ to $d \times \infty d$	$1 \times \text{diameter}$
<i>Tube Banks:</i>	
7. As in convection sections	$L \text{ (ft)} = 0.4P_T - 0.567 \text{ OD, in.}$

* Length, width, height in any order.

work. In industrial-furnace design the term *rate* is used synonymously with the term *flux* in preceding chapters and individual film coefficients are not considered. It is convenient to have charts which give the values of the radiant-heat-transfer flux q_c and q_w as functions of pL and T where

$$q_c = 0.173\epsilon_c \left(\frac{T}{100} \right)^4 \text{ at } p_{CO_2} \cdot L \text{ and } T$$

$$q_w = 0.173\epsilon_w \left(\frac{T}{100} \right)^4 \text{ at } p_{H_2O} \cdot L \text{ and } T$$

$$q_b = 0.173\epsilon_b \left(\frac{T}{100} \right)^4 \text{ and } \epsilon_b = 1.00$$

¹ Egbert, R. B., Sc. D. Thesis in *Chem. Eng.*, MIT, 1941.

Such charts are presented in Figs. 19.12 and 19.13 from data by Hottel and Egbert. The correction for interference is also given in the insert graphs in per cent, and the corrected emissivity is equal to

$$\epsilon_g = \left[\frac{(q_c + q_w)_{T_g} - (q_c + q_w)_{T_s}}{(q_b)_{T_g} - (q_b)_{T_s}} \right] \frac{100 - \%}{100} \quad (19.5)$$

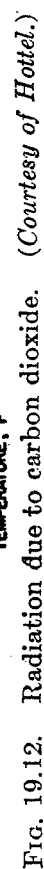
In many simple cases of radiant-heat transfer from gases to nonblack bodies no complications of angle factors need be introduced and the heat-transfer rate can be calculated simply by

$$q = \frac{Q}{A} = \epsilon_s [(q_c + q_w)_{T_g} - (q_c + q_w)_{T_s}] \frac{100 - \%}{100} \quad (19.6)$$

where the subscript S refers to the cold surface and the emissivity of the cold surface is ϵ_s . Such is the case in tube bundles as in furnace convection sections where radiant-heat transfer is often quite important. In this particular case the tubes are surrounded by the gas and no angle factor is required. The surface to be used is the circumferential tube area. (For banks of tubes there is additional radiant-heat transfer from the refractory side walls which add to the average rate or flux.)

Some conclusions may be drawn from the dependency of the gas emissivity upon pL . For furnaces of the same physical proportions but of differing sizes, one would expect the larger to have a higher rate of heat transfer with a given gas temperature (because of the greater value of L). The effect of increasing excess air is to decrease the value of p , hence the emissivity, and the radiant rate for a given gas temperature is thereby decreased. Though not strictly related to a discussion of the effects of excess air, it has been found by experience that refinery furnaces do not operate under optimum conditions because of the use of undue amounts of excess air. The combustion conditions of boilers are usually controlled more rigidly than refinery furnaces, because of the greater value of fuel in boiler plants.

Enclosing Surfaces. The part played by the refractory walls, arch, floor, etc., of the furnace in transferring heat from the gas to the cold surface is often difficult to visualize. The gas mass radiates in all directions. The emissivity of the gas evaluated from p and L , as discussed previously, is directional in that it denotes the radiation impinging on a point of the cold surface on the furnace enclosure. All of this radiation is directed from various sections of the gas mass *toward that particular point*. However, the various sections of the gas mass also radiate in other directions. Some of this radiation may be directed toward refractory surface (which is not cold), and the refractory in turn reradiates or



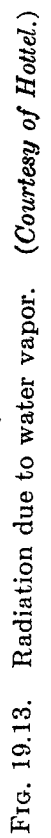


Fig. 19.13. Radiation due to water vapor. (Courtesy of Hottel.)

reflects the impinging radiation, some of it being redirected toward the point first considered. The gas mass is quite transparent to the reradiation (more so than to the reflection, since the spectral characteristics of the reradiation depend upon the characteristics of the surface of the refractory); hence the point on the cold surface receives more radiation than would be evaluated from the emissivity of the gas. The refractory is somewhat analogous to a reflector placed behind a light source.

An effective means of illustrating the effect of the refractory is to start with an enclosure containing no cold surface. It contains a gas of a certain emissivity at a given temperature. If a small aperture is made in the enclosure, the radiation streaming through the aperture will be equal to that coming from a black body at the gas temperature, no matter what the emissivity of the gas may be but provided the enclosure walls are well insulated and the system has reached thermal equilibrium. Now, if instead of the aperture one places a small section of cold surface within the enclosure, the radiation falling upon it will be equal to that which would come from black surroundings at the gas temperature. The effect of a very high ratio of refractory surface to cold surface is to produce a furnace emissivity of 1.0, even though the gas emissivity is low.

On the other hand, if the entire enclosing surface is cold and black, each point on the surface will receive only the radiation initially directed toward it, since radiation in other directions is completely absorbed and the furnace emissivity is equal to the gas emissivity. If the emissivity of the cold surface is less than 1, some of the radiation will be reflected, and though the net effect of a lower cold surface emissivity will be a decrease in heat transfer for a given temperature difference, the decrease will not be proportional to the decrease in emissivity. The reflections will be absorbed only partly by the gas, and the unabsorbed portion will be added to the primary radiation to some other portion of the cold surface. The lower the gas emissivity (absorptivity) the less will be the effect of a change in the emissivity of the cold surface. It might be mentioned that in a fire-tube boiler the radiant-heat transfer can be evaluated by application of these principles, and it is undoubtedly safe to assume that the emissivity of the tube surface is 1.0.

Qualitatively it has been demonstrated that the net radiant-heat transmission will be increased for a given gas emissivity, cold-surface emissivity, and gas and cold-surface temperatures by an increase in the ratio of refractory surface to cold surface. The random addition of refractory (in the form of a partition or bridge wall) does not provide a means of increasing the flux without increasing the furnace gas temperature, however. While the ratio of refractory to cold surface may thus be increased, it is increased at the expense of decreasing the mean beam

length for the furnace. In so-called double radiant-section furnaces, each section is best handled as a separate furnace.

Quantitatively a rigorous evaluation of the effect of the refractory surface is very difficult and beyond the range of practicability for most engineering work. Hottel has developed an equation for the overall exchange factor which is included in Eq. (4.26).

$$F_{\epsilon} = \frac{1}{(1/\epsilon_F) + (1/\epsilon_S) - 1} \quad (19.7)$$

in which ϵ_S is the emissivity of the cold surface, and ϵ_F , the effective emissivity of the furnace cavity (which will be greater than ϵ_G if any portion of the enclosure is not cold). Hottel further defines ϵ_F as a function of the gas emissivity, the ratio of refractory surface to cold surface, and an angle factor accounting for the geometrical relationships between various sections of cold and refractory surfaces.

In summary, the tube surface must be evaluated as equivalent plane surface. The emissivity of the gas mass is a function of its temperature, the temperature of the cold surface, the mean beam length of the furnace, and the partial pressures of the radiating constituents. The effective emissivity of the furnace is a function of the gas emissivity and the ratio (and relative arrangement) of refractory to cold surfaces. The overall exchange factor may be obtained from the furnace and cold surface emissivities for use in an equation of the Stefan-Boltzmann type for calculating the radiant heat transfer.

$$Q = 0.173\mathfrak{T} \left[\left(\frac{T_G}{100} \right)^4 - \left(\frac{T_S}{100} \right)^4 \right] \alpha A_{cp} \quad (19.8)$$

Theoretically the proper average value of T_G must be used or the calculation of long furnaces must be carried out in a stepwise manner. Actually it is often satisfactory to consider the temperature of the gases leaving the furnace radiant section as the average temperature when the degree of turbulence of the gases is high. When highly luminous flames or stoker firing are employed additional data are required. It should be noted that methods for calculating the *average* heat transfer rates or fluxes in the radiant section give no measure of the uniformity of the rates on various tubes. Either experience or highly analytical calculations are required to estimate specific rates.

DESIGN METHODS

The common methods for calculating heat absorption in furnace radiant sections are surveyed below. Several are illustrated by calculations at the conclusion.

1. Method of Lobo and Evans.¹ This method makes use of the overall exchange factor $\mathcal{F}$ and a Stefan-Boltzmann type equation. It has a good theoretical basis and is used extensively in refinery-furnace design work. It is also recommended for oil- or gas-fired boilers. The average deviation between the predicted and observed heat absorption for 85 tests on 19 different refinery furnaces varying widely in physical and operating characteristics was 5.3 per cent. The maximum deviation was 16 per cent. This method is illustrated by Example 19.1.

2. Method of Wilson, Lobo, and Hottel.² This is an empirical method which can be used for box-type furnaces fired with oil or refinery gas when fluxes lie between 5000 and 30,000 Btu/(hr)(ft²) of circumferential tube surface. Other limitations are that the percentage of excess air be from 5 to 80 per cent and that tube-surface temperatures be at least 400°F lower than the radiant-section exit-gas temperature. The mean beam length should not be less than 15 ft. This method is widely used in industry and is recommended under the above limitations when the accuracy of the Lobo-Evans equation is not demanded. For most of the tests referred to under the Lobo-Evans method the average deviation was 6 per cent and the maximum deviation 33 per cent. This method is illustrated by Example 19.2.

3. The Orrok-Hudson³ Equation. This is an early empirical equation for calculating heat absorption in the radiant section of a water-tube boiler. It has been replaced by more accurate expressions and is of limited value for design work. It can be used to estimate the effects of changes in firing rate or air-fuel ratio for an existing boiler fired with coal or oil if it is known that there will be no appreciable change in either the character or extent of the slagging of the furnace tubes. In such applications it may be necessary to adjust the constant in the equation to suit the known operating conditions. The use of this equation is illustrated in Example 19.3.

4. Wohlenberg⁴ Simplified Method. This is an empirical method, though undoubtedly much sounder than the Orrok-Hudson equation for calculating radiant-heat absorption. It is useful only for coal firing. It is again repeated that a knowledge of the anticipated slagging is a prerequisite to the application of a heat-transfer equation to a boiler. Tests on seven large boilers indicated an average deviation of about 10 per cent when the slag factor was estimated from the furnace appearance. The maximum deviation was about 50 per cent for stoker firing, but better accuracy was obtained in furnaces fired with pulverized coal.

¹ Lobo, W. E., and J. E. Evans, *Trans. AICHE*, **35**, 743 (1939).

² Wilson, D. W., W. E. Lobo, and H. C. Hottel, *Ind. Eng. Chem.*, **24**, 486 (1932).

³ Orrok, G. A., *Trans. ASME*, **47**, 1148 (1925).

⁴ Wohlenberg, W. J., and H. F. Mullikin, *Trans. ASME*, **57**, 531 (1935).

APPLICATIONS

Method of Lobo and Evans. The equation given previously for radiant-heat transfer to the cold surface was

$$Q = 0.173\mathfrak{F} \left[\left(\frac{T_g}{100} \right)^4 - \left(\frac{T_s}{100} \right)^4 \right] \alpha A_{cp} \quad (19.8)$$

In addition some heat will be transferred by convection, and the total heat transfer to the cold surface is

$$\Sigma Q = 0.173\mathfrak{F} \left[\left(\frac{T_g}{100} \right)^4 - \left(\frac{T_s}{100} \right)^4 \right] \alpha_{cp} A_{cp} + h_c A (T_g - T_s) \quad (19.9)$$

where A = total tube surface, ft^2

A_{cp} = equivalent cold plane surface, ft^2

$\mathfrak{F}$ = overall exchange factor, dimensionless

h_c = convection coefficient, $\text{Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$

ΣQ = total hourly heat transfer to the cold surface, Btu/hr

T_g = temperature of flue gas leaving the radiant section, $^\circ\text{R}$

T_s = tube surface temperature, $^\circ\text{R}$

α = factor by which A_{cp} must be reduced to obtain effective cold surface, dimensionless

The convection term can be simplified by assuming that $h_c = 2.0$ and that for this term alone A is approximately $2.0\alpha A_{cp}$. Since it is desired to divide all terms by $\mathfrak{F}$, a value of 0.57 will be used in its stead when the convection term is considered. Then

$$\frac{\Sigma Q}{\alpha A_{cp} \mathfrak{F}} = 0.173 \left[\left(\frac{T_g}{100} \right)^4 - \left(\frac{T_s}{100} \right)^4 \right] + 7(T_g - T_s) \quad (19.10)$$

This relationship is shown graphically in Fig. 19.14. In addition to the above flux equation, a heat balance is necessary for the solution of the heat-absorption problem. The heat balance is as follows:

$$Q = Q_F + Q_A + Q_R + Q_S - Q_W - Q_G \quad (19.11)$$

where Q = total radiant-section duty, Btu/hr

Q_A = sensible heat above 60°F in combustion air, Btu/hr

Q_F = heat liberated by fuel, Btu/hr (lower heating value)

Q_G = heat leaving the furnace radiant sections in the flue gases, Btu/hr

Q_R = sensible heat above 60°F in recirculated flue gases, Btu/hr

Q_S = sensible heat above 60°F in steam used for oil atomization, Btu/hr

Q_w = heat loss through furnace walls, Btu/hr (1 to 10 per cent of Q_F , depending upon the size, temperature, and construction of the furnace. 2 per cent is a good design figure)

As a further simplification Q_s can be neglected, and the net heat liberation is

$$Q_F + Q_A + Q_R - Q_w = Q_{\text{net}} \quad (19.12)$$

The heat leaving in the flue gases at the exit-gas temperature T_G is

$$Q_G = W(1 + G')C_{av}(T_G - 520)$$

T_G = Flue gas temperature over bridgewall, deg.fahr.

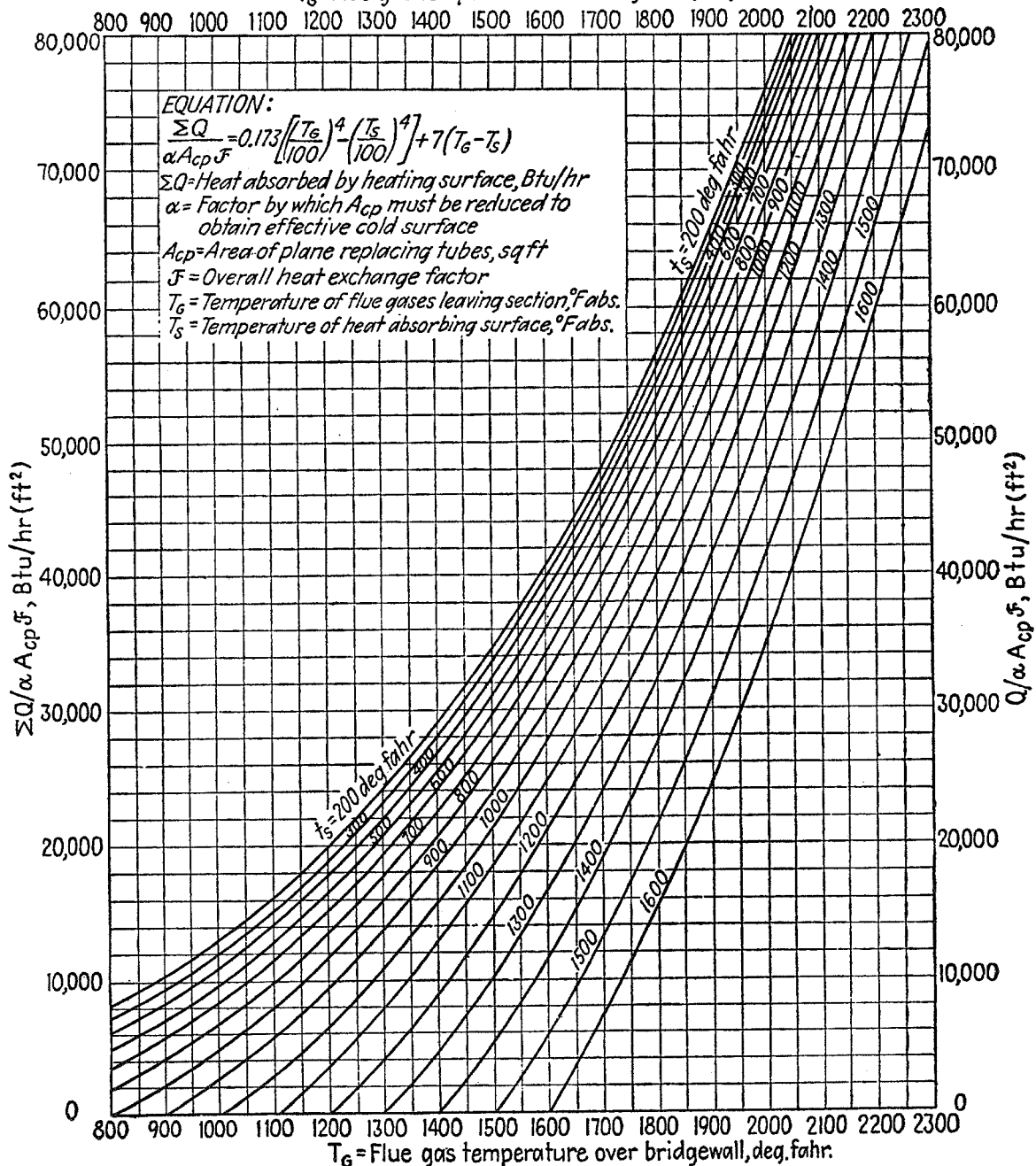


Fig. 19.14. Radiant-section heat flux.

where W = fuel rate, lb/hr

$(1 + G')$ = ratio of flue gas leaving the radiant section to fuel fired, lb/lb

G' = ratio of air to fuel, lb/lb

C_{av} = average specific heat of flue gases between $T_g^\circ\text{R}$ and 520°R ,
Btu/(lb)($^\circ\text{F}$)

In applying the equations, the equivalent cold plane surface is evaluated with the aid of Fig. 19.11. As mentioned earlier, A_{cp} is the surface of a

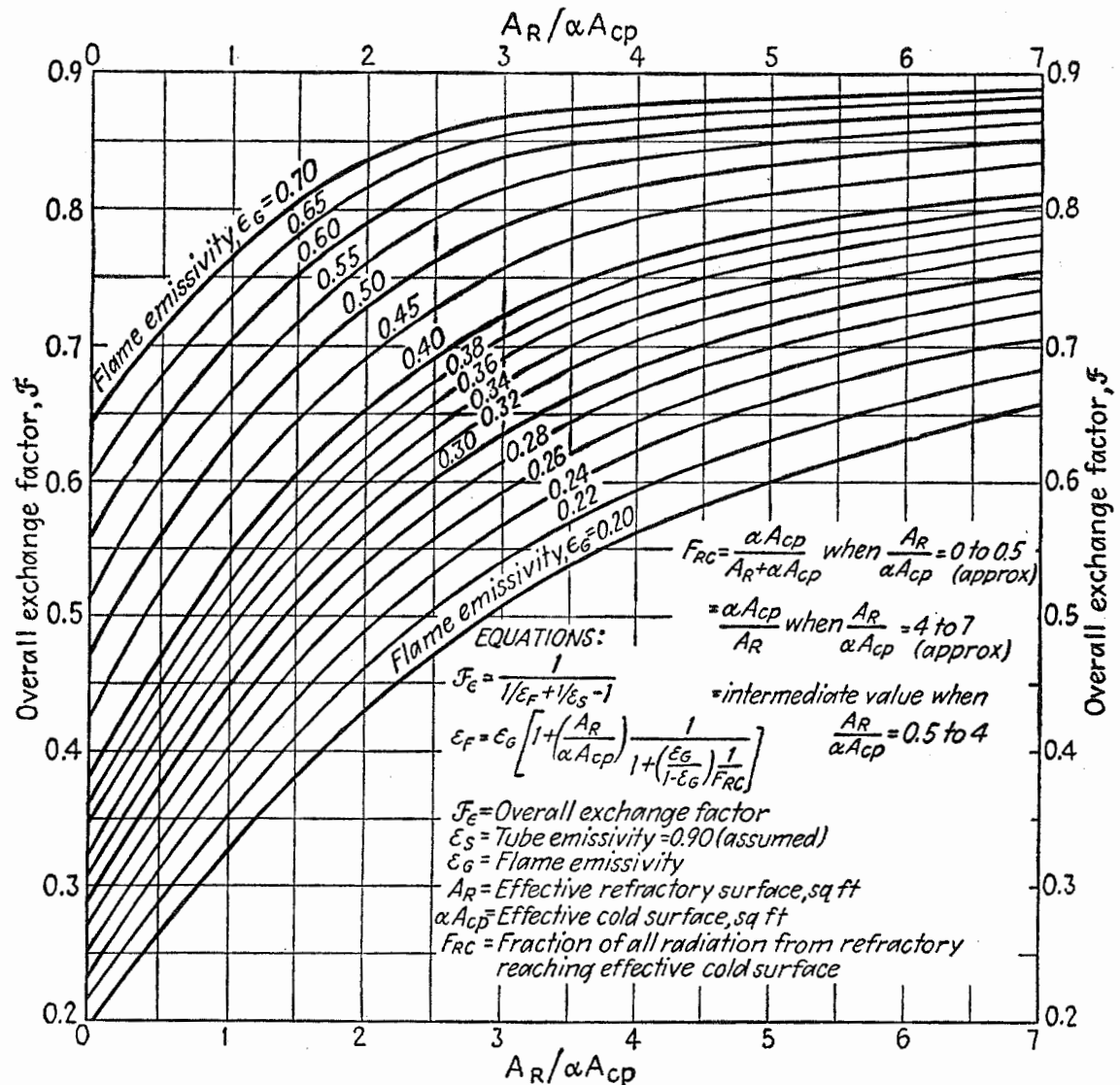


FIG. 19.15. Overall heat-exchange factor in radiant sections.

plane replacing the tube row and corresponds to the product of the number of tubes times their exposed length in feet times center-to-center spacing in feet. When the convection section is so located that it receives the benefit of direct radiation from the radiant section, it should be included in the equivalent cold plane surface. For a nest of tubes more than two rows deep, α may be taken as 1.0 and αA_{cp} is merely the product of the length times width of the opening of the convection bank.

When the convection bank is isolated from the radiant section, it is not included in the radiant-section calculations.

The gas emissivity is calculated from the mean beam length, partial pressures of the radiating constituents, tube temperature, and gas temperature (which usually has to be assumed in the first trial calculation). The overall exchange factor $\mathcal{F}$ is shown in Fig. 19.15 as a function of gas ("flame") emissivity and the ratio of effective refractory surface A_R , where

$$A_R = A_T - \alpha A_{cp} \quad (19.13)$$

where A_R = effective refractory surface, ft²

A_T = total area of furnace surfaces, ft²

αA_{cp} = equivalent cold plane surface, ft²

The exit-gas temperature is obtained by a trial-and-error calculation until it satisfies both the heat-transfer and the heat-balance equation. To lessen the amount of calculation required in applying the method of Lobo and Evans, Fig. 19.14 gives the value of $\Sigma Q / \alpha A_{cp} \mathcal{F}$ for various combinations of T_g and T_s .

It is recommended that, in furnaces where the path of the gas is more than 1.5 times the minimum dimension of the cross-sectional area for gas flow, a stepwise calculation be employed; typical of such a case would be a vertically fired, vertical cylindrical furnace whose height is twice its diameter. The top and bottom halves of the furnace would be calculated as separate furnaces except that the flue gases from the lower half would provide the heat liberation to the second.

In practice the total furnace duty is calculated first, including the sensible heat, heat of vaporization, and any heat of reaction. The efficiency of the furnace e is given by

$$e = \frac{\Sigma Q}{Q_F} \times 100 \quad \% \quad (19.14)$$

It is determined from a balance between fuel costs and the first cost of the furnace (plus air preheater if one is used). The quantity of excess air to be used depends upon the type of fuel, type of burners, type of draft, and the temperature of the combustion air. In practice, however, 40 per cent excess air is used for designing natural- or induced-draft furnaces, 25 per cent excess air for forced-draft furnaces.

The use of air preheat is dictated by the temperature of the coolest material to be heated, the cost of fuel, and the cost of furnace surface and to some degree by conventional practices. Air preheaters are easily justified by the savings in fuel cost, although they complicate the operation of the furnace and require additional maintenance.

When the heat liberation has been determined, the design of a petroleum furnace is then established on the basis of the permissible *average* radiant-section rate as defined in Table 19.2 appearing at the conclusion of the chapter. The tube diameter is dependent upon consideration of the film coefficient, pressure drop, and radiant rate. The spacing of the tubes depends upon the characteristics of the headers or return bends. The closest possible spacing is used except when special requirements arise such as the need for an improved uniformity of flux on the tube circumference.

The most economical furnace design uses the maximum tube length which is compatible with the furnace cross section and which provides adequate clearance between the tubes and burners. In some refinery units tubes 60 ft long are used, although 40 ft is the more usual limit. In boilers tube lengths may even be longer.

Example 19.1. Calculation of a Furnace by the Method of Lobo and Evans. A furnace is to be designed for a total duty of 50,000,000 Btu/hr. The overall efficiency is to be 75 per cent (lower heating value basis). Oil fuel with a lower heating value of 17,130 Btu/lb is to be fired with 25 per cent excess air (corresponding to 17.44 lb air/lb fuel), and the air preheated to 400°F. Steam for atomizing the fuel is 0.3 lb/lb of oil. The furnace tubes are to be 5 in. OD on 8½-in. centers, in a single-row arrangement. The exposed tube length is to be 38'6". The average tube temperature in the radiant section is estimated to be 800°F.

Design the radiant section of a furnace having a radiant-section average flux of 12,000 Btu/(hr) ft².

Solution. As in all trial-and-error solutions, a starting point must be assumed and checked. With experience, the first choice may come very close to meeting the desired conditions. For orientation purposes, one can make an estimate of the number of tubes required in the radiant section by assuming that

$$\frac{Q}{\alpha A_{cp}} = 2 \times \text{average flux} = 24,000 \text{ Btu}/(\text{hr})(\text{ft}^2)$$

If the overall exchange factor is 0.57, $\Sigma Q/\alpha A_{cp} F = 24,000/0.57 = 42,000$; from Fig. 19.14 it can be seen that with a tube temperature of 800°F, an exit-gas temperature of 1730°F will be required to effect such a flux. The duty in cooling the furnace gases to 1730°F can be calculated, and from it the required number of tubes determined for the first approximation of the design.

$$\text{Heat liberated by the fuel, } Q_F = \frac{50,000,000}{0.75} = 66,670,000 \text{ Btu/hr}$$

$$\text{Fuel quantity} = \frac{66,670,000}{17,130} = 3890 \text{ lb/hr}$$

$$\text{Air required} = 3890 \times 17.44 = 67,900 \text{ lb/hr}$$

$$\text{Steam for atomizing} = 3890 \times 0.3 = 1170 \text{ lb/hr}$$

$$Q_F = 66,670,000 \text{ Btu/hr}$$

$$Q_A = 67,900 \times 82 \text{ Btu/lb at } 400^\circ\text{F} = 5,560,000 \text{ Btu/hr (above } 60^\circ\text{F)}$$

$$Q_S = \text{negligible } (1170 \times 0.5 \times 190^\circ\text{F}) \text{ Btu/hr}$$

$$Q_F + Q_A = 72,230,000 \text{ Btu/hr}$$

$$Q_W = 2\% \text{ of } Q_F = 1,330,000 \text{ Btu/hr}$$

$$Q_{\text{net}} = Q_F + Q_A - Q_W = 70,900,000 \text{ Btu/hr}$$

Heat out in gases at 1730°F, 25 per cent excess air, 476 Btu/lb of flue gas,

$$Q_G = 476(3890 + 67,900 + 1170) = 34,500,000$$

$$Q = Q_{\text{net}} - Q_G = 70,900,000 - 34,500,000 = 36,400,000 \text{ Btu/hr (first estimate)}$$

$$\text{The surface/tube, } A = 38.5 \text{ ft} \times \pi \times \frac{5}{12} = 50.4 \text{ ft}^2$$

$$\text{The estimated number of tubes } N_t \text{ is } N_t = \frac{36,400,000}{12,000 \times 50.4} = 60.1$$

Try 60 tubes.

The layout of the cross section of the furnace may be as shown in Fig. 19.16.

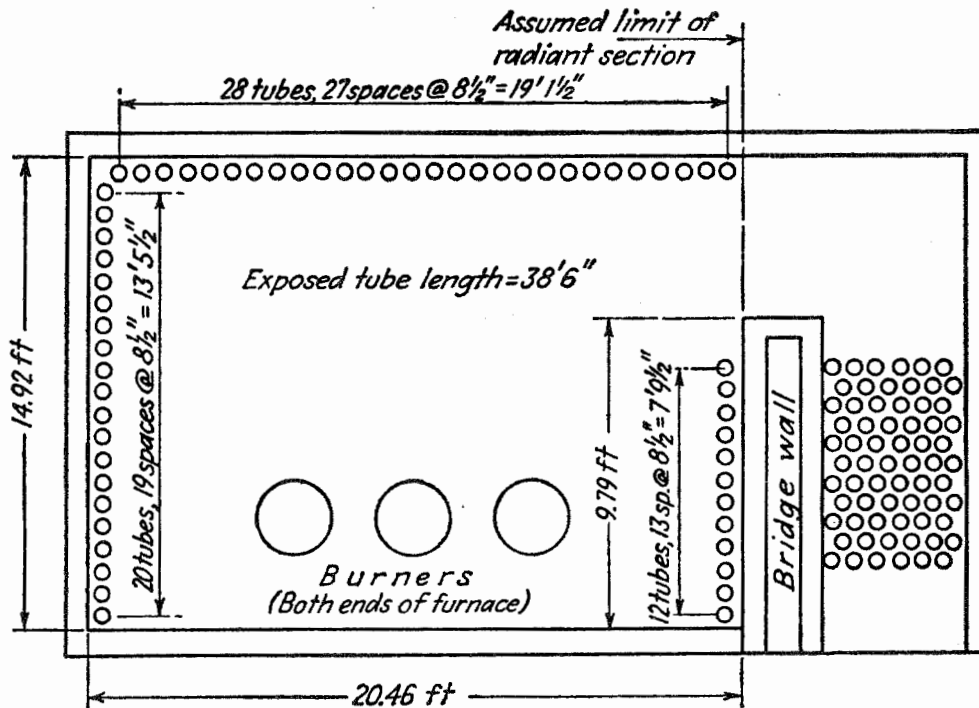


FIG. 19.16. Furnace of Examples 19.1 and 19.2.

Equivalent cold plane surface A_{cp} :

$$\text{Center-to-center distance} = 8\frac{1}{2} \text{ in.}$$

$$A_{cp} \text{ per tube} = \frac{8\frac{1}{2}}{12} \text{ in.} \times 38.5 = 25.7 \text{ ft}^2$$

Total α to a single row, refractory backed, from Fig. 19.11:

$$\text{Ratio of center-to-center/OD} = 8\frac{1}{2}/5 = 1.7 \quad \alpha = 0.937 \quad (\text{Fig. 19.11})$$

$$\alpha A_{cp}/\text{tube} = 25.7 \times 0.937 = 25 \text{ ft}^2$$

$$\alpha A_{cp} = 60 \times 25 = 1500 \text{ ft}^2$$

Refractory surface:

$$\text{End walls} = 2 \times 20.46 \times 14.92 = 611 \text{ ft}^2$$

$$\text{Side wall} = 14.92 \times 38.5 = 575 \text{ ft}^2$$

$$\text{Bridge wall} = 9.79 \times 38.5 = 377 \text{ ft}^2$$

$$\text{Floor and arch} = 2 \times 20.46 \times 38.5 = 1575 \text{ ft}^2$$

$$A_T = 3138 \text{ ft}^2$$

$$A_R = A_T - \alpha A_{cp} = 1638 \quad \frac{A_R}{\alpha A_{cp}} = \frac{1638}{1500} = 1.09$$

Mean beam length:

$$\text{Dimension ratio} = 38.5 \times 20.46 \times 14.92 = 3:2:1 \text{ (approximately)}$$

$$L = \frac{2}{3} \sqrt[3]{\text{volume}} = \frac{2}{3} \sqrt[3]{38.5 \times 20.46 \times 14.92}$$

$$L = 15 \text{ ft}$$

Gas emissivity: From the analysis of the fuel, the steam quantity, and the assumption that the humidity of the air is 50 per cent of saturation at 60°F, the partial pressures of CO₂ and H₂O in the combustion gases with 25 per cent excess air are

$$p_{\text{CO}_2} = 0.1084 \quad p_{\text{H}_2\text{O}} = 0.1248 \quad p_{\text{CO}_2}L = 0.1084 \times 15 = 1.63 \quad p_{\text{H}_2\text{O}}L = 1.87$$

From Fig. 19.12 and 19.13, the emissivity of the gas can be evaluated.

$$\epsilon_G = \left[\frac{(q_{\text{CO}_2} \text{ at } P_{\text{CO}_2}L + q_{\text{H}_2\text{O}} \text{ at } P_{\text{H}_2\text{O}}L)_{T_G} - (q_{\text{CO}_2} \text{ at } P_{\text{CO}_2}L + q_{\text{H}_2\text{O}} \text{ at } p_{\text{H}_2\text{O}}L)_{T_S}}{(q_b)_{T_G} - (q_b)_{T_S}} \right] \frac{100 - \%}{100} \quad (19.5)$$

$$\% \text{ correction at } \frac{p_{\text{CO}_2}}{p_{\text{CO}_2} + p_{\text{H}_2\text{O}}} = \frac{0.1084}{0.2332} = 0.465$$

$$p_{\text{CO}_2}L + p_{\text{H}_2\text{O}}L = 3.56$$

$$\% \text{ correction} = 8\% \text{ (estimated)}$$

$$\epsilon_G = \left[\frac{(6500 + 14,500) - (650 + 1950)}{39,000 - 4400} \right] \left(\frac{100 - 8.0}{100} \right)$$

$$\epsilon_G = 0.489$$

Overall exchange factor $\mathcal{F}$:

$$\mathcal{F} \text{ at } \epsilon_G = 0.496 \text{ and } \frac{A_R}{A_{cp}} = 1.09$$

$$\mathcal{F} = 0.635 \text{ from Fig. 19.15}$$

Check of gas temperature required to effect assumed duty on assumed surface:

$$\Sigma Q = 36,400,000 \text{ Btu/hr assumed} \quad \alpha A_{cp} = 1500 \text{ ft}^2 \text{ assumed}$$

$$\frac{\Sigma Q}{\alpha A_{cp} \mathcal{F}} = \frac{36,400,000}{1500 \times 0.635} = 38,200$$

$$T_G \text{ required (at } T_S = 800^\circ) = 1670^\circ\text{F compared with } 1730^\circ\text{F assumed in heat balance)}$$

The trial indicates that more duty than 36,400,000 Btu/hr will be performed, since this duty cools the flue gases to only 1730°F, while the flux corresponding to this duty could be effected by a gas temperature of 1670°F. Actually this is a fairly close check, and the number of tubes in the furnace need not be changed, since the final balance will be struck at about 1700°F for the assumed furnace. The duty would be 37,050,000 Btu/hr at this exit temperature, and assuming that $\mathcal{F}$ does not change (it will go up slightly) $\Sigma Q / \alpha A_{cp} \mathcal{F} = 39,000$, requiring a "driving" temperature of 1695°F, which is a close enough approximation. The circumferential flux will be $37,050,000 / 60 \times 50.4 = 12,280 \text{ Btu/(hr)(ft}^2\text{)}$ as compared with the 12,000 flux specified. Such a difference is negligible.

In general, if the gas temperature required to effect the duty used in the heat balance falls below the temperature shown by the heat balance, the number of tubes assumed is too few or the actual flux will be higher than the assumed rate

Method of Wilson, Lobo, and Hottel. The limitations of this method have been pointed out. In the original publication several equations were presented, but the most useful is the following:

$$\frac{Q}{Q_F} = \frac{1}{1 + (G'/4200) \sqrt{Q_F/\alpha A_{cp}}} \quad (19.15)$$

where again $G' = \text{lb air/lb fuel}$ and the other nomenclature is similar to that given for the Lobo and Evans method. While the cold surface is properly evaluated so that the equation is applicable to furnaces having double as well as single rows of tubes, the effect of refractory surfaces is neglected. The effect of excess air on the radiant-section efficiency is measured only by G' , the air-fuel ratio, and consequently the equation does not hold for fuels having heating values which are either very low or very high. The equation is recommended for rapid calculation (within its limitations) and for predicting the effects of changes in furnace operating conditions.

Example 19.2. Calculation of a Furnace by the Method of Wilson, Lobo, and Hottel. The furnace of Example 19.1 is to be fired with cracked gas fuel at 40 per cent excess air, using no air preheat. If the capacity of the burners limits the heat liberation to 50,000,000 Btu/hr, what will be the radiant-section duty? (The air-fuel ratio is 22.36 lb air/lb fuel.)

Solution:

$$\begin{aligned} Q &= 50,000,000 \times \frac{1}{1 + (22.36/4200) \sqrt{50,000,000/1500}} \\ &= 25,300,000 \text{ Btu/hr} \end{aligned} \quad (19.15)$$

The radiant-section average rate will be 8350 Btu/(hr)(ft²), and the exit-flue-gas temperature 1540°F by heat balance.

Wohlenberg Simplified Method. While this method is an empirical method, its derivation is of interest. Wohlenberg¹ developed a complex theoretical method for evaluating the heat absorption in boiler furnaces which took into account the many variables already discussed and in addition those factors unique to coal firing. The simplified method, however, relates the radiant-section absorption efficiency of any boiler furnace to a base or standard design by means of factors which correct for the differences in all of the characteristics between the two.

The base design is as follows:

Furnace volume = 8000 ft³

Release rate = 25,000 Btu/ft³ for pulverized coal firing
= 40,000 Btu/ft³ for stoker firing

¹ Wohlenberg, W. J., and D. G. Morrow, *Trans. ASME*, **47**, 177 (1925). Wohlenberg, W. J., and E. L. Lindseth, *Trans. ASME*, **48**, 849 (1926).

Excess air = 20 per cent for pulverized coal firing

= 40 per cent for stoker firing

Coal = Illinois bituminous

Fineness = 75 per cent through 200 mesh

Fraction cold = unity

The fraction cold is defined as the effective exposed radiant-heating surface in the furnace divided by the total exposed furnace surface (exclusive of the grate in stoker firing). The evaluation of the effective heating surface is by the method of Mullikin discussed previously. The absorption efficiency of the base furnace is 0.452 for pulverized fuel firing and 0.311 for stoker firing; the absorption efficiency e of any furnace is the ratio of heat absorbed to the heat liberated by the fuel (higher heating value) plus the heat in the combustion air.

Wohlenberg's equation is

$$e = FK_1K_2K_3K_4K_5K_6K_7K_8 + C' \quad (19.16)$$

where the various factors have the following significance: F is chosen according to type of firing, pulverized coal or stoker, and the influences of the other factors are

K_1 = furnace volume

K_2 = heat liberation rate, Btu/ft³

K_3 = fraction cold

K_4 = excess air

K_5 = heating value of coal (HHV)

K_6 = pulverizing fineness

K_7 = furnace volume beyond K_1

K_8 = heat liberation rate beyond K_2

C' = air preheat

The equation does not apply to gas- or oil-fired boilers, although the methods proposed by Mullikin or Lobo and Evans may be used. The application of the simplified equation is straightforward and gives values which are in good agreement with the more complex method of Wohlenberg. It has been mentioned several times that the ash or slag on the boiler surfaces presents problems beyond those of ordinary heat-transfer calculations. Accordingly a basis of evaluating the influence of ash or slag under operating conditions must be established before these methods can be applied to rational design.

Orrok-Hudson Equation. This is very similar in form to the Wilson, Lobo, and Hottel equation, and as already pointed out, its greatest use is for comparing furnace performance under various operating conditions. The fractional heat absorption is

$$\frac{Q}{Q_F} = \frac{1}{1 + G' \sqrt{C_R/27}} \quad (19.17)$$

where G' , Q , and Q_F are defined as before and C_R is the pounds of fuel per hour per square foot of projected radiant-heating surface. For one tube, $C_R = (\text{OD in.}/12) \times \text{exposed length}$.

Example 19.3. Calculation of Performance by the Orrok-Hudson Equation. What per cent increase in the radiant-section heat absorption may be expected in a boiler when the firing rate is increased 50 per cent?

The initial ratio of absorption to liberation is 0.38, and the excess air is expected to increase from 25 to 40 per cent as a result of the increased firing rate.

Solution:

$$\frac{Q_{F2}}{Q_{F1}} = 1.50 \quad \frac{C_{R2}}{C_{R1}} = 1.5$$

$$\frac{G'_2}{G'_1} = \frac{140}{125}$$

Q_2/Q_1 is to be determined.

$$\frac{Q_1}{Q_{F1}} = \frac{1}{1 + G' \sqrt{C_R/27}} = 0.38$$

$$1 = 0.38 + 0.38 G'_1 \sqrt{C_{R1}/27}$$

$$G'_1 \sqrt{\frac{C_{R1}}{27}} = \frac{0.62}{0.38} = 1.63$$

$$G'_2 \sqrt{\frac{C_{R2}}{27}} = \frac{140}{125} G'_1 \sqrt{\frac{1.5 C_{R1}}{27}} = 1.12 \times 1.223 G'_1 \sqrt{\frac{C_{R1}}{27}}$$

$$G'_2 \sqrt{\frac{C_{R2}}{27}} = 1.37 G'_1 \sqrt{\frac{C_{R1}}{27}} = 1.37 \times 1.63 = 2.23$$

$$Q_2 = \frac{Q_{F2}}{1 + 2.23} = 0.31 Q_{F2}$$

$$\frac{Q_2}{Q_1} = \frac{0.31 Q_{F2}}{0.38 Q_{F1}} = \frac{0.31 \times 1.5 Q_{F1}}{0.38 Q_{F1}} = 1.22$$

Hence the radiant absorption will be increased only 22 per cent for an increase of 50 per cent in the heat liberated. In such a case the effects of the higher exit-flue-gas temperature on superheater tubes would have to be investigated.

Miscellaneous Applications. Radiant-heat transfer is of importance in sections of a furnace other than the radiant section proper. Radiant-heat transfer to the "shield" tubes (top two rows) of the convection bank of a furnace such as that shown in Fig. 19.16 may be evaluated by the method of Lobo and Evans. The procedure is the same as in the case of the calculations of the radiant section proper with the exception that the gas temperature to be used is that of the gases coming over the bridge wall. One would expect that the temperature to use would be the gas temperature *after* heat has been lost to the shield tubes by radiation. In this particular case, however, the mean radiating temperature is closer to that of the gases entering the cavity immediately above the shield tubes. Hence no trial and error is involved in calculating this heat

transfer, since the radiant-section exit temperature is already known. It should be noted that, since the convection-heat transfer to the shield tubes will be evaluated independently, the value of $\Sigma Q/\alpha A_{cp} F$ corresponding to the gas temperature and the shield-tube-metal temperature should be reduced by the amount $7(T_g - T_s)$.

Once the gases have actually made their entrance into the rows of tubes, they will continue to lose heat by radiation, and in spite of the rather small mean beam length this radiation may account for 5 to 30 per cent of the total heat transfer in the entire convection section. This radiation can be evaluated by the equation

$$Q_{RC} = 0.173 \left[\left(\frac{T_g}{100} \right)^4 - \left(\frac{T_s}{100} \right)^4 \right] A \frac{1}{(1/\epsilon_g) + (1/\epsilon_s) - 1} \quad (19.18)$$

where Q_{RC} = radiant-heat flow at a point in the convection section in Btu per hour. Other terms were defined in the Lobo and Evans method. It should be noted that no correction for reradiation from the refractory side walls is made. (It is more convenient instead to use the method of Monrad¹ to apply a correction factor to the combined heat-transfer coefficient for convection and radiation.) With negligible loss in accuracy the last term can be rewritten

$$\frac{1}{(1/\epsilon_g) + (1/\epsilon_s) - 1} = \epsilon_g \epsilon_s \quad (19.19)$$

and the equation becomes, with further substitution,

$$\frac{Q_{RC}}{A} = \epsilon_s [(q_c + q_w)_{T_g} - (q_c + q_w)_{T_s}] \left(\frac{100 - \%}{100} \right) \quad (19.20)$$

The use of this equation is illustrated by Example 19.4. Another type of problem frequently encountered is one in which a kettle or tank is to be used for boiling a liquid and the vessel either is direct-fired or is heated by waste flue gases from another unit. It can be calculated by applying some of the principles already illustrated. Example 19.5 shows such an application.

Example 19.4. Calculation of the Equivalent Radiant Coefficient. In the convection section of a refinery furnace, tubes are 5 in. OD on 8½ in. centers, spaced on equilateral triangular pitch. The flue gases at the row of tubes under consideration are at 1500°F; the tube temperature is 650°F. The flue gases contain 10.84% CO₂ and 12.48% H₂O by volume. Calculate the radiant-heat transfer between the gas and tubes in terms of a coefficient which can be added to the convection-heat-transfer coefficient.

¹ Monrad, C. C., *Ind. Eng. Chem.*, **24**, 505 (1932).

Solution:

$$\begin{aligned}
 \epsilon_s &= 0.90 \quad (\text{assumed}) \\
 L &= 0.4 \text{ (center to center)} - 0.567 \text{ (OD)} \quad (\text{from 7 in Table 19.1}) \\
 &= 0.4 (8.5) - 0.567 (5) = 3.400 - 2.835 \\
 &= 0.565 \text{ ft} \\
 p_{\text{H}_2\text{O}}L &= 0.1248 \times 0.565 = 0.0704 \text{ atm-ft} \\
 p_{\text{CO}_2}L &= 0.1084 \times 0.565 = 0.0611 \text{ atm-ft} \\
 q_{\text{H}_2\text{O}} \text{ at } T_G &= 1050 & q_{\text{H}_2\text{O}} \text{ at } T_S &= 165 \\
 q_{\text{CO}_2} \text{ at } T_G &= 1700 & q_{\text{CO}_2} \text{ at } T_S &= 160 \\
 &2750 &&325 \\
 \frac{Q_{RC}}{A} &= 0.9(2750 - 325) \\
 \frac{P_{\text{CO}_2}}{P_{\text{CO}_2} + P_{\text{H}_2\text{O}}} &= \frac{0.1084}{0.1084 + 0.1248} = 0.465 \\
 P_{\text{CO}_2}L + P_{\text{H}_2\text{O}}L &= 0.1315 \\
 \% \text{ correction} &= 2\% \\
 \frac{Q_{RC}}{A} &= 0.9 \times 2425 \times 0.98 = 2140
 \end{aligned}$$

The equivalent radiation coefficient is, then,

$$h_r = \frac{Q_{RC}}{A \Delta T} = \frac{2140}{1500 - 650} = 2.51 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

which is a very appreciable part of the total coefficient. To correct the total coefficient $h_c + h_r$ for radiation from the walls of the convection bank the method of Monrad mentioned previously is recommended.

Example 19.5. Calculation of a Heated Vessel. Design a simple vessel for continuously concentrating a solution whose boiling point is 480°F . The duty required is 500,000 Btu/hr. The source of heat available is 3050 lb/hr of flue gas at 1500°F . A quantity of 48 in. OD pipe formerly used in some discarded vessels is available.

Solution. The required unit might take the shape of that shown in Fig. 19.17. Flue gases flow parallel to the axis of the vessel in a tunnel beneath it. Although the cold surface is segregated from the refractory surface to a greater extent than in a furnace radiant section, the Lobo and Evans exchange factor can be used without affecting the accuracy of the solution nearly so much as the poor choice of a mean gas temperature.

Since the vessel will be very long compared with the cross-sectional area of the flue-gas path, a stepwise calculation is indicated unless the mean gas temperature can be evaluated properly. In a stepwise calculation, the heater could be broken up into sections of varying length, each having the same duty. Inasmuch as the heat trans

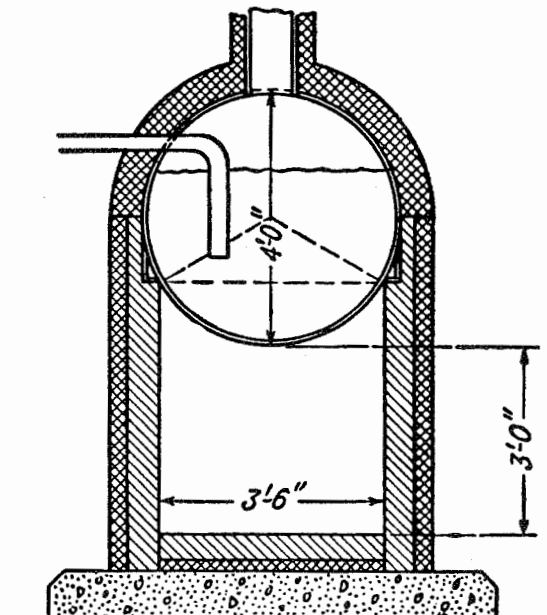


FIG. 19.17. Simple concentrator of Example 19.5.

fer is predominantly radiant and the cold surface is at a constant temperature. an adequate mean temperature is defined by

$$T_G = \sqrt[4]{T_{G,in}^2 T_{G,out}^2} \quad (19.21)$$

The solution is worked on this basis. Assuming that overall heat losses equal 10 per cent of the duty, the exit-flue-gas temperature is found to be 900°F. The cold surface is assumed to have an emissivity of 0.9 and temperature of 500°F. The flue gases contain 7.1% CO₂ and 14.3% H₂O. The gas emissivity is found to be 0.259 at the hot end and 0.27 at the cold end; hence an average value of 0.265 can be used. Calculation of the gas emissivity was illustrated in Example 19.1. The equivalent surface per foot of length obviously lies between the values corresponding to the arc and the chord of the heated portion of the circumference; hence

$$\frac{\alpha A_{cp}}{\text{ft}} = \frac{3.5 + 120\%_{360} \times \pi \times 4.0}{2} = 3.85 \text{ ft}^2/\text{ft}$$

$$\frac{A_R}{\text{ft}} = 3.0 + 3.6 + 3.0 = 9.6 \text{ ft}^2/\text{ft}$$

Arbitrarily neglecting end walls and also the side wall refractory over 3'0" above the floor,

$$\frac{A_R}{\alpha A_{cp}} = \frac{9.6}{3.85} = 2.49$$

$$\bar{\epsilon} = 0.56 \text{ at } \epsilon_G = 0.265 \quad \text{and} \quad \frac{A_R}{\alpha A_{cp}} = 2.49$$

$$\frac{Q}{\alpha A_{cp} \bar{\epsilon}} = 15,300 \text{ Btu}/(\text{hr})(\text{ft}^2) \text{ at } T_G = 1174^\circ\text{F} \quad \text{and} \quad T_S = 500^\circ\text{F}$$

However, the convection coefficient is small, $1.0 \pm \text{Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$, and $A/\alpha A_{cp}$ is not 2.0 as in the assumptions for the Lobo and Evans equation.

$$\frac{Q}{\alpha A_{cp} \bar{\epsilon}} \text{ (i.e., radiation only)} = 15,300 - 7(T_G - T_S) = 10,610 \text{ Btu}/(\text{hr})(\text{ft}^2)$$

$$\frac{Q}{\alpha A_{cp}} = 10,610 \times 0.56 = 5940 \text{ Btu}/(\text{hr})(\text{ft}^2) \text{ of } \alpha A_{cp}$$

Convection rate basis, αA_{cp} :

$$\frac{Q_c}{\alpha A_{cp}} = 1.0 \times (1174 - 500) \times \frac{4.2}{3.85} = 640$$

$$\frac{\Sigma Q}{\alpha A_{cp}} = 5940 + 640 = 6580 \text{ Btu}/(\text{hr})(\text{ft}^2)$$

$$\alpha A_{cp} \text{ required} = \frac{500,000}{6580} = 76.0 \text{ ft}^2$$

$$\text{Length required} = \frac{76.0}{3.85} = 19.7 \text{ ft}$$

Inasmuch as the heat available is fixed in quantity and temperature level, the actual design should employ a liberal safety factor (say 25-ft length) and provision should be made to control the heat transfer either by means of a flue-gas by-pass or by admitting tempering air at the inlet to the unit.

Some Practical Aspects of Refinery Furnaces. While heat-transfer calculations are very important in the design of refinery furnaces, some of

the other factors which enter into their design (and operation) require mention. The furnace must be designed around the burners. If the furnace is fired from the end wall so that the flames travel parallel to the tubes, the cross section of the furnace must be great enough to provide adequate clearance between the burners and the tubes. When such clearance is not provided, flames impinge on the tubes and may cause failures even when moderate pressures and temperatures are employed in the tubes.

When the burners are placed in the floor, side wall, or roof of a box-type furnace, a large number of small burners may be used and the clearances need not be so great. Since the space for locating the burners increases with the tube length, the maximum practical tube length may be used thereby decreasing the required number of return bends. Tube length is limited by the capacity of modern tube mills unless butt-welded tubes are used. The method of cleaning the tubes by turbinizing or the steam-air decoking method may also be a factor in determining the tube length. Ordinarily the design of the convection section requiring a reasonable flue-gas velocity for a good convection coefficient determines the tube length when the burner spacing is not controlling. The distribution of flue-gas flow in the convection section becomes more difficult when long tubes are used, and for furnaces with 50- to 60-ft tubes at least two flue-gas take-offs from the convection section should be provided. The care and operation of the burners and allied equipment have a pronounced effect on the cost of operating and maintaining the furnace. A furnace having a duty of 100,000,000 Btu/hr will burn in the neighborhood of a quarter of a million dollars worth of fuel per year, and such a furnace will use from 5,000,000 to 15,000,000 lb/year of steam for atomizing oil fuel depending upon whether or not the steam-oil ratio is carefully controlled. Oil burners should function properly when atomizing with 0.3 lb of steam per pound of oil.

Most burners require a certain amount of secondary air which is admitted through the burner register. It should be pointed out that, in general, the burner register should be used to control the air-fuel ratio (per cent excess air) while the stack damper should be used to control the draft at the top of the furnace. A slight draft (0.01 to 0.05 in. H_2O) should be maintained at that point at all times, so that leakage through the furnace roof or walls is inward. Outward leakage of flue gases damages the castings and steel supporting the brickwork, warps peep doors and explosion doors, and damages the furnace severely. The furnace should be provided with a draft-gage connection in the roof to guide the operator, but in the absence of such an instrument the appearance of the roof tile may be a guide. Dark lines at the joints between the bricks

indicate an infiltration of air; bright lines indicate outward flow of flue gases.

The use of more excess air than necessary may often make it impossible to maintain the proper draft at the top of the furnace. Furthermore the higher the excess air the greater will be the fuel requirement for a given duty, and Orsat analyses of the flue gases should be made a part of the routine operation to ensure economical furnace performance. Excess oxygen in the flue gases increases the rate of oxidation of the tubes and tube supports. In some types of furnaces (depending upon the ratio of radiant- to convection-section surface) higher excess air results in a higher temperature of the flue gases entering the convection section and may cause premature failure of the tube sheets in the convection section.

The higher the radiant flux the lower the first cost of a furnace for a given duty. At average fluxes above 15,000 Btu/(hr)(ft²), however, the savings diminish rapidly; and the problems of control and maintenance of the furnace increase disproportionately. The larger the furnace and the greater the ratio of refractory to cold surface, however, the lower will be the radiant-section flue-gas temperature for a given radiant flux. The average radiant flux permissible in a furnace depends upon the charge stock, extent of cracking, the heat-transfer coefficient within the tube, the ratio of $A/\alpha A_{cp}$ (which is the ratio of the rate on the face of the tube to the circumferential rate), and the distribution of fluxes along the length of the tubes and for tubes in various locations. All these details require consideration, and in Table 19.2 are presented some representative values of permissible average rates.

TABLE 19.2. PERMISSIBLE AVERAGE RADIANT RATES

<i>Type of furnace</i>	<i>Allowable rate, Btu/(hr)(ft² of circumferential tube area)</i>
Crude.....	10,000-16,000
Vacuum.....	5,000-10,000
Naphtha reforming.....	1,000-18,000
Gas oil cracking:	
Heating.....	10,000-15,000
Soaking.....	10,000
Visbreaking.....	10,000-12,000

It should be realized that average permissible rates are rule-of-thumb affairs and that, actually, the maximum point rate on a tube or tubes is the basic factor which should be prescribed for given conditions. Accordingly, the average permissible rate may be increased when tubes are spaced farther apart than normal, when the tubes are fired from both sides, or when any other specific steps are taken to improve the rate distribution. Tubes are generally spaced from one to one and one-half tube diameters from the face of a wall to the center line of the tubes.

One diameter is preferable from a practical standpoint, since it decreases the weight of the intermediate tube supports.

In the past, air preheaters have not been used so widely on refinery furnaces as on boiler furnaces. The use of air preheat is particularly desirable when the inlet temperature of the oil to be heated is quite high. When air preheat is employed, a greater proportion of the furnace duty is done in the radiant section than when the air is not preheated if a given radiant rate is to be maintained.

In furnace test work flue-gas temperatures should be determined with a high-velocity thermocouple, preferably of the multishielded type. Tube-metal temperatures can be measured by means of an optical pyrometer or by tube-skin thermocouples. It should be noted that the optical pyrometer may indicate temperatures higher than the true skin temperature; this is particularly true when the refractory surfaces are much hotter than the tubes, in which case some of the radiation received by the pyrometer consists of reflected radiation originating at the higher temperature refractory.

PROBLEMS

19.1. Using the Orrok-Hudson equation, calculate the efficiency of a boiler radiant furnace firing 1.0 ton/hr of coal at an air/fuel ratio of 16.0 lb air/lb fuel. The circumferential area of the boiler tubes is 7000 ft².

19.2. Calculate the maximum radiant rate on the circumference of a tube in the back (second) row of a group of double-row tubes in front of a furnace wall. The average circumferential rate on a tube in the front row is 10,000 Btu/(hr)(ft²). All tubes are 4 in. OD on 7-in. centers, spaced on equilateral triangular pitch.

19.3. Using the Wilson, Lobo, and Hottel equation for a box-type heater, calculate the average circumferential heat-transfer rate in a furnace radiant section when (a) heat liberation is 142,000,000 Btu/hr, excess air = 30 per cent (20.75 lb air/lb fuel), $\alpha A_{cp} = 1970$ ft², $A = 4710$ ft², (b) as in (a) except 60 per cent excess air, (c) as in (a) except tubes (same outside dimension and C-C) are added to make $A = 6000$ ft², (d) as in (a) except heat liberation is 100,000,000 Btu/hr.

19.4. Using the Lobo-Evans method, calculate the fuel required (pounds per hour) in a furnace, having the following characteristics if an average radiant-section rate of 12,000 Btu/(hr)(ft²) of circumferential tube area is to be developed:

Dimension of combustion chamber.....	15 by 30 by 40 ft.
Tube OD.....	5 in.
Tube center-to-center spacing.....	10 in.
Number of tubes (arranged in single row).....	90
Circumferential tube surface.....	4710 ft ²
Total wall area A_T	4300 ft ²
Fuel oil.....	Oil
Atomization medium.....	0.3 lb steam/lb oil
Heating value of fuel (LHV).....	17,130 Btu/lb
Excess air.....	25%
Lb air/lb fuel.....	17.44
Estimated tube temperature.....	1000°F

Note. Assume the average specific heat [Btu/(lb)(°F)] of flue gases to be 0.28 between 60°F (H₂O as vapor) and the exit flue-gas temperature. For water-vapor and CO₂ concentrations in the flue gases, see Example 19.1.

19.5. Calculate the fuel requirement for a furnace in which 2000 lb/hr of aluminum chips are to be melted continuously. The chips are charged at 60°F. The furnace is 12 by 12 ft inside cross-sectional dimensions. The height between the level of the molten aluminum and the roof is 9 ft. The emissivity of the mass of aluminum may be taken as 0.30. Fuel oil is fired as in Prob. 19.4 with 17.44 lb air/lb fuel and 0.3 lb steam/lb fuel. Assume same average specific heat for flue gas as in Problem 19.4.

Note. Use the Lobo-Evans equation, correcting from a tube emissivity of 0.9 used in the derivation of the overall exchanger factor to an emissivity of 0.3. The value of F_e is found using the chart. ϵ_F is found by substituting 0.9 for ϵ_s in Eq. (19.7).

NOMENCLATURE FOR CHAPTER 19

A	Heat-transfer surface for convection, ft ²
A_{cp}	Area of a cold plane replacing a bank of tubes, ft ²
A_R	Area of effective refractory surfaces, ft ²
A_T	Total area of furnace surfaces, ft ²
A'	Effective or "seen" heat-transfer area of sink or cold body, ft ²
a_G	Absorptivity of gas, dimensionless
C	Specific heat, Btu/(lb)(°F)
C_R	Lb fuel/(hr)(ft ² projected radiant-heating surface)
C'	Influence of air preheat, dimensionless
e	Furnace efficiency defined by Eq. (19.14), per cent
F	Factor for type of coal firing, dimensionless
$\mathcal{F}$	The overall exchange factor to allow for the geometry and emissivities, dimensionless
$\mathcal{F}_{bA}$	Factor to allow for the geometry of a system with a black-body sink, dimensionless
F_c	Conductivity factor, dimensionless
F_s	Slag factor, dimensionless
F_e	Emissivity factor, dimensionless
G'	Ratio of air to fuel, lb/lb; $(1 + G')$, ratio flue gas to fuel, lb/lb
h_c	Convection coefficient, Btu/(hr)(ft ²)(°F)
h_r	Equivalent radiation coefficient, Btu/(hr)(ft ²)(°F)
K_1, K_2	Constants
L	Mean length of beam, ft
N_t	Number of tubes
p	Pressure, atm
P_T	Tube pitch or center-to-center distance, in.
Q	Radiant-heat flow, Btu/hr
Q_{RC}	Radiant-heat flow at a point in the convection section, Btu/hr
Q_s	Sensible-heat flow, Btu/hr
Q_w	Heat loss through furnace wall, Btu/hr
ΣQ	Combined radiation and convection heat flow, Btu/hr
q	Heat flux, Btu/(hr)(ft ²)
T_s	Mean temperature of receiver, °F
T_1	Temperature of source, °R
T_2	Temperature of receiver, °R
W	Fuel rate, lb/hr

α	Effectiveness factor, dimensionless
β	Angle, deg
ϵ	Emissivity, dimensionless
ϵ_F	Effective emissivity of the furnace cavity, dimensionless
σ	Stefan-Boltzmann constant, 0.173×10^{-8} Btu/(hr)(ft ²)(°R ⁴)

Subscripts (except as noted above)

A	Air
b	Black body
C	Carbon dioxide
F	Fuel
G	Gas
R	Recirculated
S	Surface
W	Water

CHAPTER 20

ADDITIONAL APPLICATIONS

Introduction. There are a number of collateral uses for heat-transfer equipment which have not appeared in any of the preceding chapters. Some of these include the commonest and least expensive forms of heat-transfer surface such as coils, submerged pipes in boxes, and trombone coolers. For the most part the heat-transfer elements treated here are not very closely related to those discussed in earlier chapters, nor can their performances be calculated with equal accuracy. This is an important limitation when attempting to calculate the surface requirements for a close temperature approach. The following are treated here:

1. Jacketed vessels
2. Coils
3. Submerged-pipe coil
4. Trombone cooler
5. Atmospheric cooler
6. Evaporative condenser
7. Bayonet
8. Falling-film exchanger
9. Granular materials in tubes
10. Electric resistance heaters

1. JACKETED VESSELS

Vessels without Agitation. Few data are available in the literature for predicting coefficients within a jacket or between a jacket and a liquid inside a vertical cylindrical vessel in which no mechanical agitation occurs. During heating, the latter rely upon free convection which has not been correlated except as given in Chap. 10. Free convection-heating coefficients can be approximated for vessels of large diameter by Eqs. (10.8) through (10.11). No consistent data are available for cooling by free convection, although the coefficients are undoubtedly lower.

Colburn¹ has tabulated the results of a number of studies from which several broad generalizations can be drawn. For the transfer of heat from steam condensing in a jacket to water boiling within the vessel the clean overall coefficient is about 250 Btu/(hr)(ft²)(°F) for a copper vessel

¹ Perry, J. H., "Chemical Engineers' Handbook," 3d ed., pp. 481-482, McGraw-Hill Book Company, Inc., New York, 1950.

and 175 for a steel vessel. The difference is due to the conductivities and equivalent structural thicknesses of the two metals, respectively. The same coefficients might also be expected for boiling dilute aqueous solutions. For water-to-water heating or cooling, an overall coefficient of 100 Btu/(hr)(ft²)(°F) appears to be in order provided neither stream is refrigerated. The coefficient is perhaps between 75 and 80 for aqueous solutions whose properties do not differ greatly from those of pure water. For heating or cooling nonviscous hydrocarbons the overall coefficient should be reduced to about 50. For fluids classified as medium organics in Table 8 a more probable range of coefficients is from 10 to 20. The jacket of a vessel may be baffled helically to assure positive circulation.

A heat-transfer coefficient selected from the groups above cannot be incorporated into the Fourier equation $Q = UA \Delta t$ except when the vessel operates under steady state. A jacketed vessel can be adapted for operation under steady-state conditions when there is a continuous influx and removal of the liquid entering it. Since a jacketed vessel is primarily a batch vessel, the temperature difference during batch heating or cooling is not constant. The coefficient must consequently be substituted into a suitable *unsteady*-state equation, such as Eqs. (18.5), (18.7), (18.9), or (18.11), which takes into account the time required to change the temperature of the batch while employing a temperature difference which varies with time.

Mechanically Agitated Vessels. Chilton, Drew, and Jebens¹ have published an excellent correlation on both jacketed vessels and coils under batch and steady-state conditions and employing the Sieder-Tate heat-transfer factor j with a Reynolds number modified for mechanical agitation. They employed a flat paddle.

Although much of the work was carried out on a vessel 1 ft in diameter, checks were also obtained on vessels five times those of the experimental setup. The deviations on runs with water were the highest for the fluids tested, which included lube oils and glycerol, and were in some instances off by 17.5 per cent. Additional applications of this method have been discussed by Mack and Uhl² along with calculations. In jacketed vessels it was found that the correlation held up to the agitator rate at which air was beaten into the liquid. For water this corresponded to an agitator rate of 200 rpm, and for other liquids the rate was higher. In Fig. 20.1 a typical jacketed-vessel

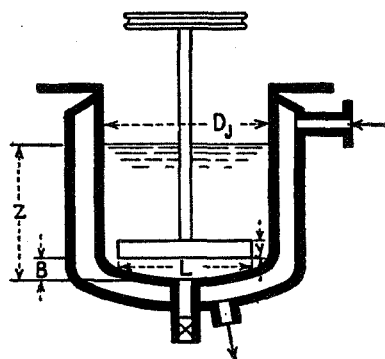


FIG. 20.1. Jacketed vessel.

¹ Chilton, T. H., T. B. Drew, and R. H. Jebens, *Ind. Eng. Chem.*, **36**, 510-516 (1944).

² Mack, D. E., and V. W. Uhl, *Chem. Eng.*, **54**, No. 9, 119-125; No. 10, 115-116 (1947).

assembly is shown. It consists of a vessel and jacket with suitable means for circulating the jacket fluid and a flat two-bladed paddle agitator.

The dimensions essential to the calculation are the height of the wetted portion of the vessel z , the diameter of the vessel D_j , the length of the paddle L , and the height of the bottom of the paddle above the bottom of the vessel B . Studies made by White and coworkers¹ indicate that power requirements can be determined as a function of the modified Reynolds number, $Re_j = L^2 N \rho / \mu$, where L is the length of the paddle in

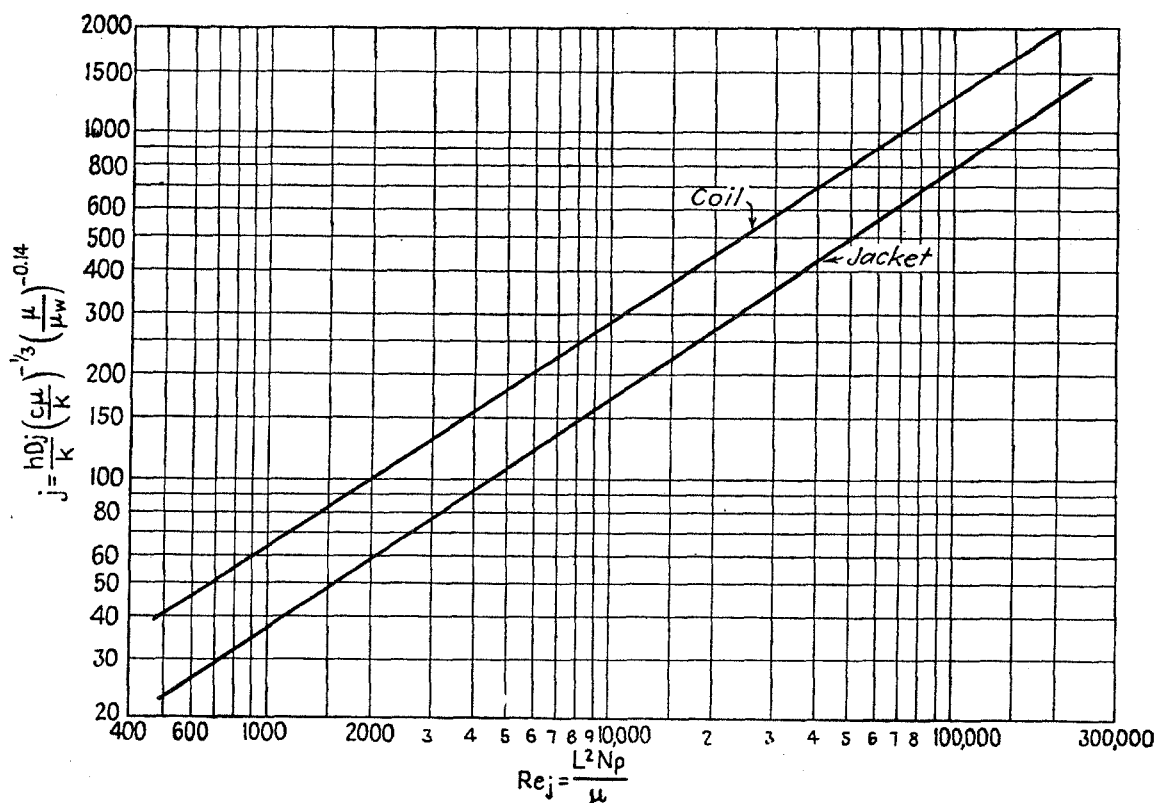


FIG. 20.2. Heat-transfer coefficients from jackets and coils. (Chilton, Drew, and Jebens, *Industrial and Engineering Chemistry*.)

feet, N the number of revolutions per hour, ρ the average density, and μ the viscosity of the liquid. Chilton, Drew, and Jebens have employed the same modified Reynolds number for heat transfer. Their results are given by the consistent equation

$$\frac{h_j D_j}{k} = 0.36 \left(\frac{L^2 N \rho}{\mu} \right)^{2/3} \left(\frac{c_p \mu}{k} \right)^{1/3} \left(\frac{\mu}{\mu_w} \right)^{0.14} \quad (20.1)$$

Equation (20.1) is plotted as the lower line in Fig. 20.2. It is seen that to scale up from pilot to plant size the change in the coefficient is given by

$$\frac{h_{j1}}{h_{j2}} = \left(\frac{L_1}{L_2} \right)^{2/3} \left(\frac{N_1}{N_2} \right) \quad (20.2)$$

¹ White, A. M., E. Brenner, G. A. Phillips, and M. S. Morrison, *Trans. AIChE*, **30**, 585 (1934).

The power requirements are given by the dimensional equation of White and coworkers.

$$\text{hp} = 1.29 \times 10^{-4} D_i^{1.1} L^{2.72} N'^{2.86} y^{0.32} \mu'^{0.14} \rho^{0.86} \quad (20.3)$$

where y = width of the agitator, ft

N' = speed, rps

μ' = the viscosity, lb/ft $\times$ sec

These equations hold for paddle agitators centrally located with $L > 0.3D_i$ and with a height $< L/6$. The coefficient in the jacket itself has not been determined, although it can be readily estimated for both steam and water. The use of this method for the solution of a typical problem is given below.

Example 20.1. Calculation of a Jacketed Vessel. A jacketed vessel has the following dimensions: A 12-in. IPS pipe is jacketed by a 14-in. IPS pipe both having dished bottoms. It is equipped with a paddle agitator 7.2 in. long and 1.2 in. high located at a distance of 1.8 in. from the bottom. The drive speed is 125 rpm. It is to be filled to a height of 10 in. with a batch of aqueous reacting liquor at 150°F which requires the addition of 32,600 Btu/hr to supply the endothermic heat of reaction and maintain temperature. A dirt factor of 0.005 should be provided. At what temperature must steam be available for the jacket?

Solution. This may be regarded as a steady-state problem, since Δt is constant.

$$L = \frac{7.2}{12} = 0.6 \text{ ft}$$

$$N = 125 \times 60 = 7500 \text{ rev/hr}$$

$$\rho = 62.5 \text{ lb/ft}^3 \text{ (approximately)}$$

$$\text{At } 150^\circ\text{F}, \mu = 0.44 \times 2.42 = 1.06 \text{ lb/(ft)(hr)} \quad (\text{Fig. 14})$$

$$k = 0.38 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F/ft)} \quad (\text{Table 4})$$

$$c = 1.0 \text{ Btu/(lb)(}^\circ\text{F)}$$

$$Re_i = \frac{L^2 N \rho}{\mu} = \frac{0.6^2 \times 7500 \times 62.5}{1.06} = 160,000$$

$$j = 1100 = \frac{h_i D_i}{k} \left(\frac{c\mu}{k} \right)^{-1/3} \left(\frac{\mu}{\mu_w} \right)^{-0.14} \quad (\text{Fig. 20.2})$$

$$\left(\frac{\mu}{\mu_w} \right)^{0.14} \text{ may be regarded as 1.0 for water.}$$

$$D_i = \frac{12.09}{12} = 1.01 \text{ ft} \quad (\text{Table 11})$$

$$\left(\frac{c\mu}{k} \right)^{1/3} = \left(1 \times \frac{1.06}{0.38} \right)^{1/3} = 1.41$$

$$h_i = j \frac{k}{D_i} \left(\frac{c\mu}{k} \right)^{1/3} \left(\frac{\mu}{\mu_w} \right)^{0.14}$$

$$= 1100 \times \frac{0.38}{1.01} \times 1.41 \times 1.0 = 588 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

For the steam in the jacket referred to the inside diameter of the vessel,

$$h_{oi} = 1500 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

Using the inside surface as reference,

$$U_c = \frac{h_i h_{oi}}{h_i + h_{oi}} = \frac{588 \times 1500}{588 + 1500} = 422 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad (6.38)$$

$$R_d = 0.005 \quad h_d = \frac{1}{R_d} = \frac{1}{0.005} = 200$$

$$\frac{1}{U_D} = \frac{1}{U_c} + R_d$$

or

$$U_D = \frac{U_c h_d}{U_c + h_d} = \frac{422 \times 200}{422 + 200} = 136$$

For calculation of the heat-transfer surface consider the bottom to be a flat plate or use tables giving the surface as a function of diameter for elliptical heads.

$$A = \pi \times 1.01 \times 0.83 + \frac{\pi}{4} \times 1.01^2 = 3.43 \text{ ft}^2$$

The temperature difference is the LMTD, since both streams are isothermal.

$$Q = U_D A \Delta t$$

$$\Delta t = \frac{32,600}{136 \times 3.43} = 70^\circ\text{F}$$

Since the reaction occurs at 150°F , the temperature of the steam must be

$$T_s = 150 + 70 = 220^\circ\text{F}$$

Note: If a higher temperature of steam were required, the jacket wall temperature must be checked by Eq. (6.8) to assure that it is not higher than the boiling temperature of the vessel fluid. If vapor bubbles form at the inside surface, the film coefficient may be reduced by an unpredictable amount.

2. COILS

Introduction. Tube coils afford one of the cheapest means of obtaining heat-transfer surface. They are usually made by rolling lengths of copper, steel, or alloy tubing into helixes or double helical coils in which the inlet and exit are conveniently located side by side. Helical coils of either type are frequently installed in vertical cylindrical vessels with or without an agitator, although free space is provided between the coil and the vessel wall for circulation. When such coils are used with mechanical agitation, the vertical axis of the agitator usually corresponds to the vertical axis of the cylinder. Double helical coils may be installed in shells with the coil connections passing through the shell or shell cover. Such an apparatus is similar to a tubular exchanger, although limited to smaller surfaces. Another type of coil is the pancake coil, which is a spiral rolled in a single plane so as to lie horizontally near the bottom of a vessel to transfer heat by free convection. Examples of the single helix and pancake are shown in Fig. 20.3. The rolling of coils, particularly

with diameters above 1 in., requires a special winding technique to prevent the tube from flattening into an elliptical cross section, since distortion reduces the flow area.

Tube-side Coefficients. Because of the increased turbulence it is to be expected that the tube-side coefficients for a coiled tube will be greater for a given weight flow than that for a straight tube. For a double pipe helical water-to-water exchanger Richter¹ obtained overall coefficients which are about 20 per cent greater than those computed for straight pipes using similar flow velocities. Jeschke² obtained data on the cooling of air in 1¼-in. steel coiled tubing. For ordinary use McAdams³ suggests that straight tube data such as Eqs. (6.1) and (6.2) or Fig. 24 can be used when the value of h so obtained is multiplied by $1 + 3.5(D/D_H)$, where D is the inside diameter of the tube in feet and D_H is the diameter of the helix in feet. McAdams also suggests that in the absence of data on specific liquids the same correction can be applied to them. Precise corrections are not important, since in most instances it is customary to use either cold water or steam in the tube, neither of which is controlling. For water flowing in the tubes it is suggested that the uncorrected coefficients be obtained from Fig. 25.

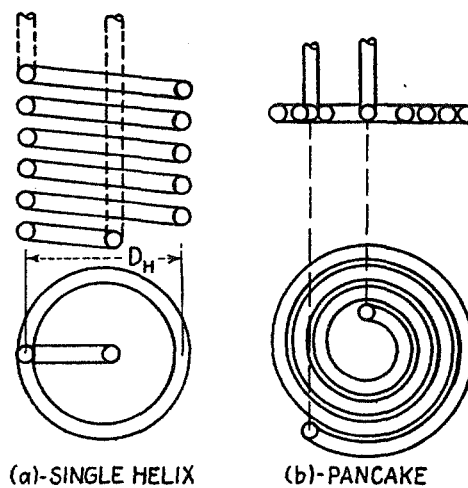


FIG. 20.3. Types of tube coils.

Outside Coefficients to Fluids without Mechanical Agitation. There is a dearth of data in the literature on the transfer of heat to helical coils by free convection. Colburn⁴ has prepared a table of the available overall coefficients. The vertical helical coil is poorly adapted for free-convection heating, since the same liquid rises from the lowest to the highest turns successively, thereby reducing the effectiveness of the upper turns. Coefficients for pancake coils can be approximated from Eq. (10.7) or (10.11). To date, however, no standard method has appeared for calculating film coefficients for the outside of a single or double helical shell-and-coil type of exchanger. When employed for cooling fluids in vessels the effect of free convection is small.

¹ Richter, G. A., *Trans. AIChE*, **12**, Part II, 147-185 (1919).

² Jeschke, D., *Z. Ver. deut. Ing.*, **69**, 1526 (1925); *Z. Ver. deut. Ing., Ergänzungsheft*, **24**, 1 (1925).

³ McAdams, W. H., "Heat Transmission," 2d ed., pp. 177, 184, McGraw-Hill Book Company, Inc., New York, 1942.

⁴ Colburn, A. P., in Perry, *op. cit.*, p. 481.

Outside Coefficients to Fluids with Mechanical Agitation. Chilton, Drew, and Jebens¹ also obtained a correlation for heat transfer to fluids in vessels with mechanical agitation heated or cooled by submerged tube coils as shown in Fig. 20.4. Their equation for the coil is similar to that for the jacketed vessel with the same deviation and is given by

$$\frac{h_c D_j}{k} = 0.87 \left(\frac{L^2 N \rho}{\mu} \right)^{\frac{2}{3}} \left(\frac{c \mu}{k} \right)^{\frac{1}{3}} \left(\frac{\mu}{\mu_w} \right)^{0.14} \quad (20.4)$$

Equation (20.4) is represented by the upper line in Fig. 20.2. Its use is also similar. As with jacketed vessels, caution is again directed to the fact that for batch applications the value of h_c cannot be used with the Fourier equation. Instead, an appropriate equation must again be selected from Chap. 18. However, if the vessel is operated with continuous feed and continuous overflow, the value of h_c and U_D obtained from Eq. (20.4) can be substituted into the Fourier equation. If possible, the dirt factor should be made the limiting resistance.

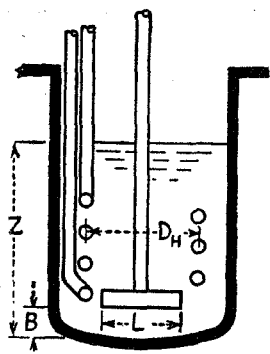


FIG. 20.4. Tube coil in a vessel.

Outside Coefficients Using Vertical Pipes. One of the disadvantages resulting from the use of a paddle-agitator and helical-coil arrangement is its low mixing efficiency. For good mixing and correspondingly higher transfer coefficients the agitator must impart vertical as well as horizontal flow lines. When using an agitator of the paddle or vertical-blade turbine

types and radial tube banks with tubes arranged vertically in the vessel, the tubes act as baffles. Rushton, Lichtmann, and Mahony² investigated this type of arrangement employing a 4-ft tank and a liquid level 4 ft high. Four banks of four 1-in. IPS vertical pipes were arranged at right angles about both a 16-in. six-blade turbine and a 12-in. four-blade turbine. The maximum coefficients were obtained when the mixing turbine was located at a height 2 ft above the bottom. The heat-transfer coefficients have been reported for water as functions of the Reynolds number by the following dimensional equations:

For the 16-in. six-blade turbine:

$$\text{Heating: } h_c = 0.00285 \frac{L^2 N \rho}{\mu} \quad (20.5a)$$

$$\text{Cooling: } h_c = 0.00265 \frac{L^2 N \rho}{\mu} \quad (20.5b)$$

¹ Chilton, T. H., T. B. Drew, and R. H. Jebens, *loc. cit.*

² Rushton, J. H., R. S. Lichtmann, and L. H. Mahony, *Ind. Eng. Chem.*, **40**, 1082-1087 (1948).

For the 12-in. four-blade turbine:

$$\text{Heating: } h_c = 0.00235 \left(\frac{L^2 N \rho}{\mu} \right)^{0.7} \quad (20.6a)$$

$$\text{Cooling: } h_c = 0.00220 \left(\frac{L^2 N \rho}{\mu} \right)^{0.7} \quad (20.6b)$$

where L is the diameter of the turbine.

Example 20.2. Calculation of a Tube Coil. The thermal conditions of Example 20.1 will be used. 32,600 Btu/hr will be introduced into an isothermal liquid at 150°F using steam at 220°F. The coils will consist of turns of ½ in. OD copper tubing with a mean coil diameter of 9.6 in. How many turns are required?

Solution:

$$L = 0.6 \text{ ft} \quad N = 7500 \text{ rev/hr} \quad \rho = 62.5 \text{ lb/ft}^3 \quad \mu = 1.06 \text{ lb/(ft)(hr)}$$

$$k = 0.38 \text{ Btu/(hr)(ft}^2\text{)(°F/ft)} \quad c = 1.0 \text{ Btu/(lb)(°F)}$$

$$Re_i = \frac{L^2 N \rho}{\mu} = 160,000$$

$$j = 1700 \text{ (upper line, Fig. 20.2)}$$

$$D_i = 1.01 \text{ ft} \quad \left(\frac{c\mu}{k} \right)^{1/3} = 1.41, \quad \left(\frac{\mu}{\mu_w} \right)^{0.14} = 1.0 \text{ for water}$$

$$h_c = j \frac{k}{D} \left(\frac{c\mu}{k} \right)^{-1/3} \left(\frac{\mu}{\mu_w} \right)^{-0.14}$$

$$= 1700 \times \frac{0.38}{1.01} \times 1.41 = 900 \text{ Btu/(hr)(ft}^2\text{)(°F)}$$

For steam,

$$h_{oi} = 1500$$

$$U_C = \frac{h_c h_{oi}}{h_c + h_{oi}} = \frac{900 \times 1500}{900 + 1500} = 562 \text{ Btu/(hr)(ft}^2\text{)(°F)}$$

$$R_d = 0.005 \quad h_d = \frac{1}{0.005} = 200$$

$$U_D = \frac{U_C h_d}{U_C + h_d} = \frac{562 \times 200}{562 + 200} = 147.5$$

$$A = \frac{Q}{U_D \Delta t} = \frac{32,600}{147.5 \times (220 - 150)} = 3.16 \text{ ft}^2$$

$$\text{External surface/lin ft} = 0.1309 \text{ ft}^2/\text{ft}$$

$$\text{Per turn} = \pi \times 0.8 \times 0.1309 = 0.328 \text{ ft}^2$$

$$\text{Turns} = \frac{3.16}{0.328} = 9.6$$

(Table 10)

3. SUBMERGED-PIPE COIL

Introduction. This is one of the simplest and cheapest methods of obtaining both cooling and condensing surface. A series of pipes are connected by standard fittings and are submerged in a concrete or wood trough with water circulating about the pipes as shown in Fig. 20.5. Coolers of this type are of considerable value when the hot fluid is corrosive or erosive such as when carrying suspended abrasive particles. The calculations of the trough side are naturally only approximate, but

since water flows in the trough, it is not the limiting resistance except when the coil is employed to condense steam.

Temperature Difference in the Submerged-pipe Coil Cooler. Since the flow of water outside the coil is mostly along the axes of the tubes, the true temperature difference depends upon the arrangement of the tube passes. The trough is usually arranged for one pass. If the pipes are connected by a header at each end with a single pass so that the pipe fluid is in counterflow with the water, the true temperature difference is given by the LMTD. If the pipes are connected by return elbows in a multi-pass arrangement, the flow pattern may be treated as parallel flow-counterflow, with the 1-2 exchanger correction applying provided the trough liquid is reasonably mixed at every point along the length of the pipes. For crossflow arrangements a corresponding correction can

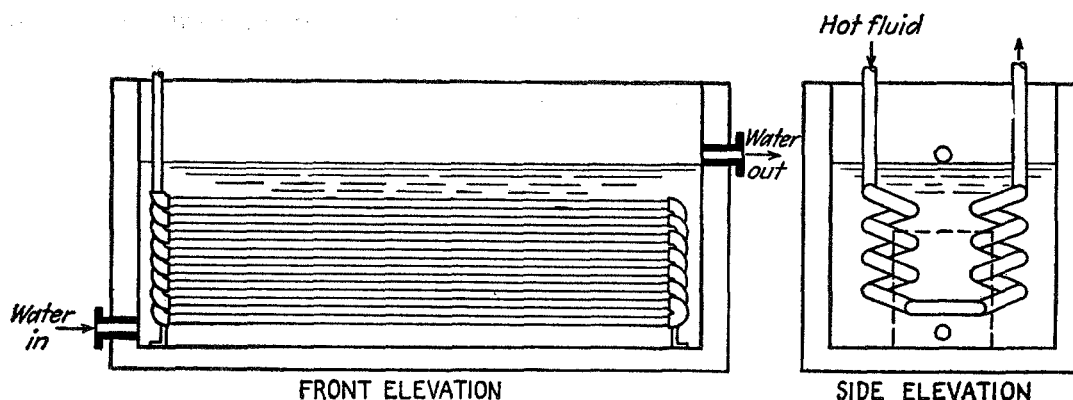


FIG. 20.5. Submerged-pipe coil.

be obtained from Fig. 16.17 or 20.7b in the event that any of these flow patterns apply. Coefficients for the pipe side can be obtained from Fig. 24.

Trough Heat-transfer Coefficients. Trough heat-transfer coefficients are usually difficult to evaluate. There are no conventional arrangements and few published data for this type of equipment. In installations with little or no baffling, a large portion of the cooling water by-passes the surface between the coil and the trough enclosure. Owing to the low water velocities usually encountered, pipe coils tend to foul at a rapid rate because of the growth of algae slime and other scales. The use of large fouling factors is requisite not only from the standpoint of dirt but as a means of providing additional design safety. Fouling factors of less than 0.01 should be avoided, in which case the maximum value of the overall design coefficient will be less than 100.

Ordinarily there is considerable free space in the cross section of the trough, so that the actual linear velocity of the water over the coil may be extremely small. At the extreme of low velocities the trough-side

coefficient will approach the free convection coefficient from a pipe to water. To estimate the minimum possible coefficient Fig. 10.4 can be used. However, if there is any sort of distribution of the water, there need be no hesitancy in using some multiple of the values obtained from Fig. 10.4.

Slurries and Suspensions. Pipe coils are becoming increasingly common in modern catalytic processes. The catalyst is often a finely divided dust which forms a slurry or a suspension with the carrying liquid. Because of the possibility that the solid will settle, slurries flow with high velocities to maintain turbulence, and the possibility of solids being thrown out of the liquid at elbows can be lessened by using streamline fittings. Many slurries are extremely erosive, the solids having an abrasive action on the metal, and in this respect the pipe coil is ideally suited, since it is possible to use extra heavy or double extra heavy steel pipe in assembling the coil. If the continuous erosion causes serious contamination to the product or poisons the catalyst, it is possible to resort to harder alloy pipes fabricated to IPS dimensions.

Very often data are available separately on the carrying liquid and the solid and not on the combined slurry dispersion. With a slurry it is fairly common to consider the controlling film coefficient to be that which exists between the liquid and the tube wall. The transfer of heat from the liquid to the catalyst particles is not considered to offer an appreciable resistance. The film coefficient can consequently be calculated for the combined heat load in the conventional manner using Eq. (6.2) or Fig. 24 and based upon the properties of the liquid alone except for the viscosity. The presence of the solid changes the viscosity of the liquid to an unpredictable extent, since at low concentrations some fine dusts tend to adsorb large quantities of liquid and greatly increase the viscosity. Other solids appear to be only superficially wetted and do not change the viscosity significantly. Using clays and similar catalysts it may be assumed that a concentration of 2 or 3 lb of solid per gallon of mixture does not increase the viscosity of the liquid by more than 100 per cent, and this higher value is suggested in the absence of actual viscosity data on a slurry.

Example 20.3. Calculation of a Submerged-pipe Coil Slurry Cooler. A spent clay catalyst comes from a catalytic reaction vessel as a slurry dispersed in an oil whose properties correspond to 28°API gas oil. The clay is dispersed to the extent of 1 lb/gal. It enters a submerged pipe coil at 675°F and leaves at 200°F. The viscosity of the mixture at 400°F is 2.3 centipoises. The combined flow is 33,100 lb/hr, and cooling will be effected by water of low mineral content from 120 to 140°F. Pipe lengths will consist of 3-in. extra heavy steel pipe 24 ft long.

What will be the size of the pipe coil?

Solution:

$$T_{av} = \frac{1}{2}(675 + 200) = 437.5^\circ\text{F} \quad c = 0.64 \text{ Btu/lb}$$

$$\text{Oil, } Q = 33,100 \times 0.64(675 - 200) = 10,200,000 \text{ Btu/hr}^*$$

$$\text{Water, } Q = 510,000 \times 1.0(140 - 120) = 10,200,000 \text{ Btu/hr}$$

Δt : It may be assumed that one stream carries all of the pipe liquid. Whether in crossflow or parallel flow-counterflow the true temperature-difference relations have not been derived. A case closely allied to the pipe coil will be treated for the two- (or more) pass trombone cooler. In any event the true temperature difference will be very nearly the same as the LMTD, since the outlet temperatures of both streams do not approach each other closely and the average temperatures are far apart.

$$\text{LMTD} = 230^\circ\text{F} \quad (5.14)$$

T_c and t_c : The use of average temperatures will be satisfactory, since the outside coefficient cannot be determined very accurately.

<i>Hot fluid: tube side, oil</i>	<i>Cold fluid: trough side, water</i>
$a'_t = 6.61 \text{ in.}^2/\text{pipe}$ [Table 11]	Assume poor circulation and a minimum coefficient.
Try all pipes in series.	For a trial assume $h_o = 150 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$
Flow area, $a_t = 6.61/144 = 0.0458 \text{ ft}^2$	
Mass vel, $G_t = W/a_t$	
$= 33,100/0.0458 = 723,000 \text{ lb}/(\text{hr})(\text{ft}^2)$	$t_w = t_{av} + \frac{h_{io}}{h_{io} + h_o} (T_{av} - t_{av})$ [Eq. (6.8)]
The viscosity is given at 400°F , which is near enough for this calculation.	$= 130 + \frac{83.7}{83.7 + 150} (437.5 - 130)$
$\mu = 2.3 \times 2.42 = 5.56 \text{ lb}/(\text{ft})(\text{hr})$	$= 240^\circ\text{F}$
$D = 2.9/12 = 0.242 \text{ ft}$ [Table 11]	$t_f = \frac{1}{2}(t_w + t_{av})$
$Re_t = DG_t/\mu$	$= \frac{1}{2}(240 + 130) = 185^\circ\text{F}$
$= 0.242 \times 723,000/5.56 = 31,300$	Refer to Fig. 10.4:
$j_H = 100$	$\Delta t = 240 - 130 = 110^\circ\text{F}$
$k(c\mu/k)^{1/3} = 0.245 \text{ Btu}/(\text{hr})(\text{ft})(^\circ\text{F}/\text{ft})$ [Fig. 16]	$d_o = 3.5 \text{ in.}$
$h_i = j_H[k(c\mu/k)^{1/3}]/D$ [Eq. (6.15)]	$(\Delta t/d_o) = 110/3.5 = 31.4$
$= 100 \times 0.245/0.242 = 101$	$h_o = 150 \text{ (trial checks)}$
OD = 3.5 in. [Table 11]	
$h_{io} = h_i \times \text{ID}/\text{OD} = 101 \times 2.9/3.5$	
$= 83.7 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$ [Eq. (6.5)]	
$U_C = \frac{h_o h_{io}}{h_o + h_{io}} = \frac{150 \times 83.7}{150 + 83.7} = 53.8 \text{ Btu}/(\text{hr})(\text{ft})(^\circ\text{F})$	
$R_a = 0.010 \quad h_a = \frac{1}{.010} = 100$	
$U_D = \frac{U_C h_a}{U_C + h_a} = \frac{53.8 \times 100}{53.8 + 100} = 35.0$	
$A = \frac{Q}{U \Delta t} = \frac{10,200,000}{35.0 \times 230} = 1265 \text{ ft}^2$	
External surface/lin ft = 0.917 ft ² /ft	(Table 11)
Pipe lengths required = $\frac{1265}{0.917 \times 24} = 58$	

* This is safe, since the specific heat of the solids is only about 0.2 Btu/(lb) (°F) and 33,100 lb/hr represents the combined flow.

These must be arranged in series either in a single helical stack 29 tubes high or, if headroom is not available, in several lower stacks. Actually some boiling occurs outside part of the pipe surface where $t_w > 212^\circ\text{F}$ so as greatly to increase the trough coefficient.

4. TROMBONE COOLERS

Introduction. Trombone coolers are sometimes called *trickle coolers*, *cascade coolers*, *horizontal-film coolers*, *S-type coolers*, and *serpentine coolers*. A sketch is shown in Fig. 20.6. Trombone coolers consist of a bank of standard pipes one above the other in series and over which water trickles downward, partly evaporating as it travels. They have been used extensively in the heavy chemical, brewing, coke, petroleum, and ice-making industries. They are frequently made of ceramic materials for cooling corrosive gases at atmospheric pressure, such as HCl and NO_2 . When calculating heat flow through ceramics the resistance of the pipe wall must be included. The trombone cooler presents two problems: (1) the evaluation of the outside film coefficient and (2) calculation of the crossflow true temperature difference.

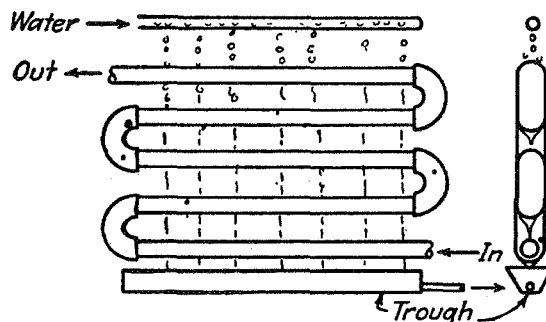


FIG. 20.6. Trombone cooler.

Temperature Difference in the Trombone Cooler. The flow arrangement of Fig. 20.6 in crossflow differs from any of the cases in Fig. 16.17 in that the fluid flowing outside the multipass pipes is not mixed over the length of the pipes whereas the multipass pipe fluid is mixed. Bowman, Mueller, and Nagle¹ have prepared correction factors F_T by which the true temperature difference Δt can be obtained as the product

$$F_T \times \text{LMTD}$$

for both the return-bend and helical types of trombone arrangement. These are given in Fig. 20.7a and b and are based, respectively, on the following:

Return bend arrangement:

$$\frac{1}{1-K} = e^{K_1 R} [\cosh K_1 R + (1 - K_1) \sinh K_1 R]$$

where $K_1 = 1 - e^{-S/2(\tau)}$.

Helical arrangement:

$$\frac{1}{1-K} = e^{K_1 R} (e^{K_1 R} + K_1^2 R)$$

¹ Bowman, Mueller, and Nagle, *Trans. ASME*, **62**, 291 (1940).

where again $K_1 = 1 - e^{-S/2(r)}$ and

$$K = \frac{T_1 - T_2}{T_1 - t_1}, \quad R = \frac{T_1 - T_2}{t_2 - t_1}, \quad (r) = \frac{\text{LMTD}}{T_1 - t_1}, \quad S = \frac{t_2 - t_1}{T_1 - t_1}$$

While these apply specifically to units having but two pipe passes only, a small error arises when the corrections are applied to units having a greater number of pipe passes.

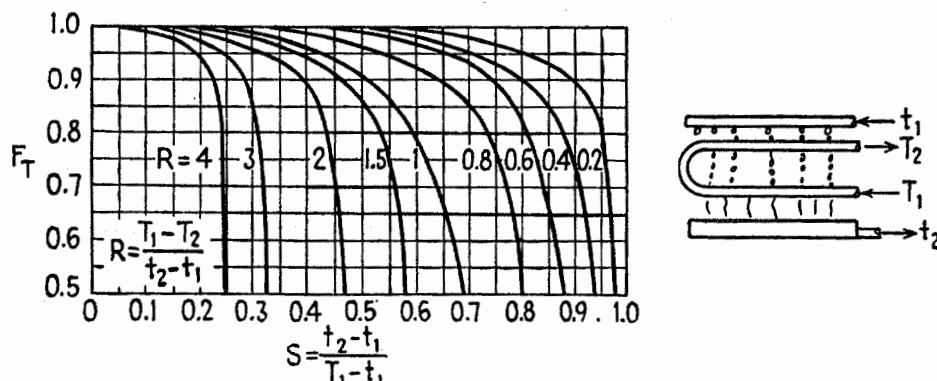


FIG. 20.7a. Mean-temperature-difference correction factor for two-pass trombone.

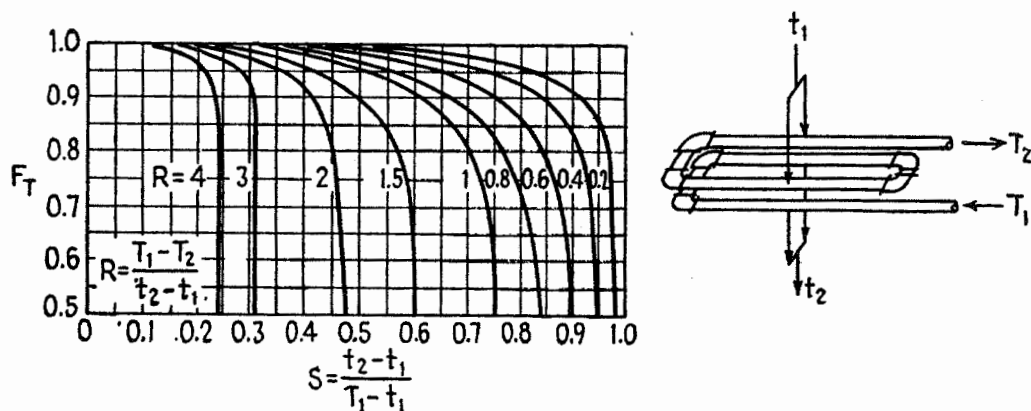


FIG. 20.7b. Mean-temperature-difference correction factor for two-pass helical trombone. (Bowman, Mueller, and Nagle, *Transactions ASME*.)

Outside Film Coefficients. The main published work on this type of apparatus was carried out by McAdams and coworkers at MIT and summarized by McAdams, Drew, and Bays.¹ (1) It is assumed that no evaporation occurs from the surface of the water although it is exposed to the atmosphere. (2) It is assumed that half of the liquid flows down each side of each pipe in streamline flow. The criterion of streamline flow is a Reynolds number, $4G'/\mu$, of less than 2100, where $G' = w/2L$,

¹ McAdams, W. H., T. B. Drew, and G. S. Bays, Jr., *Trans. ASME*, **62**, 627-631 (1940).

w is the water rate in pounds per hour, and L is the length of each pipe in the bank in feet. The equation for the transfer coefficients within ± 25 per cent is given by the dimensional equation

$$h = 65 \left(\frac{G'}{D_0} \right)^{1/4} \quad (20.7)$$

where D_0 is the outside diameter of the pipe in feet. When the value of the Reynolds number exceeds 2100, it is to be expected that the rates will be somewhat higher. Any appreciable evaporation will also increase the film coefficient. Large fouling factors and low outlet-water temperatures are recommended, particularly when the water has a large mineral content.

Example 20.4. Calculation of a Trombone SO_2 Cooler. 3360 lb/hr of SO_2 gas issues from a sulfur burner at 450°F and is to be precooled to 150°F in a high-conductivity pipe-trombone cooler. 3-in. IPS pipes will be used. Owing to space limitations the straight length of the pipe may not exceed 8'0". Cooling water is available at 85°F and should not be heated above 100°F because of scale and corrosion from the outside.

How many pipe lengths are required for the bank when the overall fouling factor is 0.010?

Solution:

$$T_{av} = 300^\circ\text{F} \quad C = 0.165 \text{ Btu}/(\text{lb})(^\circ\text{F})$$

(Fig. 3)

$$\text{SO}_2, Q = 3360 \times 0.165(450 - 150) = 166,500 \text{ Btu/hr}$$

$$\text{Water, } Q = 11,100 \times 1.0(100 - 85) = 166,500 \text{ Btu/hr}$$

Hot Fluid		Cold Fluid	
450	Higher Temp	100	350
150	Lower Temp	85	65
300	Differences	15	285

$$\text{LMTD} = 169^\circ\text{F}$$

(5.14)

R will be $300/15 = 20.0$, which is beyond the values plotted in Fig. 20.7b. The graph can be used by replacing R by $1/R$ and S by RS . This interchanges the two streams without affecting the temperature relationships.

$$R = 20 \quad \frac{1}{R} = 0.05$$

$$S = \frac{(t_2 - t_1)}{T_1 - t_1} = \frac{15}{365} = 0.0412 \quad RS = 20 \times 0.0412 = 0.824$$

$$F_T = 0.98$$

$$\Delta t = 0.98 \times 169 = 166^\circ\text{F}$$

T_c and t_c : Average values of the temperatures will suffice since the gas-film coefficient does not vary greatly from inlet to outlet. $T_{av} = 300^\circ\text{F}$ and $t_{av} = 92.5^\circ\text{F}$.

<i>Hot fluid: tube side, SO₂</i>	<i>Cold fluid outside, water</i>
$a'_t = 7.38 \text{ in.}^2$ [Table 11]	$G = w/2L$
All pipes will be in series.	$= 11,100/2 \times 8 = 694 \text{ lb}/(\text{hr})(\text{ft})$
Flow area, $a_t = 7.38/144 = 0.0512 \text{ ft}^2$	At $t_a = 92.5^\circ\text{F}$,
$G_t = W/a_t$	$\mu = 0.80 \times 2.42 = 1.94 \text{ lb}/(\text{ft})(\text{hr})$
$= 3360/0.0512 = 65,600$	$Re = 4G'/\mu$
$D = 3.068/12 = 0.256 \text{ ft}$ [Table 11]	$= 4 \times 694/1.94 = 1430 \text{ (streamline)}$
At $T_a = 300^\circ\text{F}$,	$D_o = 3.5/12 = 0.292 \text{ ft}$
$\mu = 0.017 \times 2.42 = 0.041 \text{ lb}/(\text{ft})(\text{hr})$	$h_o = 65(G'/D_o)^{1/3}$
[Fig. 15]	$= 65(694/0.292)^{1/3}$
$Re_t = DG_t/\mu$	$= 868 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$
$= 0.256 \times 65,600/0.041 = 410,000$	
$j_H = 790$	
$k = 0.0069$ (nearest value available in Table 5)	
$(c\mu/k)^{1/3} = (0.165 \times 0.041/0.0069)^{1/3}$	
$= 0.99$	
$h_i = j_H \frac{k}{D} \left(\frac{c\mu}{k}\right)^{1/3}$ [Eq. (6.15)]	
$= 790 \times \frac{0.0069}{0.256} \times 0.99 = 21.1$	
$h_{io} = h_i \times \text{ID}/\text{OD}$	
$= 21.1 \times 3.068/3.50$	
$= 18.5 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$	
$U_C = \frac{h_o h_{io}}{h_o + h_{io}} = \frac{868 \times 18.5}{868 + 18.5} = 18.1 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$	
$R_d = 0.010 \quad h_d = \frac{1}{0.010} = 100$	
$U_D = \frac{U_C h_d}{U_C + h_d} = \frac{18.1 \times 100}{18.1 + 100} = 15.3$	
$A = \frac{Q}{U \Delta t} = \frac{166,500}{15.3 \times 169} = 65.6 \text{ ft}^2$	
External surface/lin ft = 0.917	
Number of pipe lengths = $\frac{65.6}{0.917 \times 8} = 8.95$ (use 9)	

5. ATMOSPHERIC COOLERS

Introduction. The atmospheric cooler is an improvement over the trombone cooler particularly for large services. Atmospheric coolers are located in cooling towers beneath the fill (see Chap. 17). They are also known as *spray coolers*, *bare-tube coolers*, and *open* or *atmospheric coolers* and provide tubular heat-transfer surface for certain services at about one-third the cost of conventional shell-and-tube equipment. A typical arrangement in an induced-draft cooling tower is shown in Fig. 20.8. Perhaps the broadest use of atmospheric coolers lies in the cooling of the

jacket water of internal-combustion engines. In a survey of 106 plants in the natural gasoline industry, Kallam¹ found the atmospheric cooler to be the most widely used type, accounting for 44.5 per cent of all installations. Other methods of jacket-water cooling are the direct circulation of the jacket water over the cooling tower, induced-draft air-cooled extended-surface units, and the use of a conventional exchanger between the cold recirculated cooling tower water and the jacket water. Calculations applicable to these methods have already been discussed.

When piped directly from the atmospheric cooler to the water jacket, the water travels in a closed circuit and the same water is continuously heated and cooled. Naturally this minimizes corrosion and the deposition of slime in both the jacket and the tube bundle. Since jacket water is usually recirculated over a short temperature range of 10 to 20°F and

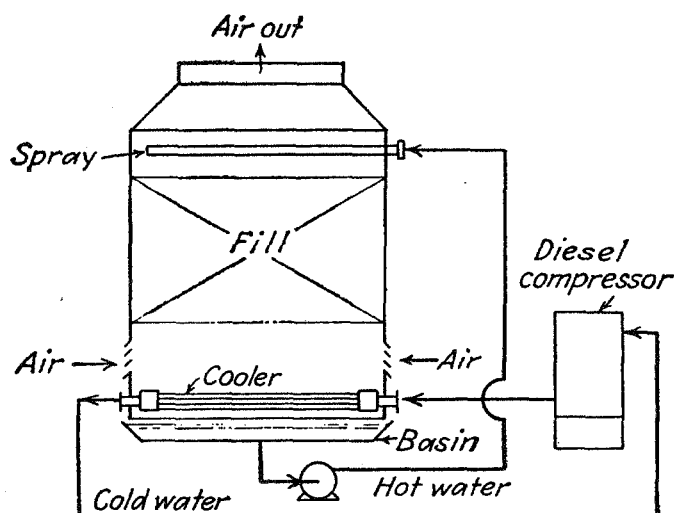


FIG. 20.8. Atmospheric cooler showing an arrangement for jacket water cooling.

at a temperature to the cooler above 120°F, the advantages of the closed circuit are important.

Other methods of cooling jacket water are less advantageous. The high temperature of the recirculated jacket water prohibits its direct circulation over the cooling-tower fill without an atmospheric cooler because the air-saturation loss would be very great. When jacket water is cooled by cooling-tower water in an external exchanger, the pumping cost of the cooling-tower water is increased by the added pressure drop through the exchanger. The equipment cost is also increased. When using an atmospheric cooler it is possible to circulate less cooling water than required for sensible-heat transfer, since some evaporation occurs on the outside of the tubes of the atmospheric cooler. When used for the condensation of exhaust steam from turbines or engines, the atmospheric

¹ Kallam, F. L., *Petroleum Refiner*, 27, 371-378 (1948).

cooler is then called an *evaporative condenser*. This modification will be discussed in the next section.

An atmospheric cooler is a bank of tubes in a supported frame between two cast headers with removable bolted covers. A sketch including alternate nozzle connections is shown in Fig. 20.9. The headers are usually cast in one piece including the nozzles. They vary in length and width to conform with the internal structure dimensions of the cooling-tower supporting structure. Generally the tubes are from 8 to 20 ft long with cooler widths of 3 to 6 ft. The width is small, so that the major tower-supporting beams can be erected between adjoining cooler

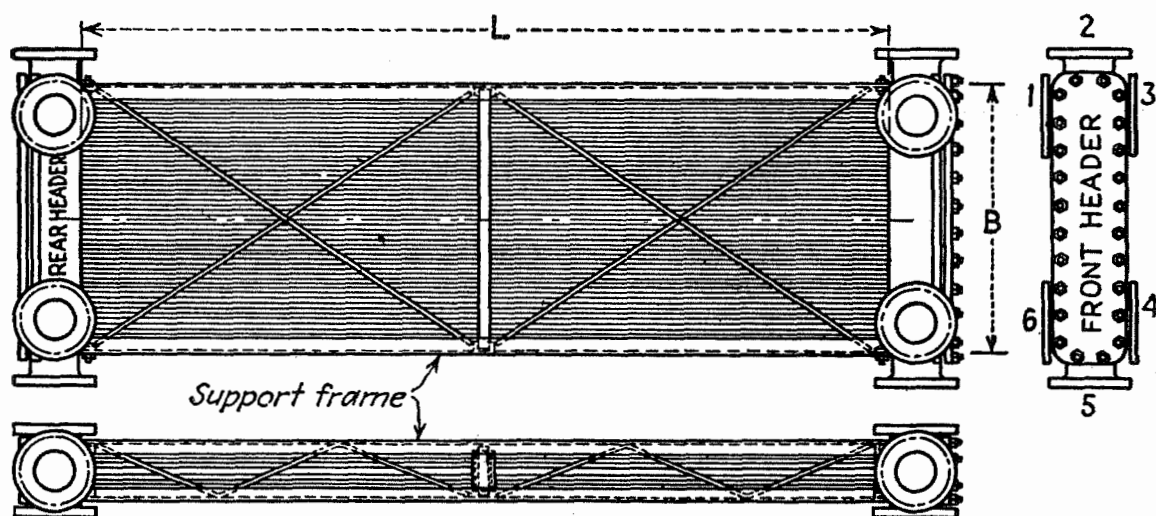


FIG. 20.9. Atmospheric cooler with alternate nozzle arrangements. (Foster Wheeler Corporation.)

sections. Thus a 24- by 24-ft cooling tower might be equipped with four sections having 20-ft tube lengths and 5-ft widths. The tubes are rolled directly into the headers and can be replaced when corroded by simply unbolting the header covers and reaming out the corroded tube. The outside structural framework of the cooler also supports the tubes. If the headers are of a weldable material, the framework is welded directly to them. If the material cannot be welded, the framework is bolted to the headers.

The tubes are laid out in horizontal rows with about three to seven superimposed horizontal rows per header. The vertical pitch between the horizontal rows is usually less than the horizontal pitch, since an adequate water-flow area must be provided to permit the free drainage of water into the basin. Where a larger number of rows is required, it is customary to use two sections one above the other. Tube passes may be orientated in any convenient manner.

The advantages of the atmospheric cooler are lower first cost and lower

pumping cost. The great disadvantage lies in the rapid rate of scale formation on the outside of the tubes due to evaporation. The rate is such that succeeding horizontal rows are always superimposed rather than staggered for accessibility during cleaning and scale removal. While scale may be loosened by thermal shock, the shock does not necessarily cause it to drop off the tubes and waters of different mineral contents present separate scale problems. Scale removal by scraping is often the only solution, and the number of horizontal tube rows per section is kept small in anticipation of the scale. Notwithstanding these precautions, pieces of scale frequently fall from the upper tube rows and lie across the lower rows, interfering with the even distribution of water over the tubes.

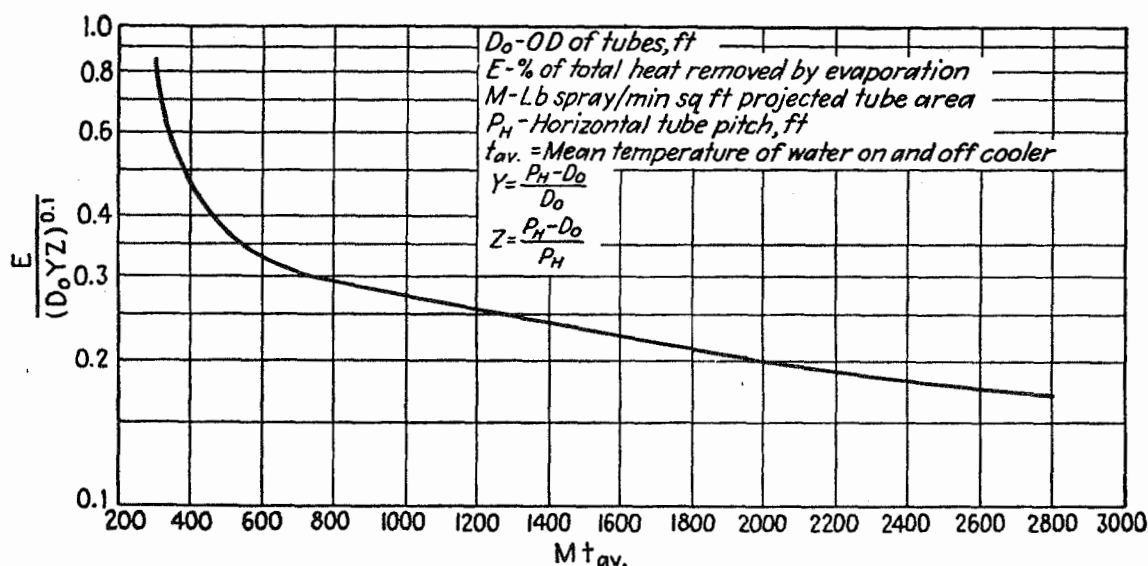


FIG. 20.10. Atmospheric coolers: Heat removed by evaporation. (Kallam, Refiner.)

Kallam recommends fouling factors for design which are identical with those of Table 12 although the service period is very much shorter.

The Calculation of Atmospheric Coolers. The method of calculation employed here is based on the correlation of Kallam.¹ While this method has been cited by Nelson,² the original article presents results without experimental data. Kallam states that 45 tests were run on three coolers embracing the following variations: cooling water rate, 3 to 38 lb/(min) (ft² of projected tube surface); water velocity in tubes, 2 to 14 fps; wind velocity at cooling tower, 10 to 1000 fpm; temperature of cooling water to atmospheric cooler (off the fill), 60 to 78°F; temperature of cooling water off cooler, 88 to 120°F (to basin); temperature of water to tubes, 109 to 153°F; temperature of water from tubes, 90 to 141°F; dry-bulb

¹ Kallam, F. L., *Petroleum Refiner*, **19**, 371-382 (1940).

² Nelson, W. L., "Petroleum Refinery Engineering," 3d ed., McGraw-Hill Book Company, Inc., New York, 1949.

temperature, 60 to 76°F; wet-bulb temperature, 54 to 66°F; heat removed by evaporation of cooling water, 0.4 to 50.0 per cent. No deviations have been given, and Kallam states that the relative humidity of the air above the atmospheric cooler did not influence the percentage of evaporation off the cooler. While the method is entirely empirical, the author knows of two instances in which coolers calculated by Kallam's method proved to be satisfactory in practice. In order to set the temperature range for the cooling-tower water over the cooler, an economic balance must be carried out between the cost of both the cooling tower and the atmospheric cooler for several temperatures to determine the economic optimum.

For the outside film coefficient Kallam employs two dimensionless configuration terms, $Y = (P_H - D_o)/D_o$ and $Z = (P_H - D_o)/P_H$, where P_H

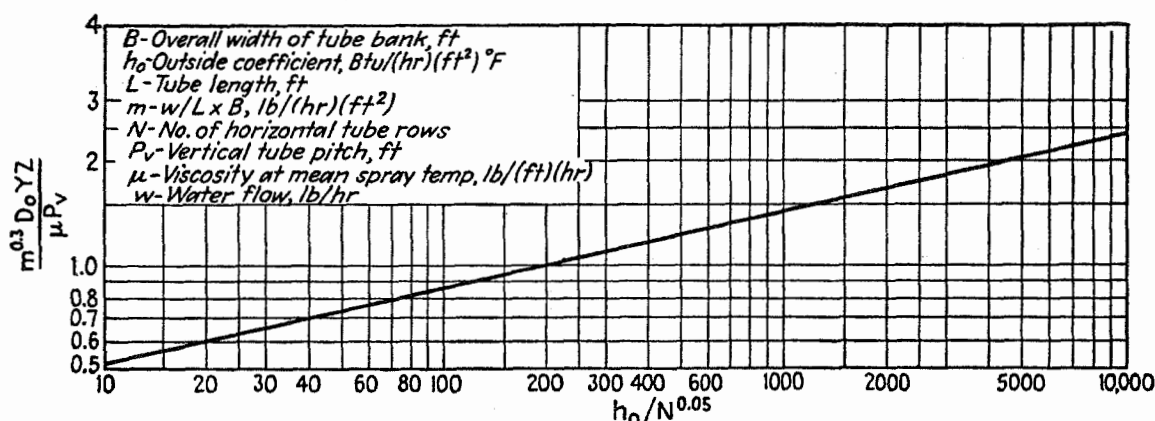


Fig. 20.11. Atmospheric coolers: Outside film coefficients. (Kallam, Refiner.)

is the horizontal pitch and D_o is the outside diameter of the tubes. Calculation curves are given in Figs. 20.10 and 20.11, although the actual groups employed are both arbitrary and dimensional. There are, in addition, two flow terms which are based on different cooler outside areas. These are the *projected tube area* of a horizontal row used for computing the water spray M in pounds per minute per square foot of projected tube area and the overall ground area of the section calculated as the product of the tube length L and the distance between the outer sides of the two extreme tubes in a horizontal row B . The latter is used to compute the spray in pounds per hour per square foot of *base area* between tube sheets m . Figure 20.10 gives the percentage of the total spray water evaporated and is plotted as a function of the spray quantity and its mean temperature over the atmospheric section. This permits adjustment of the spray quantity so that the combined sensible- and latent-heat effects will balance the tube-side heat load. This will be demonstrated in the following illustration:

Suppose that it is desired to remove 1,000,000 Btu/hr from a bare tube cooler consisting of two sections with 20 ft long tubes and a 4 ft wide bundle and that the design of the cooling tower is such that the water temperature will rise from 70 to 80°F. Assume 1 in. OD tubes with 2-in. horizontal pitch and 1¾-in. vertical pitch. If no evaporation occurred, it would be necessary to circulate 100,000 lb/hr or 200 gpm.

To determine the quantity of spray with evaporation:

$$\text{Number of tubes per horizontal row} = \frac{4 \times 12}{2} = 24$$

$$\text{Projected area per tube} = \frac{1}{12} \times 20 = 1.67 \text{ ft}^2$$

$$\text{Total projected area} = 2 \times 24 \times 1.67 = 80 \text{ ft}^2$$

$$D_o = \frac{1}{12} = 0.0833 \text{ ft}$$

$$P_H = \frac{2}{12} = 0.167 \text{ ft}$$

$$Y = \frac{P_H - D_o}{D_o} = \frac{0.167 - 0.0833}{0.0833} = 1.0$$

$$Z = \frac{P_H - D_o}{P_H} = \frac{0.167 - 0.0833}{0.167} = 0.5$$

$$(D_o Y Z)^{0.1} = (0.0833 \times 1.0 \times 0.5) = 0.7278$$

Since the average water temperature over the bundle is 75°F, $M_{t_{av}} = 75M$.

It is now necessary to assume values of the percentage evaporated E as a function of the spray loading, both of which determine the heat load.

Trial 1:

Assume $E = 20$ per cent.

$$\frac{E}{(D_o Y Z)^{0.1}} = \frac{0.20}{0.7278} = 0.275$$

From Fig. 20.10

$$M_{t_{av}} = 1040$$

$$M = \frac{1040}{75} = 13.9 \text{ lb}/(\text{min})(\text{ft}^2)$$

$$\text{Heat removed from cooler sensibly} = 13.9 \times 80 \times 60 \times 10 = 667,000 \text{ Btu}$$

$$\text{Evaporation} = \frac{1,000,000 - 667,000}{1,000,000} \times 100 = 33\%$$

The computed evaporation of 33 per cent does not check the assumed value of 20 per cent, and a new trial must be made. An assumption of 18.5 per cent checks with a spray of 17.05 lb/(min)(ft²). The heat removed sensibly will be 818,000 Btu/hr, and the spray to be circulated will be

$$\frac{818,000}{8.33 \times 60} = 163.6 \text{ gpm}$$

compared with a total of 200 gpm when vaporization is not considered. The outside film coefficient can be obtained from Fig. 20.11 or the dimensional equation

$$h_o = 190N^{0.05} \left(\frac{m^{0.3} D_o Y Z}{\mu P_v} \right)^{4.4} \quad (20.8)$$

where P_v = vertical pitch, ft

$m = w/LB$,

w = total hourly flow over the section

μ = viscosity, lb/(ft)(hr)

N = number of horizontal rows

Temperature Difference in the Atmospheric Cooler. The correct evaluation of the true temperature difference is extremely important in this type of apparatus, since the approach of both outlet streams is usually quite close. As an example of crossflow it is necessary to decide whether the spray liquid is mixed or unmixed in accordance with the derivations covered by Eqs. (16.71) through (16.103). The spray fluid in commercial towers can usually be considered unmixed, and the true temperature difference can be obtained through the use of Fig. 16.17b.

Example 20.5. Calculation of an Atmospheric Jacket Water Cooler. 295 gpm of jacket water from a diesel compressor are to be cooled from 143° to 130°F. It is possible to convert a 12 × 16 ft cooling tower to permit the insertion of two atmospheric sections side-by-side consisting of 1 in. OD × 16 BWG tubes 12 0" long on 1 7/8-in. horizontal pitch and 1 1/2-in. vertical pitch. The distance between the extreme horizontal tubes in each section will be 4'0".

Individual dirt factors of 0.003 for the cooling tower water and 0.001 for the jacket water should be provided.

What will the spray requirement be and how many horizontal rows of tubes should be provided in the sections?

Solution:

Outside film coefficient:

Number of tubes per row/section = $4 \times 12/1.875 = 25.6$ (say 25)

Total projected area = $2(1\frac{1}{2} \times 1 \times 25) = 50 \text{ ft}^2$

Assume a spray of 28 lb/(min)(ft²)

Average temperature over section = $\frac{1}{2}(90 + 110) = 100^\circ\text{F}$

$Mt_{av} = 100 \times 28 = 2800$

From Fig. 20.10

$$\frac{E}{(D_o Y Z)^{0.1}} = 0.171$$

$$D_o = \frac{1}{12} = 0.0833 \text{ ft} \quad P_H = \frac{1.875}{12} = 0.1562 \text{ ft}$$

$$Y = \frac{0.1562 - 0.0833}{0.0833} = 0.874$$

$$Z = \frac{0.1562 - 0.0833}{0.1562} = 0.466$$

$$(D_oYZ)^{0.1} = (0.0339)^{0.1} = 0.712$$

$$E = 0.171 \times 0.712 = 0.12\% \text{ evaporation}$$

$$\text{Heat load} = 295 \times 500(143 - 130) = 1,920,000 \text{ Btu/hr}$$

$$\text{Sensible heat} = 1,920,000(1 - 0.12) = 1,690,000 \text{ Btu/hr}$$

$$\text{Final spray temperature: } 1,690,000 = 28 \times 60 \times 50(t_2 - 90)$$

$$t_2 = 110^\circ\text{F} \text{ Checks for } 28 \text{ lb}/(\text{min})(\text{ft}^2)$$

$$\text{Total spray} = 28 \times 60 \times 50 = 84,000 \text{ lb/hr}$$

$$m = \frac{w}{LB} = \frac{84,000}{2 \times 4 \times 12} = 875 \text{ lb}/(\text{hr})(\text{ft}^2)$$

$$m^{0.3} = 7.62$$

$$\mu_{\text{water}} = 0.76 \text{ cp}$$

$$\frac{m^{0.3}D_oYZ}{\mu P_v} = \frac{7.62 \times 0.0339}{2.42 \times 0.76 \times 0.125} = 1.12$$

From Figure 20.11,

$$\frac{h_o}{N^{0.05}} = 300$$

Assume three horizontal rows/section

$$N^{0.05} = 3^{0.05} = 1.057$$

$$h_o = 300 \times 1.057 = 317 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$$

Tube-side coefficient

Assume an even number of passes (using vertical partitions) arranged to give a tube-side velocity of about 8 fps. Try four tube passes per section or eight total passes.

$$\text{Flow area/pass, } a_t = 3 \times 25 \times \frac{0.594}{4 \times 144} = 0.0775 \text{ ft}^2$$

$$\text{Mass velocity, } G_t = \frac{W}{a_t} = 295 \times \frac{500}{0.0775} = 1,900,000 \text{ lb}/(\text{hr})(\text{ft}^2)$$

$$\text{Velocity} = \frac{G_t}{3600\rho} = \frac{1,900,000}{3600 \times 62.5} = 8.45 \text{ fps}$$

$$\text{ID} = 0.87 \text{ in.}$$

$$h_i = 2300 \times 0.93 = 2140 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$$

(Fig. 25)

$$h_{io} = h_i \times \frac{\text{ID}}{\text{OD}} = 2140 \times \frac{0.87}{1.0} = 1860$$

(6.5)

Overall coefficients:

$$U_c = \frac{h_o h_{io}}{h_o + h_{io}} = \frac{317 \times 1860}{317 + 1860} = 271 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$$

$$\text{External surface/lin ft} = 0.2618 \text{ ft}^2/\text{ft}$$

$$\text{Total surface} = 2(3 \times 25 \times 12 \times 0.2618) = 471 \text{ ft}^2$$

(Table 11)

Δt :

Hot Fluid		Cold Fluid	
143	Higher Temp	110	33
130	Lower Temp	90	40
13	Differences	20	7

$$\text{LMTD} = 36.7^\circ\text{F} \quad (5.14)$$

$$R = \frac{1\frac{3}{2}}{20} = 0.65 \quad S = \frac{20}{53} = 0.377$$

$$F_T = 0.97$$

(Fig. 16.17b)

$$\Delta t = F_T \times \text{LMTD} = 0.97 \times 36.7 = 35.6^\circ\text{F}$$

$$U_D = \frac{Q}{A \Delta t} = \frac{1,920,000}{471 \times 35.6} = 114.5$$

Dirt factor:

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{271 - 114.5}{271 \times 114.5} = 0.0051 \text{ (hr)(ft}^2\text{)(}^\circ\text{F)/Btu}$$

The assumption of three horizontal rows is satisfactory, since a dirt factor of 0.004 was required.

The pressure drop on the tube side can be computed from Eqs. (7.45) through (7.47). The characteristics which the cooling tower must possess to permit the heat removal from the cooling tower water can be determined by the methods of Chap. 17.

6. EVAPORATIVE CONDENSERS

Atmospheric coolers were formerly used for the condensation of steam from turbines and noncondensing steam engines. They are less frequently used for this purpose now. Although Kallam's method is probably satisfactory for the condensation of organic vapors where the condensing coefficient is small by comparison, it is unsatisfactory in the case of steam condensation. The condensation of steam requires the use of high liquid loadings in the cooling tower because of the large heat flux associated with the condensation of a pound of steam. In this range the outside coefficients predicted from Fig. 20.11 are probably unreliable. It is safer to employ a conventional clean value for the overall coefficient than to attempt its calculation. A value of U_C of from 350 to 450 Btu/(hr)(ft²)(°F) should give a reasonable condenser.

7. BAYONET EXCHANGERS

Introduction. A bayonet consists of a pair of concentric tubes, the outer of which has one end sealed as shown in Fig. 20.12. Bayonets were formerly known as *field* tubes. Both the outer and inner tubes extend from separate stationary tube sheets and extend either into shells or directly into vessels. The surface of the outer tube is the principal heat-transfer surface. Bayonet exchangers are excellently adapted to the condensation of vapors under moderate and very low vacuum, outside the bayonets and in exchanger shells, since the tubes are larger and heavier, usually 1 in. OD inside 2 in. OD, and the individual bayonets are generally self-supporting without the use of support plates. These qualities combine to cause a negligible pressure drop to the flow of vapors under vacuum outside bayonets. An example of a vacuum condenser arrangement is shown in Fig. 20.13, while the exchanger shown in Fig. 20.14 is known as a *suction* heater or *suction tank* heater.

Suction heaters are installed in tanks used for the storage of viscous liquids and semiplastics such as molasses, heavy lube oils, fuel oils, and asphalt. If a liquid is viscous, it would be necessary to maintain all the liquid in the tank at a temperature corresponding to a feasible pump-

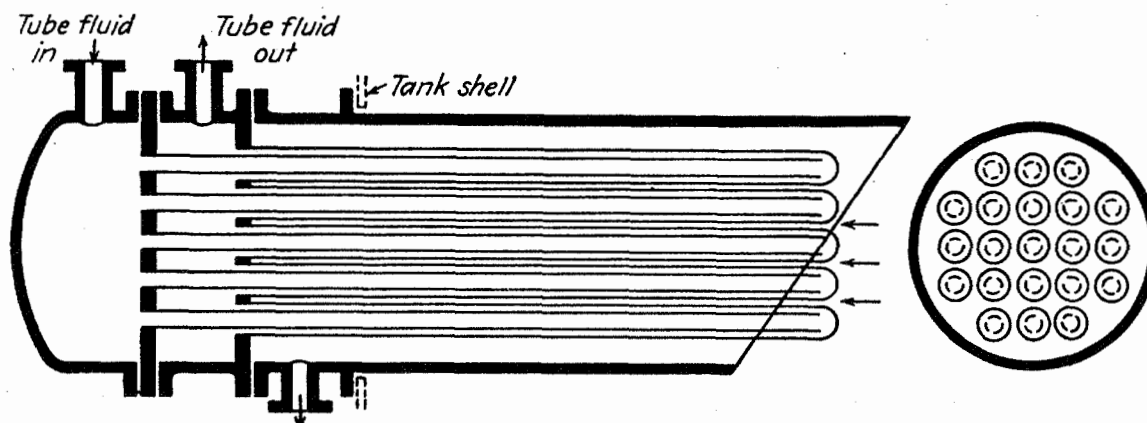


FIG. 20.12. Section through a bayonet (suction-type) exchanger.

ing viscosity. This could be achieved by means of a steam coil, which would be required to supply sufficient heat to offset the tank losses. Since viscous fluids afford little free convection, the coil would probably be supplemented by continuous agitation and if the stored material should

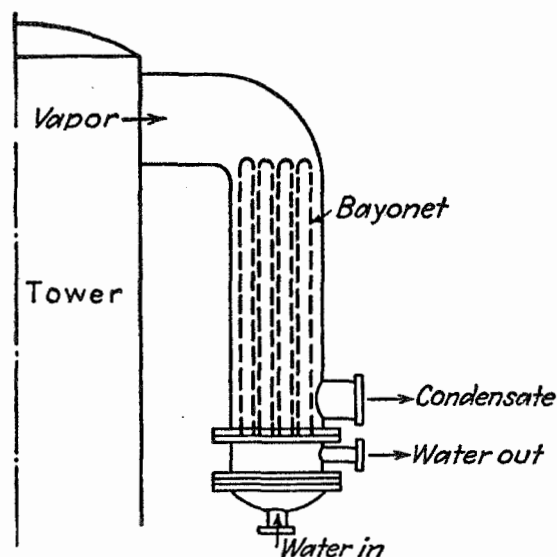


FIG. 20.13. Bayonet exchanger as vacuum condenser.

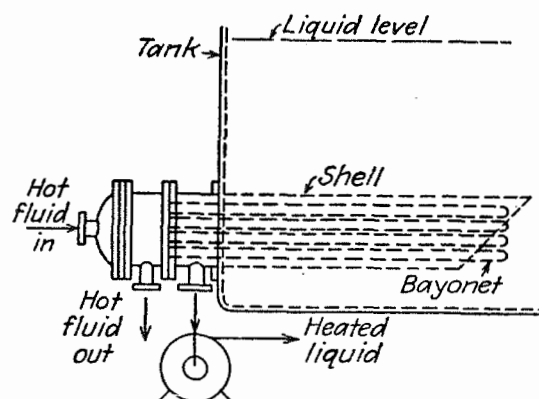


FIG. 20.14. Tank suction heater with bayonet tubes.

cool to too low a temperature, its ultimate removal might cause a serious problem. When a suction heater is installed as shown in Fig. 20.14, heat need be supplied only while the pumps are operating. As hot liquid is removed by the pump, it is replaced by the hydrostatic pressure of the liquid remaining in the tank. Bayonets are extremely effective for this type of service, but they are also rather expensive, and ordinarily the job

can be accomplished more cheaply by substituting an unbaffled U-tube bundle for the bayonets.

Temperature Difference in the Bayonet Exchanger. Bayonet exchangers introduce some interesting temperature differences which have

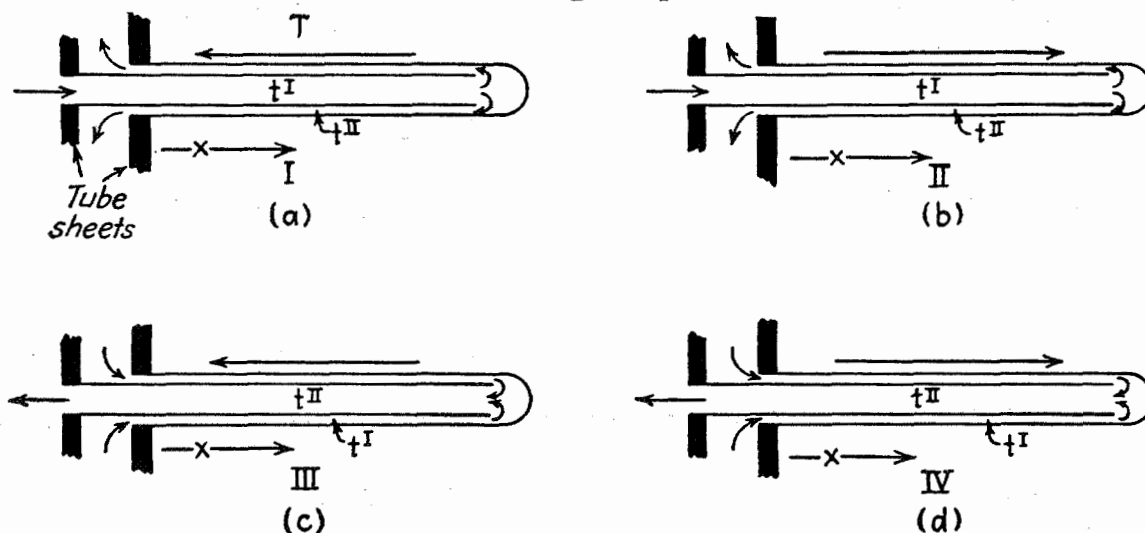


FIG. 20.15. Arrangements of parallel flow and counterflow in bayonet exchangers.

not been encountered in any of the flow patterns discussed thus far. Consider the single bayonet shown in Fig. 20.15a with hot fluid on the shell side traveling in the direction shown. The temperature diagram is shown in Fig. 20.16. Cold liquid flows into the bayonet by the inner tube and is rejected through the annulus in the opposite direction.

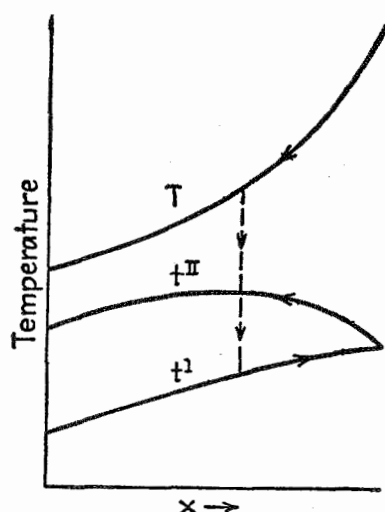


FIG. 20.16. Temperature diagram for Case I. (Hurd, *Industrial Engineering Chemistry*.)

There is a film coefficient on the shell side of the bayonet and a coefficient on the inside of the outer tube yielding the overall coefficient U_c . There is additional heat transfer between the cold medium entering the inner tube and the heated cold medium in counterflow with it in the annulus. The flow areas of the inner tube and annulus differ, although the weight flow is the same in each and the bayonet fluid has different film coefficients in the inner tube and in the annulus. These two coefficients produce a second overall coefficient u . The temperature distribution in Fig. 20.15a is but one of four possibilities.

The remainder are shown in Fig. 20.15b, c, and d. The true temperature difference is seen to be related not only to the flow pattern but also to the ratio of the two overall coefficients U and u .

The derivation employed here is that of Hurd¹ and agrees with an

¹ Hurd, N. L., *Ind. Eng. Chem.*, **38**, 1266-1271 (1946).

unpublished derivation of K. A. Gardner. As reference surfaces the outside diameters of the outer and inner tubes are used to define the cylindrical areas. The area at the sealed end itself is neglected. Letting P be the perimeter of the outer tube and p the perimeter of the inner tube, the differential heat balance on each tube-side stream gives

$$U(T - t^{\text{II}})P dx - u(t^{\text{II}} - t^{\text{I}})p dx = -wc dt^{\text{IV}} \quad (20.9)$$

$$up(t^{\text{II}} - t^{\text{I}}) dx = wc dt^{\text{I}} \quad (20.10)$$

Adding

$$UP(T - t^{\text{II}}) dx = wc(dt^{\text{I}} - dt^{\text{II}}) \quad (20.11)$$

and

$$wc(t^{\text{II}} - t^{\text{I}}) = WC(T_1 - T) \quad (20.12)$$

Taking a heat balance on the right-hand end,

$$\frac{dT}{dx} = \frac{UP}{WC} (T - t^{\text{II}}) \quad (20.13)$$

Differentiating

$$\frac{d^2T}{dx^2} = \frac{UP}{WC} \left(\frac{dT}{dx} - \frac{dt^{\text{II}}}{dx} \right) \quad (20.14)$$

Differentiating Eq. (20.12) with respect to x and rearranging,

$$\frac{dt^{\text{II}}}{dx} = \frac{dt^{\text{I}}}{dx} - \frac{WC}{wc} \frac{dT}{dx} \quad (20.15)$$

From Eqs. (20.10) and (20.12)

$$\frac{dt^{\text{I}}}{dx} = \frac{up}{wc} \frac{WC}{wc} (T_1 - T) \quad (20.16)$$

Substituting Eqs. (20.15) and (20.16) in Eq. (20.14) and letting

$$(T_1 - T) = \tau, \\ -\frac{d^2\tau}{dx^2} + \left(\frac{UP}{WC} - \frac{UP}{wc} \right) \frac{d\tau}{dx} + \frac{UPup}{(wc)^2} \tau = 0 \quad (20.17)$$

Substituting $d\tau/dx = X$ and factoring,

$$-X^2 + \left(\frac{UP}{WC} + \frac{UP}{wc} \right) X + \frac{UPup}{(wc)^2} = 0 \quad (20.18)$$

Solving the quadratic

$$X = \frac{1}{2} \frac{UP}{WC} \left[1 + \frac{WC}{wc} \pm \sqrt{\left(1 + \frac{WC}{wc} \right)^2 + 4 \frac{up}{UP} \left(\frac{WC}{wc} \right)^2} \right] \quad (20.19)$$

Designate the radical by the letter ψ , and calling the plus and minus solutions X_1 and X_2 , respectively,

$$X_1 = \frac{1}{2} \frac{UP}{WC} \left(1 + \frac{WC}{wc} + \psi \right) \quad (20.20)$$

$$X_2 = \frac{1}{2} \frac{UP}{WC} \left(1 + \frac{WC}{wc} - \psi \right) \quad (20.21)$$

Equation (20.17) is a second-order differential equation whose solution is

$$\tau = T_1 - T = C_1 e^{X_1 x} + C_2 e^{X_2 x} \quad (20.22)$$

From the terminal conditions

$$T = T_1 \quad \text{when} \quad x = L$$

and

$$T = T_2 \quad \text{when} \quad x = 0$$

from which

$$C_1 = - \frac{(T_1 - T_2) e^{X_2 L}}{e^{X_1 L} - e^{X_2 L}} \quad \text{and} \quad C_2 = \frac{(T_1 - T_2) e^{X_1 L}}{e^{X_1 L} - e^{X_2 L}} \quad (20.23)$$

Differentiating Eq. (20.22) with respect to x and substituting in Eq. (20.13),

$$C_1 X_1 e^{X_1 x} + C_2 X_2 e^{X_2 x} = - \frac{UP}{WC} (T - t^w) \quad (20.24)$$

Evaluated at $x = 0$ it becomes

$$C_1 X_1 + C_2 X_2 = - \frac{UP}{WC} (T_2 - t_2) \quad (20.25)$$

Substituting Eqs. (20.20), (20.21), and (20.23) in Eq. (20.25),

$$\frac{1}{2} (T_1 - T_2) \left[1 + \frac{WC}{wc} - \psi \left(\frac{e^{X_1 L} + e^{X_2 L}}{e^{X_1 L} - e^{X_2 L}} \right) \right] = t_2 - T_2 \quad (20.26)$$

Solving for $e^{X_1 L}/e^{X_2 L}$ and taking logarithms

$$(X_1 - X_2)L$$

$$= \ln \left[\frac{(T_2 - t_2) + \frac{1}{2} (T_1 - T_2) \left(1 + \frac{WC}{wc} \right) + \frac{1}{2} \psi (T_1 - T_2)}{(T_2 - t_2) + \frac{1}{2} (T_1 - T_2) \left(1 + \frac{WC}{wc} \right) - \frac{1}{2} \psi (T_1 - T_2)} \right] \quad (20.27)$$

Eliminating X_1 and X_2 by Eq. (20.20) and eliminating L by introduction of the Fourier equation $WC(T_1 - T_2) = UPL \Delta t$, where Δt is the true temperature difference between the shell side and tube side,

$$\Delta t = \frac{(T_1 - T_2)\psi}{\ln \left[\frac{(T_2 - t_2) + \frac{1}{2}(T_1 - T_2) \left(1 + \frac{WC}{wc}\right) + \frac{1}{2}\psi(T_1 - T_2)}{(T_2 - t_2) + \frac{1}{2}(T_1 - T_2) \left(1 + \frac{WC}{wc}\right) - \frac{1}{2}\psi(T_1 - T_2)} \right]} \quad (20.28)$$

Substituting the value of ψ and eliminating wc/WC by introducing its equivalent $(T_1 - T_2)/(t_2 - t_1)$,

$$\Delta t = \frac{\sqrt{(T_1 - T_2 + t_2 - t_1)^2 + 4 \frac{up}{UP} (t_2 - t_1)^2}}{\ln \left[\frac{T_2 - t_2 + T_1 - t_1 + \sqrt{(T_1 - T_2 + t_2 - t_1)^2 + 4 \frac{up}{UP} (t_2 - t_1)^2}}{T_2 - t_2 + T_1 - t_1 - \sqrt{(T_1 - T_2 + t_2 - t_1)^2 + 4 \frac{up}{UP} (t_2 - t_1)^2}} \right]} \quad (20.29)$$

which gives the true temperature difference in terms of the terminal differences. In order to establish a graphical representation, dimensionless parameters will again be used.

$$R = \frac{T_1 - T_2}{t_2 - t_1} = \frac{wc}{WC} \quad V = \frac{1}{2} \left(\frac{T_1 + T_2 - t_1 - t_2}{t_2 - t_1} \right)$$

where V is the arithmetic mean temperature difference per unit temperature change in the tube fluid.

$$F = \frac{up}{UP}$$

where F is the ratio of the conductances per unit of tube length and the true temperature difference per unit change in tube side temperature is $\Delta t_D = \Delta t/(t_2 - t_1)$. Using these ratios Eq. (20.29) can be written

$$\Delta t_D = \frac{\sqrt{(R + 1)^2 + 4F}}{\ln \left[\frac{2V + \sqrt{(R + 1)^2 + 4F}}{2V - \sqrt{(R + 1)^2 + 4F}} \right]} \quad (20.30)$$

This equation also expresses Δt_D for Case IV in Fig. 20.15d. For Cases II and III the equivalents of Eqs. (20.29) and (20.30) are

$$\Delta t = \frac{\sqrt{(T_1 - T_2 - t_2 + t_1)^2 + 4 \frac{up}{UP} (t_2 - t_1)^2}}{\ln \left[\frac{T_2 - t_2 + T_1 - t_1 + \sqrt{(T_1 - T_2 - t_2 + t_1)^2 + 4 \frac{up}{UP} (t_2 - t_1)^2}}{T_2 - t_2 + T_1 - t_1 - \sqrt{(T_1 - T_2 - t_2 + t_1)^2 + 4 \frac{up}{UP} (t_2 - t_1)^2}} \right]} \quad (20.31)$$

and

$$\Delta t_D = \frac{\sqrt{(R-1)^2 + 4F}}{\ln \left[\frac{2V + \sqrt{(R-1)^2 + 4F}}{2V - \sqrt{(R-1)^2 + 4F}} \right]} \quad (20.32)$$

Equations (20.30) and (20.32) can be rearranged to the form

$$\frac{\Delta t_D}{V} = \frac{2(E/V)}{\ln \left(\frac{1 + \frac{E}{V}}{1 - \frac{E}{V}} \right)} \quad (20.33)$$

where for Cases I and IV

$$E = \frac{1}{2} \sqrt{(R+1)^2 + 4F} \quad (20.34)$$

and where for Cases II and III

$$E = \frac{1}{2} \sqrt{(R-1)^2 + 4F} \quad (20.35)$$

The solution of Eq. (20.33) is given in Fig. 20.17, and the solution for Eqs. (20.34) and (20.35) is given by Fig. 20.18a and b, respectively. Because this method of obtaining the true temperature difference

differs so greatly from the methods of preceding chapters, the procedure is summarized.

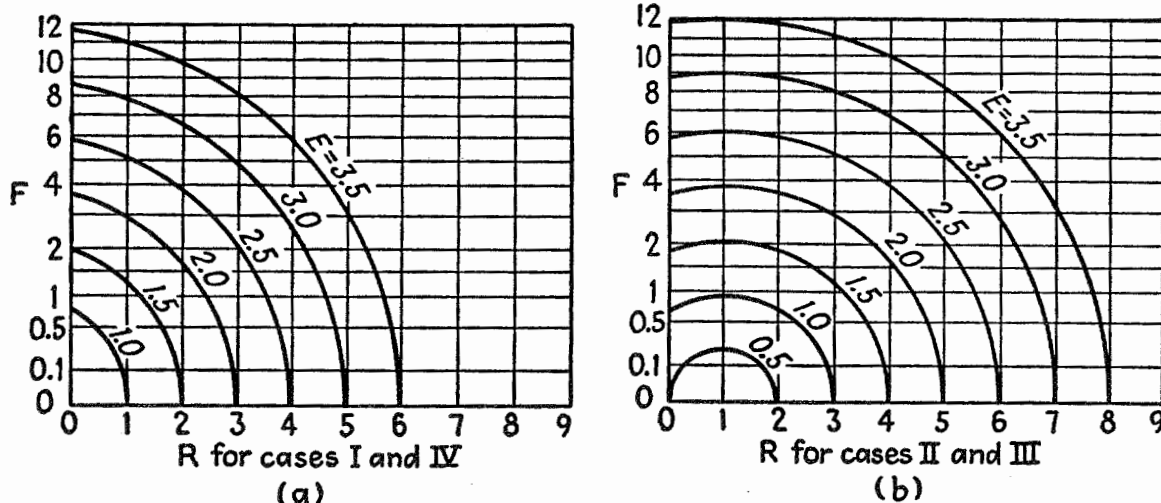


FIG. 20.18. Solution of Eqs. (20.34) and (20.35). (Hurd, *Industrial Engineering Chemistry*.)

1. Determine R and V , and after having determined U and u , determine F .

2. From R and F determine E in Fig. 20.18a or b for the arrangement identified in Fig. 20.15. If V is equal to or greater than $3E$, proceed directly to step 4, using V as identical with the Δt_D .

3. Calculate E/V , and read $\Delta t_D/V$ from Fig. 20.17. Multiply by V to obtain Δt_D .

4. Multiply Δt_D by $(t_2 - t_1)$ to obtain Δt for use in the Fourier equation.

Example 20.6. Calculation of the True Temperature Difference. 1000 lb/hr of fluid of 0.5 specific heat is to be heated from 50 to 100°F by means of 250 lb/hr of hot water in the shell which will be cooled from 200 to 100°F. The bayonet consists of 1 in. OD tubes within 2 in. OD tubes. U and u are 60 and 120 Btu/(hr) (ft²)(°F) as computed by methods described below. Assume the arrangement of Case III.

$$R = \frac{200 - 100}{100 - 50} = 2.0$$

$$V = \frac{1}{2} \left(\frac{200 + 100 - 50 - 100}{100 - 50} \right) = 1.50$$

$$F = \frac{120 \times 1}{60 \times 2} = 1.0$$

In Fig. 20.18b for $R = 2.0$ and $F = 1.0$, the abscissa and ordinate intersect at $E = 1.10$. Thus $E/V = 1.10/1.50 = 0.739$, and from Fig. 20.17 $\Delta t_D/V = 0.783$ and

$$\Delta t_D = 0.783 \times 1.50 = 1.175.$$

$$\Delta t_D = \frac{\Delta t}{t_2 - t_1} \quad \Delta t = 1.175 \times 50 = 58.8^\circ\text{F}$$

The LMTD is 72°F for counterflow.

Bayonet-exchanger Heat-transfer Coefficients. The bayonet exchanger presents no new problem in the calculation of overall heat-transfer coefficients. The overall coefficient u for the outside diameter of the inner tube is found in the same manner employed for the double pipe exchanger in Chap. 6. The equivalent diameter for the annulus is based on the perimeter at the outside diameter of the inner tube and considered effective at the outside diameter of the inner tube. To compute the annulus coefficient at the inside diameter of the outer tube, the equivalent diameter is based on the perimeter at the inside diameter of the outer tube and is considered effective at the inside diameter of the outer tube. The flux is then corrected to the outside diameter of the outer tube by the simple ratio of the inside diameter to outside diameter.

Shell-side coefficients are treated in the same manner as former prototypes. Condensation and vaporization are treated by the methods of Chaps. 12 through 15. The use of a bayonet exchanger for sensible-heat transfer in a shell without baffles usually simulates longitudinal flow along the axis of the bayonet and can be computed by the methods of Example 7.5. For segmentally baffled bayonet bundles use Fig. 28.

8. FALLING-FILM EXCHANGERS

The falling-film exchanger is usually a conventional 1-1 exchanger designed to operate vertically as shown in Fig. 20.19a. Liquid enters the channel at the top at such a rate that the tubes do not flow full of liquid, but instead, liquid descends by gravity along the inner walls of the

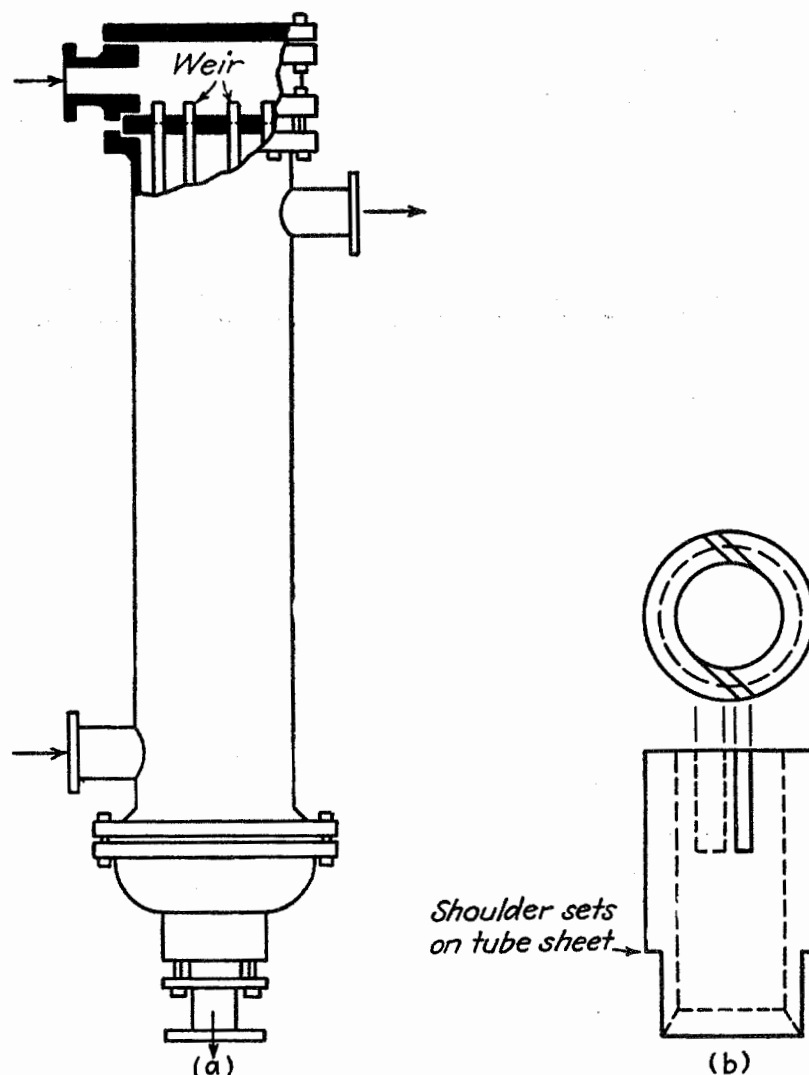


FIG. 20.19. (a) True counterflow falling-film exchanger with circular flow weirs. (b) Detail of a circular flow weir for insertion into top of tubes.

tubes as a thin film. Obviously this produces a much greater linear velocity for a given weight flow than could be obtained if the tube flowed full. The vertical tubes could be made to flow full if desired by feeding the liquid upward or through the use of a loop seal as discussed in Chap. 15. McAdams¹ quotes Bays² and coworkers as having determined the coefficients for heating water in film flow in 1.5 to 2.5 in. ID pipes 0.4 to

¹ McAdams, *op. cit.*, p. 203.

² McAdams, Drew, and Bays, *loc. cit.*

6.08 ft long. The coefficients are given within ± 18 per cent by the dimensional equation

$$h_i = 120 \left(\frac{w}{\pi D} \right)^{1/3} \quad (20.36)$$

For other liquids McAdams cites a dimensionless empirical equation attributed to T. B. Drew for turbulent flow

$$\frac{h_i}{(k^3 \rho^2 g / \mu_f^2)^{1/3}} = 0.01 \left[\left(\frac{c \mu}{k} \right) \left(\frac{4G'}{\mu_f} \right) \right]^{1/3} \quad (20.37)$$

where $G' = w/\pi D$ and μ_f is evaluated at the average film temperature. For streamline flow where the Reynolds number $4G'/\mu_f$ is less than 2000, Bays and McAdams¹ give

$$h_i = 0.67 \left[\left(\frac{k^3 \rho^2 g}{\mu_f^2} \right) \left(\frac{c \mu_f^{5/3}}{k L \rho^{2/3} g^{1/3}} \right) \right]^{1/3} \left(\frac{4G'}{\mu_f} \right)^{1/6} \quad (20.38)$$

Very often a swirl is imparted to the falling film by means of overflow pieces having nearly tangential notches which protrude above the upper tube sheet. A typical weir is shown in Fig. 20.19b.

The shell-side coefficient can be computed in the usual manner, using Fig. 28 for sensible coefficients or by the methods of Chaps. 12 or 15 for condensation or evaporation.

9. GRANULAR MATERIALS IN TUBES

The transfer of heat to or from fluidized solids has become increasingly important in modern catalytic and adsorptive processes. The transfer of heat to granular materials in beds was discussed as an unsteady-state problem in Chap. 18. When a dry granular material travels through a vertical or horizontal tube, the flow mechanism is akin to rodlike flow as discussed in Chap. 3. Based upon an earlier derivation of Graetz,² Drew³ established the theoretical equation for a fluid in rodlike flow to be

$$\frac{t_2 - t_1}{T_s - t_1} = 1 - 0.692e^{-5.78 \frac{\pi k L}{w c}} - 0.131e^{-30.4 \frac{\pi k L}{w c}} - 0.0536e^{-74.8 \frac{\pi k L}{w c}} \quad (20.39)$$

where T_s = constant temperature of the wall

w = granular flow rate, lb/hr

For values of $(t_2 - t_1)/(T_s - t_1)$ greater than 0.55 only the first term of the series need be considered. Equation (20.39) reduces to

$$\frac{t_2 - t_1}{T_s - t_1} = 1 - 0.692e^{-5.78 \pi (k L / w c)} \quad (20.40)$$

¹ Bays, G. S., and W. H. McAdams, *Ind. Eng. Chem.*, **29**, 1240-1246 (1937).

² Graetz, L., *Ann. Physik*, **25**, 337 (1885).

³ Drew, T. B., *Trans. AIChE*, **26**, 26-80 (1931).

or

$$\frac{T_s - t_2}{T_s - t_1} = 0.692e^{-5.78\pi(kL/wc)} \quad (20.41)$$

In a study of heat transfer to sand, Brinn and coworkers¹ experimentally verified the rodlike nature of the flow of several granular materials in tubes. They also made the following additional observations:

1. The transfer of heat to a fluid in rodlike flow is identical with the *unsteady-state* heating of an infinitely long cylinder with negligible resistance between the heat source and the cylinder as defined by an equation of the form of Eq. (18.52). The series expression for the infinitely long cylinder and Eq. (18.52) can be shown to be equivalent when the values of the exponents in the infinite series of Eq. (20.39) are identical. This requires that the Graetz number wc/kL and the group $4\alpha\theta/D^2$ * be equal. In these groups α is the thermal diffusivity, θ the time in hours, and D the inside diameter of the cylinder in feet. Since the linear velocity of the material in the tubes is given by

$$V = \frac{4w}{\rho\pi D^2} \quad \text{ft/hr}$$

and the volume of granular material by

$$v_s = \frac{\pi D^2}{4} L$$

The time is

$$\theta = \frac{4v_s}{V} \pi D^2$$

The transformation is given by

$$\frac{4\alpha\theta}{D^2} = \left(\frac{4k}{c\rho D^2} \right) \left(\frac{\pi D^2 L}{4w/\rho} \right) = \frac{\pi kL}{wc}$$

2. For a given weight flow through the tube the temperature change is independent of the diameter. This leads to the interesting point that a desired value can be obtained for the heat-transfer coefficient by changing the tube diameter.

3. If there are other resistances in series with the rodlike flow, it is not possible to obtain the overall resistance and hence U by the arithmetic addition of the resistances. The solutions for such cases are given graphically in Fig. 20.20.

¹ Brinn, M. S., S. J. Friedman, F. A. Guckert, and R. L. Pigford, *Ind. Eng. Chem.*, 40, 1050-1061 (1948).

* The inside diameter of the tube D here replaces the general length l in Eq. (18.52), since only cylinders are under consideration.

4. It is not necessary to obtain the film coefficient h_i for the granular material flowing inside the tube in calculating the size or performance of the system. It may be obtained as a matter of interest, however, from Eq. (20.41) and the following, which defines the arithmetic mean film coefficient,

$$h_i = \frac{2k}{\pi D} \left[\frac{1 - (T_s - t_2)/(T_s - t_1)}{1 + (T_s - t_2)/(T_s - t_1)} \right] \frac{wc}{kL} \quad (20.42)$$

While the preceding has applied to granular materials in rodlike flow in which the temperature of the wall is uniform, Brinn *et al.* have also solved the equations and given charts for the cases in which the temperature of the medium in the tube jacket is not uniform and also for the cases where the film coefficient from the jacket fluid to the tube is finite and infinite. To this extent they have employed the two temperature parameters R' and S' as an index of the temperature variation of both the rodlike and jacket flow and a parameter β which is the reciprocal of the Nusselt number, so that

$$R' = \frac{wc}{WC} \quad S' = \frac{t_2 - t_1}{T_1 - t_1} \quad \beta = \frac{2k}{h_{oi}D}$$

where h_{oi} is the jacket coefficient referred to the inside diameter of the tube. R is replaced by R' and S by S' , since wc , t_2 and t_1 always refer to the granular material whether or not it is the colder of the two. Obviously when $R' = 0$, it is equivalent to a constant jacket temperature, and when $\beta = 0$, it corresponds to a negligible contact resistance between the tube and the jacket fluid or an infinite value of h_{oi} . Graphs for values of $\beta = 0$ and with $R' = 0, 0.5$, and 1.5 are shown in Fig. 20.20a, b, c, and d. The reciprocals of these graphs correspond to the Gurney-Lurie chart of Fig. 18.13.

Data are usually lacking on the thermal conductivity of the granular material. Its value differs from the thermal conductivity of the solid, since the conduction of heat is obviously affected by granulation. Brinn *et al.* have found an excellent agreement in the value of the thermal conductivity for a granular solid when determined by heating or cooling a stationary bed similar to that of Chap. 18 or heating or cooling a granular solid in a tube under rodlike flow. For Ottawa sand and crushed ilmenite ore they report values of 0.172 and 0.132 Btu/(hr)(ft²)(°F/ft), respectively, for stationary beds and 0.155 and 0.141, respectively, for vertical rodlike flow. The densities of the granular materials were, respectively, 101 and 103 lb/ft³ for the Ottawa sand and 149 and 168 for the ilmenite ore, although the density of the moving material is usually increased by movement. Specific heats for granular solids have been reported for

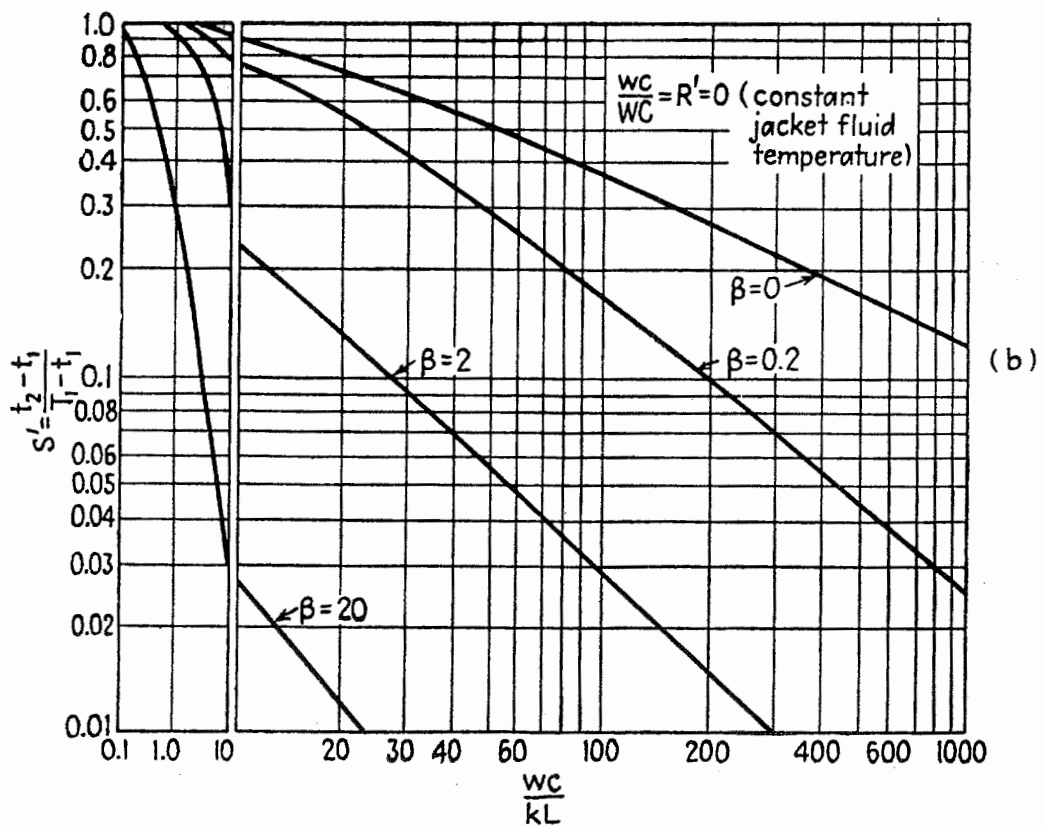
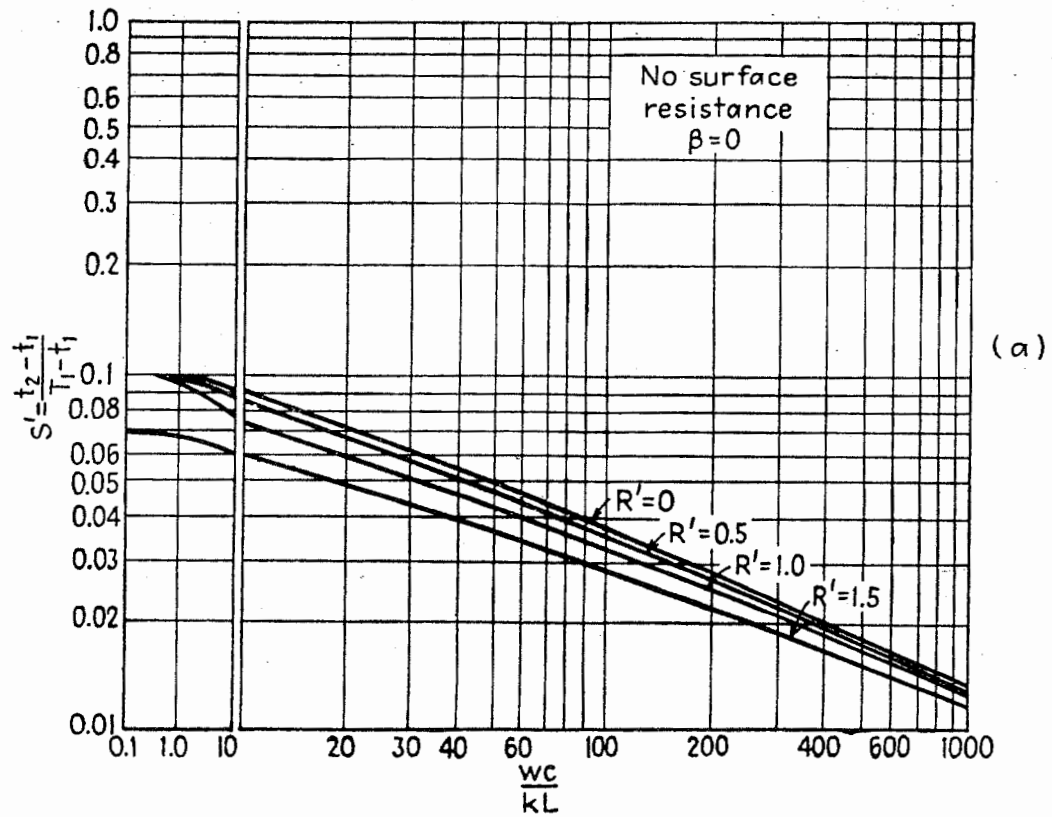
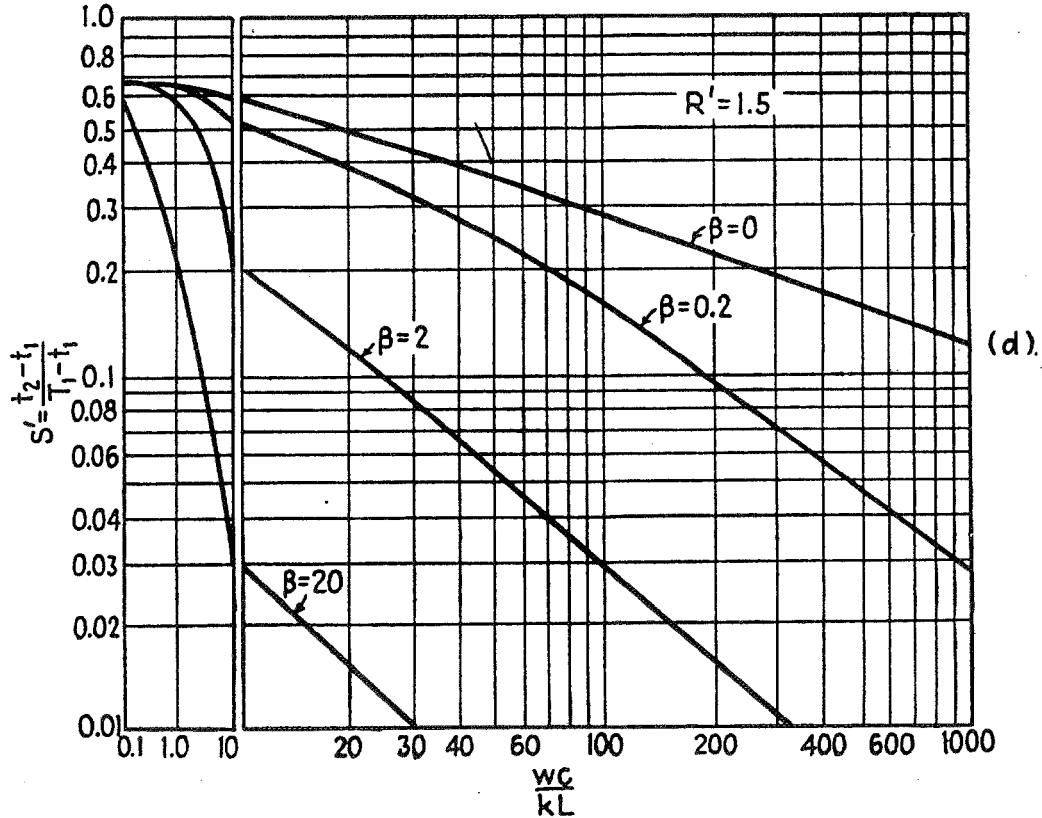
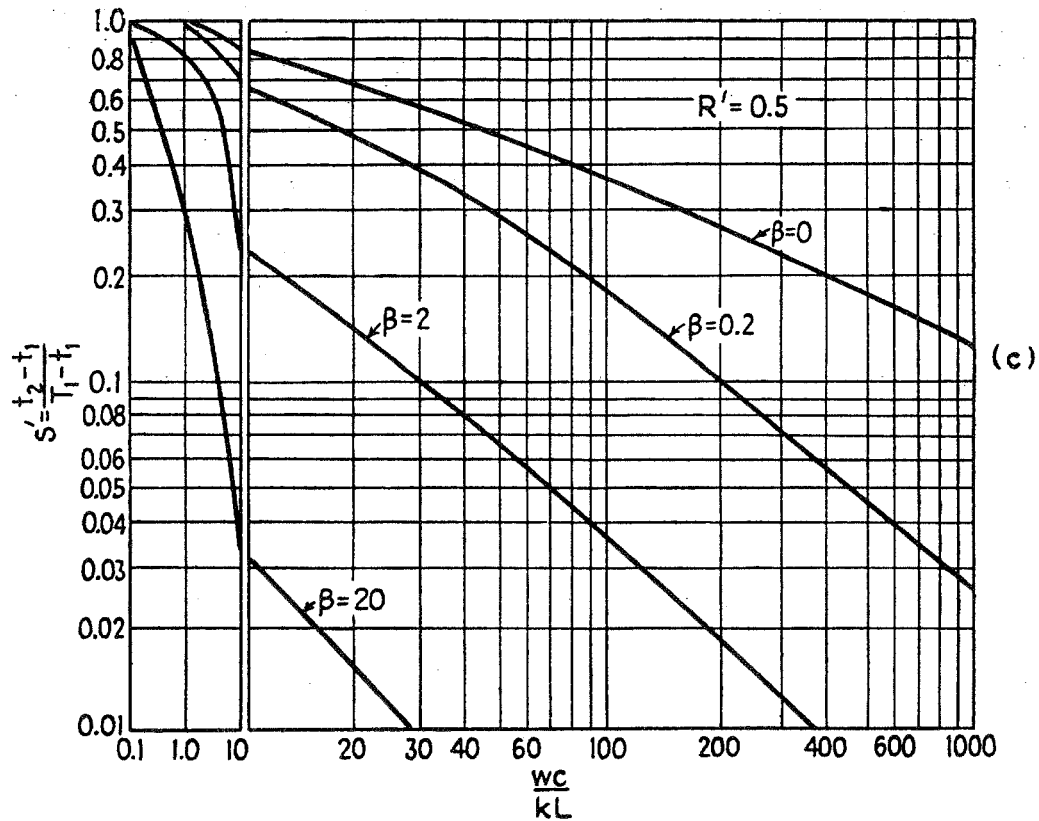


FIG. 20.20. Theoretical curves for heat transfer to fluids in rodlike flow in



cylindrical pipes for various values of R' and β . (Brinn, et al., *Industrial and Engineering Chemistry*.)

several materials, and these are somewhat higher than those of the solid. For quartz, silica, and Ottawa sand they range from 0.175 to 0.20 Btu/(lb)(°F). Examples given below have been taken from the original reference.

Example 20.7. Calculation of Sand Cooling with Negligible Resistance. A sand has a specific heat 0.20 Btu/(lb)(°F) and a thermal conductivity of 0.15 Btu/(hr)(ft²)(°F/ft). It is to be cooled from 284 to 104°F at the rate of 1000 lb/hr by water from 86 to 104°F. A bundle will consist of vertical tubes 10'0" long and with inside diameters of 2.0 in.

(a) How many tubes are required?

(b) How many tubes are required if air from 86 to 104°F is the cooling medium?

Solution. (a) Assume h_{oi} for water is 500 Btu/(hr)(ft²)(°F).

$$\beta = \frac{2k}{h_{oi}D} = \frac{2 \times 1.05}{500 \times \frac{2}{12}} = 0.0036 \quad (\text{nearly zero})$$

The water cooling rate is

$$\frac{1000 \times 0.2(284 - 104)}{104 - 86} = 2000 \text{ lb/hr}$$

$$R' = \frac{wc}{WC} = \frac{1000 \times 0.2}{2000 \times 1} = 0.1$$

$$S' = \frac{t_2 - t_1}{T_1 - t_1} = \frac{104 - 284}{86 - 284} = 0.91$$

From Fig. 20.20b

$$\beta = 0 \quad R' = 0.1 \quad S' = 0.91$$

$$\frac{wc}{kL} = 8.33$$

$$w = \frac{8.33 \times 0.15 \times 10}{0.20} = 62.5 \text{ lb/hr}$$

This is the rate per tube.

$$\text{Total number of tubes} = \frac{1000}{62.5} = 16$$

(b) For air, assume $h_{oi} = 9$.

$$\beta = 0.2$$

The air rate using 0.25 for the specific heat of air is

$$\frac{1000 \times 0.2(284 - 104)}{0.25(104 - 86)} = 8000 \text{ lb/hr}$$

Interpolating between Fig. 20.20b and c,

$$\beta = 0.2 \quad R' = 0.1 \quad S' = 0.91$$

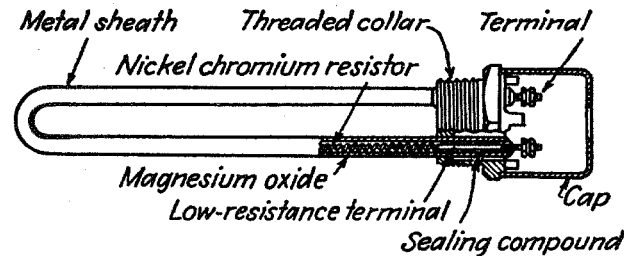
$$\frac{wc}{kL} = 5.23$$

$$w = 39 \text{ lb/hr}$$

and 26 tubes are needed. The film resistance increases the size of the unit by 62 per cent.

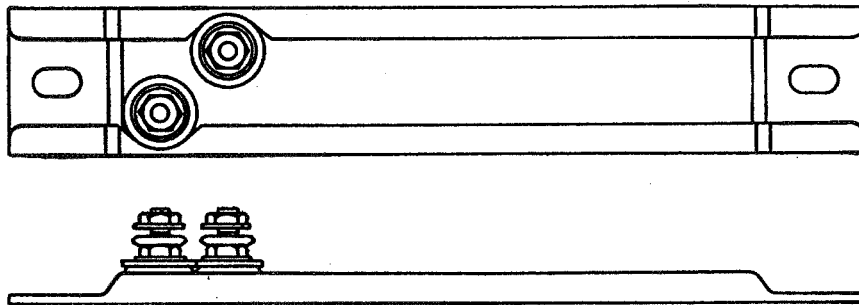
10. ELECTRIC-RESISTANCE HEATING

Introduction. The transformation of fuel energy into work (and then into electricity) is an inefficient process, only a portion of the fuel energy

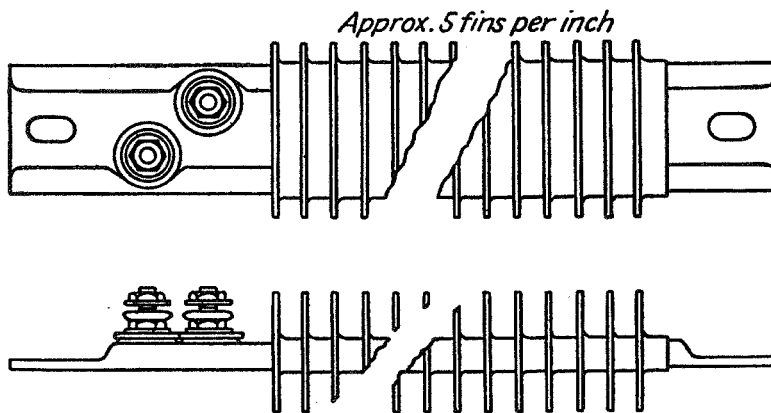


Cutaway view of a two-terminal (single heat) immersion heater

(a)-IMMERSION HEATER



(b)-STRIP HEATER



(c)-FINNED STRIP HEATER

FIG. 20.21. Common process resistance heating elements. (Westinghouse Electric Corporation.)

finally being available as work. The transformation of fuel energy into steam can be accomplished theoretically with 100 per cent efficiency. These are the practical deductions of the Carnot cycle, and electricity costs ten to fifteen times as much as steam in localities which are not served by hydroelectric power. For this reason electric resistance heating has not been considered a part of process heat transfer, although the

comparison of the cost of electricity and coal need not necessarily preclude its use. There are innumerable applications in which electric-resistance heating can be made to operate more effectively than fluid heat transfer particularly in batch operations. There are still other advantages which may be derived from the compactness of the standard types of electric heating elements: the ease with which they provide high temperatures, the elimination of combustion hazards, and their ready application to automatic regulation and control. It is fairly safe to conclude, however, that most of the applications of electric-resistance heating are small compared with those of the majority of industrial processes.

The three commonest resistance heating elements are the *immersion heater*, the *strip heater*, and the *finned-strip heater*. These are shown in Fig. 20.21. Others are available such as the cartridge heater which are treated by the same general method. Electric heaters consist of resistance wires embedded in a refractory which is then encased in a metal sheath. Immersion heaters are used for direct introduction into liquids and are made with copper sheathing for water heating and steel sheathing for oil and organics. The temperature of the copper sheathing may not exceed 350°F, while the steel-sheathed heaters may operate at a maximum temperature of 750°F. The strip heater is used most for the conduction heating of solids, and it can also be used for heating air and gases. Strip heaters are available with two maximum operating temperatures: 750°F for steel-sheathed heaters and 1200°F for chrome-steel-sheathed heaters. By far the best suited for air and gas heating is the finned-strip heater combining electric heating and extended surface and available with maximum operating temperatures of 750 and 950°F.

The technique of handling electric-heating calculations has been developed by the electrical industry and is not identical with the methods employed in either the mechanical or chemical industries. This is doubtless due to the greater standardization of electric equipment.

The laws governing electric-resistance heating, neglecting the effect of the resistance on the circuit, are among the simplest in elementary physics. These are Ohm's law, $I = E/R^\omega$ and Joule's law, $P = EI$, where I is the current in amperes, E the voltage, R^ω the resistance in ohms, and P the power in watts. The rate at which electrical energy is converted into heat is given by I^2R^ω . The rate of energy delivery is the watt or kilowatt, and this is equivalent to the *Btu per hour*. Similarly the total quantity of energy is the *kilowatt-hour* or *watt-hour*, and this is equivalent to Btu without a time unit.

In electric-resistance heating the element is capable of attaining a very high temperature. The ultimate temperature attained by a resistance heater is that which causes the heat energy to be dissipated at the same

rate as it is produced. To prevent the element from burning out, the cold fluid or solid must be capable of receiving the heat at a sufficiently rapid rate to keep the sheath from exceeding its maximum allowable

TABLE 20.1. IMMERSION HEATERS

Water heaters, copper sheath, max temp 350°F						Oil heaters, steel sheath, max temp 750°F					
Watts	Watts/ in. ² im- mersion	Im- mersed length, in.	Collar, in. IPS	No. heats	No. tubes	Watts	Watts/ in. ² im- mersion	Im- mersed length, in.	Collar, in. IPS	No. heats	No. tubes
1,000	35	8	1¼	1	1	1000	20	9	1¼	1	1
2,000	35	10	2	3	2	2000	20	17	2	3	2
3,000	35	15	2	3	2	3000	20	25	2	3	2
4,000	35	20	2	3	2	4000	20	34	2	3	2
5,000	35	24	2	3	2	5000	20	41	2	3	2
7,500	35	30	2	3	4	Forced Circulation					
10,000	35	35	2	3	4						
						5000*	35	25	2	3	2
						6000*	35	30	2	3	2
						8000*	30	33	2	3	4

* 230 volts only. All others 115 or 230 volts.

TABLE 20.2. STRIP HEATERS

Steel sheath, max surface temp 750°F				Chrome steel sheath, max surface temp 1200°F			
Watts	Watts/ in.	Total length, in.	Heated length, in.	Watts	Watts/ in.	Total length, in.	Heated length, in.
150	37	8	4	250	62	8	4
250	31	12	8	350	44	12	8
350	25	18	14	500	36	18	14
500	25	24	20	750	37	24	20
750	28	31	27	1000	38	31	27
1000	31	36	32	1500	38	43	39
1250	32	43	39				

temperature. In other words, the design of electric elements is dictated by the flux which must be dissipated to the cold material. The flux is the deliverable energy in Btu per hour per square foot of surface. In

electrical units it is expressed in watts per square inch of element surface or watts per linear inch of element corresponding to watts per square inch for elements of uniform surface per inch of length.

Flux data for the various types of heater are given in Tables 20.1 through 20.4. The wattage indicates the output of the element but does

TABLE 20.3. FINNED-STRIP HEATERS

Steel sheath, max surface temp 750°F				Chrome steel sheath, max surface temp 950°F			
Watts	Watts/in.	Total length, in.	Heated length, in.	Watts	Watts/in.	Total length, in.	Heated length, in.
250	31	12	8	250	62	8	4
350	25	18	14	350	44	12	8
500	25	24	30	500	36	18	14
750	25	31	27	750	37	24	30
1000	31	36	32	1000	38	31	27
1250	32	43	39	1500	47	36	32

TABLE 20.4. STRIP HEATERS

Clamp-on service watts per square inch of contact surface

Steel sheath, max surface temp 750°F		Chrome steel sheath, max surface temp 1200°F	
Total length, in.	Watts/in. ²	Total length, in.	Watts/in.
12*	21	8*	42
18	16	12	29
24	16	18	24
31	19	24	25
36	21	31	25
43	22	43	26

* Passes.

not indicate its temperature in so doing. Included is the watts per square inch or watts per inch which the various elements may dissipate to the cold fluid or solid. For finned- and ordinary strip heaters the watts per inch removed by air as a function of the air velocity and initial temperature are given in Figs. 20.22 and 20.23. At low air velocities it is seen that there is little advantage to the use of finned-strip heaters in preference to bare-strip heaters. Since the outlet-air temperature also affects

the sheath temperature, Fig. 20.24 relates the surface temperature of strip heaters as a function of the flux and the ambient-air temperature. When dealing with elements operating at such high temperatures, the

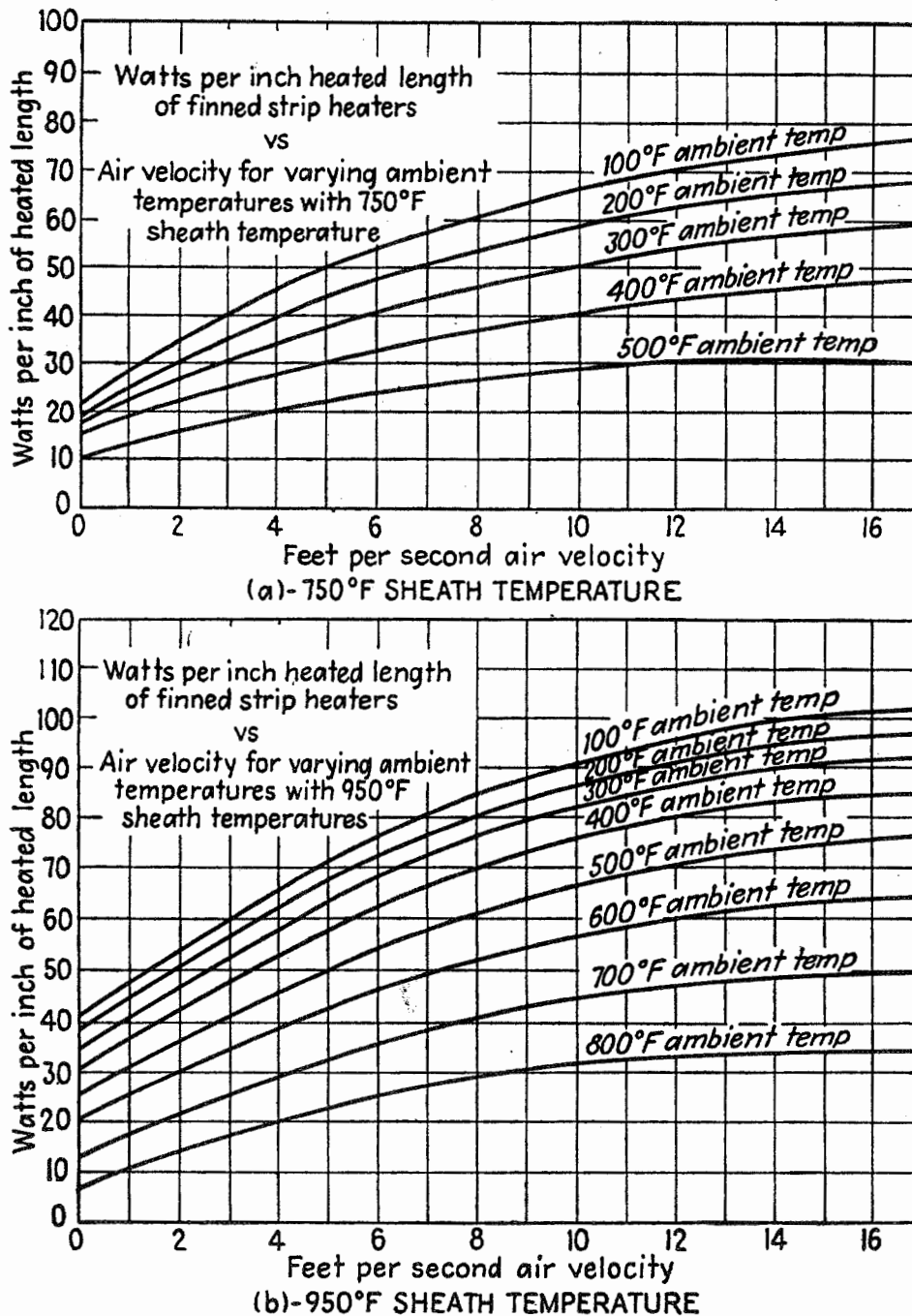
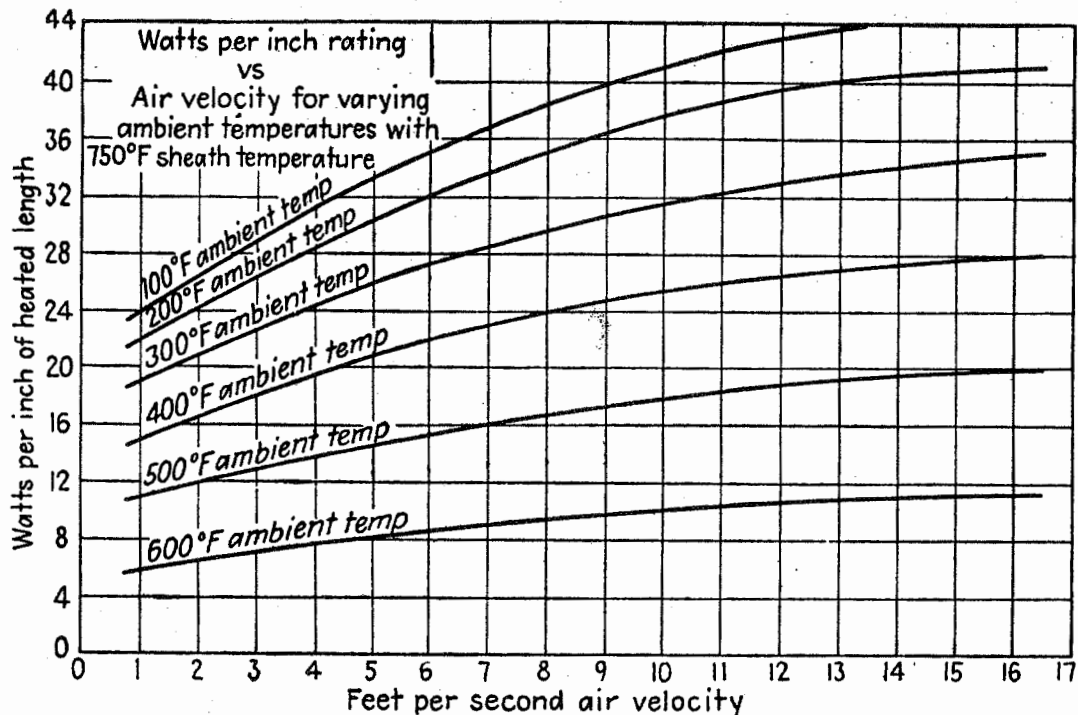
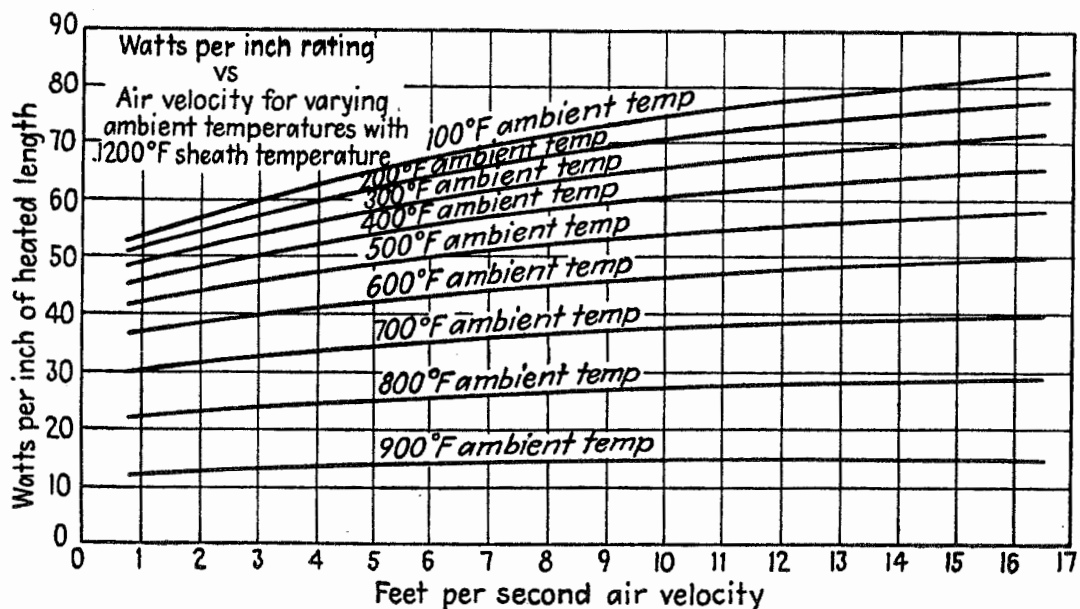


FIG. 20.22. Finned-strip heater air-heating curves. (Westinghouse Electric Corporation.)

effect of radiation is of greater importance than in ordinary steam-heated equipment. Figure 20.25 relates the losses obtained experimentally under varying conditions. The use of these graphs and tables is shown explicitly by solving typical problems to which they apply.



(a)- 750°F SHEATH TEMPERATURE



(b)-1200°F SHEATH TEMPERATURE

FIG. 20.23. Strip-heater air-heating curves. (Westinghouse Electric Corporation.)

Example 20.8a. Immersion Water Heater. A tank 2 by 3 by 2 ft high is filled with water to a height of 18 in. It is required to heat the water from a tap temperature of 50°F to a demand temperature of 150°F within 2 hr. Thereafter, the tank must supply 8 gal/hr of water at 150°F.

What immersion elements are required?

Solution:

Unsteady state.

$$\text{Lb water} = 2 \times 3 \times 1\frac{3}{4} \times 62.5 = 562 \text{ lb}$$

$$Q' = 562.5(150 - 50)1.0 = 56,250 \text{ Btu}$$

$$Q = \frac{56,250}{2} = 28,125 \text{ Btu/hr} = \frac{28,125}{3412} = 8.25 \text{ kwhr}$$

Losses:

$$\text{From surface of water (Fig. 20.25c)} = 2 \times 3 \times 260 = 1.56 \text{ kwhr}$$

$$\text{From sides of vessel (Fig. 20.25c)} = 5.5(2 \times 2 \times 2 + 2 \times 2 \times 3)/1000 = 0.11$$

From bottom = negligible

$$\text{Total losses} = 1.67 \text{ kwhr}$$

$$\text{Total requirement} = 8.25 + 1.67 = 9.92 \text{ kwhr}$$

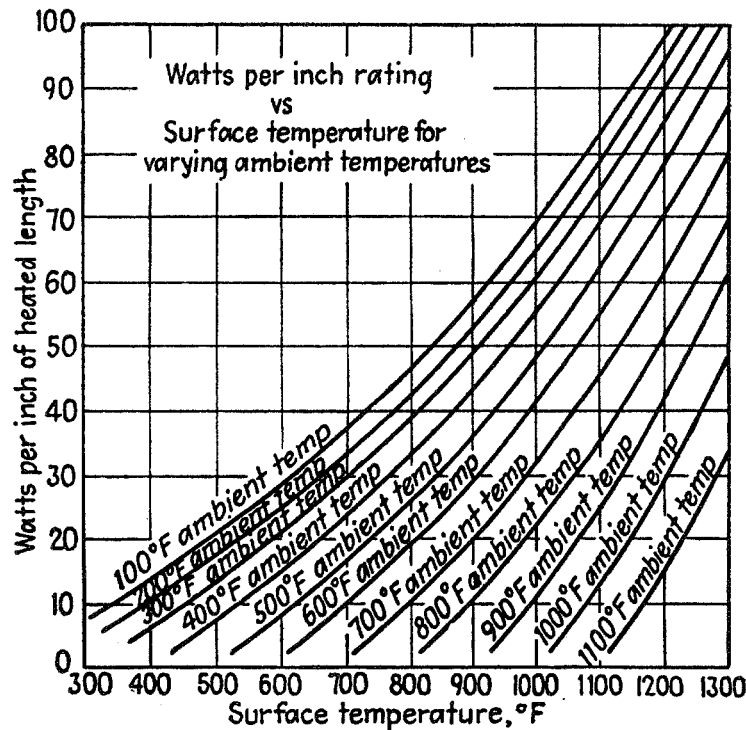


FIG. 20.24. Surface temperature vs. flux for strip heaters. (Westinghouse Electric Corporation.)

Steady state: This is the same as heating 8 gal/hr of water.

$$Q = 8 \times 8.33(150 - 50)1.0 = 6670 \text{ Btu/hr or } 1.95 \text{ kwhr}$$

$$\text{Losses} = \frac{1.67}{\text{Total requirement} = 3.62 \text{ kwhr}}$$

The unsteady-state requirement controls the selection of the immersion heater requiring 10 kwhr. The nearest size in Table 20.1 is the 4000-watt or 4-kw immersion element with a 20-in. immersed length and wired for three heats. Three are required. The rate of absorption of heat by water is high, and there is little need ever to check the sheath temperature although the same rule is not universally true for organics.

Example 20.8b. Strip Heater for Air Heating. An oven is charged hourly with 100 lb of steel which is air dried at 370°F by means of strip heaters located inside it. Four air changes per hour are required for the removal of the moisture. The oven is 3 by 4 by 2 ft insulated with 2 in. of rock wool. The inlet air and steel are at 70°F.

What arrangement of strip heaters is required?

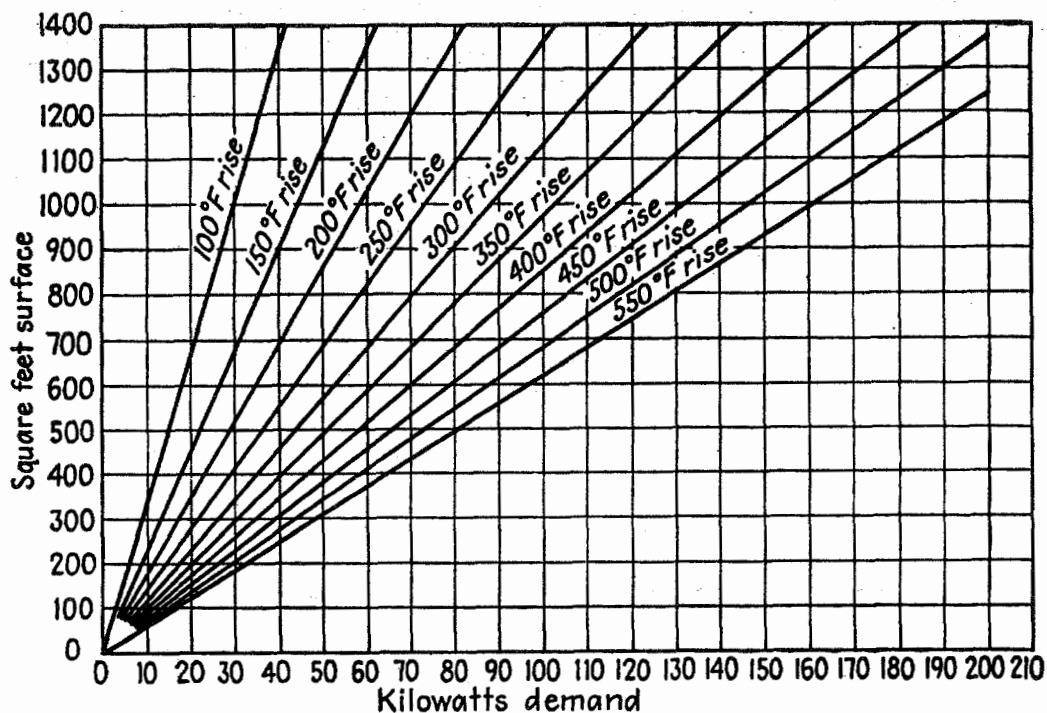


FIG. 20.25a. Heat losses through insulated oven walls. Curve based on an insulation 1 in. thick of standard high-grade material, such as 85 per cent magnesia, rock wool, Filinsul, etc. If insulation is 2 in. thick, divide curve values by 2; if 4 in. thick, divide by 4; etc.

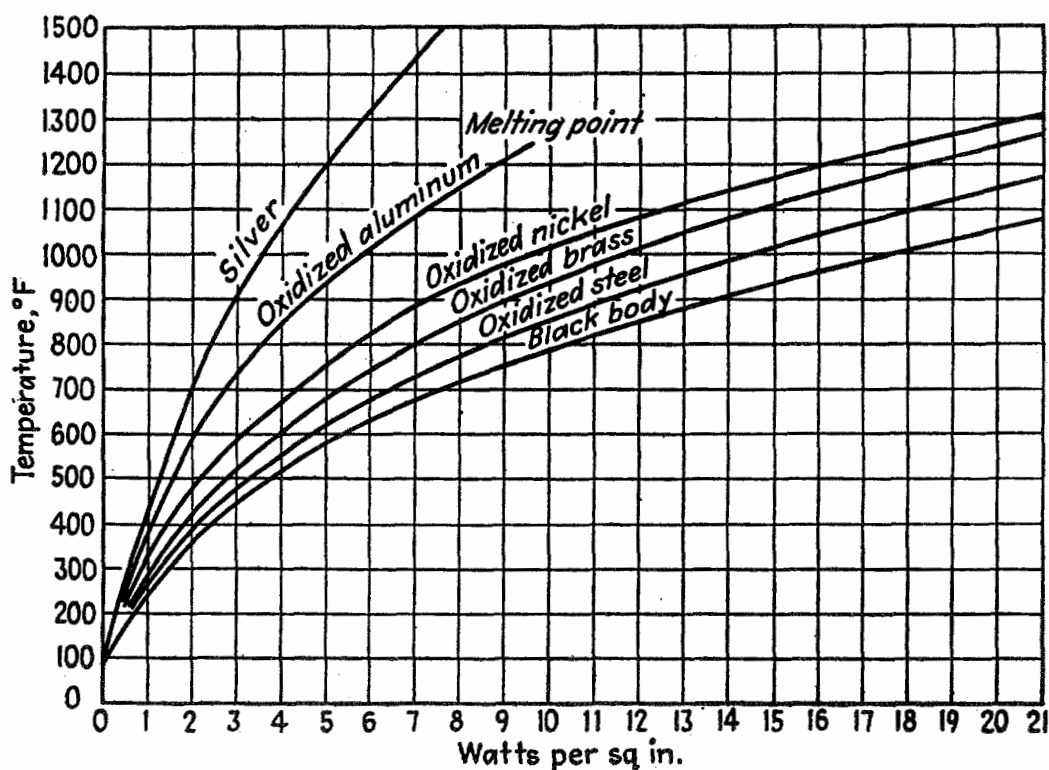


FIG. 20.25b. Heat losses from smooth solid surfaces. Heat losses from a horizontal surface laid flat on one side are top surface, 110 per cent of curve values; bottom surface, 55 per cent of curve values; averages of losses from top and bottom surfaces, $82\frac{1}{2}$ per cent of curve values.

Solution:

$$\text{Heat to steel charge} = 100 \times 0.12(370 - 70) = 3600 \text{ Btu/hr}$$

$$\text{Heat to air} = 4 \times 2 \times 3 \times 4 \times 0.075 \times 0.25(370 - 70) = 540$$

$$\text{Total} = 4140 \text{ Btu/hr or } 1.22 \text{ kw}$$

Losses:

From Fig. 20.25a for 52 ft² of oven outside surface and a temperature rise of 300°F the loss is 5 kw for 1 in. thick insulations. For 2 in. thick insulation the loss is $\frac{5}{2}$

$$\text{Total} = \frac{2.50}{3.72 \text{ kw}}$$

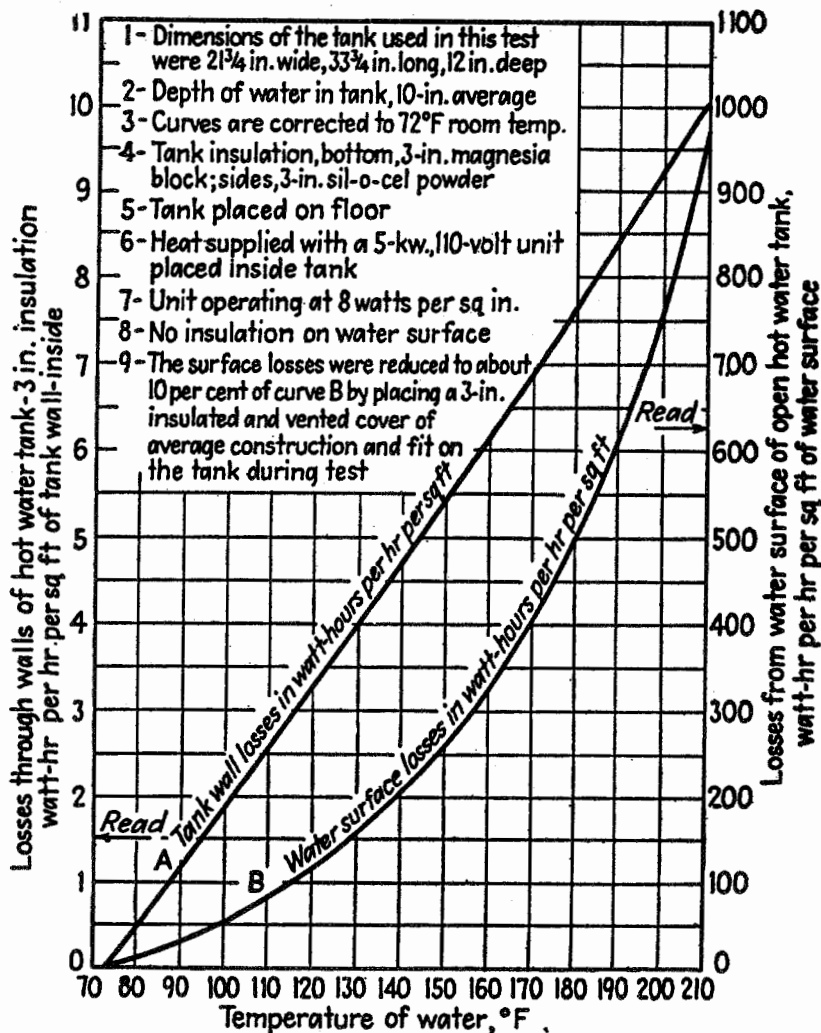


FIG. 20.25c. Heat losses from open hot-water tanks. (Westinghouse Electric Corporation.)

From Table 20.2 it is seen that the energy and space requirement can be fulfilled by four heaters of 1000 watts capacity each. The distribution will be poor, however, because so few elements are employed, and the use of eight elements of 500 watts is preferable. It has been assumed that steel-sheathed elements are used. The surface temperature must be checked to assure that this element will not exceed a surface temperature of 750°F, since air absorbs heat relatively slowly.

From Table 20.2 the required dissipation is 25 watts/in. From Fig. 20.24 for 370°F air the surface of the sheath will be 680°F and hence is satisfactory. If the temperature had exceeded 750°F, a chrome-steel element would have been necessary.

Example 20.8c. Finned-strip Heater. Determine the number of finned-strip heaters required to raise the temperature of 270 cfm of air flowing in an 18- by 18-in. duct from 70 to 120°F. Density of air at 70°F is 0.075 lb/ft³.

$$Q = 270 \times 0.075 \times 60 \times 0.25(120 - 70) = 15,200 \text{ Btu/hr at 4.47 kw}$$

$$\text{Velocity} = 270/1.5 \times 1.5 \times 60 = 2.0 \text{ fps}$$

Refer to Fig. 20.22a. The air is capable of removing 33 watts/in which is the maximum dissipation which may be expected. Any group of heaters providing 5 kw which do not require a dissipation of more than 33 watts/in. and which will fit into the duct will be satisfactory. Thus in Table 20.3 elements of 350 watts with a total length each of 18 in. each are satisfactory.

Consider the same service handled by conventional strip heaters without fins. From Fig. 20.23a the air will remove 26 watt/in. of heated length. From Table 20.2 the 18 in. long element with 350 watts capacity will also be satisfactory and cheaper.

Clamp-on Applications. One of the best examples of the versatility of the strip heater is the ease with which it can be clamped-on or bound to various vessels. When used in clamp-on services, it has been found that the temperature drop across the surface-to-surface contact is 14 to 19°F per unit of flux.

Example 20.8d. Clamp-on Plastic Heating. It is desired to heat a cast-iron plate 26 by 12 by 1 in. at 70°F on which a plastic with specific heat of 0.22 Btu/(lb)(°F) is to be softened. It is necessary that one 70-lb batch of the plastic be heated on the plate after it has attained a steady state rate from 70 to 300°F in 1 hr. The plate will be heated on its underside by clamp-on strip heaters in 1 hr. How many are required?

Solution. The specific gravity of cast iron is 7.2, and its specific heat is 0.13 Btu/(lb)(°F).

Unsteady state:

$$\text{Weight of plate} = 26 \times 12 \times 1 \times 62.5 \times 7.2/1728 = 81 \text{ lb}$$

$$Q = 81(300 - 70)0.13 = 2400 \text{ Btu/hr or 0.705 kw}$$

Losses: From Figure 20.25b for a black body the radiation is 1.5 watts/in.². The radiation from the top is actually 110 per cent of this value, and from the bottom of the plate it is 55 per cent for an average of 82.5 per cent.

$$\text{Total radiation loss} = 2 \times 26 \times 12 \times 1.5 \times 0.825/1000 = 0.80 \text{ kw}$$

$$\text{Total requirement} = 0.70 + 0.80 = 1.50 \text{ kw}$$

Steady state: Input required to plate

$$Q = 70 \times 0.22(300 - 70) = 3550 \text{ Btu/hr or 1.04 kw}$$

$$\text{Losses} = \frac{0.80}{}$$

$$\text{Total requirement} = 1.84 \text{ kw}$$

The steady state is controlling. The requirements are satisfied by four 24-in. strip heaters, but the sheath temperature must now be checked. Since the temperature drop per unit flux density is 14 to 19°F, assume an average of 16.5°F. For clamp-on strips 24 in. long the watts per square inch deliverable are 16. The total temperature drop is the product of the two.

$$\Delta t = 16.5 \times 16 = 264^\circ\text{F}$$

The sheath temperature is then $300 + 264 = 564^{\circ}\text{F}$, which is satisfactory for steel-sheathed elements with a 750°F maximum.

NOMENCLATURE FOR CHAPTER 20

A	Heat-transfer surface, ft^2
B	Height of agitator above vessel bottom, ft; width of an atmospheric cooler, ft
C	Specific heat of hot fluid, $\text{Btu}/(\text{lb})(^{\circ}\text{F})$
C_1, C_2	Constants
D	Inside diameter of pipe or tube, ft
D_i	Diameter of inside of vessel, ft
D_o	Outside diameter of pipe or tube, ft
E	Temperature group defined by Eq. (20.34) or (20.35), dimensionless; voltage; evaporation, per cent
F	The ratio up/UP , dimensionless
G	Mass velocity, $\text{lb}/(\text{hr})(\text{ft}^2)$
G'	Liquid loading for trombone coolers $w/2L$, $\text{lb}/(\text{hr})(\text{ft})$; for falling films $w/\pi D$, $\text{lb}/(\text{hr})(\text{ft})$
g	Acceleration of gravity, ft/hr^2
h	Heat-transfer coefficient, $\text{Btu}/(\text{hr})(\text{ft}^2)(^{\circ}\text{F})$
h_i, h_o	Heat-transfer coefficient referred to the inside diameter and outside diameter of pipe or tube, $\text{Btu}/(\text{hr})(\text{ft}^2)(^{\circ}\text{F})$
h_{oi}, h_{io}	h_o referred to the inside diameter; h_i referred to the outside diameter, $\text{Btu}/(\text{hr})(\text{ft}^2)(^{\circ}\text{F})$
I	Current, amp
j	Factor for heat transfer, dimensionless
K	Temperature group in crossflow $(T_1 - T_2)/(T_2 - t_1)$, dimensionless
K_1, K_2	Constants
k	Thermal conductivity, $\text{Btu}/(\text{hr})(\text{ft}^2)(^{\circ}\text{F}/\text{ft})$
L	Tube length, ft; length of agitator paddle, ft
LMTD	Log mean temperature difference, $^{\circ}\text{F}$
M	Spray, $\text{lb}/(\text{min})(\text{ft}^2 \text{ projected tube area})$
m	Spray, $\text{lb}/(\text{hr})(\text{ft}^2 \text{ cooler cross-sectional area})$
N	Agitator speed, rev/hr ; number of horizontal tube rows
N'	Agitator speed, rpm
P	Perimeter of large pipe or tube, ft; power, watts
P_H, P_V	Horizontal tube pitch, ft; vertical tube pitch, ft
p	Perimeter of smaller pipe or tube, ft;
Q	Heat flow, Btu/hr
Q'	Heat, Btu
R	Temperature group $(T_1 - T_2)/(t_2 - t_1)$, dimensionless
R'	Temperature group in rodlike flow, dimensionless
R_d	Dirt factor, $(\text{hr})(\text{ft}^2)(^{\circ}\text{F})/\text{Btu}$
R^{ω}	Resistance, ohms
(r)	Temperature group $\text{LMTD}/(T_1 - t_1)$, dimensionless
S	Temperature group $(t_2 - t_1)/(T_1 - t_1)$, dimensionless
S'	Temperature group in rodlike flow, dimensionless
T	Temperature of the hot fluid in general, $^{\circ}\text{F}$
T_1, T_2	Inlet and outlet temperatures of the hot fluid, $^{\circ}\text{F}$
T_s	Constant hot-fluid temperature, $^{\circ}\text{F}$
t	Temperature of the cold fluid in general, $^{\circ}\text{F}$

t_f, t_w	Temperature of the film, temperature of the wall, °F
t_1, t_2	Inlet and outlet temperature of the cold fluid, °F
Δt	True temperature difference, °F
Δt_D	Temperature group $\Delta t/(t_2 - t_1)$, dimensionless
U	Overall coefficient of heat transfer, Btu/(hr)(ft ²)(°F)
U_D, U_C	Overall design coefficient; overall clean coefficient, Btu/(hr)(ft ²)(°F)
u	Overall coefficient of heat transfer, Btu/(hr)(ft ²)(°F)
V	Velocity, fps; temperature group, dimensionless
v_s	Volume of granular material, ft ³
W	Weight flow of hot fluid, lb/hr
w	Weight flow of cold fluid, lb/hr
X	A function
x	Distance, ft
Y	Space group $(P_H - D_o)/P_H$, dimensionless
y	Thickness of agitator, ft
Z	Space group $(P_H - D_o)/D_o$, dimensionless
z	Height of liquid in vessel, ft
α	Thermal diffusivity, ft ² /hr
β	Dimensionless group, $2k/h_{oi}D$
θ	Time, hr
μ	Viscosity, lb/(ft)(hr)
ρ	Density, lb/ft ³
τ	A function
ψ	A function

Subscripts (except as noted above)

c	Coil
j	Jacket

CHAPTER 21

THE CONTROL OF TEMPERATURE AND RELATED PROCESS VARIABLES

Introduction. The even operation of a process is dependent upon the control of the process variables. These are defined as conditions in the process materials or apparatuses which are subject to change. Because there may be several materials and several pertinent operating factors which may change in the simplest of processes, the maintenance of control over an entire process is an important aspect of process design. Many of the advances in processing technology during recent years have been due in part to the widespread use of automatic-control mechanisms. Naturally a comprehensive study of so broad a field of endeavor is beyond the scope of this text, and it is the intention here to introduce in a practical way only the most elementary principles of process control.

Process Variables. When the flow sheet is laid out for a process, the temperatures, pressures, and fluid-flow quantities are theoretically fixed in accordance with heat, pressure, and material balances. The translation of the flow sheet into an operable plant requires that special provision be made to assure the *relative* constancy of the various quantities and qualities. It is impossible to achieve absolute constancy in even the simplest of industrial operations and these do not include the multitude of complex operations which are normally encountered. Take the simple case of a storage tank to which one pump continuously supplies liquid and an identical pump continuously removes liquid. Because of differences in suction and discharge, both pumps, acting independently, pump at different rates and the liquid level in the storage tank cannot be expected to remain constant.

Similar factors influence nearly every other steady-state condition. Consider the utilities alone such as high- and low-pressure steam, cooling water, electricity, compressed air, and fuel supply. When any single unit process in a plant is either shut down or started up, it may affect the supply of the utilities to the other unit processes. Furthermore, when the furnaces in a powerhouse are periodically refired, the temperature, pressure, and quantity of steam throughout the plant may show some variation. Similarly a sudden change in the steam demand at some point in the plant for steaming out large vessels may cause a sufficient

speed variation to affect the performance of turbine-driven pumps, compressors, and generators including their discharge rates and pressures. Or the temperature of cooling-tower water, varying with atmospheric conditions, might affect the total heat transfer at critical points in the process. Add to these the variations resulting from changes in the composition of the feed materials such as boiling points, specific heats, or viscosities, and additional fluctuations may be anticipated in the pressure, temperature, and fluid flow of the process streams.

Automatic control is employed to measure, suppress, correct, and modify changes of the four principal types of process variation:

1. Temperature control
2. Pressure control
3. Flow control
4. Level control

There are, in addition, other¹ controllable variables such as specific gravity, thermal conductivity, speed, and composition. Because of its

importance to heat transfer particular attention is given in the following pages to temperature control and its relationship to the other principal process variables.

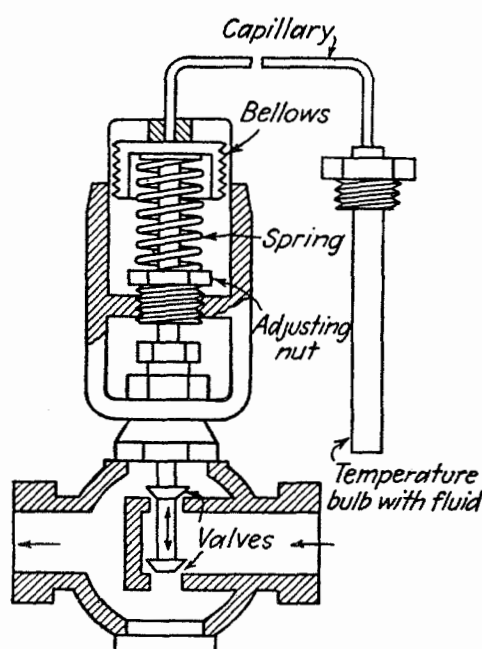


FIG. 21.1. Self-acting temperature controller.

temperature controller is shown in Fig. 21.1. A bulb filled with a fluid having a favorable coefficient of thermal expansion is placed in a stream or vessel at the point where the temperature is to be controlled. The bulb is connected to the controller by means of a capillary tube. As an example of its operation suppose that the temperature of a stream

Self-acting and Pilot-acting Controllers. It is the object of all controllers to regulate the process variables, and to do so they must be capable of first measuring the variables. Some instruments are equipped to *indicate* the variable in a continuously readable form, and others, *recorders*, are equipped with pen and ink on a traveling chart calibrated for time. These are not essential parts of the control but additional conveniences.

Controllers or regulators are either *self-acting* or *pilot-acting*. A self-acting

¹ See also Perry, J. H., "Chemical Engineers' Handbook," 3d ed., pp. 1263-1340 by R. W. Porter and D. M. Considine, McGraw-Hill Book Company, Inc., New York, 1950.

leaving the shell of a heater is to be controlled by adjusting the flow of steam to the tubes of the heater. Any increase in the temperature of the stream causes an increase in the pressure of the bulb fluid, which is transmitted in turn to the bellows of the regulator assembly. The larger area of the bellows amplifies the force acting against the spring and causes the valve to move in a direction which decreases the flow of steam. If the same type of regulator were used to control the flow of cooling water instead of steam, the valve would be arranged to open when the bulb pressure increased. The control points of self-acting controllers can be varied by means of adjusting nuts, which change the tension on the spring.

In general the temperature-sensitive element consists of a bulb and capillary filled or partly filled with a volatile fluid, an inert gas, or an expanding liquid. Thermocouple and thermal-expansion elements also find wide application in temperature control. With the exception of the liquid-filled bulb the total amount of fluid contained in the capillary is very small compared with the amount confined in the bulb. If the capillary were accidentally to lie across a steam pipe, it would not materially affect the measurement of the instrument, since the amount of fluid present where the capillary contacts the pipe is exceedingly small. Under certain conditions compensation must be included for variations in the ambient temperature at the point where the variation is registered.

From a mechanical standpoint self-acting types of instruments are the simplest. Most of these fall into the regulator rather than the instrument class, because the regulator does not actually measure the variable but acts upon deviations from the set control point. Many involve no calibration in their manufacture whatever but are adjusted to produce the desired effect in the field. Typical of this class are weight-loaded reducing or back-pressure valves, spring-loaded diaphragm valves for reducing or back-pressure regulation, and the temperature regulator discussed above.

The second class of instruments are the pilot-acting controllers in which the detection of a small variation actuates a more powerful force by which the variable is corrected. This class is capable of a greater degree of sensitivity than the regulators because they eliminate some of the lags which would be inherent in self-acting mechanisms actuated by the force of a large volume of fluid. Most pilot-acting instruments use compressed air to effect the larger controls, while some employ electric relays and motor-driven controllers. The latter are used particularly for isolated services or where compressed air is not available.

The majority of pilot-acting instruments operate on the principle shown in Fig. 21.2 or a modification of it. The controller operates

between two fluids, one being controlled and another which adds or removes heat to or from the fluid being controlled. In Fig. 21.2 compressed air at 15 to 20 psig enters the control head, part passing to a

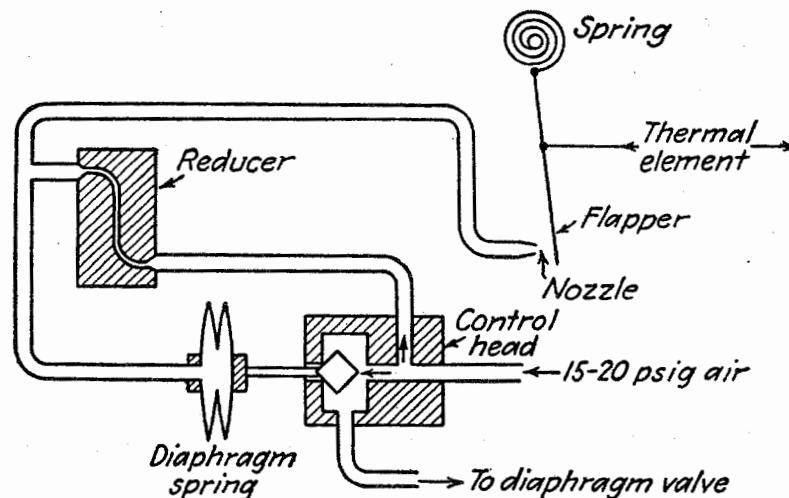


FIG. 21.2. Air-operated pilot.

diaphragm valve and part through the reducer. The discharge air from the reducer is at a low pressure and continuously escapes through the nozzle by impinging against a flat flapper. While air escapes through the nozzle, the diaphragm spring is contracted, since it is also affected by

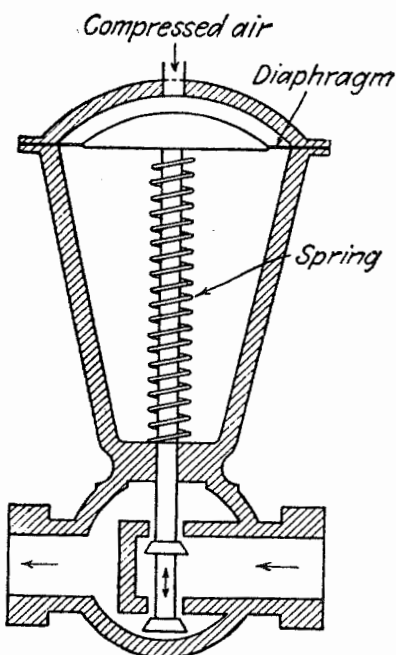


FIG. 21.3. Diaphragm valve.

the escape of air through the nozzle. This holds the valve in the control head against the left end of the control head. When a change occurs in the temperature of the fluid being controlled a capillary spiral or capillary helix connected to a temperature-sensitive element such as a bulb is displaced in response to the temperature change. The movement is used to bring the flapper toward the nozzle, thereby obstructing the escape of air and expanding the diaphragm spring. This may in turn be used to relieve the air pressure on a diaphragm valve effecting the control, thereby permitting it to close. A movement of the flapper of 0.002 in. is sufficient to actuate the control. A typical diaphragm valve is shown in Fig. 21.3, although diaphragm valves are available which either open or close when subjected to the action described.

Lags. Instrumentation would pose no problem if only a single variable were to be controlled. Usually, however, the regulation of a process

requires the control of pressures, temperatures, and flow rates and the maintenance of adequate liquid levels to assure flow continuity. The greatest impediment to precise control results from a series of time lags between the measurement of a variation and its correction. These are inherent in all automatic control systems. Haigler¹ has presented a visual picture of the time lags in a simple heat-transfer system from which Fig. 21.4 has been taken. Starting with the measurement of the variable at *a*, the lags may be traced alphabetically.

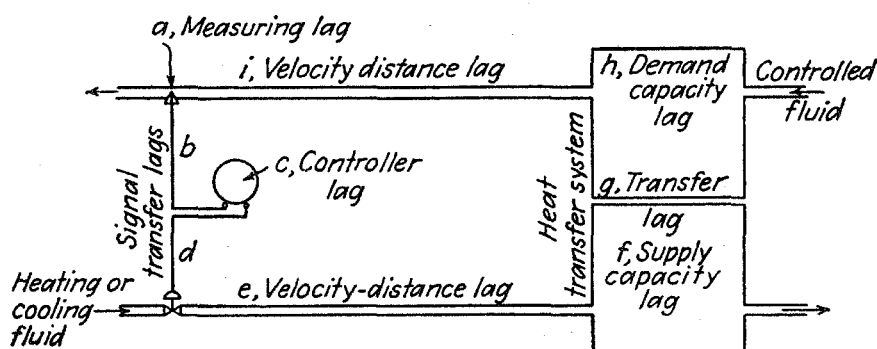


FIG. 21.4. Sources of lags in automatic control.

- a.* Measuring lag
- b.* Signal transfer from sensitive element to instrument
- c.* Controller operation lag
- d.* Signal transfer from controller to regulator
- e.* Velocity-distance lag until adjusted stream reaches apparatus
- f.* Capacity lag until the supply side fluid is adjusted
- g.* Transfer lag through tubing from supply to demand side
- h.* Capacity lag until the demand side is corrected
- i.* Velocity-distance lag until the measuring element detects the correction

All these lags are not of equal importance, but they do present difficulties in the accomplishment of close control such as is required by modern high-temperature, high-pressure, and catalytic processes. Numerous questions arise in automatic control such as the extent to which the regulating valve on the correcting fluid should be opened and whether or not the opening should vary with the size of the deviation measured. How fast should it open? How much of the time-temperature relationship of the response of the correction should an instrument be able to anticipate? These are particularly important in the larger processes in which all the process variables must be controlled simultaneously in less time than would accompany the operation of several related simple instruments.

¹ Haigler, E. D., *Trans. ASME*, 60, 633-640 (1938).

Automatic-control Mechanisms. For a general background and mathematical development the literature on instrumentation should be consulted. Available to deal with the lags inherent in process control are four basic mechanisms. These are

1. *Two-position control* (also called *on-and-off* or *open-and-shut*), in which the overall flow of fluids, Btu, watts, etc., is limited to predetermined maximum and minimum values or completely turned on or off when the measured variable exceeds or falls below the control point setting.

2. *Proportional control* (also *throttling*), in which the *magnitude* of the corrective action taken by the instrument is proportional to the extent of the deviation of the measured variable from the control point setting.

3. *Floating control*, in which the *rate* of corrective action taken by an instrument is proportional to the extent of the deviation of the measured variable from the control-point setting.

4. *Proportional control and automatic reset*, in which the corrective action taken by an instrument is a combination of proportional control plus a corrective action which is proportional to the rate of the deviation and the duration of that deviation from the control-point setting. Another basic mechanism, the *derivative function*, is one which applies a correction which is proportional to the rate of the deviation and which is unaffected by the amount or duration of the deviation.

An example of two-position control may be found in the simple constant-temperature bath. When the temperature falls below a fixed minimum, heat is supplied until the bath reaches a fixed maximum temperature. The heat supply is then turned off until the bath again reaches the minimum whence heat is again supplied (on and off). For proportional control consider a vessel of small capacity containing a heating element. Cold water continuously enters and hot water leaves the vessel. The control-point temperature has been fixed. Suppose the temperature of the cold-water supply suddenly drops very much. For each degree that the water in the vessel deviates from the control point, a *greater* amount of heat will be supplied. In floating control, the *rate* at which the valve moves is proportional to the deviation of the temperature from the control setting. If the temperature is below the control point, the valve opens at a constant *rate* or a rate which increases with the extent of deviation. As the control point is reached, the valve closes but at a rate in proportion to the deviation. Floating control is not well suited to operations requiring rapid temperature control, since the correction in the temperature near the control point (narrow band) is slow and cyclical. Some floating-control mechanisms are modified to give a rapid action outside a fixed narrow band of deviations and a slow

action inside the narrow band. A combination of floating and proportional control will make the action more rapid. *Proportional control and automatic reset* correct the action of proportional control when a major deviation may continue with additional minor deviations in the same variable. Suppose a major demand change occurs. In proportional control the valve position is fixed by the magnitude of the deviation. If the valve is correcting for a large deviation while an additional minor deviation occurs, the correction for the minor deviations would be made with a valve which is open primarily to the extent of the major deviation. The reset shifts the proportioning band so that the minor corrections can be made as if the major deviation were the control point and not in proportion to the total deviation from the control point.

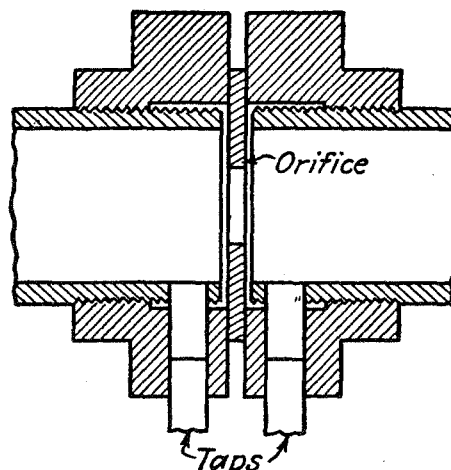


FIG. 21.5. Orifice taps.

Flow Control. The temperature control of a fluid can be accomplished by changing the flow of a heating or cooling medium. In many instances, however, the control of flow may be of greater importance than the control of temperature, and the methods of automatically maintaining flow warrant preliminary consideration. Unlike the expansion of a bulb, flow variations are detected by means of the differential pressures they cause

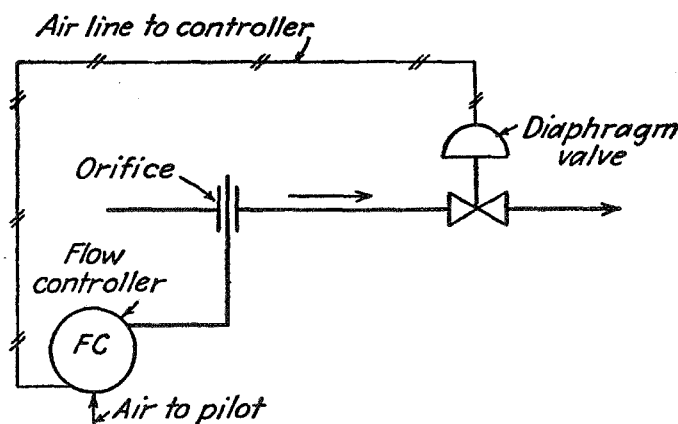


FIG. 21.6. Simple flow control.

across an orifice. Usually the flange and orifice assembly shown in Fig. 21.5 is tapped and fitted with tubes leading to a capillary helix or other device which is sensitive to the differential pressure. A displacement changes the position of the flapper, and the remainder of the control is similar to that shown in Fig. 21.2.

The simplest arrangement of flow control is shown in Fig. 21.6. The

flow of compressed air to the pilot controller mechanism is usually assumed to be present without being indicated, and very often the controller may also record the variations, thus being a *flow-recorder controller*. In establishing the primary flow control for a process it is important to consider whether the pumps will be of the reciprocating or centrifugal

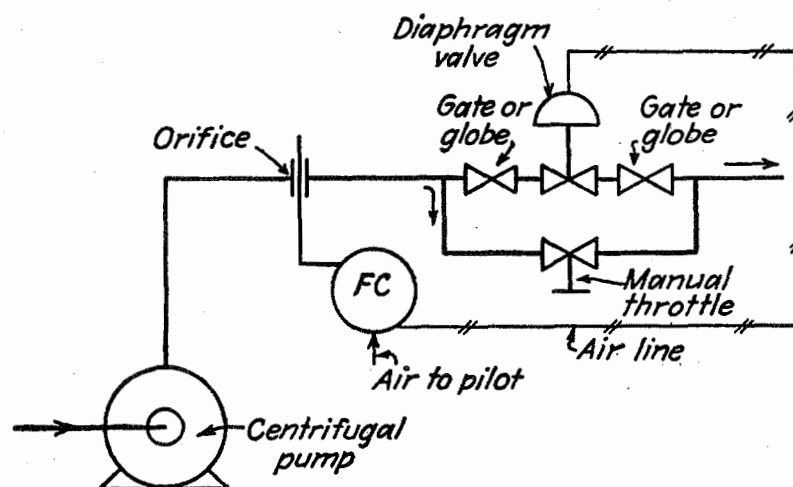


FIG. 21.7. Flow control with centrifugal pump.

types. If a centrifugal pump is employed, flow control can be effected by throttling against the pump pressure as shown in Fig. 21.7. In the pilot shown in Fig. 21.2 the compressed air leaving the control head kept the diaphragm valve open. About 70 per cent of the controllers are of this type, and about 30 per cent open if the air fails. The latter type is

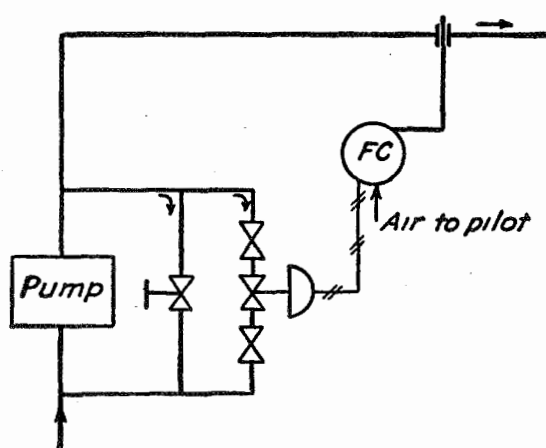


FIG. 21.8. Flow control with reciprocating pump.

particularly useful when controlling the flow of reflux to distilling towers. The use of a by-pass around the controller is a standard practice which permits the continued operation of the process with manual control if the controller or its air supply should fail. When using reciprocating pumps, throttling against the displacement of the pump may cause destructive pressures to be built up.

The flow controller in these cases is installed in a by-pass, returning to the reciprocating pump suction as shown in Fig. 21.8. In this manner constant discharge is maintained by automatically altering the amount of by-passing. The second line having but a single valve allows manual by-passing when both the valves adjoining the diaphragm valve must be closed because of controller or air failure.

When using automatic controllers, the flow may be intentionally divided so that the bulk of the fluid flows through a continual by-pass and only a small amount through the controlling line. This reduces the cost of the controller by reducing its size. However, the use of a continual by-pass as an integral part of the flow-control system introduces several unfavorable factors in the maintenance of uniform control. Except where the flow lines are extremely large and the control valve is expensive, it is undesirable to use a continual by-pass. Like temperature control, pressure and level control are often intimately associated with the control of flow.

TEMPERATURE CONTROL

Instrumentation Symbols. Most of the common arrangements of process apparatus require several interrelated instruments to assure reasonable control. To simplify representation, the instrumentation flow-plan symbols employed here are those suggested in a survey questionnaire by a symbols subcommittee of the Instrument Society of America. The legend for a number of the common symbols is given in Fig. 21.9. The first letters which appear as instrument designations are

F = flow

L = level

P = pressure

T = temperature

The second or third letters are

C = controller

G = glass (as LG)

I = indicator

R = recorder

S = safety

V = valve

W = well

Often it is desirable to assemble all the instruments at a single point away from the locality at which the measurement and control actually occur. These are referred to as *board-mounted* instruments as differentiated from locally mounted instruments and are designated by a horizontal line through the instrument. Often the instruments on a single control board are at some distance from the point of measurement. In such cases the response is amplified by a pneumatic transmitter. This is particularly requisite for the control of low-pressure fluids at a centralized board.

In the typical instrumentation diagrams which follow, the air lines to the instruments and the by-passes around the controllers have been

omitted. Except where necessary to manual control, the indicating or recording features have also been omitted, although they are essential to the instrumentation of the process as a whole.

Whenever a control valve is installed directly in the inlet or outlet line of an exchanger, cooler, or heater, the fluid passing through the

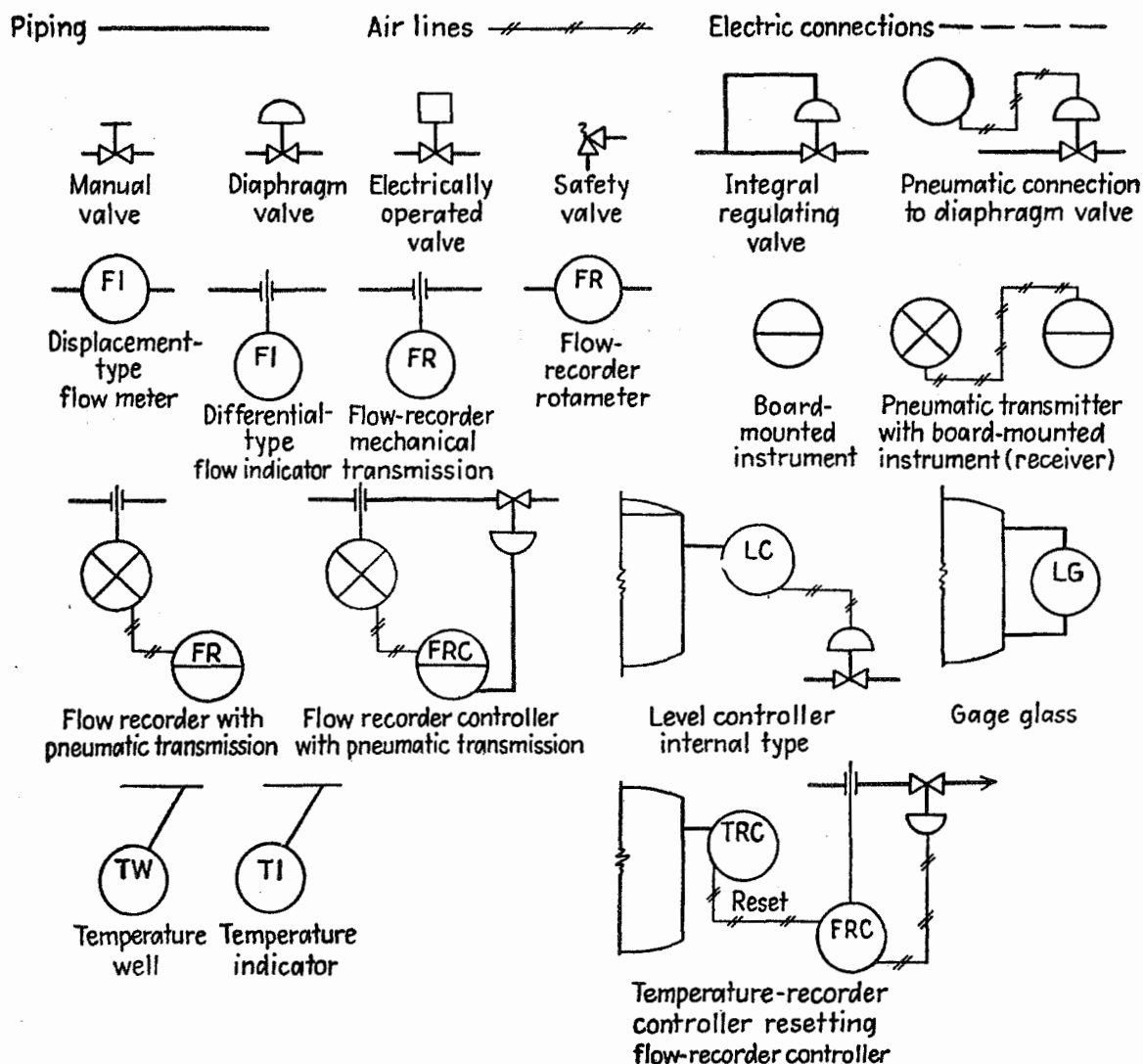


FIG. 21.9. Typical instrument symbols. Instrument lines are usually lighter than piping or equipment.

apparatus must still retain sufficient available pressure so that, when operating, the control valve is capable of increasing or decreasing the flow of the line. Thus an exchanger which at design flow utilizes nearly all of the available static pressure on a line has little pressure available for overcoming the effect of the control valve. An instrument is useless which operates under such conditions that the control valve is wide open and unable to open farther to effect control. Thus when selecting a pump for a process the total head must be not only sufficient to overcome

the sum of the static heads and pressure drops of the equipment but also enough to overcome the drop through the control valve.

Coolers. Figures 21.10 to 21.12 show three methods of providing control for coolers. Figure 21.10 is used where the maximum possible cooling is desired or where there is an abundance of cooling water. When an overhead volatile product from a distilling column passes through an aftercooler and is to be sent to storage, it is cooled as low as possible, since any decrease in the evaporation and vent loss may constitute a significant economy. In such cases only a manual throttling valve is provided on the outlet-water line. If water is abundant, the manual valve is left completely open. The only instrumentation suggested is an industrial thermometer or temperature indicator on the hot-fluid outlet. This enables the operator to make routine checks to ascertain that the temperature of the liquid is well below the maximum allowable storage temperature. Thermometers are useful at the other nozzles but only when checking the performance of the cooler or other apparatus. Thermometer wells are usually provided for this purpose as plugged bosses

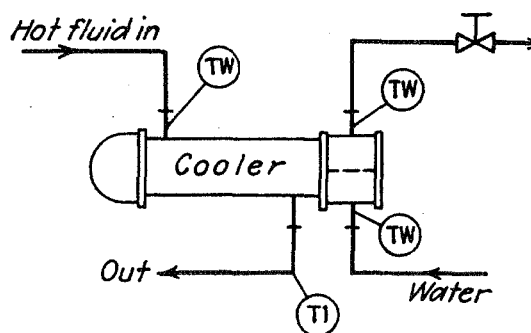


FIG. 21.10. Simple cooler.

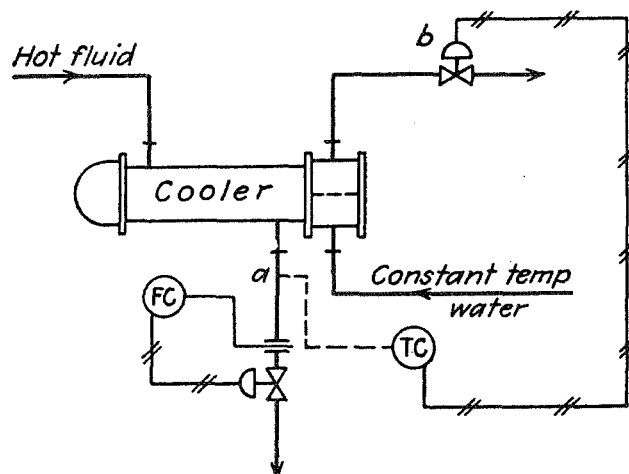


FIG. 21.11. Cooler with fixed hot fluid outlet and near-constant water temperature.

in the necks of nozzles. If the nozzle necks are not conveniently located for testing, the thermometers can be inserted in Tees as near the cooler as possible. However, when checking exchanger performance and guarantee, the temperatures should be taken at the nozzle necks if at all possible.

The arrangement in Fig. 21.11 is used when a fixed hot-fluid outlet temperature and flow quantity are desired. The flow of the hot fluid is

controlled independently by the flow controller. The cooling water is assumed to have a relatively constant temperature, so that the hot-fluid outlet temperature is controlled by the quantity of cooling water. The temperature of the hot-fluid discharge is measured by a temperature-sensitive element at *a* and transferred to the temperature controller which actuates an air-operated valve on the cold-water discharge at *b*. If the temperature at *a* varies, the valve at *b* opens or closes accordingly. This same arrangement can be used for gases provided a sufficiently sensitive element is used.

In Fig. 21.12 an arrangement is shown for cooling gases below their dew points such as occurs in compressor aftercoolers. The gas is cooled

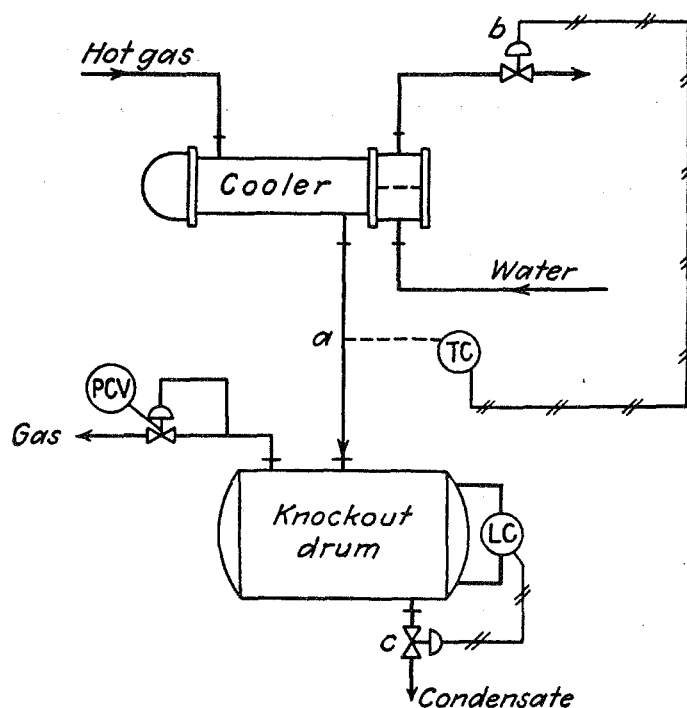


FIG. 21.12. Gas cooler with condensate removal.

and its temperature is measured at *a* and controlled at the water discharge *b*. The cooled gas and condensate pass into the knockout drum, where a liquid level is continuously maintained by the level controller. The simplest type of level controller consists of an external float whose vertical movement actuates the control. When the level rises above the control point or range of the level controller, it opens the valve at *c* and lets the liquid run out. The purpose of the level controller is to maintain a liquid seal which prevents the escape of gas from the condensate outlet and also prevents the condensate from backing up into the shell. The back-pressure regulator or pressure-control valve (PCV) releases gas from the knockout drum without upsetting the system flow, pressure, or liquid level.

Exchangers. Exchangers do not always require special temperature control. Since their purpose in a process is to provide the maximum recovery of heat, there is no reason to restrict their performance by the use of controls. It will ordinarily suffice to use manual valves which are kept wide open. The principal controls are usually provided on both the cooler and heater which are adjacent to the exchanger. For example, when a stream is to serve as the feed to a distilling column, it may enter the system through a bottoms-to-feed exchanger as in Fig. 11.1 and then through a feed preheater. The temperature control of the stream will be effected in the preheater by a flow adjustment of the steam entering the preheater. Similarly, when a fluid is cooled in an exchanger, it usually passes through a cooler and its temperature is controlled by a flow adjustment of the water. It is not possible to control both the flow quantities and outlet temperatures of both streams passing through an exchanger at the exchanger itself, since one adjustable quality must always be present. Thus, if the outlet temperatures of both streams are to be controlled and the flow or temperature of one stream may vary, the flow or outlet temperature of the other stream must also vary.

There are, however, a number of instances in which the outlet temperature of one of the streams must be controlled, and such cases are treated here. Most of the problems of exchanger instrumentation are encountered when the two streams are of unequal size, the one being very much larger than the other. By the same token the larger stream possesses a short temperature range and the smaller stream a very large temperature range. Figures 21.13 and 21.14 are typical of this application. In Fig. 21.13 the hot fluid is considered the large stream and the cold fluid the small stream. The hot fluid is to be flow- and temperature-controlled, and its inlet temperature is assumed to be subject to small variations. The cold fluid is flow-controlled, and its outlet temperature is assumed to be reasonably constant. The net Btu removal or injection in this case can best be effected on only a portion of the large stream, and to this end the by-pass is provided for temperature control. It will be seen, however, that the flow control of the hot fluid is carried out on the entire stream. This arrangement provides flexible temperature control, allowing any proportion to be by-passed. It also permits the elimination of any problems of excessive pressure drop which might result if the entire large stream were passed through the exchanger. It is also advantageous where an exchanger is repeatedly used for several different services. As an example, a particular distilling column and its auxiliaries may be used for a time with one feed stock and then changed to another stock for another period. The controls shown in Fig. 21.13 will enable the exchange of a fairly uniform heat load in the exchanger for the several

stocks. The pressure-control valve permits a constant controlled flow of the fluid being temperature-controlled, and the only varying qualities are the inlet temperature of the temperature-controlled fluid and the outlet temperature of the secondary fluid.

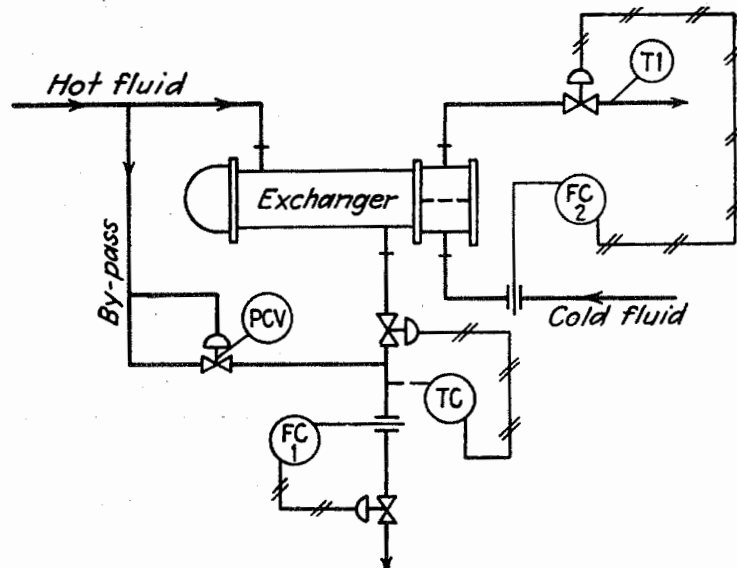


FIG. 21.13. Exchanger with temperature-controlled large stream.

Heaters. Few types of apparatus cause greater difficulty than heaters using low-pressure steam with temperature control on the cold-fluid outlet temperature. The difficulty can be eliminated, however, when proper means for removing the condensate and air are provided. All heaters

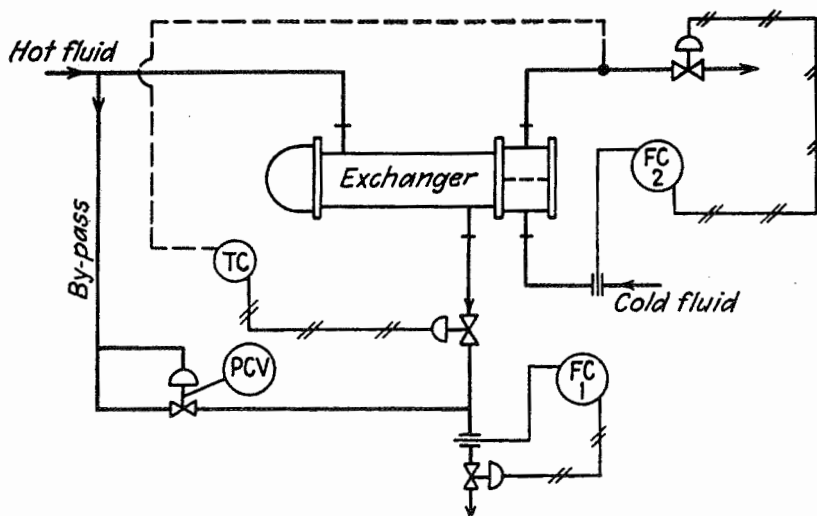


FIG. 21.14. Exchanger with temperature-controlled small stream.

using steam at superatmospheric pressures should be equipped with air vents at the highest accessible points. Heaters operating with a cold-fluid outlet temperature exceeding 212°F as in Fig. 21.15 generally offer little difficulty, because the condensate is not subcooled below the pressure necessary to force it through the outlet trap.

In practice, heaters are often designed for small temperature differences. Added to this is the fact that they are usually oversized, particularly when clean, so that the cold-fluid outlet temperature and the steam condensate temperature are nearly the same. The cold fluid and the condensate may both have outlet temperatures below 212°F (corresponding to 0 psig) indicating that more heat is being removed from the steam than is desired thereby subcooling the condensate to vacuum. Without temperature control a condensate level may build up in the heater to cover a portion of the tubes, reduce the available surface and the transferred heat load, and reduce the outlet temperature on the hot

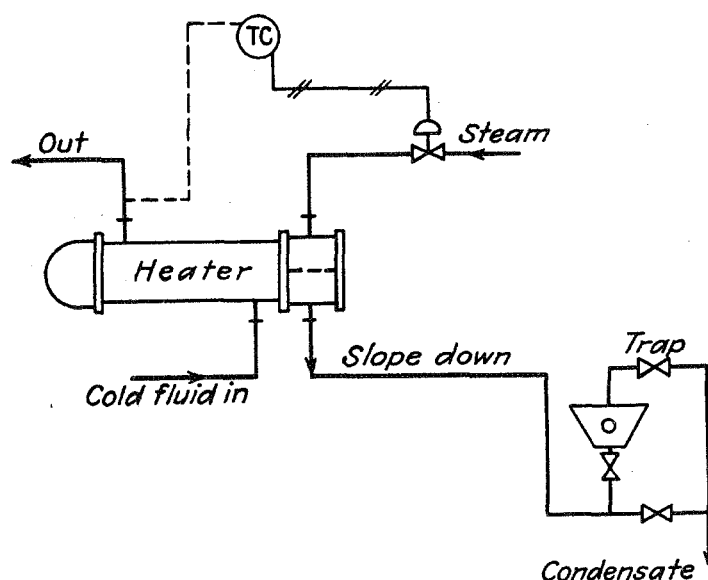


FIG. 21.15. Heater for high cold-fluid outlet temperature.

fluid. If the corrosive action resulting when a condensate level is maintained in the tubes is not excessive, the steam flow can be controlled manually or by a self-regulating flow controller with a simple trap for condensate removal provided the heat demand does not vary too frequently or rapidly for manual adjustment.

With temperature control a system for the removal of low-pressure steam embodying the principles shown in Fig. 21.16 must be employed. However, a pressure-control valve may be used to maintain the pressure in the heater. When the pressure-control valve is installed in the condensate outlet line, the temperature-control valve then varies the condensate level in the heater to compensate for varying heat loads. Very often an arrangement like that shown in Fig. 21.15 is unsuited for heaters providing a low cold-fluid outlet temperature. Without condensate covering some of the tubes, the total surface, containing some excess, is always exposed and the steam has to condense at subatmospheric pres-

sure in order to reduce the LMTD and provide the desired cold-fluid outlet temperature. On attaining an abnormally high fluid outlet temperature, the temperature-control valve throttles the steam until a condensate level builds up inside the heater. By the time the cold-fluid outlet temperature has decreased to normal, condensate covers the tubes, causing the condensate in the lower portion to be subcooled and a resultant vacuum within the heater. When the temperature controller next detects a subnormal cold-fluid outlet temperature, it must admit enough steam to raise the pressure sufficiently to remove all the subcooled condensate and decrease the condensate level in the heater. In doing

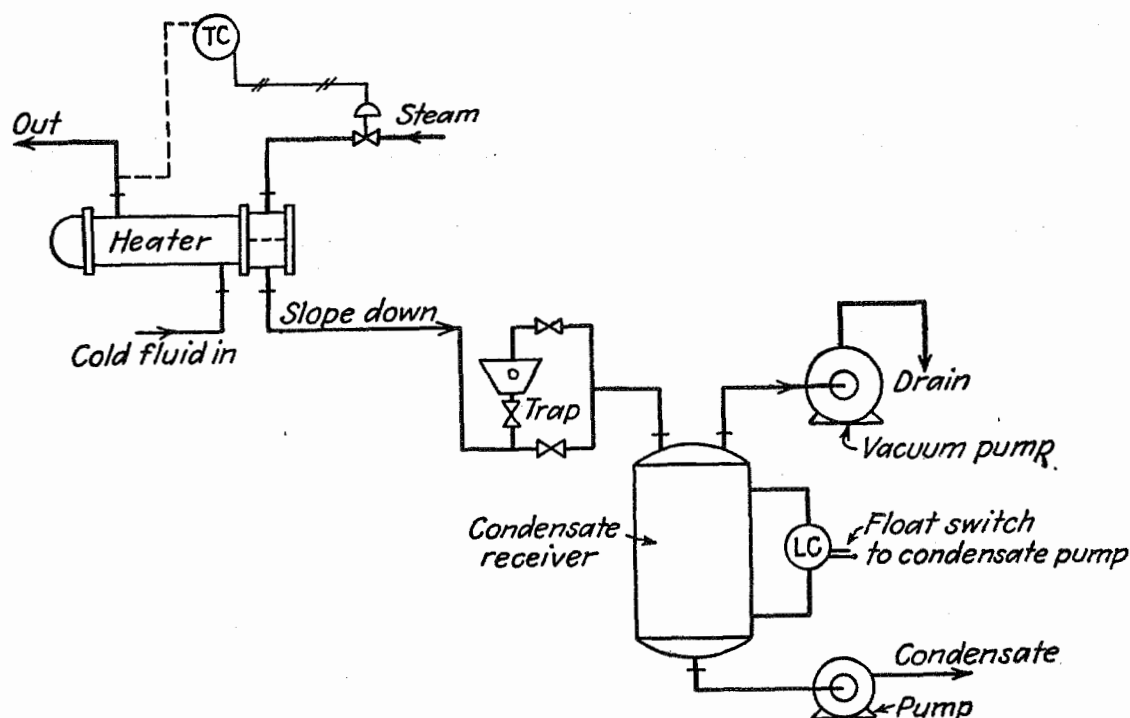


FIG. 21.16. Heater using low-pressure steam.

this too many tubes may be uncovered by the steam pressure, causing the cold-fluid outlet temperature to rise above the control point. This procedure causes cycling and a virtually uncontrolled cold-fluid outlet temperature. The arrangement in Fig. 21.15 is suitable for heaters using high-pressure steam in which the condensing temperature and pressure, after passing through the control valve, is sufficiently high to blow the condensate out.

Total Condensers. Figure 21.17 shows the arrangement for a condenser with gravity flow of reflux. The principal disadvantage to the use of gravity flow lies in the fact that the condenser and accumulator must be elevated above the tower, requiring additional structural support. The overhead condensate drains into the accumulator, which is provided with a manual vent for continuous bleeding if the operation is at elevated

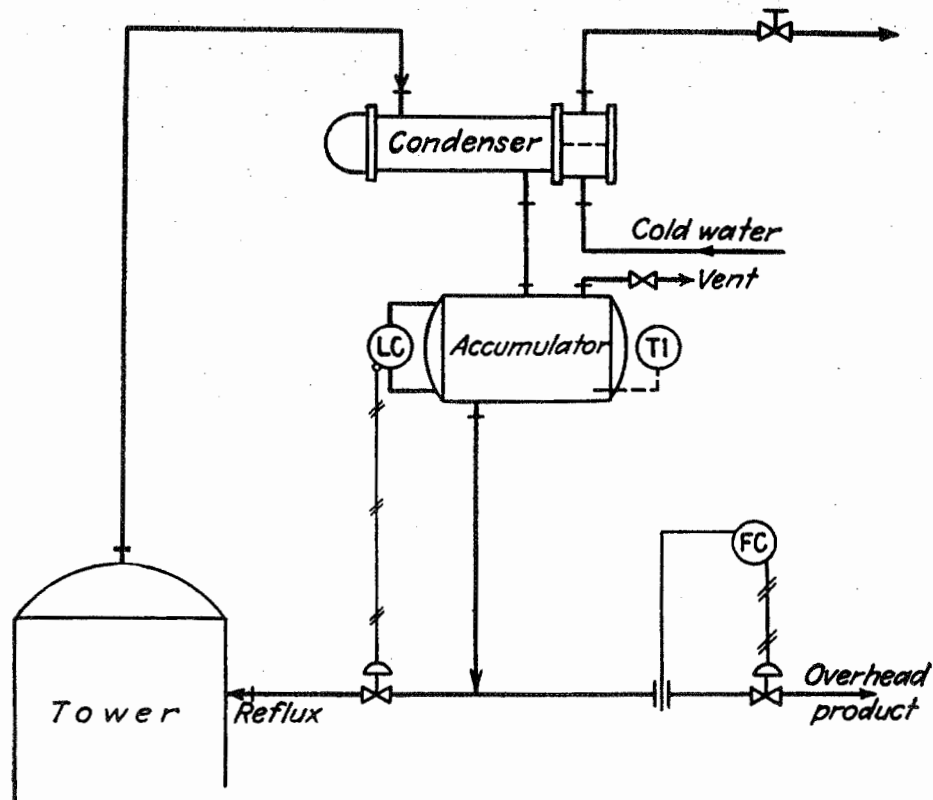


FIG. 21.17. Condenser with gravity flow of reflux.

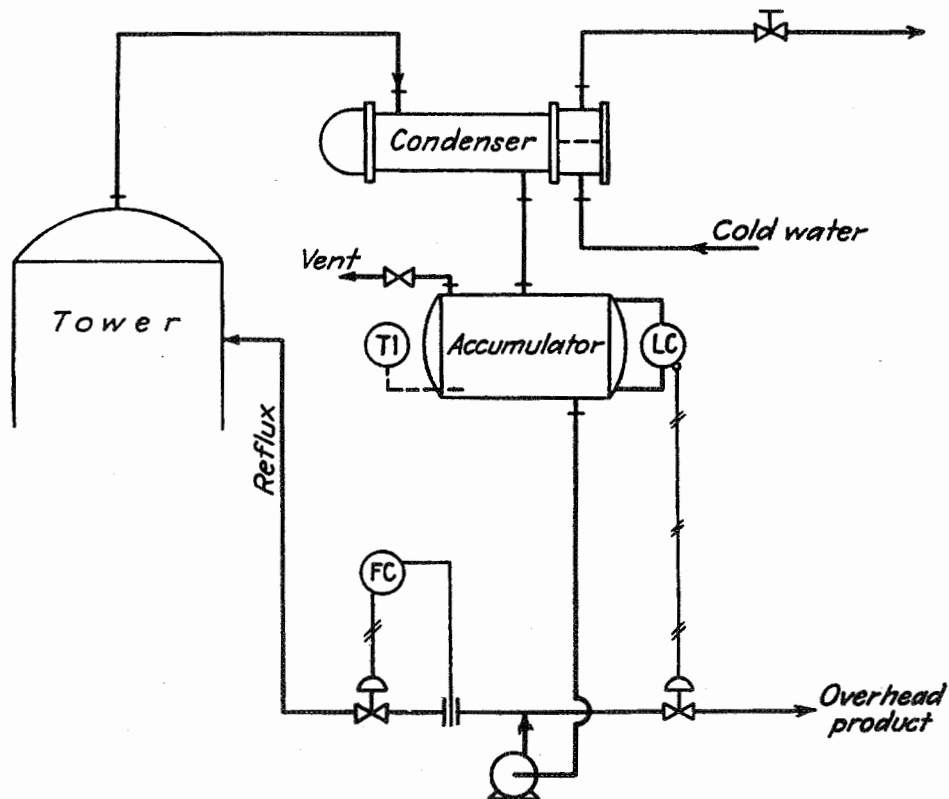


FIG. 21.18. Condenser with pumped reflux.

pressure or left wide open if operation is at atmospheric pressure. If there is any tendency for noncondensables to enter the system, it may be necessary to bleed them at the top of the condenser. The condensate builds up a level in the accumulator, and reflux is regulated by the level controller. The overhead product is removed at a fixed rate by the flow controller. It is seen that there is no positive temperature control other than manual in the event that the reflux temperature should fall appreciably below that of the top tray. This may be compensated for, how-

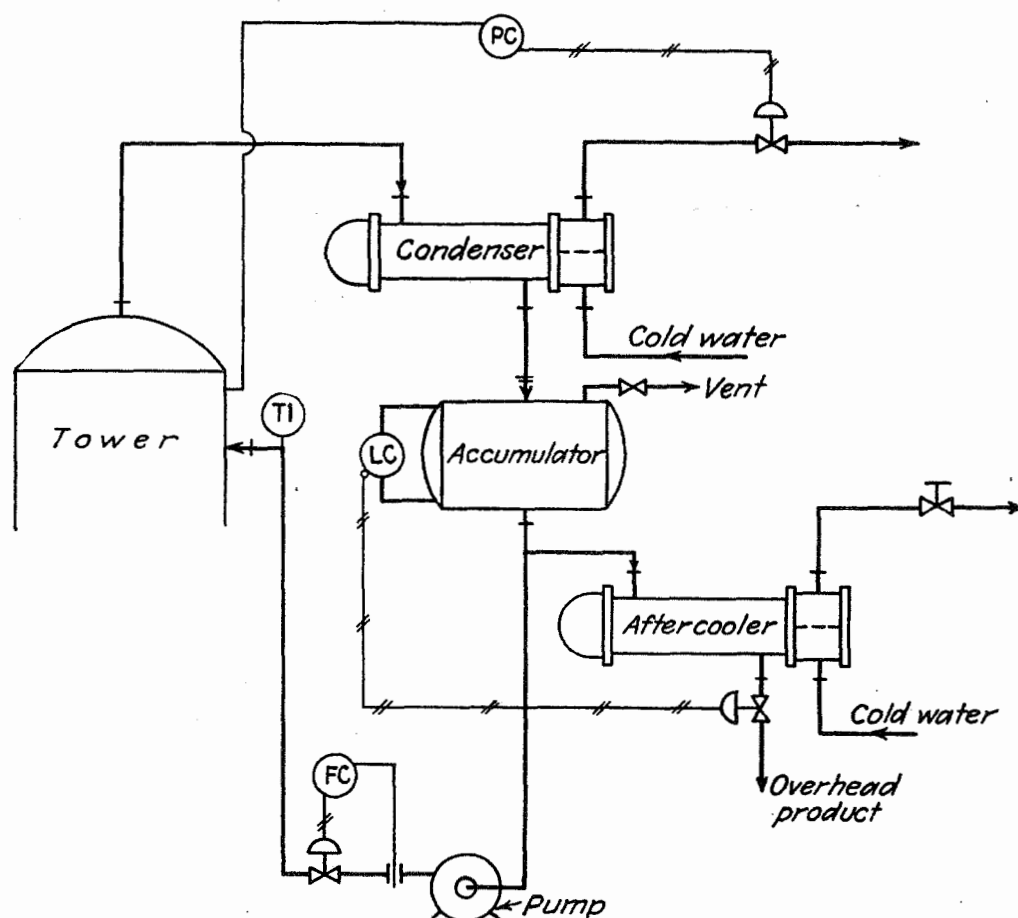


FIG. 21.19. Condensation of a pure volatile product.

ever, by introducing a temperature controller at one of the intermediate trays of the column which operates on the steam line going to the reboiler at the bottom of the column.

Figure 21.18 is a typical example of a condenser operating with pumped-back reflux. The reflux pump is often at the ground level, and the condenser and the accumulator are immediately above it. In this particular application, the reflux rate is set by the flow controller and the overhead product flow is set by the level controller. The reversal of the method of overhead product flow control between Figs. 21.17 and 21.18 is due to the smaller liquid head in the gravity-flow system. Figure 21.19

shows an arrangement for the condensation of a pure volatile product which must be subcooled in an aftercooler to prevent the flashing of the overhead product when its liquid pressure is relieved. A particular feature of this type of arrangement is the use of a pressure controller for the control of temperature by direct connection between the water outlet flow and the tower pressure. When the system contains noncondensables this arrangement cannot be employed.

Figure 21.20 shows an arrangement for a system having noncondensables. The instruments used here differ in the manner by which they

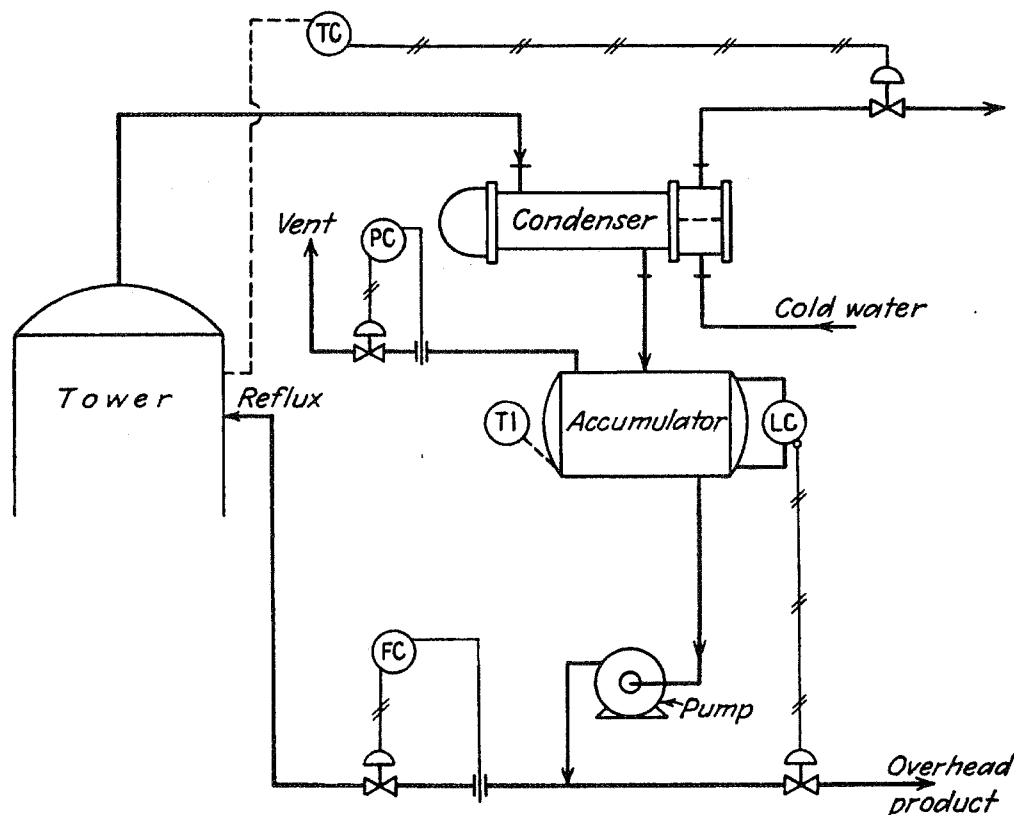


FIG. 21.20. Condensation with noncondensables.

effect control. First, the noncondensables are prevented from building up in the system by means of the pressure controller on the accumulator, which provides a continuous automatic vent to the atmosphere or to the next step in the process if the saturated noncondensables are subjected to further treatment. The tower overhead temperature is maintained by the controller adjusting the rate of water flow to the condenser and likewise adjusting the reflux temperature to maintain the overhead product purity.

Partial Condensers. A partial condenser arrangement is shown in Fig. 21.21. The partial condenser is used for volatile materials to enable reflux to be pumped back to the tower at substantially the top tray

temperature while the remainder of the overhead is condensed and cooled so as to prevent reflashing when the pressure is released in the storage tank. The elements of the control can be identified by comparison with Figs. 21.19 and 21.20.

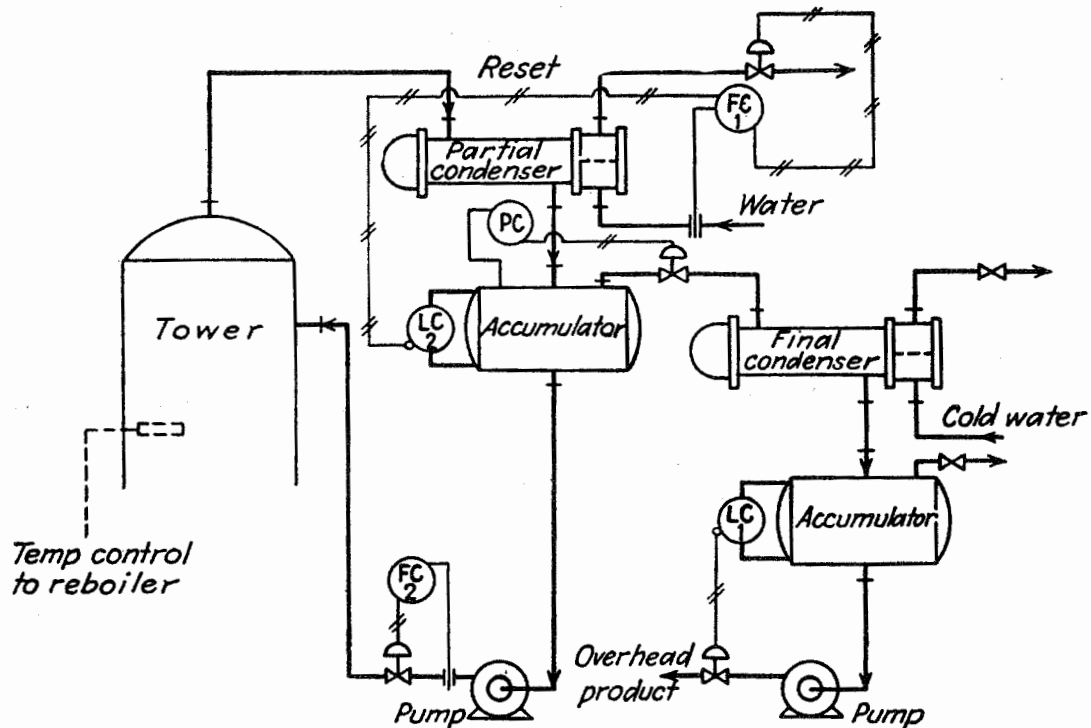


FIG. 21.21. Partial condenser.

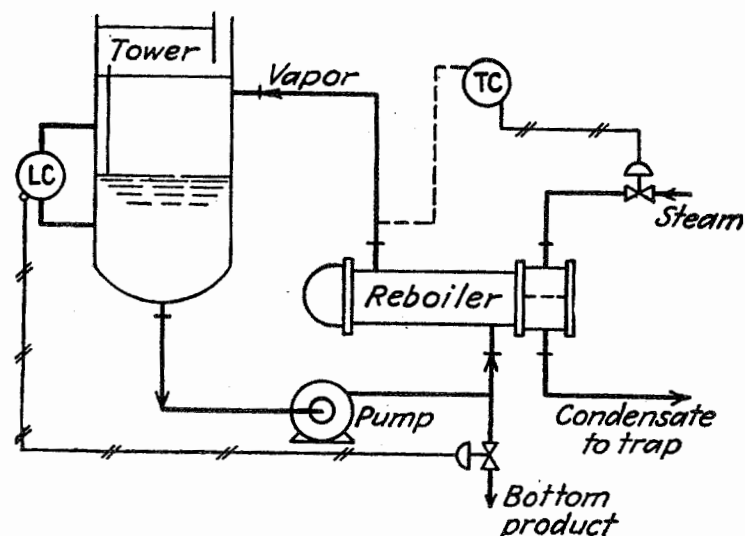


FIG. 21.22. Pump-through reboiler with small bottoms rate.

Pump-through Reboilers. A reboiler can be critical in a distillation process, and its control is extremely important. Furthermore, steam-heated reboilers are subject to the same operating difficulties encountered with heaters. Because pump-through reboilers are used only for small

services, the requirements of control are usually more sensitive than those for larger operations.

Figure 21.22 shows the arrangement for a pump-through reboiler where the amount of bottom product is small compared with the total flow through the reboiler. Figure 21.23 shows the arrangement for cases where the bottom product is an appreciable part of the total flow. In this case the bottom product rate is set separately by the level controller

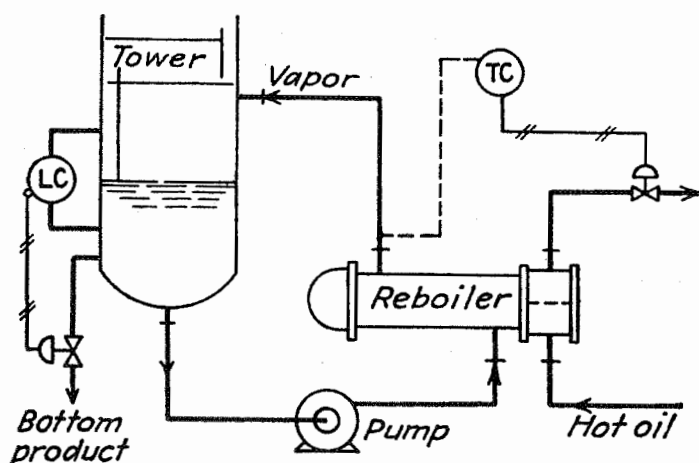


FIG. 21.23. Pump-through reboiler with high bottoms rate.

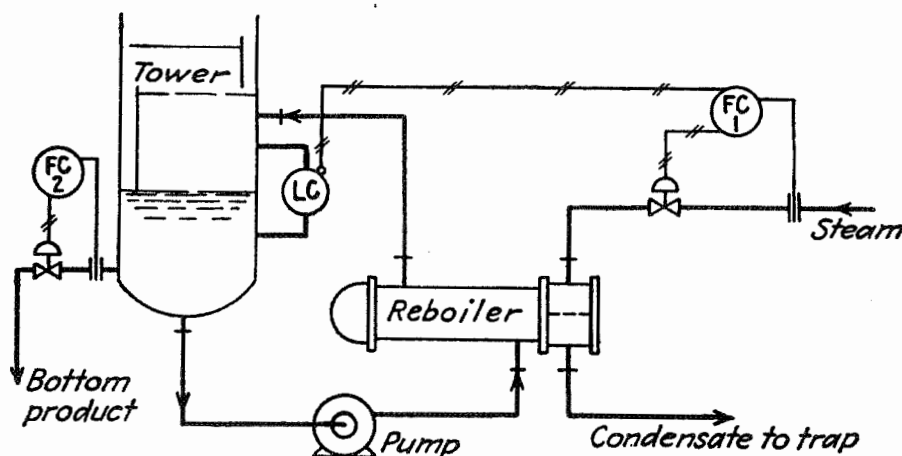


FIG. 21.24. Pump-through reboiler with pure bottoms or constant-boiling mixture.

so as not to interfere with the reboiler operation. In both Figs. 21.22 and 21.23 it has been assumed that the liquids have appreciable boiling ranges so as to make the use of a temperature controller effective. If the bottom product is substantially pure or a constant boiling mixture, as in Fig. 21.24, the use of temperature control is impractical, since the temperature will remain constant independently of the steam supply. Only the flow rate changes, and to overcome this limitation the flow of steam is adjusted by the level controller. The reset permits partial response of the flow controller so as to increase the level controller range.

Natural-circulation Reboilers and Evaporators. Because of their similarity the chiller- and kettle-type reboiler are included in this classification. The instrumentation for the chiller is shown in Fig. 21.25. The valve on the temperature controller should be sized for a small

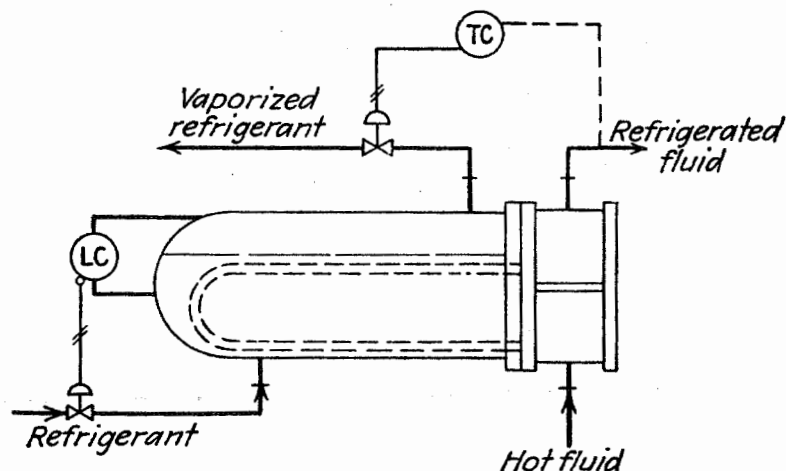


FIG. 21.25. Chiller.

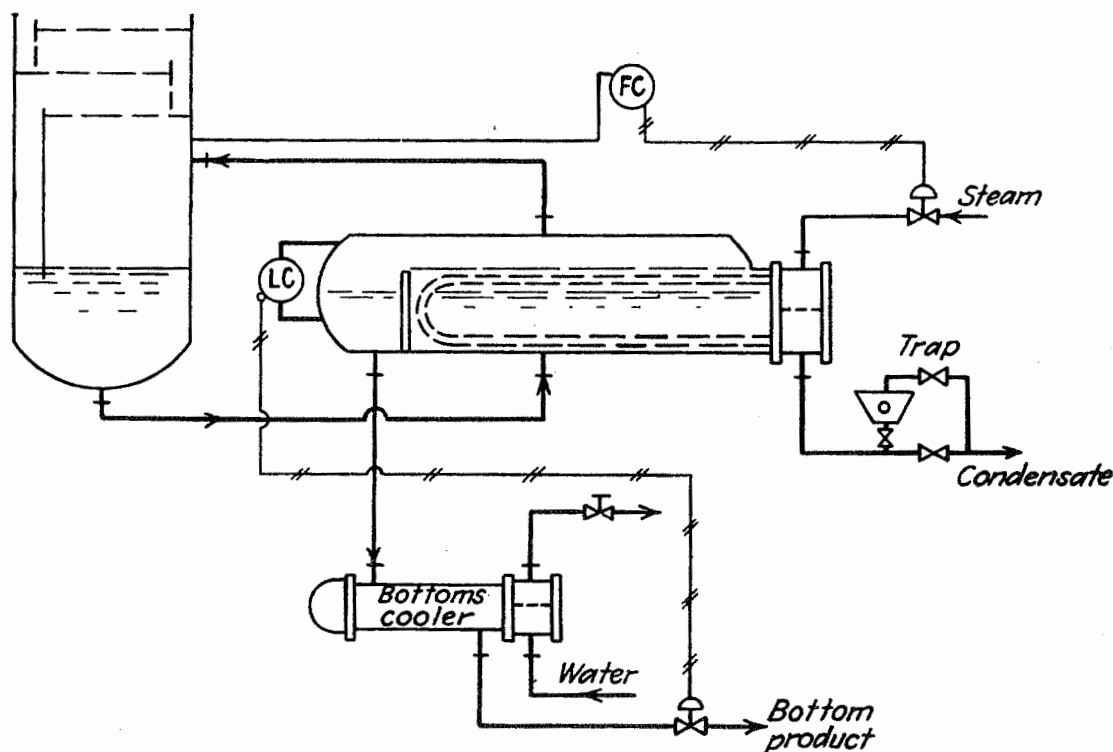


FIG. 21.26. Kettle-type reboiler.

pressure drop so as not to reduce the compressor suction or unnecessarily increase the power cost. Figure 21.26 shows the instrumentation for the kettle reboiler. The functions of the level control are evident. The horizontal thermosyphon is shown in Fig. 21.27 with once-through flow as compared with a recirculating arrangement. The arrangements of

Figs. 21.26 and 21.27 may also serve as the basis for the instrumentation of horizontal and vertical evaporators.

Batch Processes. Batch processes are usually very simple to control.¹ Most batch processes do not require automatic control of any sort.

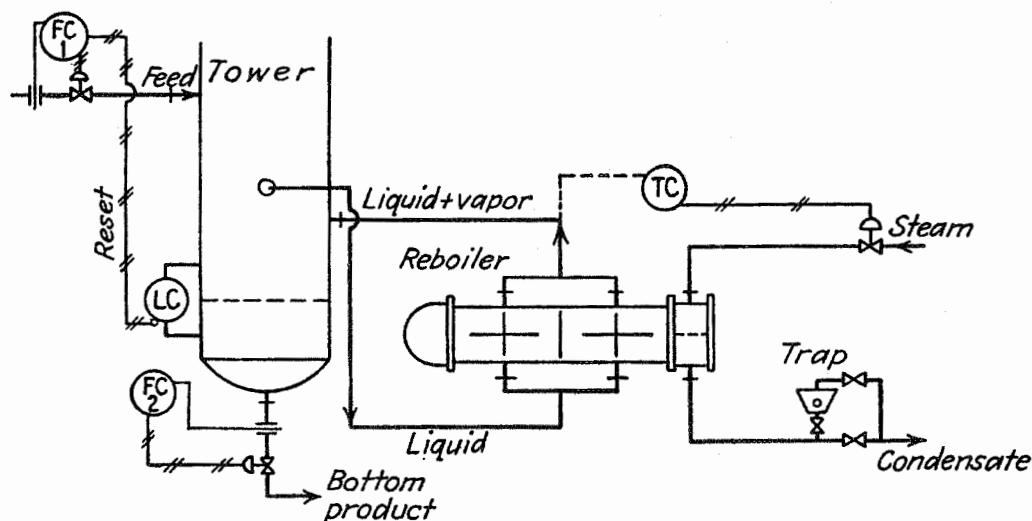


FIG. 21.27. Thermosyphon reboiler.

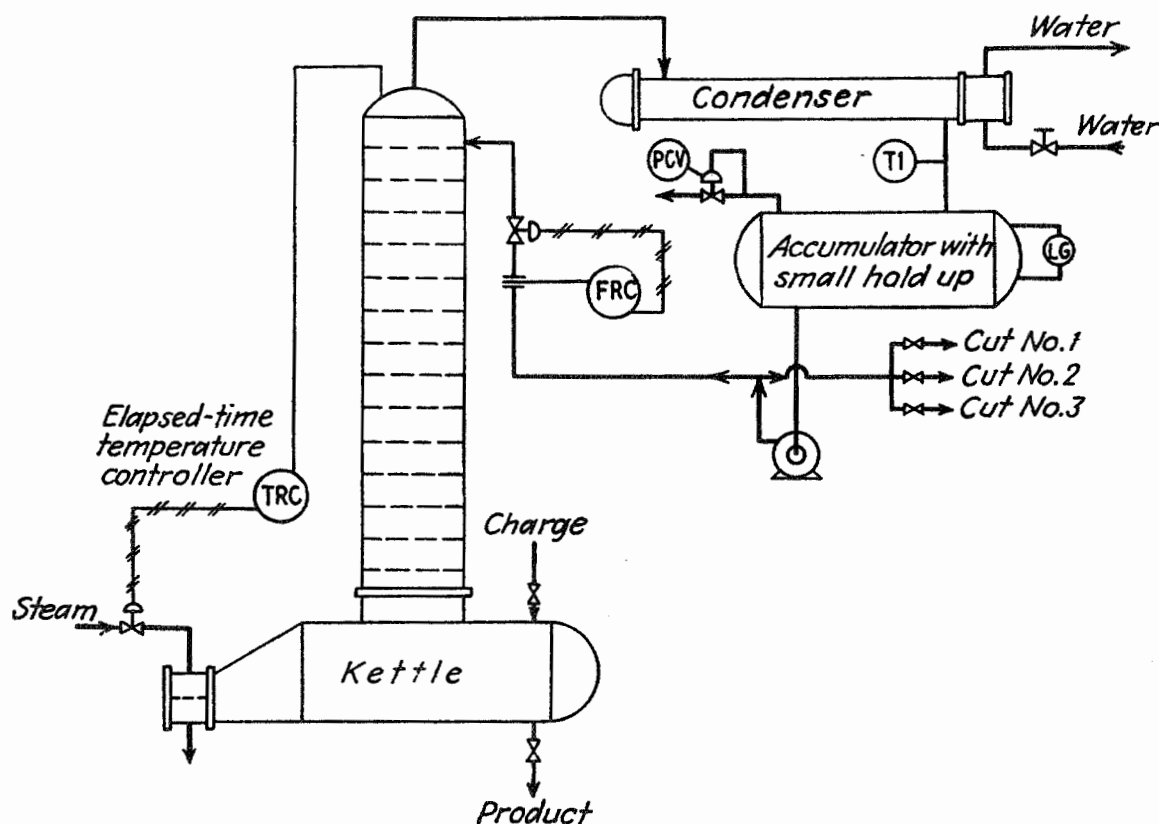


FIG. 21.28. Batch distillation with elapsed time-temperature controller.

Batch distillation presents several interesting problems which are shown in Fig. 21.28. During batch distillation the still and possibly the over-

¹ See particularly Perry, J. H., "Chemical Engineers' Handbook," 3d ed., McGraw-Hill Book Company, Inc., New York, 1950.

head temperatures change continuously as material is taken off overhead, and the related problem is then the control of temperature and pressure. Instead of using a temperature controller, an elapsed time-temperature controller or program controller is used. It sets a time period for the distillation temperatures and throttles the steam while the still tempera-

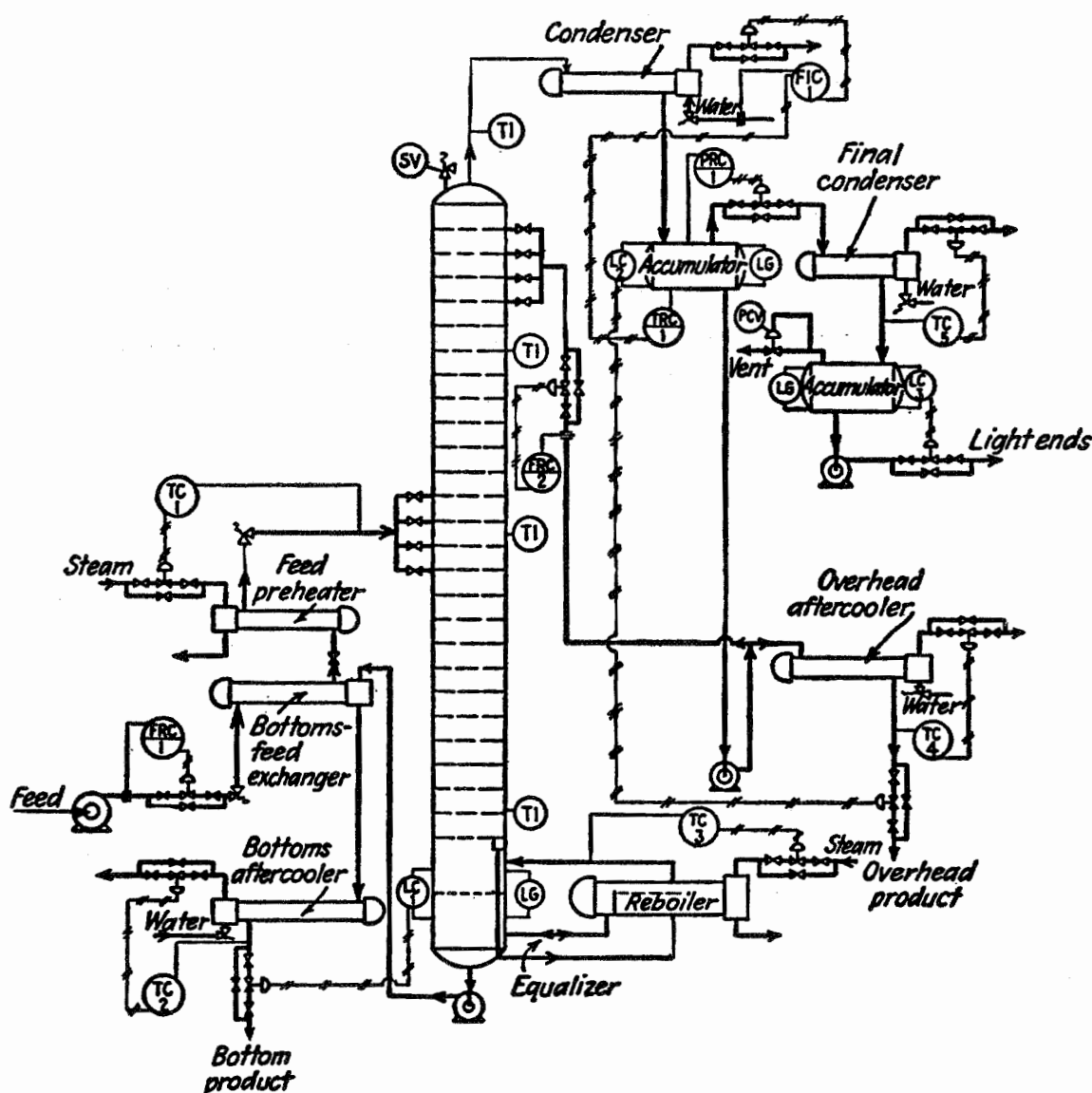


FIG. 21.29. Automatic control of a distillation process with feed containing some non-condensable gas.

ture rises through a timed cycle of temperature rises. The pressure control is maintained simply by a back-pressure regulator (pressure-control valve) on the assumption that there is some small amount of volatiles or noncondensables. The reflux is flow-controlled during the entire operation.

Continuous Distillation. Figure 21.29 shows the instrumentation for a continuous distillation charged with a feed having noncondensables.

With the exception of steam distillation which closely resembles it, other types of feed permit the instrumentation to be simplified. Included in Fig. 21.29 are the by-passes about the instrument controllers and the safety or relief valves on the exchangers. Safety valves are of the expansion-spring type. If it is possible in any way for the flow of the cold liquid to be halted because a valve has been accidentally turned off or the instrument control fails, a safety valve is placed on the inlet side of the cold fluid. It permits the expansion of the halted cold liquid to be relieved inasmuch as the hot fluid may continue to flow and prevents destructively large pressures from being built up in the liquid lines. After the small amount of liquid expansion has been relieved, the valve reseats itself. Attention should also be focused on the hook-up of the reboiler. Instead of the overflow going to the product outlet, as in the recirculating hook-up, the overflow is used to equalize the liquid level on a once-through arrangement. This is preferable in the event that the reboiler is overdesigned and more than the required amount of vaporization may occur. The temperature indicators shown in the column are multiple thermocouples connected usually to a single indicating instrument. The instruments which would make a convenient panel board for the control of the process are shown by a horizontal line for board mounting.

Conclusion. The elements of process control presented here have been among the simplest. The steppingstones by which an overall modern process can be controlled involve far more than the combination of several single effects. For these complex applications the many compound instruments of particular manufacturers are required to amplify the measurements or surpress the lags resulting from a large number of controls.

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¹ See Index for corrections.² See Index for other data.

TABLE 1. CONVERSION FACTORS AND CONSTANTS

Energy and power:

$\text{Btu} = 0.252 \text{ kg-cal}$
 $\text{Btu} = 0.293 \text{ watt-hr}$
 $\text{Btu} = 0.555 \text{ pcu (pound centigrade unit)}$
 $\text{Btu} = 778 \text{ ft-lb}$
 $\text{Btu/min} = 0.236 \text{ hp}$
 $\text{Hp} = 42.4 \text{ Btu/min}$
 $\text{Hp} = 33,000 \text{ ft-lb/min}$
 $\text{Hp} = 0.7457 \text{ kw}$
 $\text{Hp-hr} = 2543 \text{ Btu}$
 $\text{Kw} = 1.3415 \text{ hp}$
 $\text{Watt-hr} = 3.415 \text{ Btu}$

Fluid flow:

$\text{Bbl/hr} = 0.0936 \text{ cfm}$
 $\text{Bbl/hr} = 0.700 \text{ gpm}$
 $\text{Bbl/day} = 0.0292 \text{ gpm}$
 $\text{Bbl/day} = 0.0039 \text{ cfm}$
 $\text{Cfm} = 10.686 \text{ bbl/hr}$
 $\text{Gpm} = 1.429 \text{ bbl/hr}$
 $\text{Gpm} = 34.3 \text{ bbl/day}$
 $\text{Gpm} \times s \text{ (specific gravity)} = 500 \times s \text{ lb/hr}$

Heat-transfer coefficients:

$\text{Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} = 1.0 \text{ pcu/(hr)(ft}^2\text{)(}^\circ\text{C)}$
 $\text{Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} = 4.88 \text{ kg-cal/(hr)(m}^2\text{)(}^\circ\text{C)}$
 $\text{Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} = 0.00204 \text{ watts/(in.}^2\text{)(}^\circ\text{F)}$

Length, area, and volume:

$\text{Bbl} = 42 \text{ gal}$
 $\text{Bbl} = 5.615 \text{ ft}^3$
 $\text{Cm} = 0.3937 \text{ in.}$
 $\text{Ft}^3 = 0.1781 \text{ bbl}$
 $\text{Ft}^3 = 7.48 \text{ gal}$
 $\text{Ft}^3 = 0.0283 \text{ m}^3$
 $\text{M}^3 = 6.290 \text{ bbl}$
 $\text{M}^3 = 35.314 \text{ ft}^3$
 $\text{Ft} = 30.48 \text{ cm}$
 $\text{Ft} = 0.3048 \text{ m}$
 $\text{Gal} = 0.02381 \text{ bbl}$
 $\text{Gal} = 0.1337 \text{ ft}^3$
 $\text{Gal} = 3.785 \text{ liter}$
 $\text{Gal} = 0.8327 \text{ gal (Imperial)}$
 $\text{In.} = 2.54 \text{ cm}$
 $\text{Liter} = 0.2642 \text{ gal}$
 $\text{Liter} = 1.0567 \text{ qt}$
 $\text{M} = 3.281 \text{ ft}$
 $\text{Ft}^2 = 0.0929 \text{ m}^2$
 $\text{M}^2 = 10.76 \text{ ft}^2$

Pressure:

$$\text{Atm} = 33.93 \text{ ft of water at } 60^{\circ}\text{F}$$

$$\text{Atm} = 29.92 \text{ in. Hg at } 32^{\circ}\text{F}$$

$$\text{Atm} = 760 \text{ mm Hg at } 32^{\circ}\text{F}$$

$$\text{Atm} = 14.696 \text{ psi}$$

$$\text{Atm} = 2116.8 \text{ lb/ft}^2$$

$$\text{Atm} = 1.033 \text{ kg/cm}^2$$

$$\text{Ft of water at } 60^{\circ}\text{F} = 0.4331 \text{ psi}$$

$$\text{In. of water at } 60^{\circ}\text{F} = 0.0361 \text{ psi}$$

$$\text{Kg/cm}^2 = 14.223 \text{ psi}$$

$$\text{Psi} = 2.309 \text{ Ft of water at } 60^{\circ}\text{F}$$

Temperature:

$$\text{Temperature } ^{\circ}\text{C} = \frac{5}{9} (^{\circ}\text{F} - 32)$$

$$\text{Temperature } ^{\circ}\text{F} = \frac{9}{5} (^{\circ}\text{C} + 32)$$

$$\text{Temperature } ^{\circ}\text{F absolute } (^{\circ}\text{R}) = ^{\circ}\text{F} + 460$$

$$\text{Temperature } ^{\circ}\text{C absolute } (^{\circ}\text{K}) = ^{\circ}\text{C} + 273$$

Thermal conductivity:

$$\text{Btu/(hr)(ft}^2\text{)}(^{\circ}\text{F/ft}) = 12 \text{ Btu/(hr)(ft}^2\text{)}(^{\circ}\text{F/in.})$$

$$\text{Btu/(hr)(ft}^2\text{)}(^{\circ}\text{F/ft}) = 1.49 \text{ kg-cal/(hr)(m}^2\text{)}(^{\circ}\text{C/m})$$

$$\text{Btu/(hr)(ft}^2\text{)}(^{\circ}\text{F/ft}) = 0.0173 \text{ watts/(cm}^2\text{)}(^{\circ}\text{C/cm})$$

Viscosity (additional factors are contained in Fig. 13):

$$\text{Poise} = 1 \text{ g/(cm)(sec)}$$

$$\text{Centipoise} = 0.01 \text{ poise}$$

$$\text{Centipoise} = 2.42 \text{ lb/(ft)(hr)}$$

Weight:

$$\text{Lb} = 0.4536 \text{ kg}$$

$$\text{Lb} = 7000 \text{ grains}$$

$$\text{Ton (short or net)} = 2000 \text{ lb}$$

$$\text{Ton (long)} = 2240 \text{ lb}$$

$$\text{Ton (metric)} = 2205 \text{ lb}$$

$$\text{Ton (metric)} = 1000 \text{ kg}$$

Constants:

$$\text{Acceleration of gravity} = 32.2 \text{ ft/sec}^2$$

$$\text{Acceleration of gravity} = 4.18 \times 10^8 \text{ ft/hr}^2$$

$$\text{Density of a cubic foot of water} = 62.5 \text{ lb/ft}^3$$

TABLE 2. THERMAL CONDUCTIVITIES OF SOME BUILDING AND INSULATING MATERIALS*

$$k = \text{Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$$

Material	Apparent density ρ , lb/ft ³ at room temperature	$^\circ\text{F}$	k
Aerogel, silica, opacified.....	8.5	248	0.013
		554	0.026
Asbestos-cement boards.....	120	68	0.43
Asbestos sheets.....	55.5	124	0.096
Asbestos slate.....	112	32	0.087
	112	140	0.114
Asbestos.....	29.3	-328	0.043
	29.3	32	0.090
	36	32	0.087
	36	212	0.111
	36	392	0.120
	36	752	0.129
	43.5	-328	0.090
	43.5	32	0.135
Aluminum foil, 7 air spaces per 2.5 in.....	0.2	100	0.025
		351	0.038
Ashes, wood.....		32-212	0.041
Asphalt.....	132	68	0.43
Boiler scale (ref. 364).....			
Bricks			
Alumina (92-99% Al ₂ O ₃ by weight) fused...		801	1.8
Alumina (64-65% Al ₂ O ₃ by weight).....		2399	2.7
(See also Bricks, fire clay).....	115	1472	0.62
	115	2012	0.63
Building brickwork.....		68	0.4
Chrome brick (32% Cr ₂ O ₃ by weight).....	200	392	0.67
	200	1202	0.85
	200	2399	1.0
Diatomaceous earth, natural, across strata	27.7	399	0.051
	27.7	1600	0.077
Diatomaceous, natural, parallel to strata	27.7	399	0.081
	27.7	1600	0.106
Diatomaceous earth, molded and fired.....	38	399	0.14
	38	1600	0.18
Diatomaceous earth and clay, molded and fired.....	42.3	399	0.14
	42.3	1600	0.19
Diatomaceous earth, high burn, large pores.....	37	392	0.13
	37	1832	0.34

TABLE 2. THERMAL CONDUCTIVITIES OF SOME BUILDING AND INSULATING MATERIALS.*—(Continued)

Material	Apparent density ρ , lb/ft ³ at room temperature	°F	k
Bricks: (Continued)			
Fire clay, Missouri.....		392	0.58
		1112	0.85
		1832	0.95
		2552	1.02
Kaolin insulating brick.....	27	932	0.15
	27	2102	0.26
Kaolin insulating firebrick.....	19	392	0.050
	19	1400	0.113
Magnesite (86.8% MgO, 6.3% Fe ₂ O ₃ , 3% CaO, 2.6% SiO ₂ by weight).....	158	399	2.2
	158	1202	1.6
	158	2192	1.1
Silicon carbide brick, recrystallized.....	129	1112	10.7
	129	1472	9.2
	129	1832	8.0
	129	2192	7.0
	129	2552	6.3
Calcium carbonate, natural.....	162	86	1.3
White marble.....			1.7
Chalk.....	96		0.4
Calcium sulphate (4H ₂ O), artificial.....	84.6	104	0.22
Plaster, artificial.....	132	167	0.43
Building.....	77.9	77	0.25
Cambric, varnished.....		100	0.09
Carbon, gas.....		32-212	2.0
Cardboard, corrugated.....			0.037
Celluloid.....	87.3	86	0.12
Charcoal flakes.....	11.9	176	0.043
	15	176	0.051
Clinker, granular.....		32-1292	0.27
Coke, petroleum.....		212	3.4
		932	2.9
Coke, powdered.....		32-212	0.11
Concrete, cinder.....			0.20
1:4 dry.....			0.44
Stone.....			0.54
Cotton wool.....	5	86	0.024
Cork board.....	10	86	0.025
Cork, ground.....	9.4	86	0.025
Reggranulated.....	8.1	86	0.026

TABLE 2. THERMAL CONDUCTIVITIES OF SOME BUILDING AND INSULATING MATERIALS.*—(Continued)

Material	Apparent density ρ , lb/ft ³ at room temperature	°F	k
Diatomaceous earth powder, coarse	20.0	100	0.036
	20.0	1600	0.082
Fine	17.2	399	0.040
	17.2	1600	0.074
Molded pipe covering	26.0	399	0.051
	26.0	1600	0.088
4 vol. calcined earth and 1 vol. cement, poured and fired	61.8	399	0.16
	61.8	1600	0.23
Dolomite	167	122	1.0
Ebonite			0.10
Enamel, silicate	38		0.5-0.75
Felt, wool	20.6	86	0.03
Fiber insulating board	14.8	70	0.028
Fiber, red	80.5	68	0.27
With binder, baked		68-207	0.097
Gas carbon		32-212	2.0
Glass			0.2-0.73
Boro-silicate type	139	86-167	0.63
Soda glass			0.3-0.44
Window glass			0.3-0.61
Granite			1.0-2.3
Graphite, dense, commercial		32	86.7
Powdered, through 100 mesh	30	104	0.104
Gypsum, molded and dry	78	68	0.25
Hair, felt, perpendicular to fibers	17	86	0.021
Ice	57.5	32	1.3
Infusorial earth (see Diatomaceous earth)			
Kapok	0.88	68	0.020
Lampblack	10	104	0.038
Lava			0.49
Leather, sole	62.4		0.092
Limestone (15.3 vol % H ₂ O)	103	75	0.54
Linen		86	0.05
Magnesia, powdered	49.7	117	0.35
Magnesia, light carbonate	19	70	0.04
Magnesium oxide, compressed	49.9	68	0.32
Marble			1.2-1.7

TABLE 2. THERMAL CONDUCTIVITIES OF SOME BUILDING AND INSULATING MATERIALS.*—(Continued)

Material	Apparent density ρ , lb/ft ³ at room temperature	°F	k
Mica, perpendicular to planes.....		122	0.25
Mill shavings.....			0.033–0.05
Mineral wool.....	9.4	86	0.0225
	19.7	86	0.024
Paper.....			0.075
Paraffin wax.....		32	0.14
Petroleum coke.....		212	3.4
		932	2.9
Porcelain.....		392	0.88
Portland cement (see Concrete).....		194	0.17
Pumice stone.....		70–151	0.14
Pyroxylin plastics.....			0.075
Rubber, hard.....	74.8	32	0.087
Para.....		70	0.109
Soft.....		70	0.075–0.092
Sand, dry.....	94.6	68	0.19
Sandstone.....	140	104	1.06
Sawdust.....	12	70	0.03
Scale (ref. 364).....			
Silk.....	6.3		0.026
Varnished.....		100	0.096
Slag, blast furnace.....		75–261	0.064
Slag wool.....	12	86	0.022
Slate.....		201	0.86
Snow.....	34.7	32	0.27
Sulphur, monoclinic.....		212	0.09–0.097
Rhombic.....		70	0.16
Wallboard, insulating type.....	14.8	70	0.028
Wallboard, stiff pasteboard.....	43	86	0.04
Wood shavings.....	8.8	86	0.034
Wood, across grain			
Balsa.....	7–8	86	0.025–0.03
Oak.....	51.5	59	0.12
Maple.....	44.7	122	0.11
Pine, white.....	34.0	59	0.087
Teak.....	40.0	59	0.10
White fir.....	28.1	140	0.062
Wood, parallel to grain			
Pine.....	34.4	70	0.20
Wool, animal.....	6.9	86	0.021

* From L. S. Marks, "Mechanical Engineers' Handbook," McGraw-Hill Book Company, Inc., New York, 1941.

TABLE 3. THERMAL CONDUCTIVITIES, SPECIFIC HEATS, SPECIFIC GRAVITIES OF METALS AND ALLOYS
 $k = \text{Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$

Substance	Temp, $^\circ\text{F}$	k^*	Specific heat, $\dagger$ Btu/(lb)($^\circ\text{F}$)	Specific gravity
Aluminum	32	117	0.183	2.55-7.8
Aluminum	212	119	0.1824	
Aluminum	932	155	0.1872	
Antimony	32	10.6	0.0493	9.8
Antimony	212	9.7	0.0508	
Bismuth	64	4.7	0.0294	
Bismuth	212	3.9	0.0304	8.4-8.7
Brass (70-30)	32	56	0.1315 $\dagger$	
Brass	212	60	0.1488 $\dagger$	
Brass	752	67	0.2015 $\dagger$	8.8-8.95
Copper	32	224	0.1487	
Copper	212	218	0.1712	
Copper	932	207	0.2634	8.65
Cadmium	64	53.7	0.0550	
Cadmium	212	52.2	0.0567	
Gold	64	169.0	0.030	19.25-19.35
Gold	212	170.8	0.031	
Iron, cast	32	32	0.1064	
Iron, cast	212	30	0.1178	7.03-7.13
Iron, cast	752	25	0.1519	
Iron, wrought	64	34.6	See Iron	
Iron, wrought	212	27.6	See Iron	7.6-7.9
Lead	32	20	0.0306	
Lead	212	19	0.0315	
Lead	572	18	0.0335	11.34
Magnesium	32-212	92	0.255	
Mercury	32	4.8	0.0329	
Nickel	32	36	0.1050	1.74
Nickel	212	34	0.1170	
Nickel	572	32	0.1408	
Silver	32	242	0.0557	13.6
Silver	212	238	0.0571	
Steel	32	26	See Iron	
Steel	212	26	See Iron	8.9
Steel	1112	21	See Iron	
Tantalum	64	32	0.0342	
Zinc	32	65	0.0917	10.4-10.6
Zinc	212	64	0.0958	
Zinc	752	54	0.1082	

* From L. S. Marks, "Mechanical Engineers' Handbook," McGraw-Hill Book Company, Inc., New York, 1941.

$\dagger$ From K. K. Kelley, *U.S. Bur. Mine Bull.* 371 (1939).

$\ddagger$ Weighted value for copper and zinc.

TABLE 4. THERMAL CONDUCTIVITIES OF LIQUIDS*

$$k = \text{Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$$

A linear variation with temperature may be assumed. The extreme values given constitute also the temperature limits over which the data are recommended.

Liquid	$^\circ\text{F}$	k	Liquid	$^\circ\text{F}$	k
Acetic acid 100 %	68	0.099	Heptyl alcohol (n-)	86	0.094
50 %	68	0.20	Hexyl alcohol (n-)	167	0.091
Acetone	86	0.102		86	0.093
	167	0.095		167	0.090
Allyl alcohol	77-86	0.104	Kerosene	68	0.086
Ammonia	5-86	0.29		167	0.081
Ammonia, aqueous 26 %	68	0.261	Lauric acid	212	0.102
	140	0.29			
Amyl acetate	50	0.083	Mercury	82	4.83
Alcohol (n-)	86	0.094	Methyl alcohol 100 %	68	0.124
	212	0.089	80 %	68	0.154
	86	0.088	60 %	68	0.190
Aniline	167	0.087	40 %	68	0.234
	32-68	0.100	20 %	68	0.284
Benzene	86	0.092	100 %	122	0.114
	140	0.087	Chloride	5	0.111
Bromobenzene	86	0.074		86	0.089
	212	0.070	Nitrobenzene	86	0.095
Butyl acetate (n-)	77-86	0.085		212	0.088
Alcohol (n-)	86	0.097	Nitromethane	86	0.125
(iso-)	167	0.095		140	0.120
	50	0.091	Nonane (n-)	86	0.084
Calcium chloride brine 30 %	86	0.32		140	0.082
15 %	86	0.34	Octane (n-)	86	0.083
Carbon disulphide	86	0.093		140	0.081
	167	0.088	Oils		
Tetrachloride	32	0.107	Castor	68	0.104
	154	0.094		212	0.100
Chlorobenzene	50	0.083	Olive	68	0.097
Chloroform	86	0.080		212	0.095
Cymene (para)	86	0.078	Oleic acid	212	0.0925
	140	0.079	Palmitic acid	212	0.0835
Decane (n-)	86	0.085	Paraldehyde	86	0.084
	140	0.083		212	0.078
Dichlorodifluoromethane	20	0.057	Pentane (n-)	86	0.078
	60	0.053		167	0.074
	100	0.048	Perchloroethylene	122	0.092
	140	0.043	Petroleum ether	86	0.075
Dichloroethane	180	0.038		167	0.073
Dichloromethane	122	0.082	Propyl alcohol (n-)	86	0.099
	5	0.111		167	0.095
	86	0.096	Alcohol (iso-)	86	0.091
Ethyl acetate	68	0.101		140	0.090
Alcohol 100 %	68	0.105	Sodium	212	49
80 %	68	0.137		410	46
60 %	68	0.176	Sodium chloride brine 25.0 %	86	0.33
40 %	68	0.224	12.5 %	86	0.34
20 %	68	0.281	Stearic acid	212	0.0786
100 %	122	0.087	Sulfuric acid 90 %	86	0.21
Benzene	86	0.086	60 %	86	0.25
	140	0.082	30 %	86	0.30
Bromide	68	0.070	Sulfur dioxide	5	0.128
Ether	86	0.080		86	0.111
	167	0.078	Toluene	86	0.086
Iodide	104	0.064		167	0.084
	167	0.063	β -trichloroethane	122	0.077
Ethylene glycol	32	0.153	Trichloroethylene	122	0.080
Gasoline	86	0.078	Turpentine	59	0.074
Glycerol 100 %	68	0.164	Vaseline	59	0.106
80 %	68	0.189	Water	32	0.330
60 %	68	0.220		86	0.356
40 %	68	0.259		140	0.381
20 %	68	0.278		176	0.398
100 %	212	0.164	Xylene (ortho-)	68	0.090
Heptane (n-)	86	0.081	(meta-)	68	0.090
	140	0.079			
Hexane (n-)	86	0.080			
	140	0.078			

* From Perry, J. H., "Chemical Engineers' Handbook," 3d ed., McGraw-Hill Book Company, Inc., New York, 1950.

TABLE 5. THERMAL CONDUCTIVITIES OF GASES AND VAPORS*

$$k = \text{Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$$

The extreme temperature values given constitute the experimental range. For extrapolation to other temperatures, it is suggested that the data given be plotted as $\log k$ vs. $\log T$ or that use be made of the assumption that the ratio c_p/k is practically independent of temperature (or of pressure, within moderate limits).

Substance	$^\circ\text{F}$	k	Substance	$^\circ\text{F}$	k
Acetone.....	32	0.0057	Dichlorodifluoromethane.....	32	0.0048
	115	0.0074		122	0.0064
	212	0.0099		212	0.0080
	363	0.0147		302	0.0097
Acetylene.....	-103	0.0068	Ethane.....	-94	0.0066
	32	0.0108		-29	0.0086
	122	0.0140		32	0.0106
	212	0.0172		212	0.0175
Air.....	-148	0.0095	Ethyl acetate.....	115	0.0072
	32	0.0140		212	0.0096
	212	0.0183		363	0.0141
	392	0.0226	Alcohol.....	68	0.0089
	572	0.0265		212	0.0124
Ammonia.....	-76	0.0095	Chloride.....	32	0.0055
	32	0.0128		212	0.0095
	122	0.0157		363	0.0135
	212	0.0185		413	0.0152
			Ether.....	32	0.0077
Benzene.....	32	0.0052		115	0.0099
	115	0.0073		212	0.0131
	212	0.0103		363	0.0189
	363	0.0152		413	0.0209
	413	0.0176	Ethylene.....	-96	0.0064
Butane (n-).....	32	0.0078		32	0.0101
	212	0.0135		122	0.0131
(iso-).....	32	0.0080		212	0.0161
	212	0.0139			
Carbon dioxide.....	-58	0.0068	Heptane (n-).....	392	0.0112
	32	0.0085		212	0.0103
	212	0.0133	Hexane (n-).....	32	0.0072
	392	0.0181		68	0.0080
	572	0.0228	Hexene.....	32	0.0061
Disulphide.....	32	0.0040		212	0.0109
	45	0.0042	Hydrogen.....	-148	0.065
Monoxide.....	-312	0.0041		-58	0.083
	-294	0.0046		32	0.100
	32	0.0135		122	0.115
Tetrachloride.....	115	0.0041		212	0.129
	212	0.0052		572	0.178
	363	0.0065	Hydrogen and carbon dioxide...	32	
Chlorine.....	32	0.0043	0% H ₂		0.0083
Chloroform.....	32	0.0038	20%.....		0.0165
	115	0.0046	40%.....		0.0270
	212	0.0058	60%.....		0.0410
	363	0.0077	80%.....		0.0620
Cyclohexane.....	216	0.0095	100%.....		0.10

TABLE 5. THERMAL CONDUCTIVITIES OF GASES AND VAPORS.*—(Continued)

Substance	°F	<i>k</i>	Substance	°F	<i>k</i>
Hydrogen and nitrogen.....	32		Nitric oxide.....	-94	0.0103
0 % H ₂		0.0133		32	0.0138
20 %.....		0.0212	Nitrogen.....	-148	0.0095
40 %.....		0.0313		32	0.0140
60 %.....		0.0438		122	0.0160
80 %.....		0.0635		212	0.0180
Hydrogen and nitrous oxide.....	32		Nitrous oxide.....	-98	0.0067
0 % H ₂		0.0002		32	0.0087
20 %.....		0.0170		212	0.0128
40 %.....		0.0270	Oxygen.....	-148	0.0095
60 %.....		0.0410		-58	0.0119
80 %.....		0.0650		32	0.0142
Hydrogen sulphide.....	32	0.0076		122	0.0164
				212	0.0185
Mercury.....	392	0.0197	Pentane (<i>n</i> -).....	32	0.0074
Methane.....	-148	0.0100		68	0.0083
	-58	0.0145	(<i>iso</i> -).....	32	0.0072
	32	0.0175		212	0.0127
Methyl alcohol.....	122	0.0215	Propane.....	32	0.0087
	32	0.0083		212	0.0151
Acetate.....	212	0.0128	Sulphur dioxide.....	32	0.0050
	32	0.0059		212	0.0069
	68	0.0068	Water vapor.....	115	0.0120
Methyl chloride.....	32	0.0053		212	0.0137
	115	0.0072		392	0.0187
	212	0.0094		572	0.0248
	363	0.0130		752	0.0315
	413	0.0148		932	0.0441
Methylene chloride.....	32	0.0039			
	115	0.0049			
	212	0.0063			
	413	0.0095			

* From Perry, J. H., "Chemical Engineers' Handbook," 3d ed., McGraw-Hill Book Company, Inc., New York, 1950.

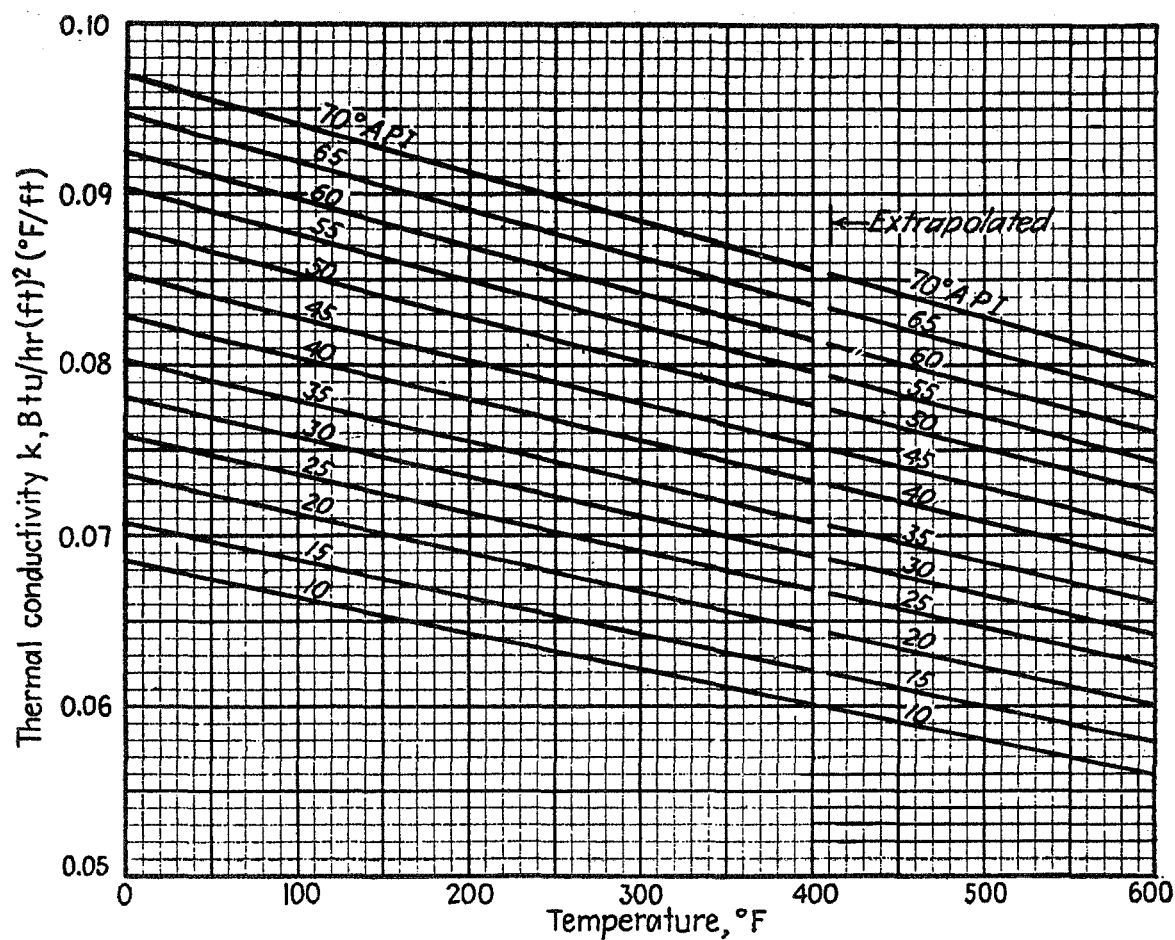


FIG. 1. Thermal conductivities of hydrocarbon liquids. (Adapted from Natl. Bur. Standards Misc. Pub. 97.)

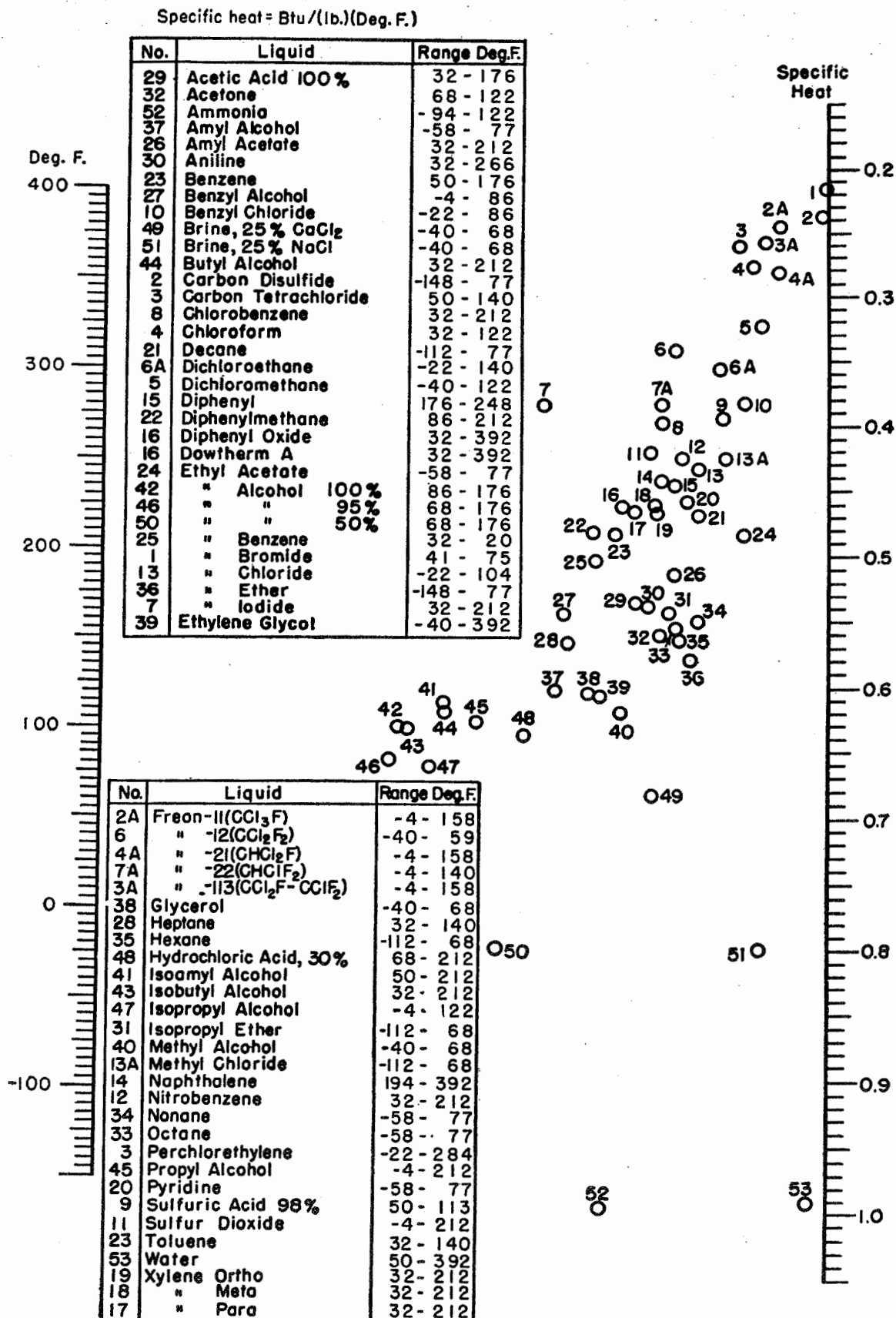


FIG. 2. Specific heats of liquids. (Chilton, Colburn, and Vernon, based mainly on data from International Critical Tables. Perry, "Chemical Engineers' Handbook," 3d ed., McGraw-Hill Book Company, Inc., New York, 1950.)

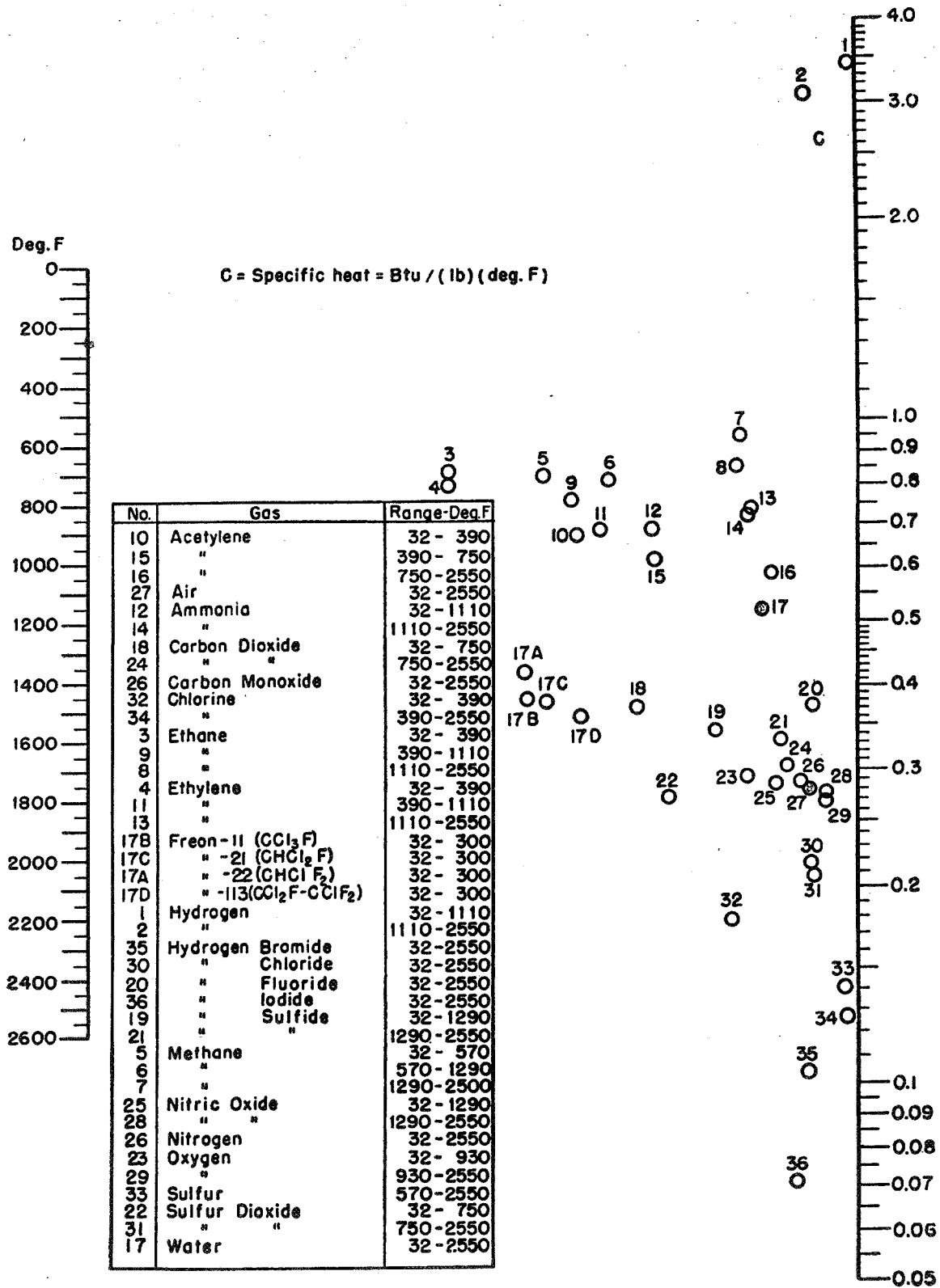


FIG. 3. Specific heats of gases at 1 atm. (Perry, "Chemical Engineers' Handbook," 3d ed., McGraw-Hill Book Company, Inc., New York, 1950.)

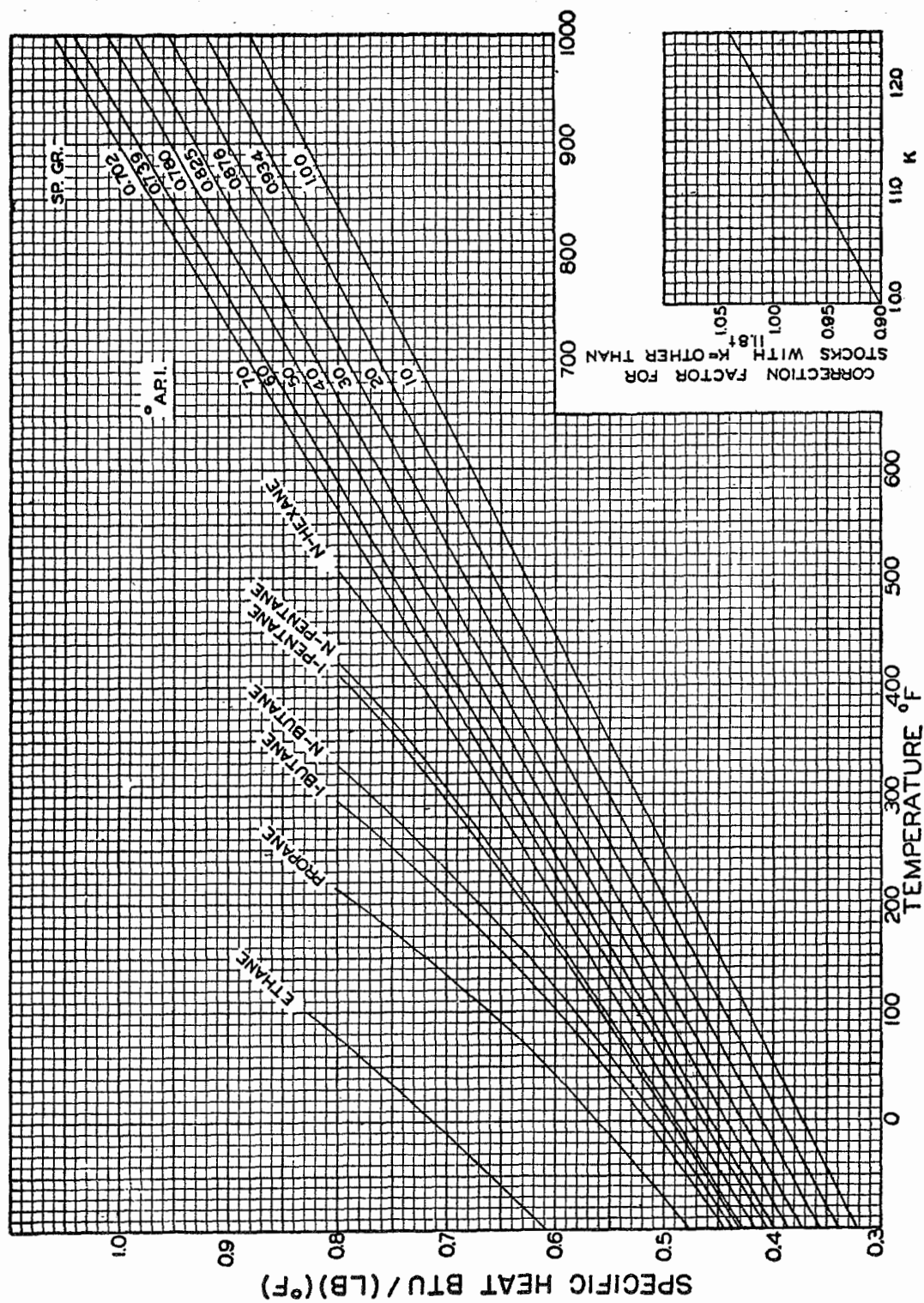


FIG. 4. Specific heats of hydrocarbon liquids. [Holcomb and Brown, *Ind. Eng. Chem.*, **34**, 595 (1942).]

$^\dagger K$ = characterization factor.

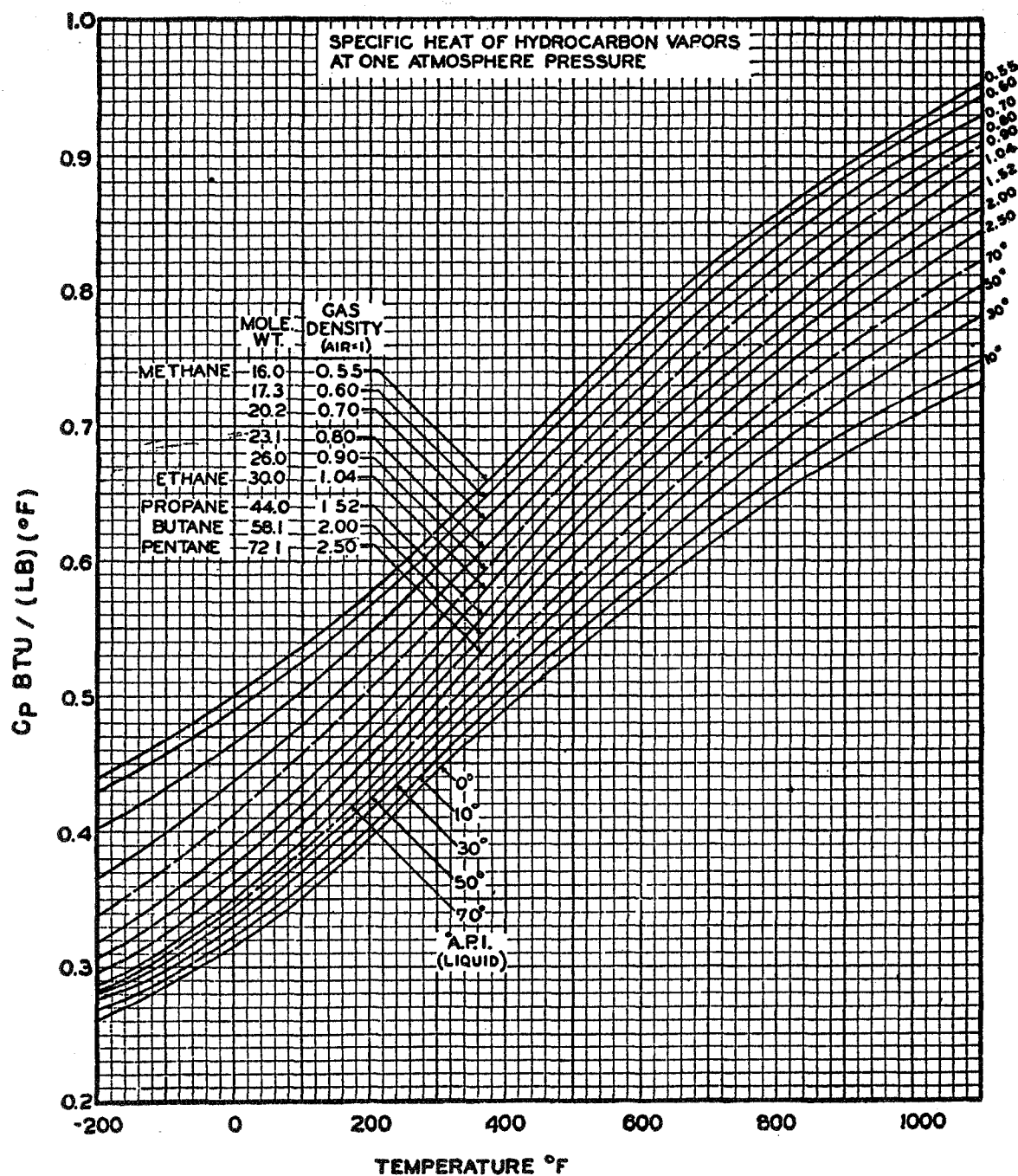


FIG. 5. Specific heats of hydrocarbon vapors at 1 atm. [Holcomb and Brown, *Ind. Eng. Chem.*, 34, 595 (1942).]

TABLE 6. SPECIFIC GRAVITIES AND MOLECULAR WEIGHTS OF LIQUIDS

Compound	Mol. wt.	s*	Compound	Mol. wt.	s*
Acetaldehyde.....	44.1	0.78	Ethyl iodide.....	155.9	1.93
Acetic acid, 100 %.....	60.1	1.05	Ethyl glycol.....	88.1	1.04
Acetic acid, 70 %.....		1.07	Formic acid.....	46.0	1.22
Acetic anhydride.....	102.1	1.08	Glycerol, 100 %.....	92.1	1.26
Acetone.....	58.1	0.79	Glycerol, 50 %.....		1.13
Allyl alcohol.....	58.1	0.86	n-Heptane.....	100.2	0.68
Ammonia, 100 %.....	17.0	0.61	n-Hexane.....	86.1	0.66
Ammonia, 26 %.....		0.91	Isopropyl alcohol.....	60.1	0.79
Amyl acetate.....	130.2	0.88	Mercury.....	200.6	13.55
Amyl alcohol.....	88.2	0.81	Methanol, 100 %.....	32.5	0.79
Aniline.....	93.1	1.02	Methanol, 90 %.....		0.82
Anisole.....	108.1	0.99	Methanol, 40 %.....		0.94
Arsenic trichloride.....	181.3	2.16	Methyl acetate.....	74.9	0.93
Benzene.....	78.1	0.88	Methyl chloride.....	50.5	0.92
Brine, CaCl ₂ 25 %.....		1.23	Methyl ethyl ketone.....	72.1	0.81
Brine, NaCl 25 %.....		1.19	Naphthalene.....	128.1	1.14
Bromotoluene, ortho.....	171.0	1.42	Nitric acid, 95 %.....		1.50
Bromotoluene, meta.....	171.0	1.41	Nitric acid, 60 %.....		1.38
Bromotoluene, para.....	171.0	1.39	Nitrobenzene.....	123.1	1.20
n-Butane.....	58.1	0.60	Nitrotoluene, ortho.....	137.1	1.16
i-Butane.....	58.1	0.60	Nitrotoluene, meta.....	137.1	1.16
Butyl acetate.....	116.2	0.88	Nitrotoluene, para.....	137.1	1.29
n-Butyl alcohol.....	74.1	0.81	n-Octane.....	114.2	0.70
i-Butyl alcohol.....	74.1	0.82	Octyl alcohol.....	130.23	0.82
n-Butyric acid.....	88.1	0.96	Pentachloroethane.....	202.3	1.67
i-Butyric acid.....	88.1	0.96	n-Pentane.....	72.1	0.63
Carbon dioxide.....	44.0	1.29	Phenol.....	94.1	1.07
Carbon disulfide.....	76.1	1.26	Phosphorus tribromide.....	270.8	2.85
Carbon tetrachloride.....	153.8	1.60	Phosphorus trichloride.....	137.4	1.57
Chlorobenzene.....	112.6	1.11	Propane.....	44.1	0.59
Chloroform.....	119.4	1.49	Propionic acid.....	74.1	0.99
Chlorosulfonic acid.....	116.5	1.77	n-Propyl alcohol.....	60.1	0.80
Chlorotoluene, ortho.....	126.6	1.08	n-Propyl bromide.....	123.0	1.35
Chlorotoluene, meta.....	126.6	1.07	n-Propyl chloride.....	78.5	0.89
Chlorotoluene, para.....	126.6	1.07	n-Propyl iodide.....	170.0	1.75
Cresol, meta.....	108.1	1.03	Sodium.....	23.0	0.97
Cyclohexanol.....	100.2	0.96	Sodium hydroxide, 50 %.....		1.53
Dibromo methane.....	187.9	2.09	Stannic chloride.....	260.5	2.23
Dichloro ethane.....	99.0	1.17	Sulfur dioxide.....	64.1	1.38
Dichloro methane.....	88.9	1.34	Sulfuric acid, 100 %.....	98.1	1.83
Diethyl oxalate.....	146.1	1.08	Sulfuric acid, 98 %.....		1.84
Dimethyl oxalate.....	118.1	1.42	Sulfuric acid, 60 %.....		1.50
Diphenyl.....	154.2	0.99	Sulfuryl chloride.....	135.0	1.67
Dipropyl oxalate.....	174.1	1.02	Tetra chloroethane.....	167.9	1.60
Ethyl acetate.....	88.1	0.90	Tetra chloroethylene.....	165.9	1.63
Ethyl alcohol, 100 %.....	46.1	0.79	Titanium tetrachloride.....	189.7	1.73
Ethyl alcohol, 95 %.....		0.81	Toluene.....	92.1	0.87
Ethyl alcohol, 40 %.....		0.94	Trichloroethylene.....	131.4	1.46
Ethyl benzene.....	106.1	0.87	Vinyl acetate.....	86.1	0.93
Ethyl bromide.....	108.9	1.43	Water.....	18.0	1.0
Ethyl chloride.....	64.5	0.92	Xylene, ortho.....	106.1	0.87
Ethyl ether.....	74.1	0.71	Xylene, meta.....	106.1	0.86
Ethyl formate.....	74.1	0.92	Xylene, para.....	106.1	0.86

* At approximately 68°F. These values will be satisfactory, without extrapolation, for most engineering problems.

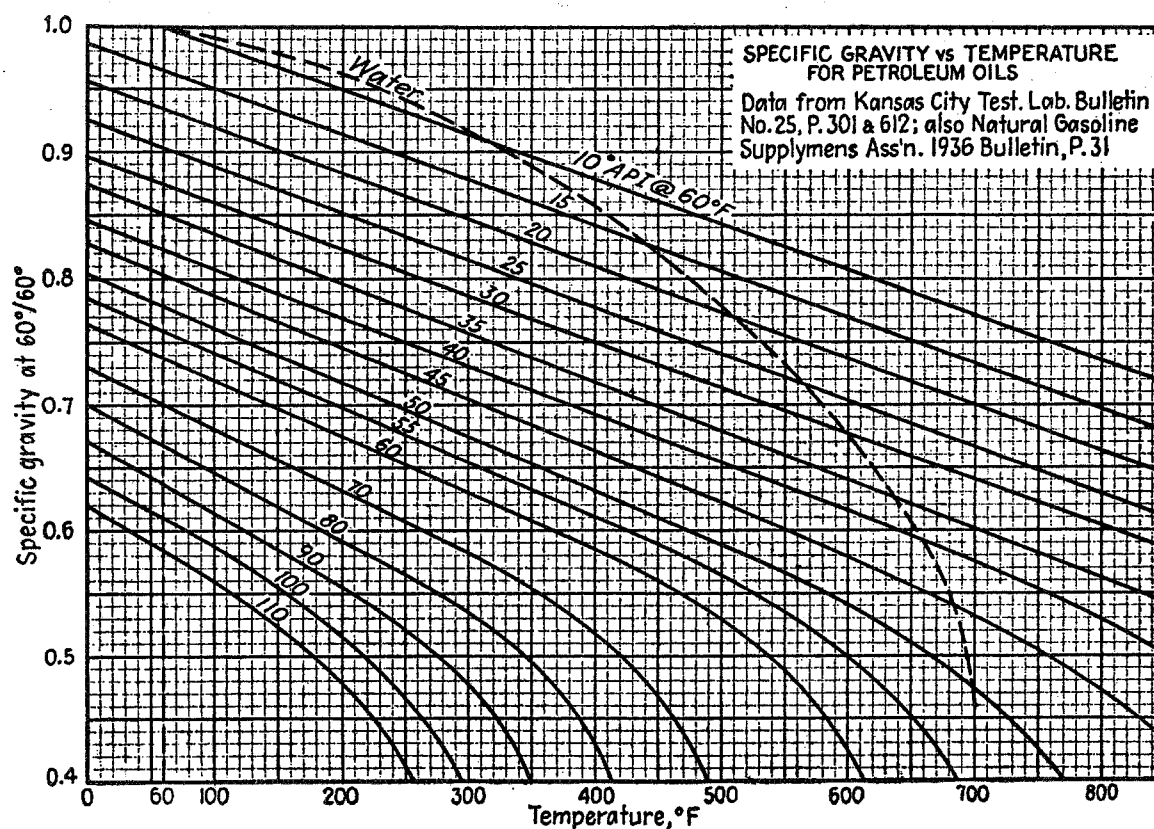


FIG. 6. Specific gravities of hydrocarbons.

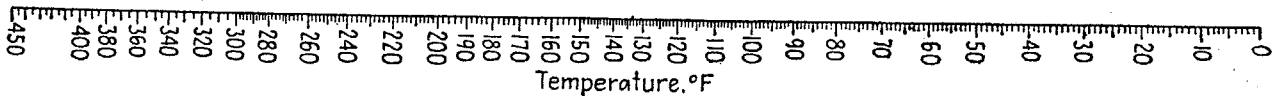


FIG. 7. Equilibrium constants for hydrocarbons. [Scheibel and Jenny, *Ind. Eng. Chem.*, **37**, 81 (1945).]

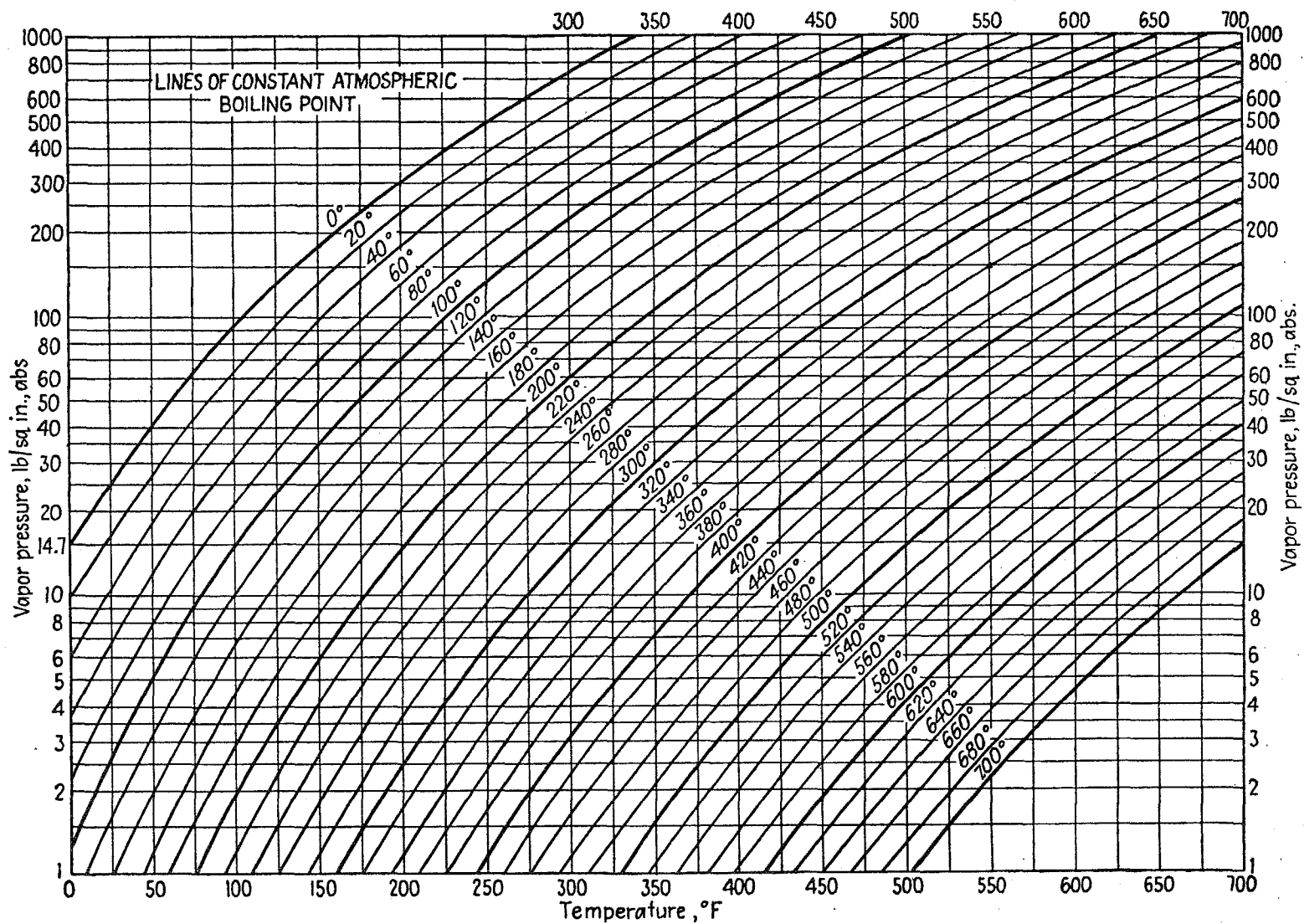


FIG. 8. Vapor pressures of hydrocarbons.

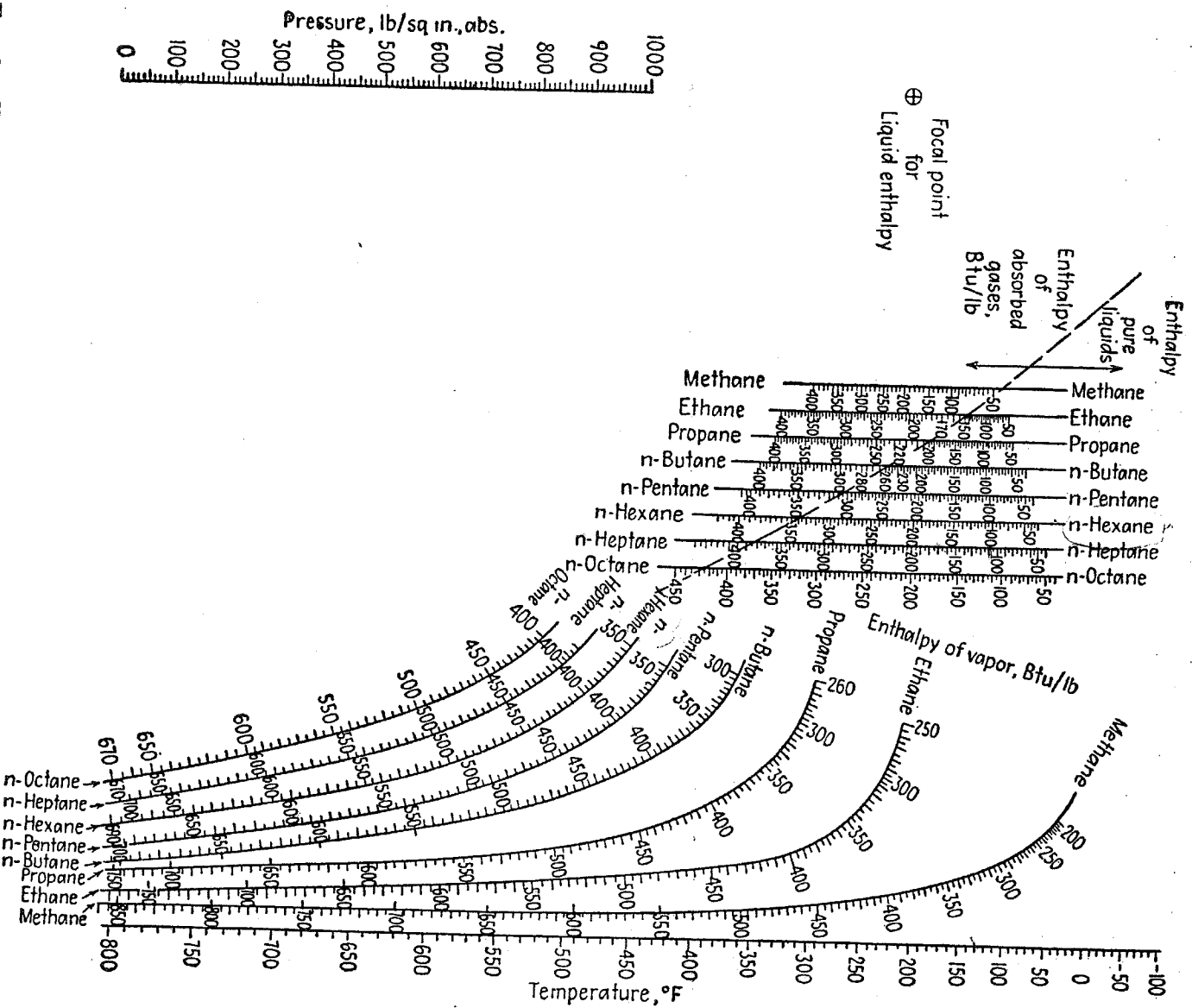


Fig. 9.

Enthalpies of pure hydrocarbons.

[Scheibel and Jenny, Ind. Eng. Chem., 37, 992 (1945).]

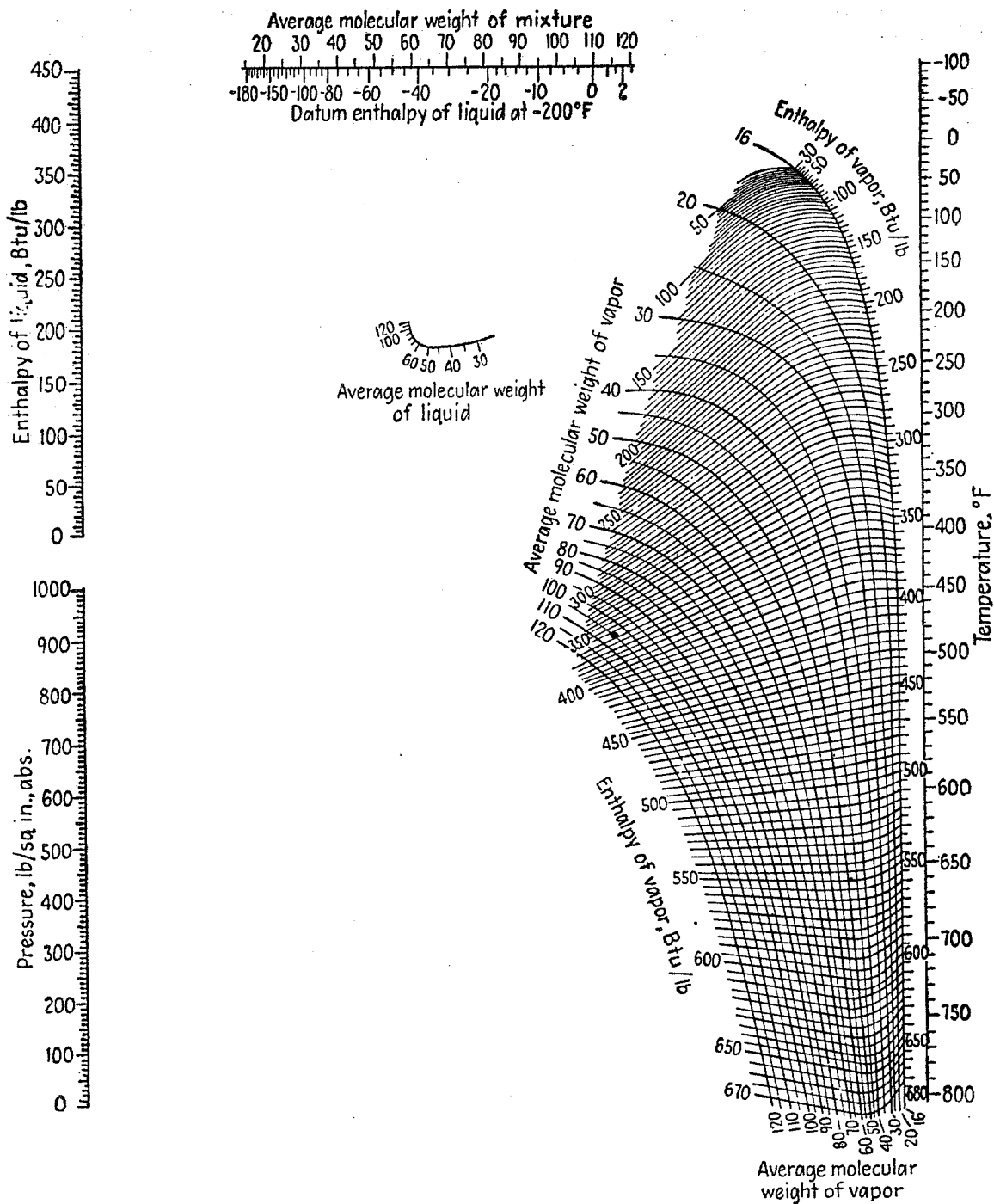


FIG. 10. Enthalpies of light hydrocarbons. [Scheibel and Jenny, *Ind. Eng. Chem.*, **37**, 993 (1945).]

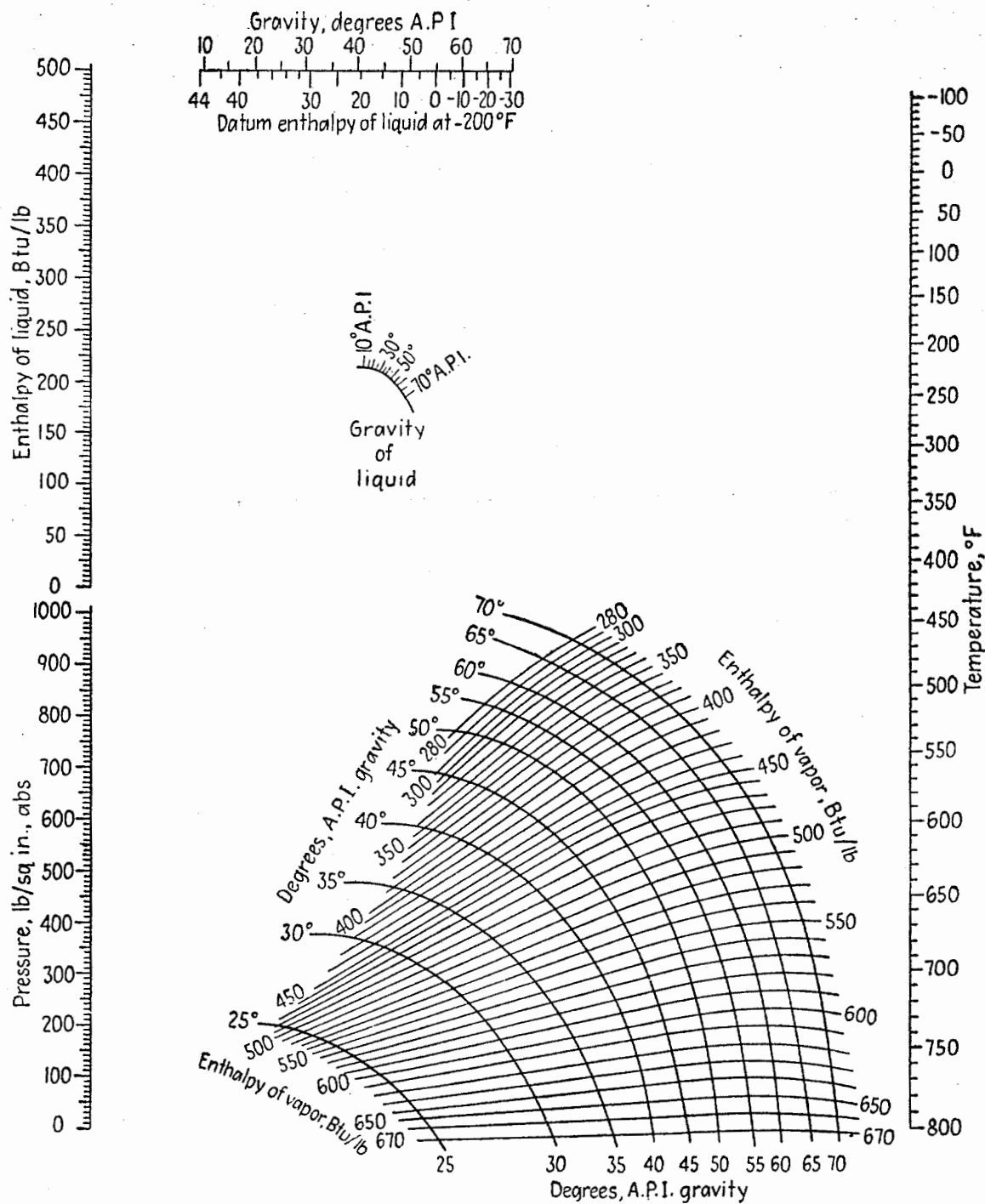


FIG. 11. Enthalpies of petroleum fractions. [Scheibel and Jenny, *Ind. Eng. Chem.*, **37**, 994 (1945).]

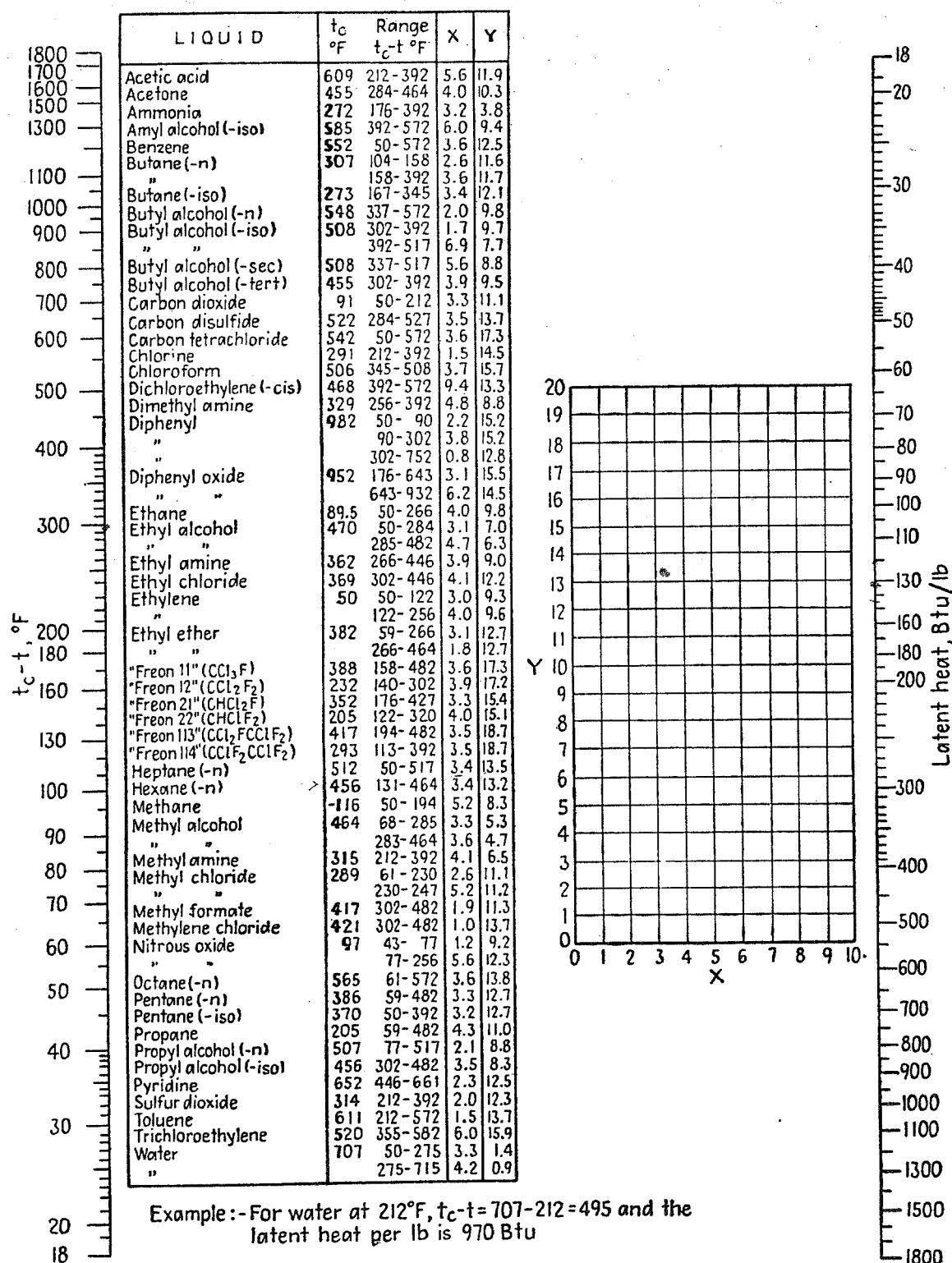


Fig. 12. Latent heats of vaporization. [Reproduced by permission of Chilton, Colburn, and Vernon, personal communication (revised) 1947.]

TABLE 7. THERMODYNAMIC PROPERTIES OF STEAM
Dry saturated steam: pressure table*

Abs press, psi	Temp, °F	Specific volume		Enthalpy			Entropy			Internal energy		Abs press., psi
		Sat. liquid v_f	Sat. vapor v_g	Sat. liquid h_f	Evap h_{fg}	Sat. vapor h_g	Sat. liquid s_f	Evap s_{fg}	Sat. vapor s_g	Sat. liquid u_f	Sat. vapor u_g	
p	t											p
1.0	101.74	0.01614	333.6	69.70	1036.3	1106.0	0.1326	1.8456	1.9782	69.70	1044.3	1.0
2.0	126.08	0.01623	173.73	93.99	1022.2	1116.2	0.1749	1.7451	1.9200	93.98	1051.9	2.0
3.0	141.48	0.01630	118.71	109.37	1013.2	1122.6	0.2008	1.6855	1.8863	109.36	1056.7	3.0
4.0	152.97	0.01636	90.63	120.86	1006.4	1127.3	0.2198	1.6427	1.8625	120.85	1060.2	4.0
5.0	162.24	0.01640	73.52	130.13	1001.0	1131.1	0.2347	1.6094	1.8441	130.12	1063.1	5.0
6.0	170.06	0.01645	61.98	137.96	996.2	1134.2	0.2472	1.5820	1.8292	137.94	1065.4	6.0
7.0	176.85	0.01649	53.64	144.76	992.1	1136.9	0.2581	1.5586	1.8167	144.74	1067.4	7.0
8.0	182.86	0.01653	47.34	150.79	988.5	1139.3	0.2674	1.5383	1.8057	150.77	1069.2	8.0
9.0	188.28	0.01656	42.40	156.22	985.2	1141.4	0.2759	1.5203	1.7962	156.19	1070.8	9.0
10	193.21	0.01659	38.42	161.17	982.1	1143.3	0.2835	1.5041	1.7876	161.14	1072.2	10
14.696	212.00	0.01672	26.80	180.07	970.3	1150.4	0.3120	1.4446	1.7566	180.02	1077.5	14.696
15	213.03	0.01672	26.29	181.11	969.7	1150.8	0.3135	1.4415	1.7549	181.06	1077.8	15
20	227.96	0.01683	20.089	196.16	960.1	1156.3	0.3356	1.3962	1.7319	196.10	1081.9	20
25	240.07	0.01692	16.303	208.42	952.1	1160.6	0.3533	1.3606	1.7139	208.34	1085.1	25
30	250.33	0.01701	13.746	218.82	945.3	1164.1	0.3680	1.3313	1.6993	218.73	1087.8	30
35	259.28	0.01708	11.898	227.91	939.2	1167.1	0.3807	1.3063	1.6870	227.80	1090.1	35
40	267.25	0.01715	10.498	236.03	933.7	1169.7	0.3919	1.2844	1.6763	235.90	1092.0	40
45	274.44	0.01721	9.401	243.36	928.6	1172.0	0.4019	1.2650	1.6669	243.22	1093.7	45
50	281.01	0.01727	8.515	250.09	924.0	1174.1	0.4110	1.2474	1.6585	249.93	1095.3	50
55	287.07	0.01732	7.787	256.30	919.6	1175.9	0.4193	1.2316	1.6509	256.12	1096.7	55
60	292.71	0.01738	7.175	262.09	915.5	1177.6	0.4270	1.2168	1.6438	261.90	1097.9	60
65	297.97	0.01743	6.655	267.50	911.6	1179.1	0.4342	1.2032	1.6374	267.29	1099.1	65
70	302.92	0.01748	6.206	272.61	907.9	1180.6	0.4409	1.1906	1.6315	272.38	1100.2	70
75	307.60	0.01753	5.816	277.43	904.5	1181.9	0.4472	1.1787	1.6259	277.19	1101.2	75
80	312.03	0.01757	5.472	282.02	901.1	1183.1	0.4531	1.1676	1.6207	281.76	1102.1	80
85	316.25	0.01761	5.168	286.39	897.8	1184.2	0.4587	1.1571	1.6158	286.11	1102.9	85
90	320.27	0.01766	4.896	290.56	894.7	1185.3	0.4641	1.1471	1.6112	290.27	1103.7	90
95	324.12	0.01770	4.652	294.56	891.7	1186.2	0.4692	1.1376	1.6068	294.25	1104.5	95
100	327.81	0.01774	4.432	298.40	888.8	1187.2	0.4740	1.1286	1.6026	298.08	1105.2	100
110	334.77	0.01782	4.049	305.66	883.2	1188.9	0.4832	1.1117	1.5948	305.30	1106.5	110
120	341.25	0.01789	3.728	312.44	877.9	1190.4	0.4916	1.0962	1.5878	312.05	1107.6	120
130	347.32	0.01796	3.455	318.81	879.9	1191.7	0.4995	1.0817	1.5812	318.38	1108.6	130
140	353.02	0.01802	3.220	324.82	868.2	1193.0	0.5069	1.0682	1.5751	324.35	1109.6	140
150	358.42	0.01809	3.015	330.51	863.6	1194.1	0.5138	1.0556	1.5694	330.01	1110.5	150
160	363.53	0.01815	2.834	335.93	859.2	1195.1	0.5204	1.0436	1.5640	335.39	1111.2	160
170	368.41	0.01822	2.675	341.09	854.9	1196.0	0.5266	1.0324	1.5590	340.52	1111.9	170
180	373.06	0.01827	2.532	346.03	850.8	1196.9	0.5325	1.0217	1.5542	345.42	1112.5	180
190	377.51	0.01833	2.404	350.79	846.8	1197.6	0.5381	1.0116	1.5497	350.15	1113.1	190
200	381.79	0.01839	2.288	355.36	843.0	1198.4	0.5435	1.0018	1.5453	354.68	1113.7	200
250	400.95	0.01865	1.8438	376.00	825.1	1201.1	0.5675	0.9588	1.5263	375.14	1115.8	250
300	417.33	0.01890	1.5433	393.84	809.0	1202.8	0.5879	0.9225	1.5104	392.79	1117.1	300
350	431.72	0.01913	1.3260	409.69	794.2	1203.9	0.6056	0.8910	1.4966	408.45	1118.0	350
400	444.59	0.0193	1.1613	424.0	780.5	1204.5	0.6214	0.8630	1.4844	422.6	1118.5	400
450	456.28	0.0195	1.0320	437.2	767.4	1204.6	0.6356	0.8378	1.4734	435.5	1118.7	450
500	467.01	0.0197	0.9278	449.4	755.0	1204.4	0.6487	0.8147	1.4634	447.6	1118.6	500
550	476.94	0.0199	0.8424	460.8	743.1	1203.9	0.6608	0.7934	1.4542	458.8	1118.2	550
600	486.21	0.0201	0.7698	471.6	731.6	1203.2	0.6720	0.7734	1.4454	469.4	1117.7	600
650	494.90	0.0203	0.7083	481.8	720.5	1202.3	0.6826	0.7548	1.4374	479.4	1117.1	650
700	503.10	0.0205	0.6554	491.5	709.7	1201.2	0.6925	0.7371	1.4296	488.8	1116.3	700
750	510.86	0.0207	0.6092	500.8	699.2	1200.0	0.7019	0.7204	1.4223	500.0	1115.4	750
800	518.23	0.0209	0.5687	509.7	688.9	1198.6	0.7108	0.7045	1.4153	506.6	1114.4	800
850	525.26	0.0210	0.5327	518.3	678.8	1197.1	0.7194	0.6891	1.4085	515.0	1113.3	850
900	531.98	0.0212	0.5006	526.6	668.8	1195.4	0.7275	0.6744	1.4020	523.1	1112.1	900
950	538.43	0.0214	0.4717	534.6	659.1	1193.7	0.7355	0.6602	1.3957	530.9	1110.8	950
1000	544.61	0.0216	0.4456	542.4	649.4	1191.8	0.7430	0.6467	1.3897	538.4	1109.4	1000
1100	556.31	0.0220	0.4001	557.4	630.4	1187.8	0.7575	0.6205	1.3780	552.9	1106.4	1100
1200	567.22	0.0223	0.3619	571.7	611.7	1183.4	0.7711	0.5956	1.3667	566.7	1103.0	1200
1300	577.46	0.0227	0.3293	585.4	593.2	1178.6	0.7840	0.5719	1.3559	580.0	1099.4	1300
1400	587.10	0.0231	0.3012	598.7	574.7	1173.4	0.7963	0.5491	1.3454	592.7	1095.4	1400
1500	596.23	0.0235	0.2765	611.6	556.3	1167.9	0.8082	0.5269	1.3351	605.1	1091.2	1500
2000	635.82	0.0257	0.1878	671.7	463.4	1135.1	0.8619	0.4230	1.2849	662.2	1065.6	2000
2500	668.13	0.0287	0.1307	730.6	360.5	1091.1	0.9126	0.3197	1.2322	717.3	1030.6	2500
3000	695.36	0.0346	0.0858	802.5	217.8	1020.3	0.9731	0.1885	1.1615	783.4	972.7	3000
3206.2	705.40	0.0503	0.0503	902.7	0	902.7	1.0580	0	1.0580	872.9	872.9	3206.2

* Abridged "Thermodynamic Properties of Steam" by Joseph H. Keenan and Frederick G. Keyes. John Wiley & Sons, Inc., New York, 1937.

TABLE 7. THERMODYNAMIC PROPERTIES OF STEAM.—(Continued)
Dry saturated steam: temperature table*

Temp., °F	Abs press., psi	Specific volume			Enthalpy			Entropy			Temp., °F
		Sat. liquid	Evap.	Sat. vapor	Sat. liquid	Evap.	Sat. vapor	Sat. liquid	Evap.	Sat. vapor	
<i>t</i>	<i>p</i>	<i>v_f</i>	<i>v_{fg}</i>	<i>v_g</i>	<i>h_f</i>	<i>h_{fg}</i>	<i>h_g</i>	<i>s_f</i>	<i>s_{fg}</i>	<i>s_g</i>	<i>t</i>
32	0.08854	0.01602	3306	3306	0.00	1075.8	1075.8	0.0000	2.1877	2.1877	32
35	0.09995	0.01602	2947	2947	3.02	1074.1	1077.1	0.0061	2.1709	2.1770	35
40	0.12170	0.01602	2444	2444	8.05	1071.3	1079.3	0.0162	2.1435	2.1597	40
45	0.14752	0.01602	2036.4	2036.4	13.06	1068.4	1081.5	0.0262	2.1167	2.1429	45
50	0.17811	0.01603	1703.2	1703.2	18.07	1065.6	1083.7	0.0361	2.0903	2.1264	50
60	0.2563	0.01604	1206.6	1206.7	28.06	1059.9	1088.0	0.0555	2.0393	2.0948	60
70	0.3631	0.01606	867.8	867.9	38.04	1054.3	1092.3	0.0745	1.9902	2.0647	70
80	0.5069	0.01608	633.1	633.1	48.02	1048.6	1096.6	0.0932	1.9428	2.0360	80
90	0.6982	0.01610	468.0	468.0	57.99	1042.9	1100.9	0.1115	1.8972	2.0087	90
100	0.9492	0.01613	350.3	350.4	67.97	1037.2	1105.2	0.1295	1.8531	1.9826	100
110	1.2748	0.01617	265.3	265.4	77.94	1031.6	1109.5	0.1471	1.8106	1.9577	110
120	1.6924	0.01620	203.25	203.27	87.92	1025.8	1113.7	0.1645	1.7694	1.9339	120
130	2.2225	0.01625	157.32	157.34	97.90	1020.0	1117.9	0.1816	1.7296	1.9112	130
140	2.8886	0.01629	122.99	123.01	107.89	1014.1	1122.0	0.1984	1.6910	1.8894	140
150	3.718	0.01634	98.06	97.07	117.89	1008.2	1126.1	0.2149	1.6537	1.8685	150
160	4.741	0.01639	77.27	77.29	127.89	1002.3	1130.2	0.2311	1.6174	1.8485	160
170	5.992	0.01645	62.04	62.06	137.90	996.3	1134.2	0.2472	1.5822	1.8293	170
180	7.510	0.01651	50.21	50.23	147.92	990.2	1138.1	0.2630	1.5480	1.8109	180
190	9.339	0.01657	40.94	40.96	157.95	984.1	1142.0	0.2785	1.5147	1.7932	190
200	11.526	0.01663	33.62	33.64	167.99	977.9	1145.9	0.2938	1.4824	1.7762	200
210	14.123	0.01670	27.80	27.82	178.05	971.6	1149.7	0.3090	1.4508	1.7598	210
212	14.696	0.01672	26.78	26.80	180.07	970.3	1150.4	0.3120	1.4446	1.7566	212
220	17.186	0.01677	23.13	23.15	188.13	965.2	1153.4	0.3239	1.4201	1.7440	220
230	20.780	0.01684	19.365	19.382	198.23	958.8	1157.0	0.3387	1.3901	1.7288	230
240	24.969	0.01692	16.306	16.323	208.34	952.2	1160.5	0.3531	1.3609	1.7140	240
250	29.825	0.01700	13.804	13.821	216.48	945.5	1164.0	0.3675	1.3323	1.6998	250
260	35.429	0.01709	11.746	11.763	228.64	938.7	1167.3	0.3817	1.3043	1.6860	260
270	41.858	0.01717	10.044	10.061	238.84	931.8	1170.6	0.3958	1.2769	1.6727	270
280	49.203	0.01726	8.628	8.645	249.06	924.7	1173.8	0.4096	1.2501	1.6597	280
290	57.556	0.01735	7.444	7.461	259.31	917.5	1176.8	0.4234	1.2238	1.6472	290
300	67.013	0.01745	6.449	6.466	269.59	910.1	1179.7	0.4369	1.1980	1.6350	300
310	77.68	0.01755	5.609	5.626	279.92	902.6	1182.5	0.4504	1.1727	1.6231	310
320	89.66	0.01765	4.896	4.914	290.28	894.9	1185.2	0.4637	1.1478	1.6115	320
330	103.06	0.01776	4.289	4.307	300.68	887.0	1187.7	0.4769	1.1233	1.6002	330
340	118.01	0.01787	3.770	3.788	311.13	879.0	1190.1	0.4900	1.0992	1.5891	340
350	134.63	0.01799	3.324	3.342	321.63	870.7	1192.3	0.5029	1.0754	1.5783	350
360	153.04	0.01811	2.939	2.957	332.18	862.2	1194.4	0.5158	1.0519	1.5677	360
370	173.37	0.01823	2.606	2.625	342.79	853.5	1196.3	0.5286	1.0287	1.5573	370
380	195.77	0.01836	2.317	2.335	353.45	844.6	1198.1	0.5413	1.0059	1.5471	380
390	220.37	0.01850	2.0651	2.0836	364.17	835.4	1199.6	0.5539	0.9832	1.5371	390
400	247.31	0.01864	1.8447	1.8633	374.97	826.0	1201.0	0.5664	0.9608	1.5272	400
410	276.75	0.01878	1.6512	1.6700	385.83	816.3	1202.1	0.5788	0.9386	1.5174	410
420	308.83	0.01894	1.4811	1.5000	396.77	806.3	1203.1	0.5912	0.9166	1.5078	420
430	343.72	0.01910	1.3308	1.3499	407.79	796.0	1203.8	0.6035	0.8947	1.4982	430
440	381.59	0.01926	1.1979	1.2171	418.90	785.4	1204.3	0.6158	0.8730	1.4887	440
450	422.6	0.0194	1.0799	1.0993	430.1	774.5	1204.6	0.6280	0.8513	1.4793	450
460	466.9	0.0196	0.9748	0.9944	441.4	763.2	1204.6	0.6402	0.8298	1.4700	460
470	514.7	0.0198	0.8811	0.9009	452.8	751.5	1204.3	0.6523	0.8083	1.4606	470
480	566.1	0.0200	0.7972	0.8172	464.4	739.4	1203.7	0.6645	0.7868	1.4513	480
490	621.4	0.0202	0.7221	0.7423	476.0	726.8	1202.8	0.6766	0.7653	1.4419	490
500	680.8	0.0204	0.6545	0.6749	487.8	713.9	1201.7	0.6887	0.7438	1.4325	500
520	812.4	0.0209	0.5385	0.5594	511.9	686.4	1198.2	0.7130	0.7006	1.4136	520
540	962.5	0.0215	0.4434	0.4649	536.6	656.6	1193.2	0.7374	0.6568	1.3942	540
560	1133.1	0.0221	0.3647	0.3868	562.2	624.2	1186.4	0.7621	0.6121	1.3742	560
580	1325.8	0.0228	0.2989	0.3217	588.9	588.4	1177.3	0.7872	0.5659	1.3532	580
600	1542.9	0.0236	0.2432	0.2668	617.0	548.5	1165.5	0.8131	0.5176	1.3307	600
620	1786.6	0.0247	0.1955	0.2201	646.7	503.6	1150.3	0.8398	0.4664	1.3062	620
640	2059.7	0.0260	0.1538	0.1798	678.6	452.0	1130.5	0.8679	0.4110	1.2789	640
660	2365.4	0.0278	0.1165	0.1442	714.2	390.2	1104.4	0.8987	0.3485	1.2472	660
680	2708.1	0.0305	0.0810	0.1115	757.3	309.9	1067.2	0.9351	0.2719	1.2071	680
700	3093.7	0.0369	0.0392	0.0761	823.3	172.1	995.4	0.9905	0.1484	1.1389	700
705.4	3206.2	0.0503	0	0.0503	902.7	0	902.7	1.0580	0	1.0580	705.4

* Abridged from "Thermodynamic Properties of Steam" by Joseph H. Keenan and Frederick G. Keyes, John Wiley & Sons, Inc., New York.

TABLE 7. THERMODYNAMIC PROPERTIES OF STEAM.—(Continued)
Properties of superheated steam*

Abs press., psi (sat temp)		Temp, °F												
		200	300	400	500	600	700	800	900	1000	1100	1200	1400	1600
1 (101.74)	v.....	392.6	452.3	512.0	571.6	631.2	690.8	750.4	809.9	869.5	929.1	988.7	1107.8	1227.0
	h.....	1150.4	1195.8	1241.7	1288.3	1335.7	1383.8	1432.8	1482.7	1533.5	1585.2	1637.7	1745.7	1857.5
	s.....	2.0512	2.1153	2.1720	2.2233	2.2702	2.3137	2.3542	2.3923	2.4283	2.4625	2.4952	2.5566	2.6137
5 (162.24)	v.....	78.16	90.25	102.26	114.22	126.16	138.10	150.03	161.95	173.87	185.79	197.71	221.6	245.4
	h.....	1148.8	1195.0	1241.2	1288.0	1335.4	1383.6	1432.7	1482.6	1533.4	1585.1	1637.7	1745.7	1857.4
	s.....	1.8718	1.9370	1.9942	2.0456	2.0927	2.1361	2.1767	2.2148	2.2509	2.2851	2.3178	2.3792	2.4363
10 (193.21)	v.....	38.85	45.00	51.04	57.05	63.03	69.01	74.98	80.95	86.92	92.88	98.84	110.77	122.69
	h.....	1146.6	1193.9	1240.6	1287.5	1335.1	1383.4	1432.5	1482.4	1533.2	1585.0	1637.6	1745.6	1857.3
	s.....	1.7927	1.8595	1.9172	1.9689	2.0160	2.0596	2.1002	2.1383	2.1744	2.2086	2.2413	2.3028	2.3598
14.696 (212.00)	v.....		30.53	34.68	38.78	42.86	46.94	51.00	55.07	59.13	63.19	67.25	75.37	83.48
	h.....		1192.8	1239.9	1287.1	1334.8	1383.2	1432.3	1482.3	1533.1	1584.8	1637.5	1745.5	1857.3
	s.....		1.8160	1.8743	1.9261	1.9734	2.0170	2.0576	2.0958	2.1319	2.1662	2.1989	2.2603	2.3174
20 (227.96)	v.....		22.36	25.43	28.46	31.47	34.47	37.46	40.45	43.44	46.42	49.41	55.37	61.34
	h.....		1191.6	1239.2	1286.6	1334.4	1382.9	1432.1	1482.1	1533.0	1584.7	1637.4	1745.4	1857.2
	s.....		1.7808	1.8396	1.8918	1.9392	1.9829	2.0235	2.0618	2.0978	2.1321	2.1648	2.2263	2.2834
40 (267.25)	v.....		11.040	12.628	14.168	15.688	17.198	18.702	20.20	21.70	23.20	24.69	27.68	30.66
	h.....		1186.8	1236.5	1284.8	1333.1	1381.9	1431.3	1481.4	1532.4	1584.3	1637.0	1745.1	1857.0
	s.....		1.6994	1.7608	1.8140	1.8619	1.9058	1.9467	1.9850	2.0212	2.0555	2.0883	2.1498	2.2069
60 (292.71)	v.....		7.259	8.357	9.403	10.427	11.441	12.449	13.452	14.454	15.453	16.451	18.446	20.44
	h.....		1181.6	1233.6	1283.0	1331.8	1380.9	1430.5	1480.8	1531.9	1583.8	1636.6	1744.8	1856.7
	s.....		1.6492	1.7135	1.7678	1.8162	1.8605	1.9015	1.9400	1.9762	2.0106	2.0434	2.1049	2.1621
80 (312.03)	v.....			6.220	7.020	7.797	8.562	9.322	10.077	10.830	11.582	12.332	13.830	15.325
	h.....			1230.7	1281.1	1330.5	1379.9	1429.7	1480.1	1531.3	1583.4	1636.2	1744.5	1856.5
	s.....			1.6791	1.7346	1.7836	1.8281	1.8694	1.9079	1.9442	1.9787	2.0115	2.0731	2.1303
100 (327.81)	v.....			4.937	5.589	6.218	6.835	7.446	8.052	8.656	9.259	9.860	11.060	12.258
	h.....			1227.6	1279.1	1329.1	1378.9	1428.9	1479.5	1530.8	1582.9	1635.7	1744.2	1856.2
	s.....			1.6518	1.7085	1.7581	1.8029	1.8443	1.8829	1.9193	1.9538	1.9867	2.0484	2.1056
120 (341.25)	v.....			4.081	4.636	5.165	5.683	6.195	6.702	7.207	7.710	8.212	9.214	10.213
	h.....			1224.4	1277.2	1327.7	1377.8	1428.1	1478.8	1530.2	1582.4	1635.3	1743.9	1856.0
	s.....			1.6287	1.6869	1.7370	1.7822	1.8237	1.8625	1.8990	1.9335	1.9664	2.0281	2.0854
140 (353.02)	v.....			3.468	3.954	4.413	4.861	5.301	5.738	6.172	6.604	7.035	7.895	8.752
	h.....			1221.1	1275.2	1326.4	1376.8	1427.3	1478.2	1529.7	1581.9	1634.9	1743.5	1855.7
	s.....			1.6087	1.6683	1.7190	1.7645	1.8063	1.8451	1.8817	1.9163	1.9493	2.0110	2.0683
160 (363.53)	v.....			3.008	3.443	3.849	4.244	4.631	5.015	5.396	5.775	6.152	6.906	7.656
	h.....			1217.6	1273.1	1325.0	1375.7	1426.4	1477.5	1529.1	1581.4	1634.5	1743.2	1855.5
	s.....			1.5908	1.6519	1.7033	1.7491	1.7911	1.8301	1.8667	1.9014	1.9344	1.9962	2.0535
180 (373.06)	v.....			2.649	3.044	3.411	3.764	4.110	4.452	4.792	5.129	5.466	6.136	6.804
	h.....			1214.0	1271.0	1323.5	1374.7	1425.6	1476.8	1528.6	1581.0	1634.1	1742.9	1855.2
	s.....			1.5745	1.6373	1.6894	1.7355	1.7776	1.8167	1.8534	1.8882	1.9212	1.9831	2.0404
200 (381.79)	v.....			2.361	2.726	3.060	3.380	3.693	4.002	4.309	4.613	4.917	5.521	6.123
	h.....			1210.3	1268.9	1322.1	1373.6	1424.8	1476.2	1528.0	1580.5	1633.7	1742.6	1855.0
	s.....			1.5594	1.6240	1.6767	1.7232	1.7655	1.8048	1.8415	1.8763	1.9094	1.9713	2.0287
220 (389.86)	v.....			2.125	2.465	2.772	3.066	3.352	3.634	3.913	4.191	4.467	5.017	5.565
	h.....			1206.5	1266.7	1320.7	1372.0	1424.0	1475.5	1527.5	1580.0	1633.3	1742.3	1854.7
	s.....			1.5453	1.6117	1.6652	1.7120	1.7545	1.7939	1.8308	1.8656	1.8987	1.9607	2.0181
240 (397.37)	v.....			1.9276	2.247	2.533	2.804	3.068	3.327	3.584	3.839	4.093	4.597	5.100
	h.....			1202.5	1264.5	1319.2	1371.5	1423.2	1474.8	1526.9	1579.6	1632.9	1742.0	1854.5
	s.....			1.5219	1.6003	1.6546	1.7017	1.7444	1.7839	1.8209	1.8558	1.8889	1.9510	2.0084
260 (404.42)	v.....				2.063	2.330	2.582	2.827	3.067	3.305	3.541	3.776	4.242	4.707
	h.....				1262.3	1317.7	1370.4	1422.3	1474.2	1526.3	1579.1	1632.5	1741.7	1854.2
	s.....				1.5897	1.6447	1.6922	1.7352	1.7748	1.8118	1.8467	1.8799	1.9420	1.9995
280 (411.05)	v.....				1.9047	2.156	2.392	2.621	2.845	3.066	3.286	3.504	3.938	4.370
	h.....				1260.0	1316.2	1369.4	1421.5	1473.5	1525.8	1578.6	1632.1	1741.4	1854.0
	s.....				1.5796	1.6354	1.6834	1.7265	1.7662	1.8033	1.8383	1.8716	1.9337	1.9912
300 (417.33)	v.....				1.7675	2.005	2.227	2.442	2.652	2.859	3.065	3.269	3.674	4.078
	h.....				1257.6	1314.7	1368.3	1420.6	1472.8	1525.2	1578.1	1631.7	1741.0	1853.7
	s.....				1.5701	1.6268	1.6751	1.7184	1.7582	1.7954	1.8305	1.8638	1.9260	1.9835
350 (431.72)	v.....				1.4923	1.7036	1.8980	2.084	2.266	2.445	2.622	2.798	3.147	3.493
	h.....				1251.5	1310.9	1365.5	1418.5	1471.1	1523.8	1577.0	1630.7	1740.3	1853.1
	s.....				1.5481	1.6070	1.6563	1.7002	1.7403	1.7777	1.8130	1.8463	1.9086	1.9663
400 (444.59)	v.....				1.2851	1.4770	1.6508	1.8161	1.9767	2.134	2.290	2.445	2.751	3.055
	h.....				1245.1	1306.9	1362.7	1416.4	1469.4	1522.4	1575.8	1629.6	1739.5	1852.5
	s.....				1.5281	1.5894	1.6398	1.6842	1.7247	1.7623	1.7977	1.8311	1.8936	1.9513

* Abridged from "Thermodynamic Properties of Steam," by Joseph H. Keenan and Frederick G. Keyes, John Wiley & Sons, Inc., New York, 1937.

TABLE 7. THERMODYNAMIC PROPERTIES OF STEAM.—(Continued)
Properties of superheated steam*

Abs Press., psi (sat temp)	Temp, °F														
	500	550	600	620	640	660	680	700	800	900	1000	1200	1400	1600	
450 (456.28)	v.....	1.1231	1.2155	1.3005	1.3332	1.3652	1.3967	1.4278	1.4584	1.6074	1.7516	1.8928	2.170	2.443	2.714
	h.....	1238.4	1272.0	1302.8	1314.6	1326.2	1337.5	1348.8	1359.9	1414.3	1467.7	1521.0	1628.6	1738.7	1851.9
	s.....	1.5095	1.5437	1.5735	1.5845	1.5951	1.6054	1.6153	1.6250	1.6699	1.7108	1.7486	1.8177	1.8803	1.9381
500 (467.01)	v.....	0.9927	1.0800	1.1591	1.1893	1.2188	1.2478	1.2763	1.3044	1.4405	1.5715	1.6996	1.9504	2.197	2.442
	h.....	1231.3	1266.8	1298.6	1310.7	1322.6	1334.2	1345.7	1357.0	1412.1	1466.0	1519.6	1627.6	1737.9	1851.3
	s.....	1.4919	1.5280	1.5588	1.5701	1.5810	1.5915	1.6016	1.6115	1.6571	1.6982	1.7363	1.8056	1.8683	1.9262
550 (476.94)	v.....	0.8852	0.9686	1.0431	1.0714	1.0989	1.1259	1.1523	1.1783	1.3038	1.4241	1.5414	1.7706	1.9957	2.219
	h.....	1223.7	1261.2	1294.3	1306.8	1318.9	1330.8	1342.5	1354.0	1409.9	1464.3	1518.2	1626.6	1737.1	1850.6
	s.....	1.4751	1.5131	1.5451	1.5568	1.5680	1.5787	1.5890	1.5991	1.6452	1.6868	1.7250	1.7946	1.8575	1.9155
600 (486.21)	v.....	0.7947	0.8753	0.9463	0.9729	0.9988	1.0241	1.0489	1.0732	1.1899	1.3013	1.4096	1.6208	1.8279	2.033
	h.....	1215.7	1255.5	1289.9	1302.7	1315.2	1327.4	1339.3	1351.1	1407.7	1462.5	1516.7	1625.5	1736.3	1850.0
	s.....	1.4586	1.4990	1.5323	1.5443	1.5558	1.5667	1.5773	1.5875	1.6343	1.6762	1.7147	1.7846	1.8476	1.9066
700 (503.10)	v.....	0.7277	0.7934	0.8177	0.8411	0.8639	0.8860	0.9077	0.9277	1.0108	1.1082	1.2024	1.3853	1.5641	1.7405
	h.....	1243.2	1280.6	1294.3	1307.5	1320.3	1332.8	1345.0	1403.2	1459.0	1515.9	1623.5	1734.8	1848.8	1.9881
	s.....	1.4722	1.5084	1.5212	1.5333	1.5449	1.5559	1.5665	1.6147	1.6573	1.6963	1.7666	1.8299	1.8881	
800 (518.23)	v.....	0.6154	0.6779	0.7006	0.7223	0.7433	0.7635	0.7833	0.8033	0.8763	0.9633	1.0470	1.2088	1.3662	1.5214
	h.....	1229.8	1270.7	1285.4	1299.4	1312.9	1325.9	1338.6	1351.1	1407.7	1462.5	1516.7	1625.5	1736.3	1850.0
	s.....	1.4467	1.4863	1.5000	1.5129	1.5250	1.5366	1.5476	1.5592	1.6407	1.6801	1.7510	1.8146	1.8729	
900 (531.98)	v.....	0.5264	0.5873	0.6089	0.6294	0.6491	0.6680	0.6863	0.7046	0.8506	0.9262	1.0714	1.2124	1.3509	
	h.....	1215.0	1260.1	1275.9	1290.9	1305.1	1318.8	1332.1	1345.8	1401.8	1456.8	1511.0	1621.4	1733.2	1847.5
	s.....	1.4216	1.4653	1.4800	1.4938	1.5066	1.5187	1.5303	1.5414	1.6257	1.6656	1.7371	1.8009	1.8595	
1000 (544.61)	v.....	0.4533	0.5140	0.5350	0.5546	0.5733	0.5912	0.6084	0.6258	0.7604	0.8294	0.9615	1.0893	1.2146	
	h.....	1198.3	1248.8	1265.9	1281.9	1297.0	1311.4	1325.3	1339.2	1448.2	1505.1	1617.3	1730.0	1845.0	
	s.....	1.3961	1.4450	1.4610	1.4757	1.4893	1.5021	1.5141	1.5260	1.6121	1.6525	1.7245	1.7886	1.8474	
1100 (556.31)	v.....	0.4532	0.4738	0.4929	0.5110	0.5281	0.5445	0.5617	0.5783	0.6866	0.7503	0.8716	0.9885	1.1031	
	h.....	1236.7	1255.3	1272.4	1288.5	1303.7	1318.3	1334.3	1344.5	1444.5	1502.2	1615.2	1728.4	1843.8	
	s.....	1.4251	1.4425	1.4583	1.4728	1.4862	1.4989	1.5114	1.5239	1.5995	1.6405	1.7130	1.7775	1.8363	
1200 (567.22)	v.....	0.4016	0.4222	0.4410	0.4586	0.4752	0.4909	0.5067	0.5223	0.6191	0.6843	0.7967	0.9046	1.0101	
	h.....	1223.5	1243.9	1262.4	1279.6	1295.7	1311.0	1327.3	1344.0	1440.7	1499.2	1613.1	1726.9	1842.5	
	s.....	1.4052	1.4243	1.4413	1.4568	1.4710	1.4843	1.5009	1.5189	1.5879	1.6293	1.7025	1.7672	1.8263	
1400 (587.10)	v.....	0.3174	0.3390	0.3580	0.3753	0.3912	0.4062	0.4214	0.4362	0.5281	0.5805	0.6789	0.7727	0.8640	
	h.....	1193.0	1218.4	1240.4	1260.3	1278.5	1295.5	1312.5	1329.7	1433.1	1493.2	1608.9	1723.7	1840.0	
	s.....	1.3639	1.3877	1.4079	1.4258	1.4419	1.4567	1.4717	1.4862	1.5666	1.6093	1.6886	1.7489	1.8083	
1600 (604.90)	v.....	0.2733	0.2936	0.3112	0.3271	0.3417	0.3567	0.3714	0.3861	0.4034	0.4553	0.5027	0.5906	0.6738	0.7545
	h.....	1187.8	1215.2	1238.7	1259.6	1278.7	1297.3	1316.1	1334.8	1425.3	1487.0	1604.6	1720.5	1837.5	
	s.....	1.3489	1.3741	1.3952	1.4137	1.4303	1.4464	1.4619	1.4774	1.5476	1.5914	1.6669	1.7328	1.7926	
1800 (621.03)	v.....	0.2407	0.2597	0.2760	0.2907	0.3050	0.3191	0.3332	0.3473	0.3614	0.3986	0.4421	0.5218	0.5968	0.6693
	h.....	1185.1	1214.0	1238.5	1260.3	1281.4	1302.5	1323.6	1344.7	1437.2	1498.8	1600.4	1717.3	1835.0	
	s.....	1.3377	1.3638	1.3855	1.4044	1.4237	1.4429	1.4621	1.4813	1.5531	1.5952	1.6752	1.7185	1.7786	
2000 (635.82)	v.....	0.1936	0.2161	0.2337	0.2489	0.2637	0.2783	0.2928	0.3074	0.3219	0.3532	0.3935	0.4668	0.5352	0.6011
	h.....	1145.6	1184.9	1214.8	1240.0	1265.5	1290.0	1314.5	1339.0	1439.5	1499.2	1608.8	1714.1	1832.5	
	s.....	1.2945	1.3300	1.3564	1.3783	1.3976	1.4170	1.4363	1.4556	1.5139	1.5603	1.6394	1.7055	1.7666	
2500 (668.13)	v.....	0.1484	0.1686	0.1886	0.2086	0.2286	0.2486	0.2686	0.2886	0.3086	0.3532	0.3935	0.4668	0.5352	0.6011
	h.....	1132.3	1176.8	1203.6	1230.6	1257.6	1284.6	1311.6	1338.6	1438.6	1498.3	1608.8	1714.1	1832.5	
	s.....	1.2687	1.3073	1.3412	1.3703	1.3994	1.4285	1.4576	1.4867	1.5458	1.5848	1.6639	1.7229	1.7819	
3000 (695.36)	v.....	0.0984	0.1760	0.2159	0.2476	0.2783	0.3086	0.3389	0.3691	0.3994	0.4421	0.5011	0.5593	0.6175	0.6757
	h.....	1060.7	1267.2	1365.0	1441.8	1517.3	1592.8	1668.3	1743.8	1819.3	1900.8	2000.4	2100.0	2200.0	
	s.....	1.1966	1.3696	1.4439	1.4984	1.5439	1.5884	1.6329	1.6774	1.7219	1.7664	1.8109	1.8554	1.8999	
3206.2 (705.40)	v.....	0.1583	0.1981	0.2288	0.2806	0.3267	0.3703	0.4139	0.4575	0.5011	0.5448	0.5884	0.6321	0.6757	0.7194
	h.....	1250.5	1355.2	1434.7	1569.8	1694.6	1817.2	1940.1	2062.4	2186.7	2311.0	2435.3	2559.6	2683.9	
	s.....	1.3508	1.4309	1.4874	1.5742	1.6452	1.7080	1.7708	1.8336	1.8964	1.9592	2.0220	2.0848	2.1476	
3500	v.....	0.0306	0.1364	0.1762	0.2058	0.2546	0.2977	0.3381	0.3785	0.4189	0.4593	0.5000	0.5406	0.5812	0.6219
	h.....	780.5	1224.9	1340.7	1424.5	1563.3	1689.8	1813.6	1938.1	2062.4	2186.7	2311.0	2435.3	2559.6	
	s.....	0.9515	1.3241	1.4127	1.4723	1.5615	1.6336	1.6968	1.7600	1.8232	1.8864	1.9496	2.0128	2.0760	
4000	v.....	0.0287	0.1052	0.1462	0.1743	0.2192	0.2581	0.2943	0.3305	0.3667	0.4029	0.4391	0.4753	0.5115	0.5477
	h.....	763.8	1174.8	1314.4	1406.8	1552.1	1681.7	1807.2	1932.7	2058.2	2183.7	2309.2	2434.7	2560.2	
	s.....	0.9347	1.2757	1.3827	1.4482	1.5417	1.6154	1.6795	1.7436	1.8077	1.8718	1.9359	1.9999	2.0640	
4500	v.....	0.0276	0.0798	0.1226	0.1500	0.1917	0.2273	0.2602	0.2931	0.3260	0.3589	0.3918	0.4247	0.4576	0.4905
	h.....	753.5	1113.9	1286.5	1388.4	1540.8	1673.5	1806.2	1938.9	2071.6	2204.3	2337.0	2469.7	2602.4	
	s.....	0.9235	1.2204	1.3529	1.4253	1.5235	1.5990	1.6640	1.7285	1.7930	1.8575	1.9220	1.9865	2.0510	
5000	v.....	0.0268	0.0593	0.1036	0.1303	0.1696	0.2027	0.2329	0.2631	0.2933	0.3235	0.3537	0.3839	0.4141	0.4443
	h.....	746.4	1047.1	1258.5	1369.5	1529.5	1665.3	1794.5	1929.8	2065.1	2200.4	2335.7	2471.0	2606.3	
	s.....	0.9152	1.1622	1.3231	1.4034	1.5066	1.5839	1.6499	1.7159	1.7819	1.8479	1.9139	1.9799	2.0459	
5500	v.....	0.0262	0.0463	0.0880	0.1143	0.1516	0.1825	0.2106	0.2387	0.2668	0.2949	0.3230	0.3511	0.3792	0.4073
	h.....	741.3	985.0	1224.1	1349.3	1518.2	1657.0	1788.1	1919.2	2050.3	2181.4	2312.5	2443.6	2574.7	
	s.....	0.9090	1.1093	1.2930	1.3821	1.4908	1.5699	1.6360	1.7021	1.7682	1.8343	1.9004	1.9665	2.0326	

* Abridged from "Thermodynamic Properties of Steam," by Joseph H. Keenan and Frederick G. Keyes, John Wiley & Sons, Inc., New York, 1937.

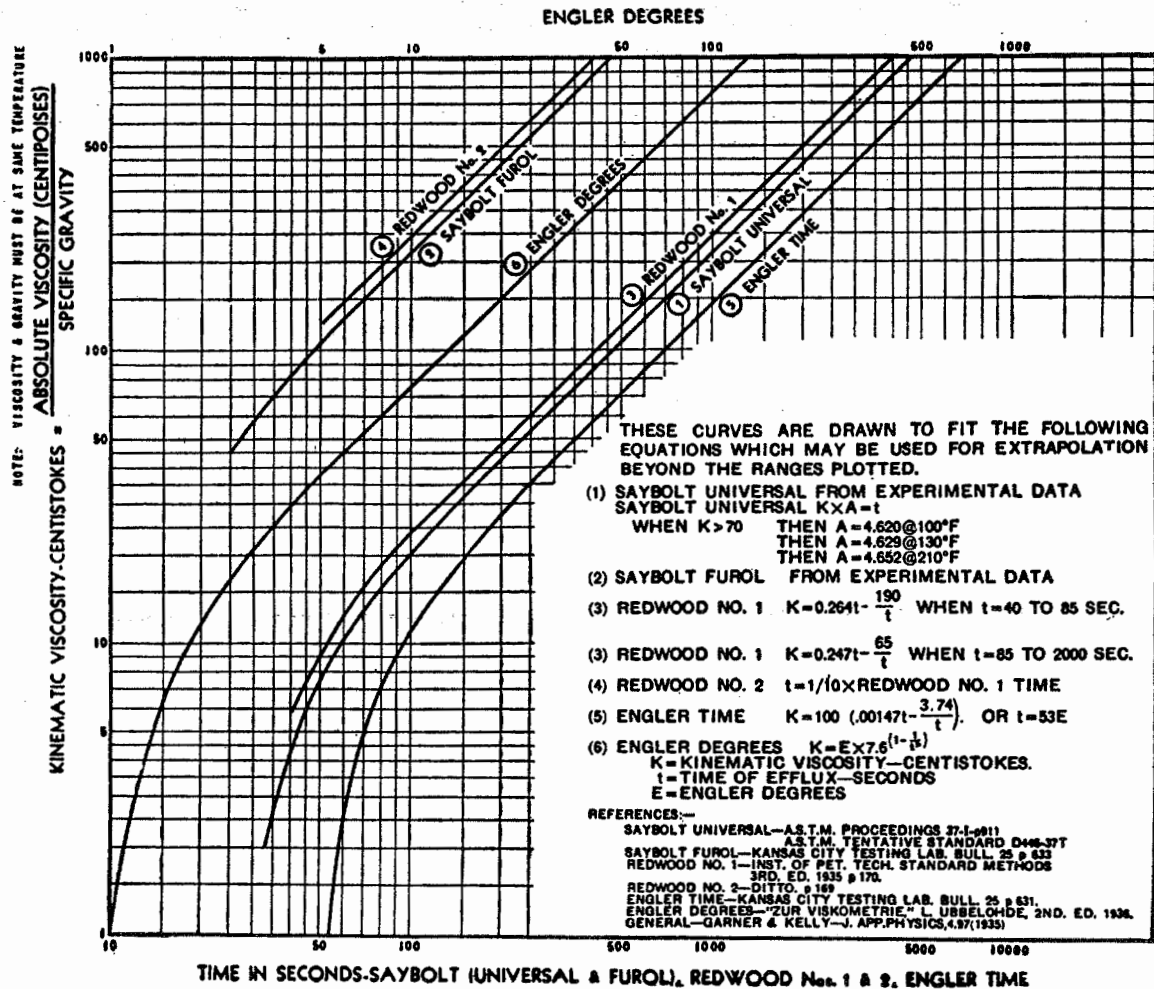
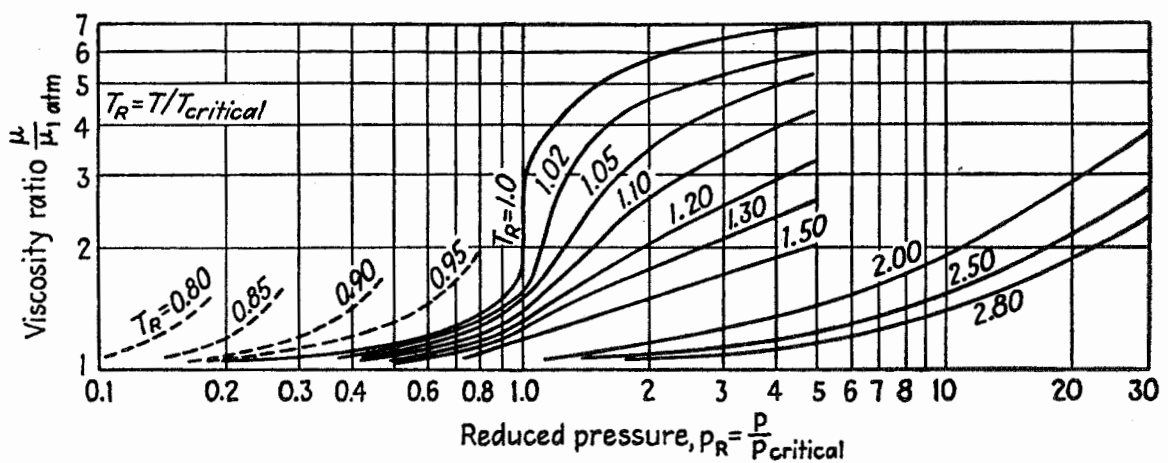


FIG. 13a. Viscosity conversion chart.

FIG. 13b. Viscosity correction chart for gases at different pressures. [Comings and Egly, *Ind. Eng. Chem.*, **32**, 715 (1940).]

APPENDIX OF CALCULATION DATA

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VISCOSITIES OF PETROLEUM FRACTIONS For temperature ranges employed in the text Coordinates to be used with Fig. 14

	X	Y
76°API natural gasoline.....	14.4	6.4
56°API gasoline.....	14.0	10.5
42°API kerosene.....	11.6	16.0
35°API distillate.....	10.0	20.0
34°API mid-continent crude.....	10.3	21.3
28°API gas oil.....	10.0	23.6

VISCOSITIES OF ANIMAL AND VEGETABLE OILS*

	Acid No.	Sp gr, 20/4°C	X	Y
Almond.....	2.85	0.9188	6.9	28.2
Coconut.....	0.01	0.9226	6.9	26.9
Cod liver.....		0.9138	7.7	27.7
Cottonseed.....	14.24	0.9187	7.0	28.0
Lard.....	3.39	0.9138	7.0	28.2
Linseed.....	3.42	0.9297	6.8	27.5
Mustard.....		0.9237	7.0	28.5
Neatsfoot.....	13.35	0.9158	6.5	28.0
Olive.....		0.9158	6.6	28.3
Palm kernel.....	9.0	0.9190	7.0	26.9
Perilla, raw.....	1.36	0.9297	8.1	27.2
Rapeseed.....	0.34	0.9114	7.0	28.8
Sardine.....	0.57	0.9384	7.7	27.3
Soybean.....	3.50	0.9228	8.3	27.5
Sperm.....	0.80	0.8829	7.7	26.3
Sunflower.....	2.76	0.9207	7.5	27.6
Whale, refined.....	0.73	0.9227	7.5	27.5

* Based on data at 100 and 210°F of A. R. Rescorla and F. L. Carnahan, *Ind. Eng. Chem.*, **28**, 1212-1213 (1936).

VISCOSITIES OF COMMERCIAL FATTY ACIDS* 250 to 400°F

	Sp gr at 300°F	X	Y
Lauric.....	0.792	10.1	23.1
Oleic.....	0.799	10.0	25.2
Palmitic.....	0.786	9.2	25.9
Stearic.....	0.789	10.5	25.5

* From data of D. Q. Kern and W. Van Nostrand, *Ind. Eng. Chem.*, **41**, 2209 (1949).

VISCOSITIES OF LIQUIDS*
Coordinates to be used with Fig. 14

Liquid	X	Y	Liquid	X	Y
Acetaldehyde.....	15.2	4.8	Freon-21.....	15.7	7.5
Acetic acid, 100%.....	12.1	14.2	Freon-22.....	17.2	4.7
Acetic acid, 70%.....	9.5	17.0	Freon-113.....	12.5	11.4
Acetic anhydride.....	12.7	12.8	Freon-114.....	14.6	8.3
Acetone, 100%.....	14.5	7.2	Glycerol, 100%.....	2.0	30.0
Acetone, 35%.....	7.9	15.0	Glycerol, 50%.....	6.9	19.6
Allyl alcohol.....	10.2	14.3	Heptane.....	14.1	8.4
Ammonia, 100%.....	12.6	2.0	Hexane.....	14.7	7.0
Ammonia, 26%.....	10.1	13.9	Hydrochloric acid, 31.5%.....	13.0	16.6
Amyl acetate.....	11.8	12.5	Isobutyl alcohol.....	7.1	18.0
Amyl alcohol.....	7.5	18.4	Isobutyric acid.....	12.2	14.4
Aniline.....	8.1	18.7	Isopropyl alcohol.....	8.2	16.0
Anisole.....	12.3	13.5	Mercury.....	18.4	16.4
Arsenic trichloride.....	13.9	14.5	Methanol, 100%.....	12.4	10.5
Benzene.....	12.5	10.9	Methanol, 90%.....	12.3	11.8
Brine, CaCl ₂ , 25%.....	6.6	15.9	Methanol, 40%.....	7.8	15.5
Brine, NaCl, 25%.....	10.2	16.6	Methyl acetate.....	14.2	8.2
Bromine.....	14.2	13.2	Methyl chloride.....	15.0	3.8
Bromotoluene.....	20.0	15.9	Methyl ethyl ketone.....	13.9	8.6
n-Butane.....	15.3	3.3	Naphthalene.....	7.9	18.1
Isobutane.....	14.5	3.7	Nitric acid, 95%.....	12.8	13.8
Butyl acetate.....	12.3	11.0	Nitric acid, 60%.....	10.8	17.0
Butyl alcohol.....	8.6	17.2	Nitrobenzene.....	10.6	16.2
Butyric acid.....	12.1	15.3	Nitrotoluene.....	11.0	17.0
Carbon dioxide.....	11.6	0.3	Octane.....	13.7	10.0
Carbon disulfide.....	16.1	7.5	Octyl alcohol.....	6.6	21.1
Carbon tetrachloride.....	12.7	13.1	Pentachloroethane.....	10.9	17.3
Chlorobenzene.....	12.3	12.4	Pentane.....	14.9	5.2
Chloroform.....	14.4	10.2	Phenol.....	6.9	20.8
Chlorosulfonic acid.....	11.2	18.1	Phosphorus tribromide.....	13.8	16.7
Chlorotoluene, ortho.....	13.0	13.3	Phosphorus trichloride.....	16.2	10.9
Chlorotoluene, meta.....	13.3	12.5	Propane.....	15.3	1.0
Chlorotoluene, para.....	13.3	12.5	Propionic acid.....	12.8	13.8
Cresol, meta.....	2.5	20.8	Propyl alcohol.....	9.1	16.5
Cyclohexanol.....	2.9	24.3	Propyl bromide.....	14.5	9.6
Dibromoethane.....	12.7	15.8	Propyl chloride.....	14.4	7.5
Dichloroethane.....	13.2	12.2	Propyl iodide.....	14.1	11.6
Dichloromethane.....	14.6	8.9	Sodium.....	16.4	13.9
Diethyl oxalate.....	11.0	16.4	Sodium hydroxide, 50%.....	3.2	25.8
Dimethyl oxalate.....	12.3	15.8	Stannic chloride.....	13.5	12.8
Diphenyl.....	12.0	18.3	Sulfur dioxide.....	15.2	7.1
Dipropyl oxalate.....	10.3	17.7	Sulfuric acid, 110%.....	7.2	27.4
Ethyl acetate.....	13.7	9.1	Sulfuric acid, 98%.....	7.0	24.8
Ethyl alcohol, 100%.....	10.5	13.8	Sulfuric acid, 60%.....	10.2	21.3
Ethyl alcohol, 95%.....	9.8	14.3	Sulfuryl chloride.....	15.2	12.4
Ethyl alcohol, 40%.....	6.5	16.6	Tetrachloroethane.....	11.9	15.7
Ethyl benzene.....	13.2	11.5	Tetrachloroethylene.....	14.2	12.7
Ethyl bromide.....	14.5	8.1	Titanium tetrachloride.....	14.4	12.3
Ethyl chloride.....	14.8	6.0	Toluene.....	13.7	10.4
Ethyl ether.....	14.5	5.3	Trichloroethylene.....	14.8	10.5
Ethyl formate.....	14.2	8.4	Turpentine.....	11.5	14.9
Ethyl iodide.....	14.7	10.3	Vinyl acetate.....	14.0	8.8
Ethylene glycol.....	6.0	23.6	Water.....	10.2	13.0
Formic acid.....	10.7	15.8	Xylene, ortho.....	13.5	12.1
Freon-11.....	14.4	9.0	Xylene, meta.....	13.9	10.6
Freon-12.....	16.8	5.6	Xylene, para.....	13.9	10.9

* From Perry, J. H., "Chemical Engineers' Handbook," 3d ed., McGraw-Hill Book Company, Inc., New York, 1950.

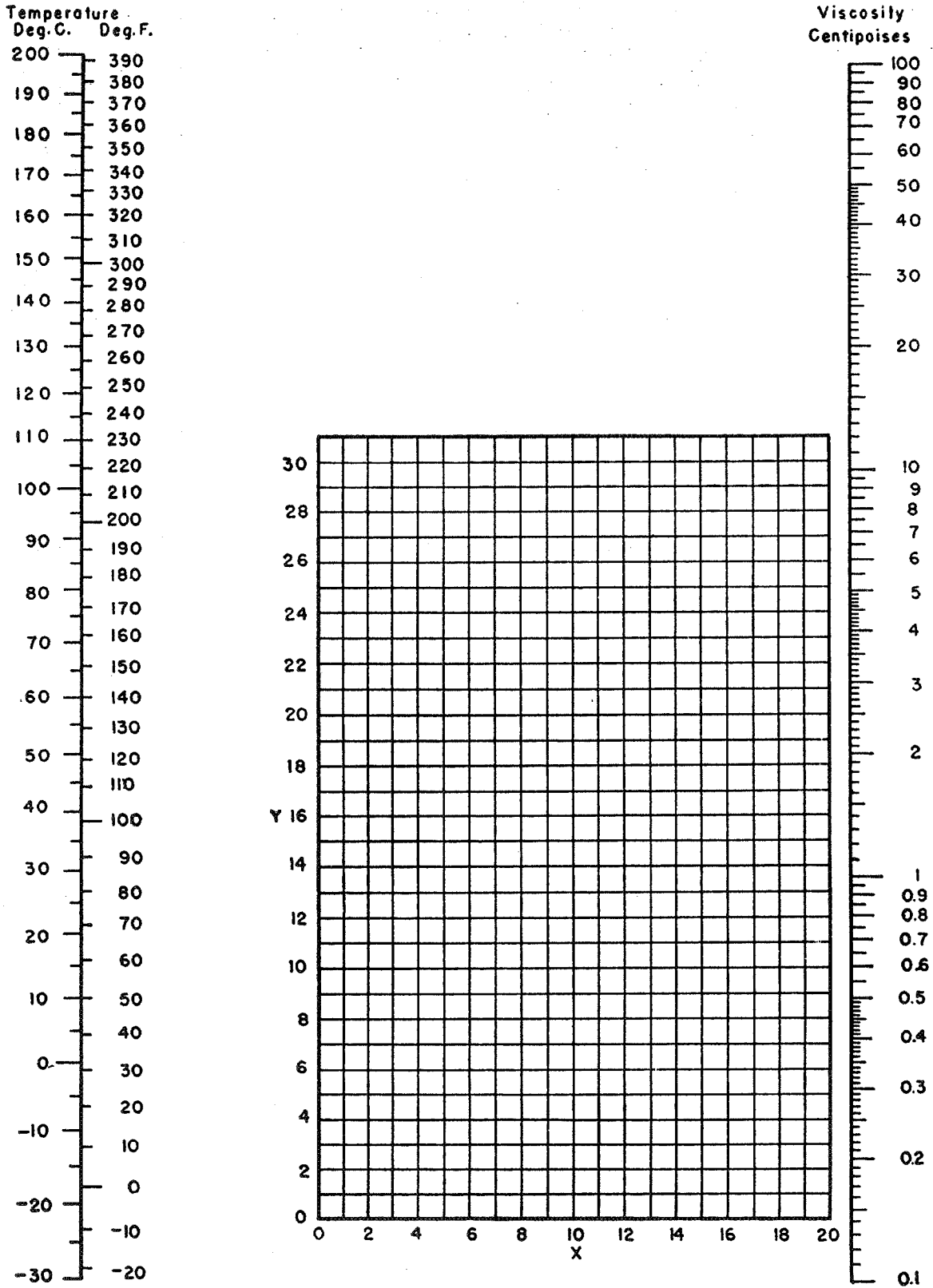


FIG. 14. Viscosities of liquids. (Perry, "Chemical Engineers' Handbook," 3d ed., McGraw Hill Book Company, Inc., New York, 1950.)

PROCESS HEAT TRANSFER

VISCOSITIES OF GASES*
Coordinates to be used with Fig. 15

Gas	X	Y
Acetic acid.....	7.7	14.3
Acetone.....	8.9	13.0
Acetylene.....	9.8	14.9
Air.....	11.0	20.0
Ammonia.....	8.4	16.0
Argon.....	10.5	22.4
Benzene.....	8.5	13.2
Bromine.....	8.9	19.2
Butene.....	9.2	13.7
Butylene.....	8.9	13.0
Carbon dioxide.....	9.5	18.7
Carbon disulfide.....	8.0	16.0
Carbon monoxide.....	11.0	20.0
Chlorine.....	9.0	18.4
Chloroform.....	8.9	15.7
Cyanogen.....	9.2	15.2
Cyclohexane.....	9.2	12.0
Ethane.....	9.1	14.5
Ethyl acetate.....	8.5	13.2
Ethyl alcohol.....	9.2	14.2
Ethyl chloride.....	8.5	15.6
Ethyl ether.....	8.9	13.0
Ethylene.....	9.5	15.1
Fluorine.....	7.3	23.8
Freon-11.....	10.6	15.1
Freon-12.....	11.1	16.0
Freon-21.....	10.8	15.3
Freon-22.....	10.1	17.0
Freon-113.....	11.3	14.0
Helium.....	10.9	20.5
Hexane.....	8.6	11.8
Hydrogen.....	11.2	12.4
3H ₂ + 1N ₂	11.2	17.2
Hydrogen bromide.....	8.8	20.9
Hydrogen chloride.....	8.8	18.7
Hydrogen cyanide.....	9.8	14.9
Hydrogen iodide.....	9.0	21.3
Hydrogen sulfide.....	8.6	18.0
Iodine.....	9.0	18.4
Mercury.....	5.3	22.9
Methane.....	9.9	15.5
Methyl alcohol.....	8.5	15.6
Nitric oxide.....	10.9	20.5
Nitrogen.....	10.6	20.0
Nitrosyl chloride.....	8.0	17.6
Nitrous oxide.....	8.8	19.0
Oxygen.....	11.0	21.3
Pentane.....	7.0	12.8
Propane.....	9.7	12.9
Propyl alcohol.....	8.4	13.4
Propylene.....	9.0	13.8
Sulfur dioxide.....	9.6	17.0
Toluene.....	8.6	12.4
2, 3, 3-Trimethylbutane.....	9.5	10.5
Water.....	8.0	16.0
Xenon.....	9.3	23.0

* From Perry, J. H., "Chemical Engineers' Handbook," 3d ed., McGraw-Hill Book Company, Inc., New York, 1950.

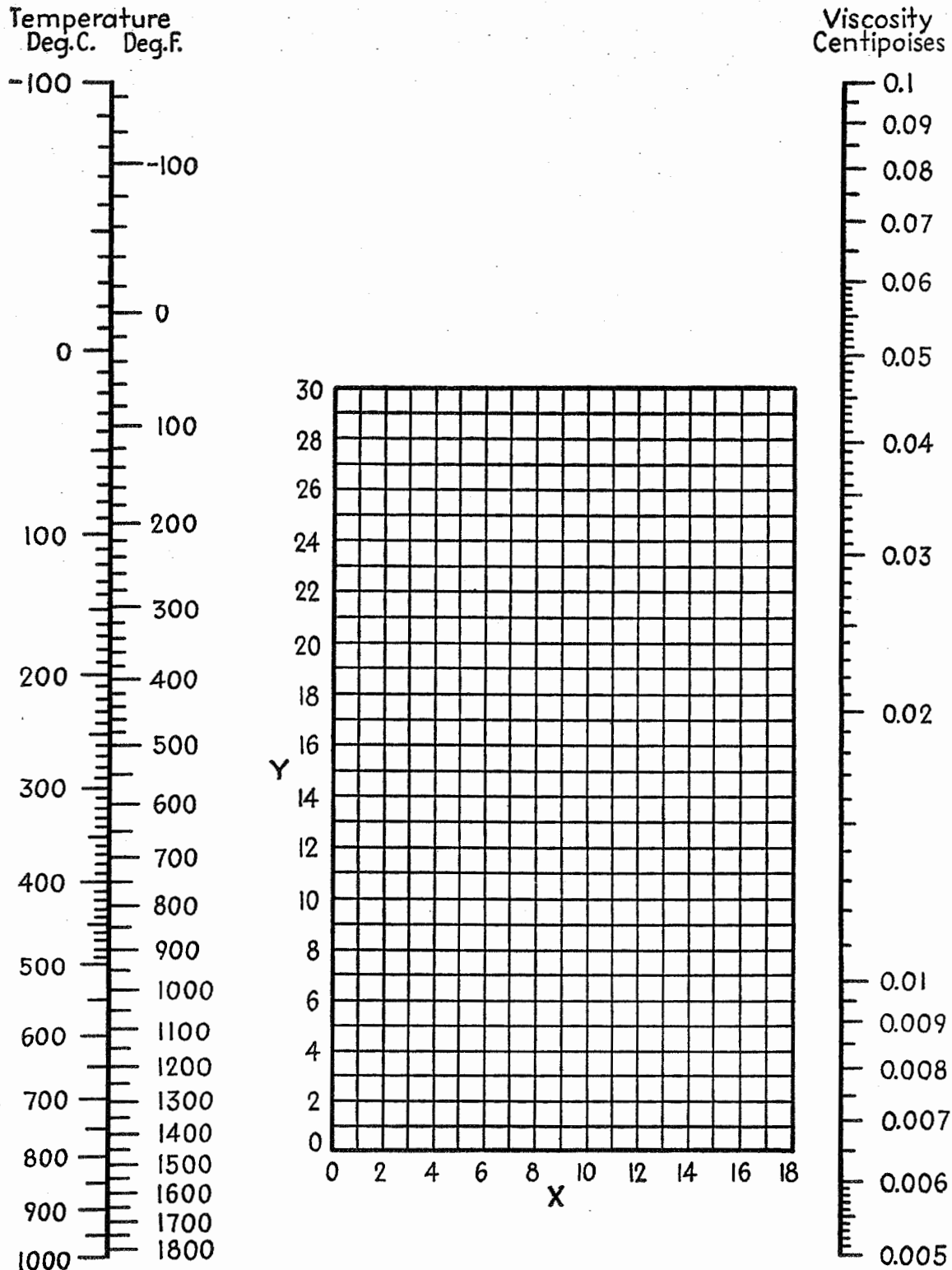


FIG. 15. Viscosities of gases. (Perry, "Chemical Engineers' Handbook," 3d ed., McGraw-Hill Book Company, Inc., New York, 1950.)

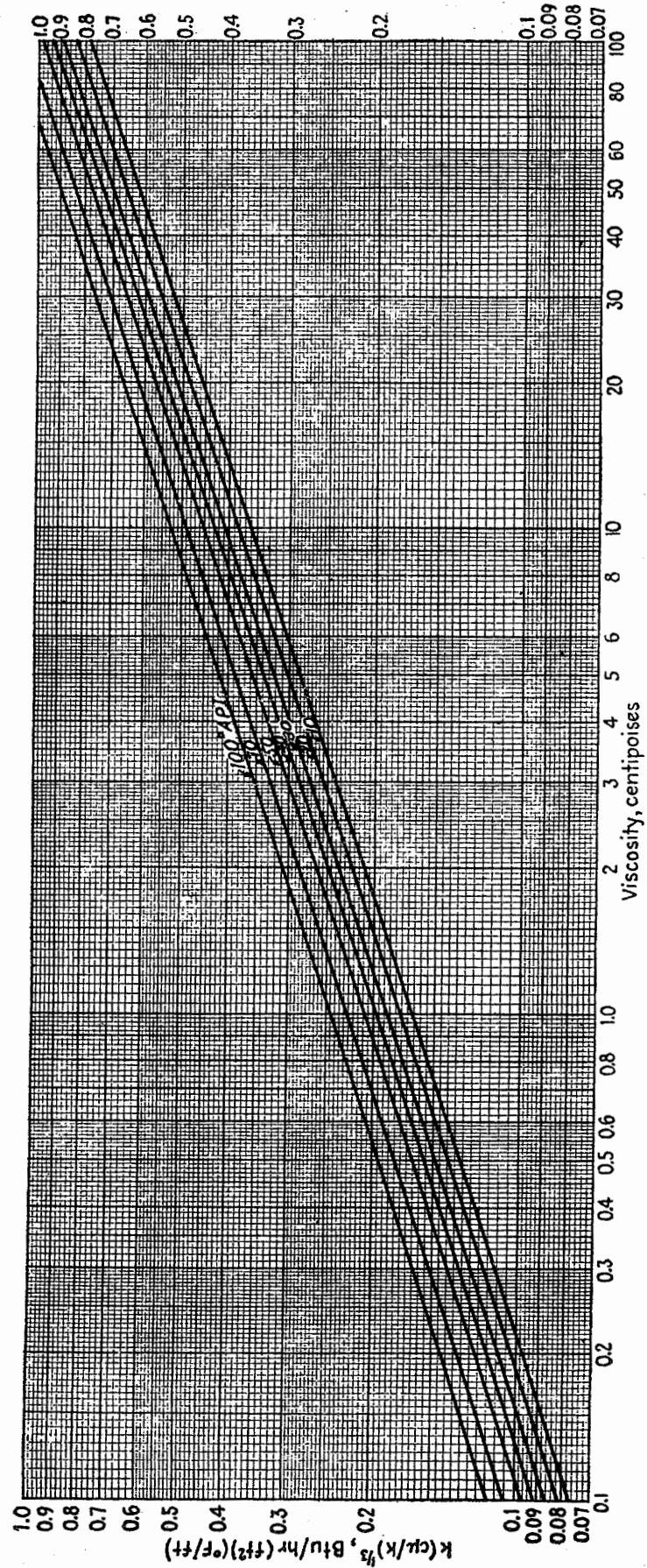


FIG. 16. Values of $k(c\mu/k)^{1/3}$ for hydrocarbons.

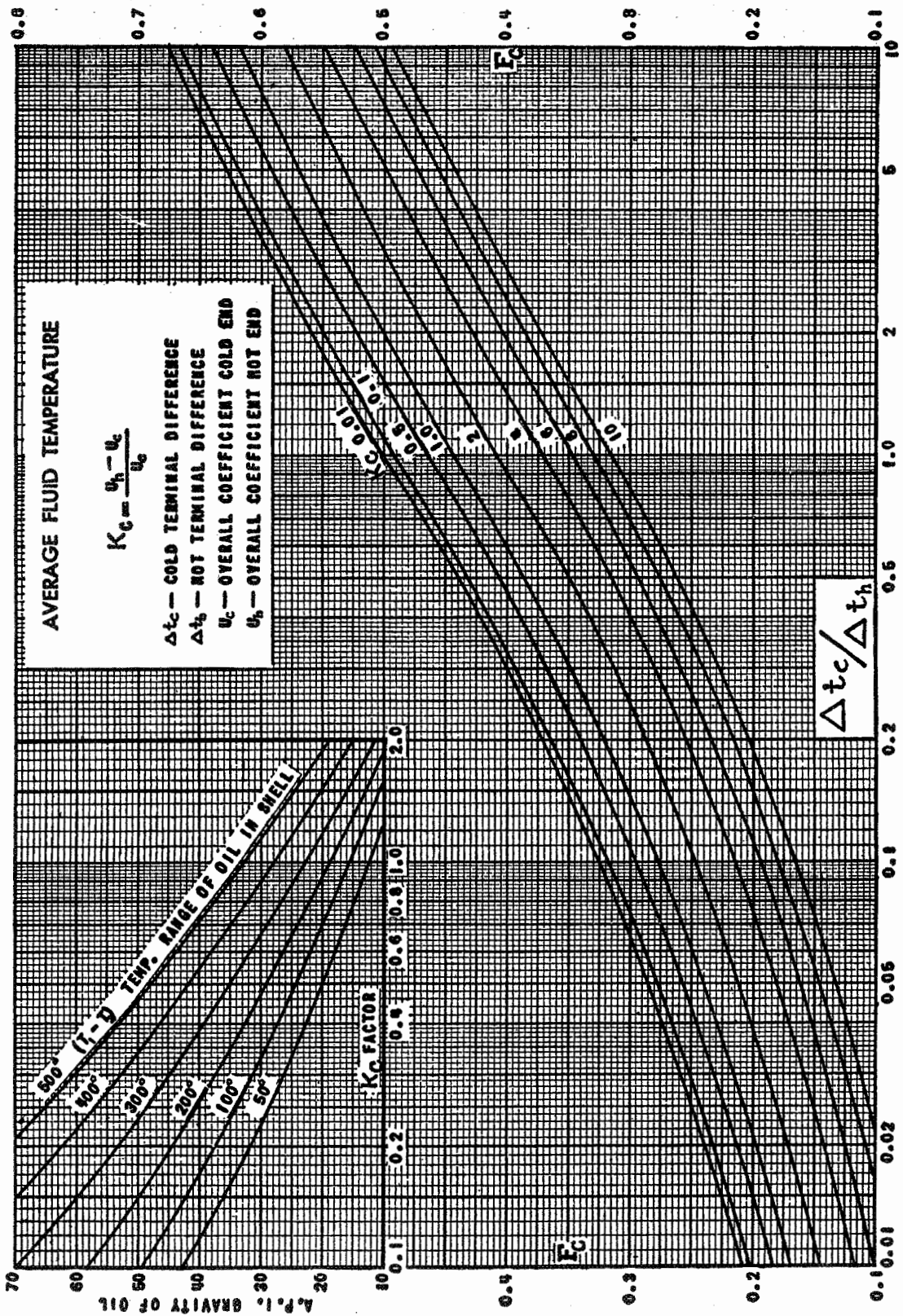


FIG. 17. The caloric temperature factor F_c . (Standards of Tubular Exchanger Manufacturers Association, 2d ed., New York, 1949.)

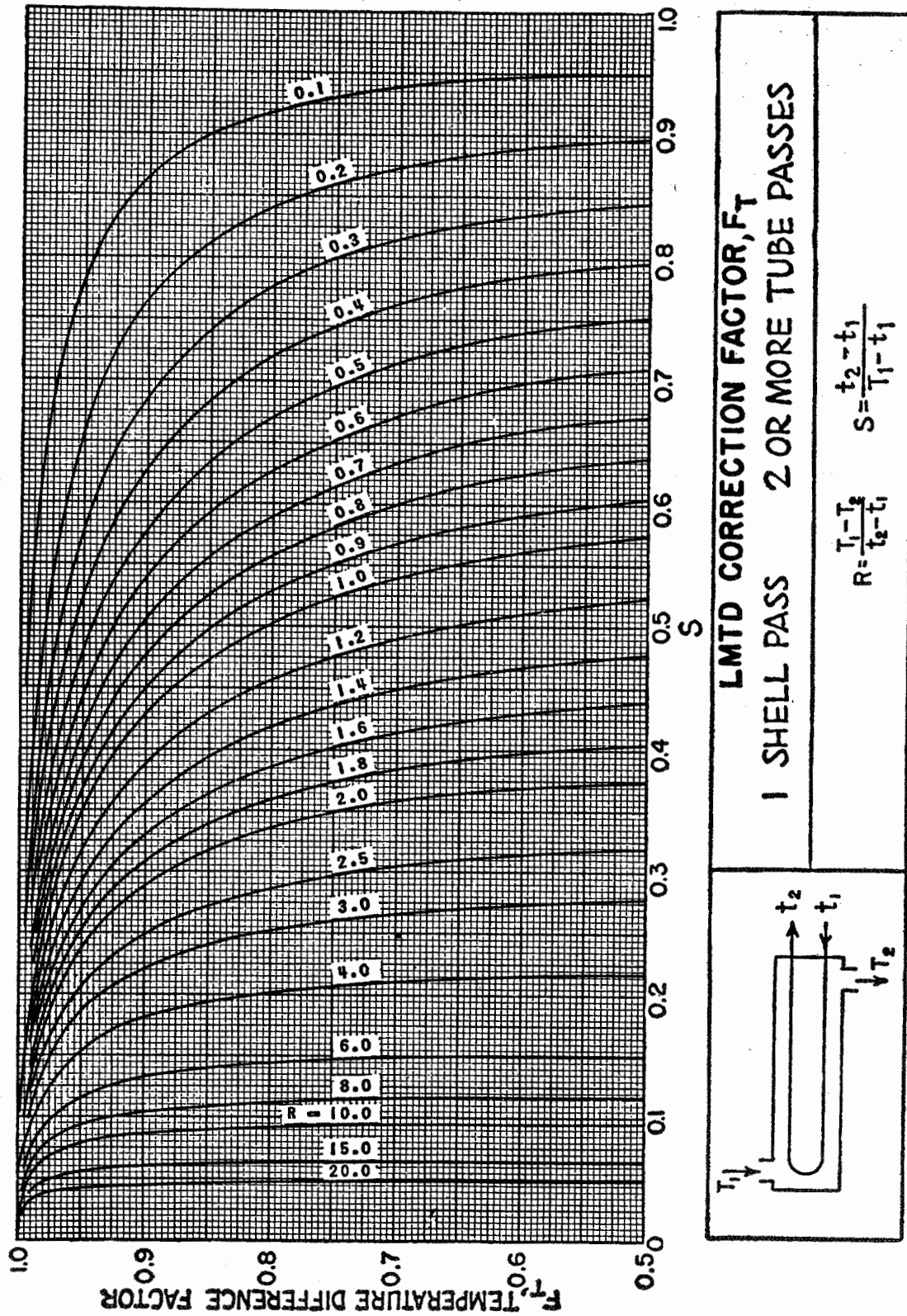


Fig. 18. LMTD correction factors for 1-2 exchangers. (Standards of Tubular Exchanger Manufacturers Association, 2d ed., New York, 1949.)

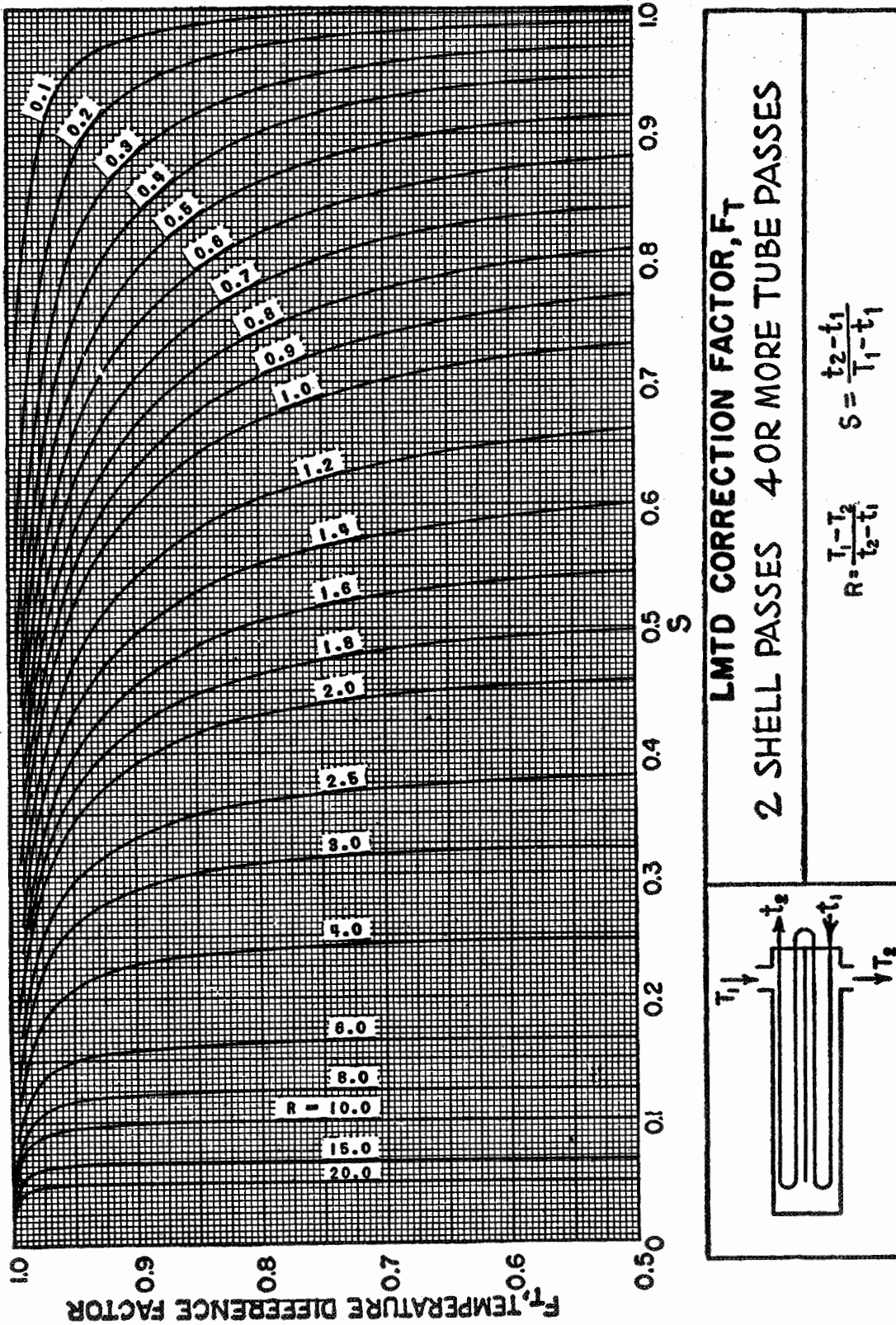


FIG. 19. LMTD correction factors for 2-4 exchangers. (Standards of Tubular Exchanger Manufacturers Association 2d ed., New York, 1949.)

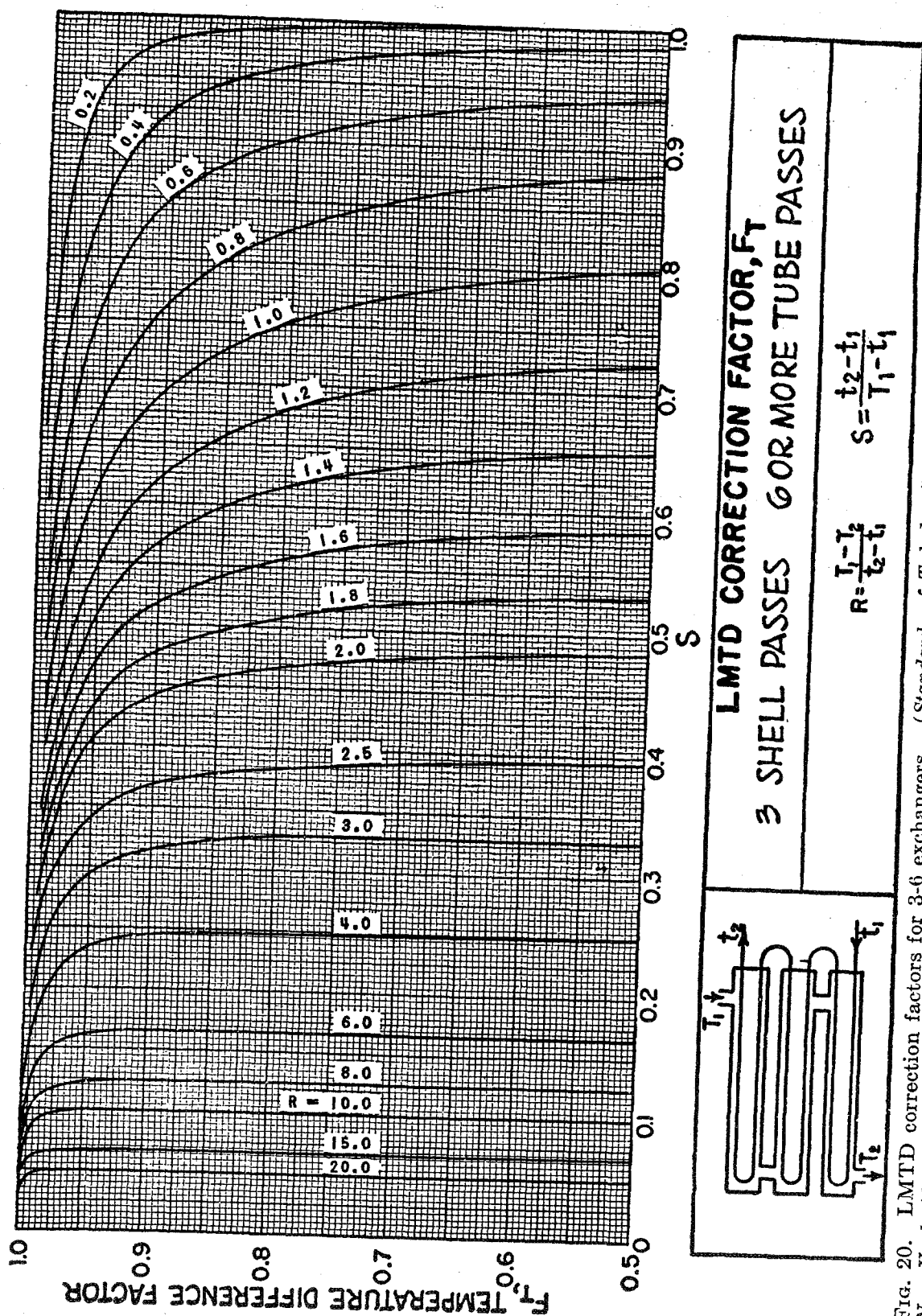
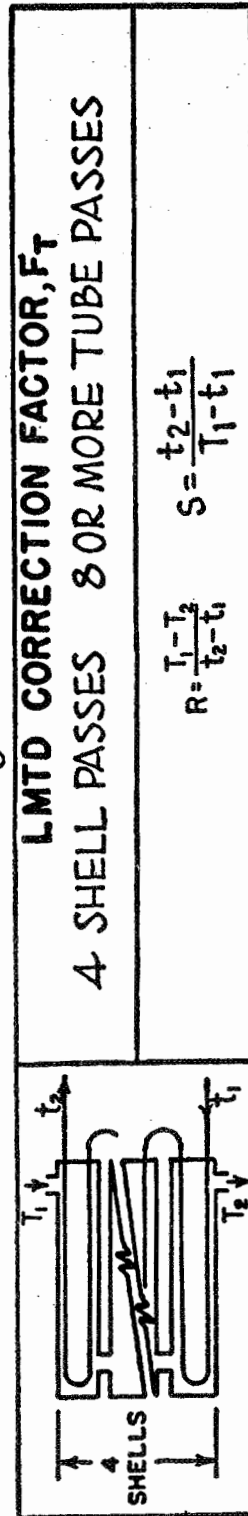
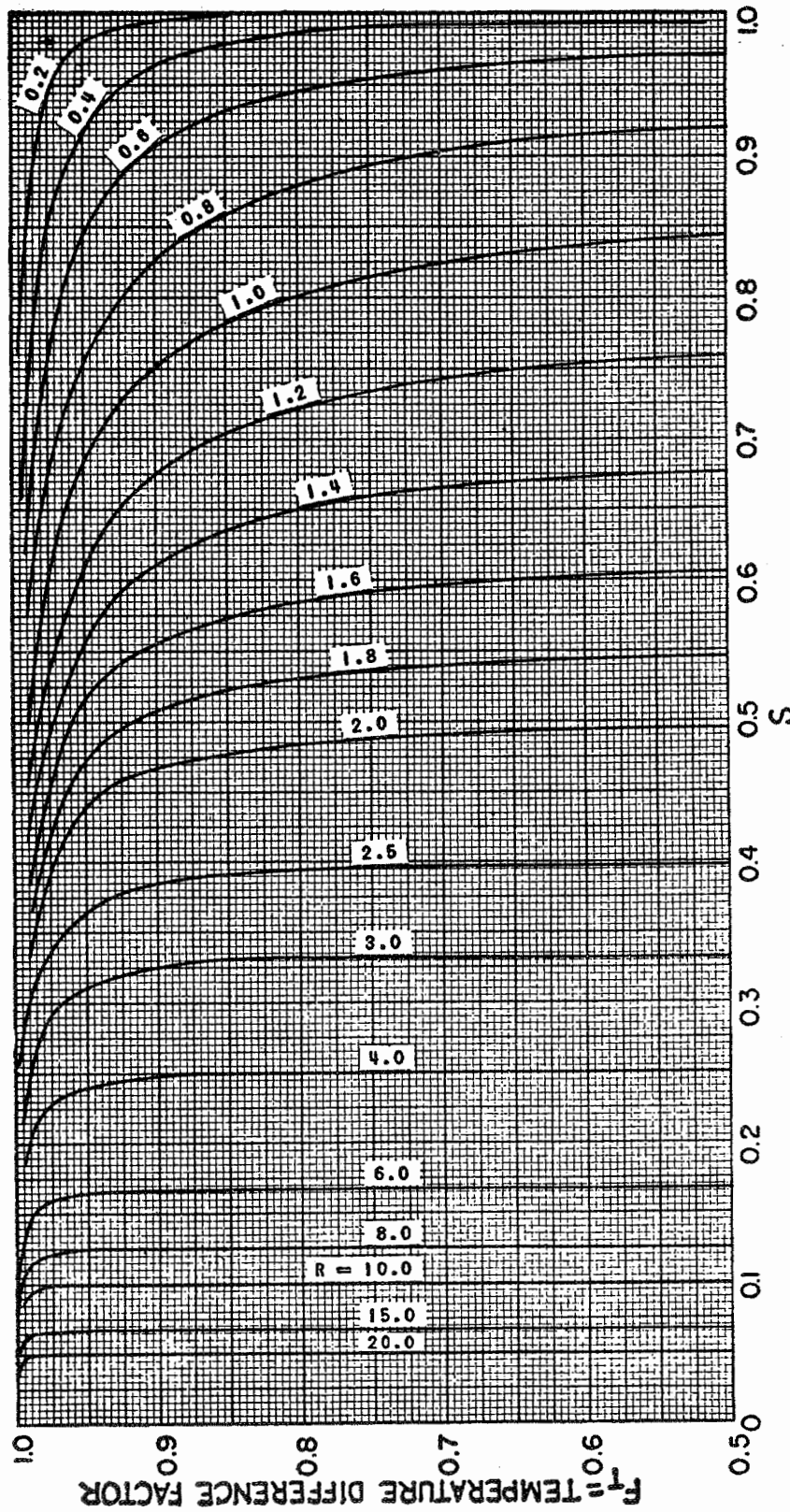


Fig. 20. LMTD correction factors for 3-6 exchangers. (Standards of Tubular Exchanger Manufacturers Association, 2d ed., New York, 1949.)



LMTD CORRECTION FACTOR, F_t
4 SHELL PASSES 8 OR MORE TUBE PASSES

$$R = \frac{T_1 - T_2}{t_2 - t_1} \quad S = \frac{t_2 - t_1}{T_1 - t_1}$$

FIG. 21. LMTD correction factors for 4-8 exchangers. (Standards of Tubular Exchanger Manufacturers Association, 2d ed., New York, 1949.)

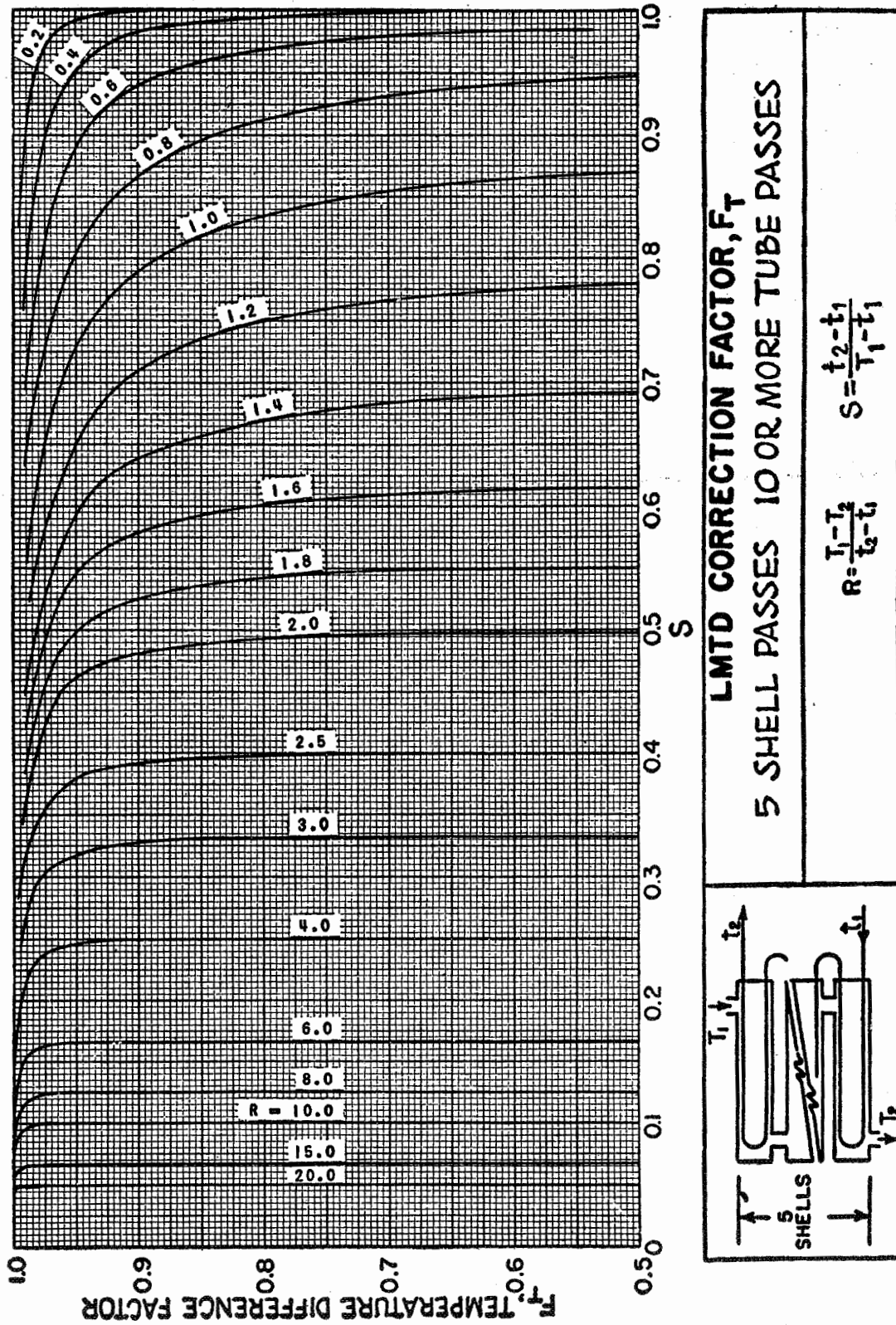


Fig. 22. LMTD correction factors for 5-10 exchangers. (Standards of Tubular Exchanger Manufacturers Association, 2d ed., New York, 1949.)

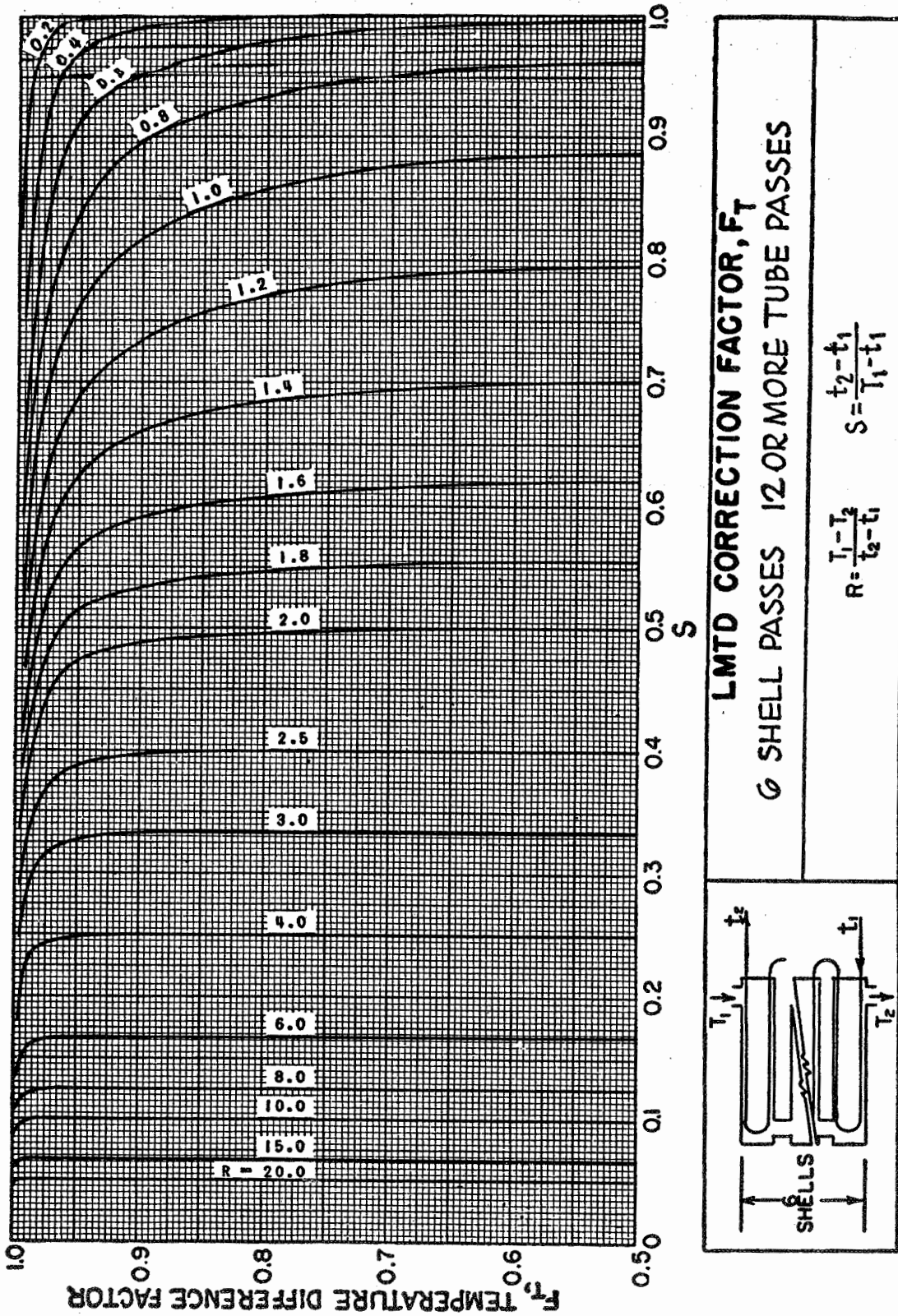


FIG. 23. LMTD correction factors for 6-12 exchangers. (Standards of Tubular Exchanger Manufacturers Association, 2d ed., New York, 1949.)

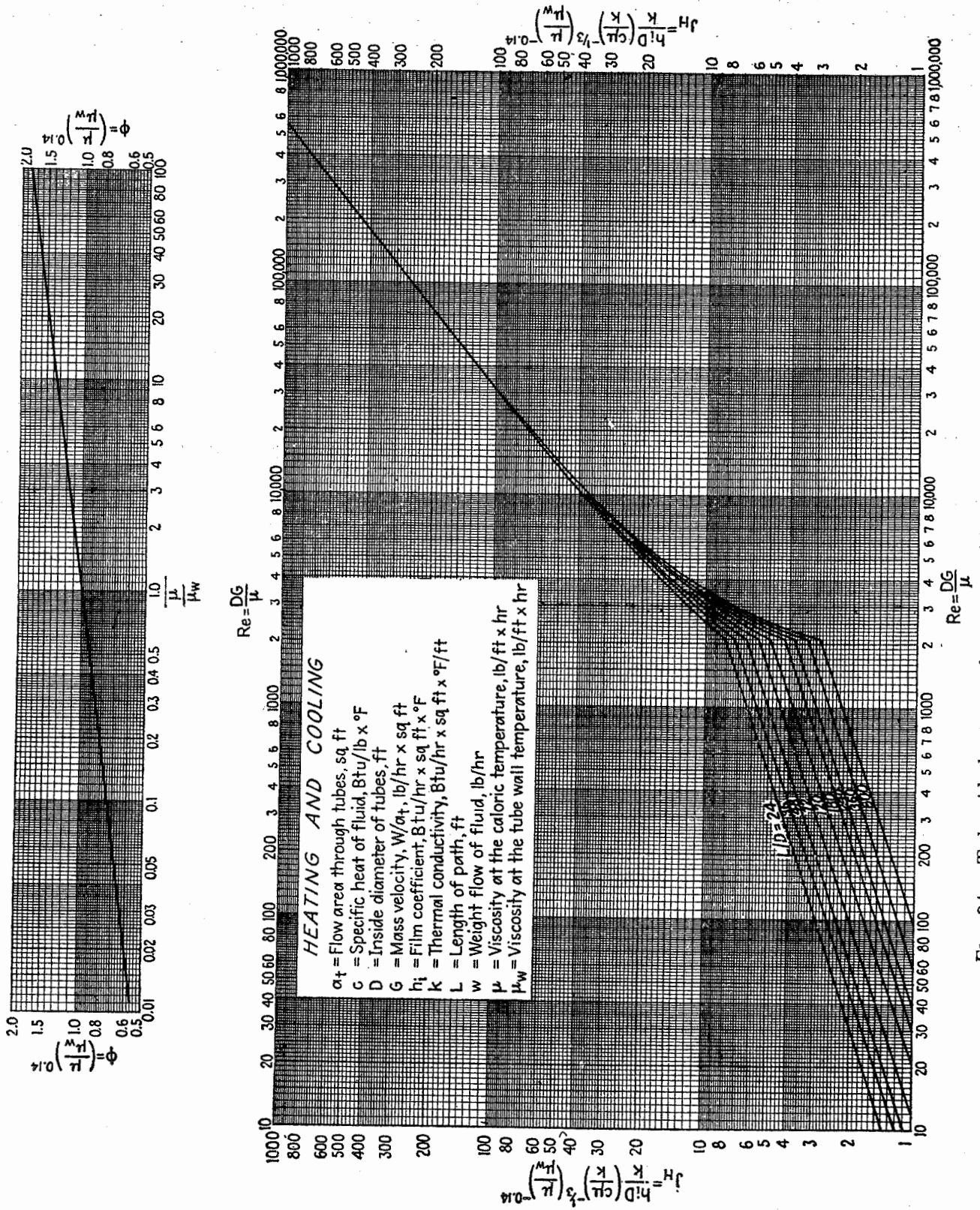


Fig. 24. Tube-side heat-transfer curve. (Adapted from Sieder and Tate.)

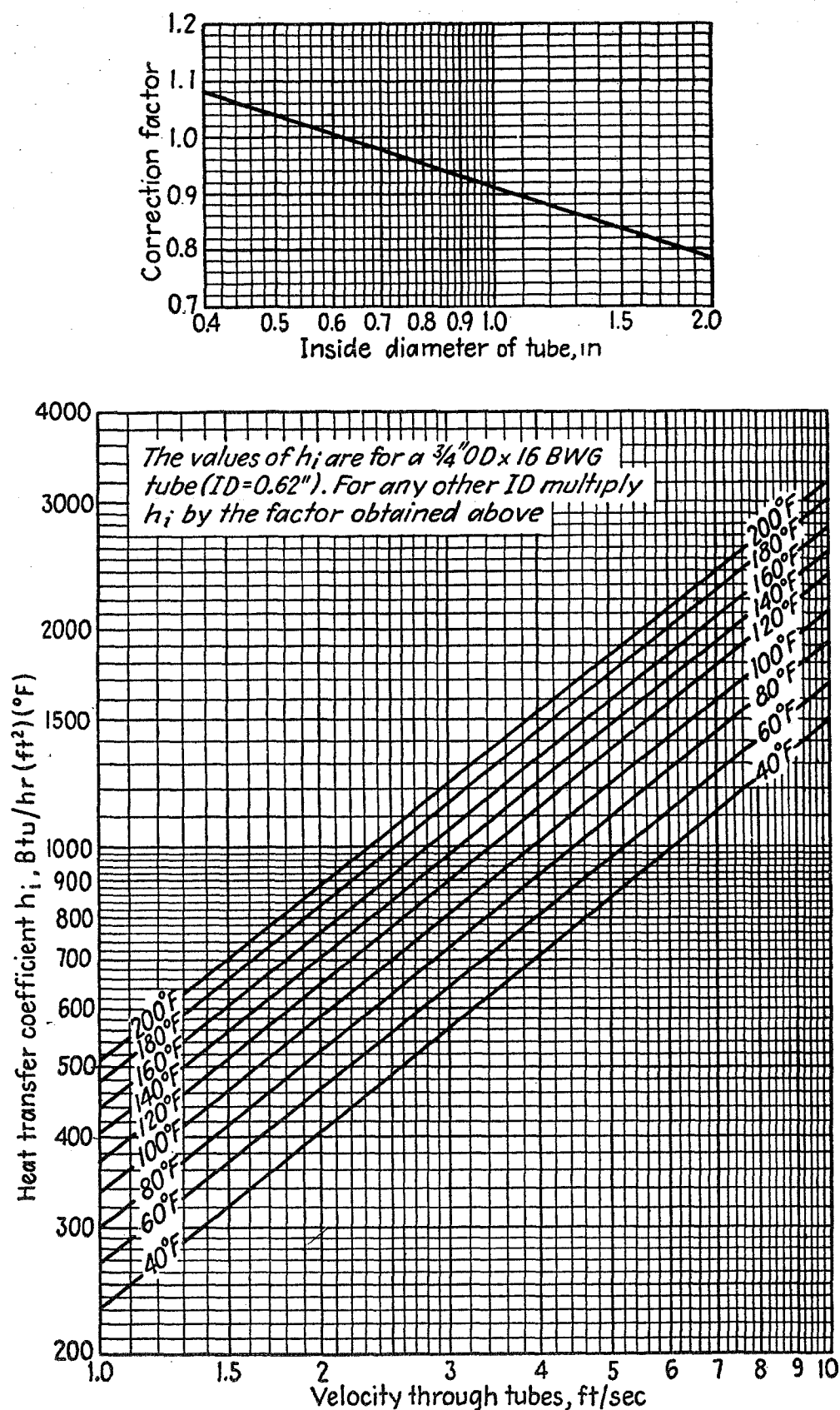


FIG. 25. Tube-side water-heat-transfer curve. [Adapted from Eagle and Ferguson, *Proc. Roy. Soc.*, A127, 540 (1930).]

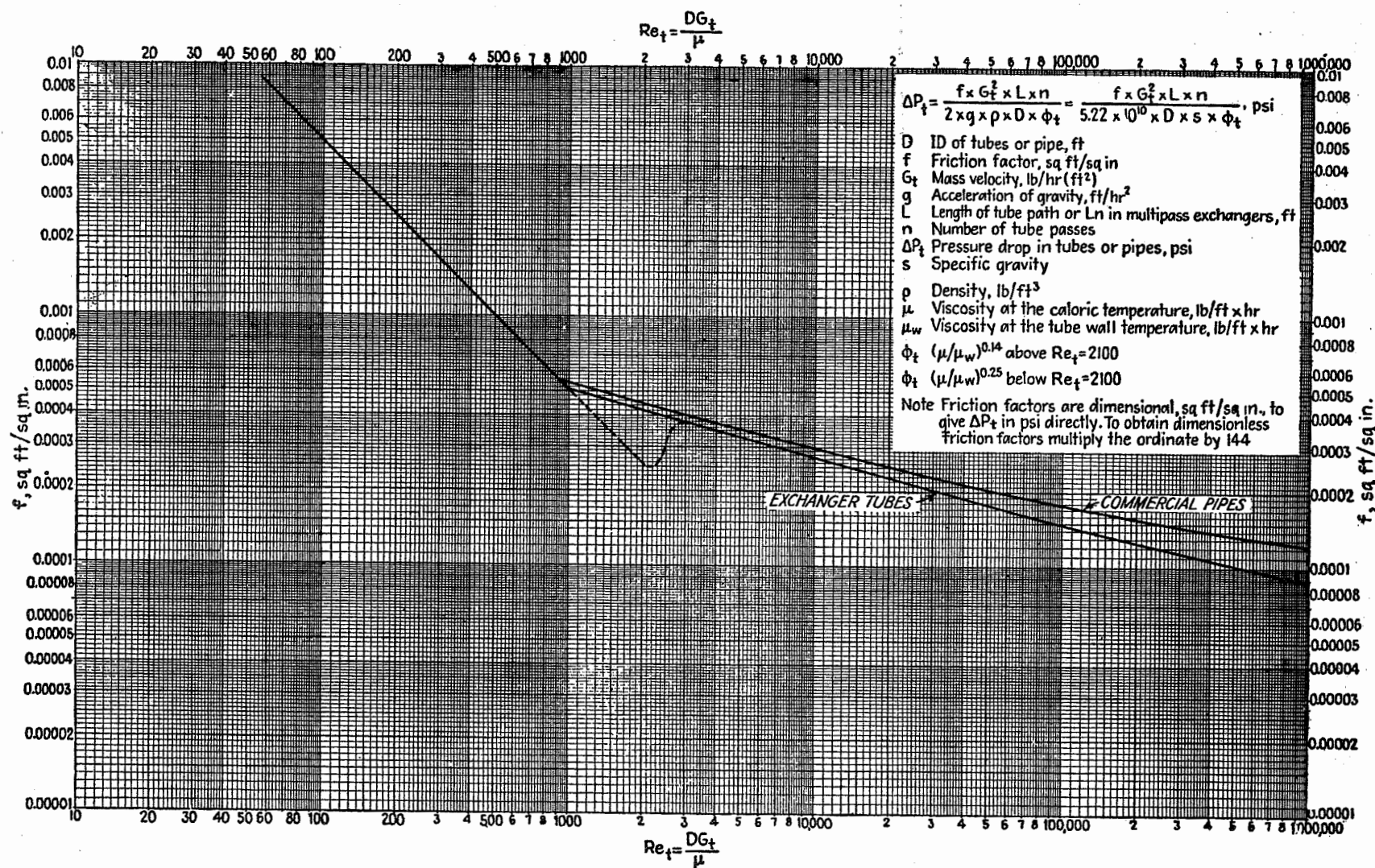


FIG. 26. Tube-side friction factors. (Standards of Tubular Exchanger Manufacturers Association, 2d ed., New York, 1949.)

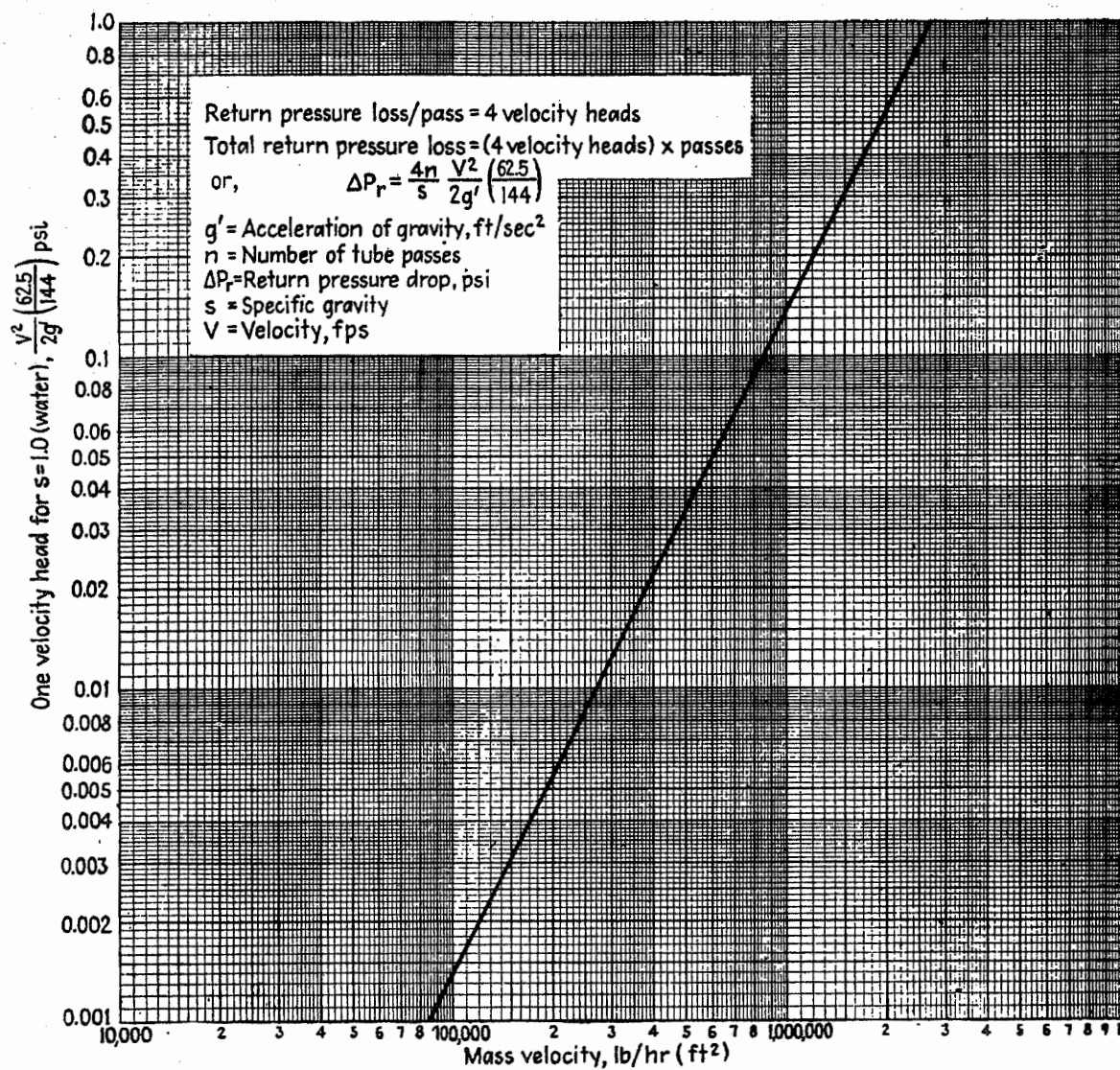


FIG. 27. Tube-side return pressure loss.

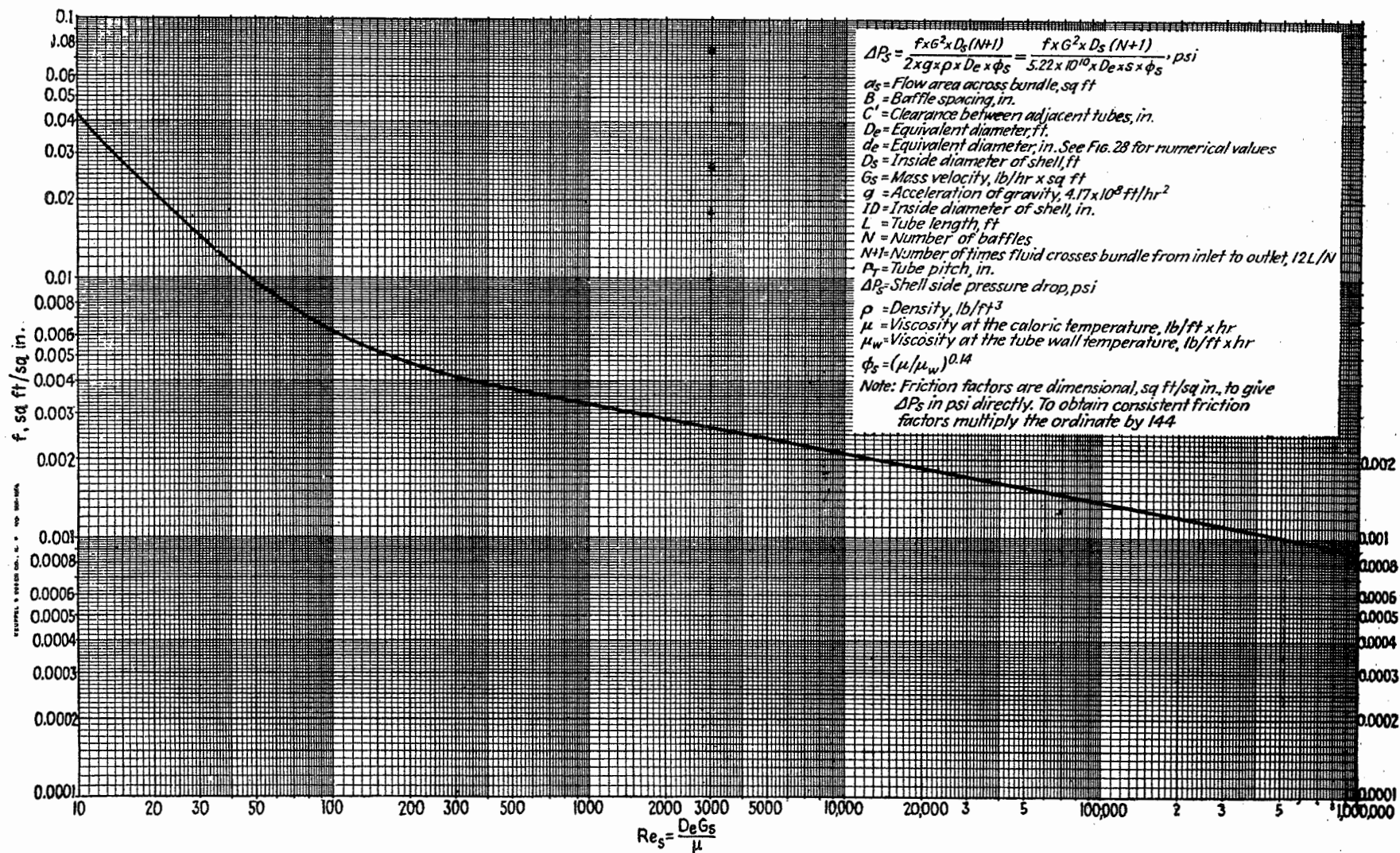


FIG. 29. Shell-side friction factors for bundles with 25 % cut segmental baffles.

TABLE 8. APPROXIMATE OVERALL DESIGN COEFFICIENTS
 Values include total dirt factors of 0.003 and allowable pressure drops of 5 to 10 psi on the controlling stream

Coolers		
Hot fluid	Cold fluid	Overall U_D
Water	Water	250–500§
Methanol	Water	250–500§
Ammonia	Water	250–500§
Aqueous solutions	Water	250–500§
Light organics*	Water	75–150
Medium organics†	Water	50–125
Heavy organics‡	Water	5–75
Gases	Water	2–50¶
Water	Brine	100–200
Light organics	Brine	40–100
Heaters		
Hot fluid	Cold fluid	Overall U_D
Steam	Water	200–700§
Steam	Methanol	200–700§
Steam	Ammonia	200–700§
Steam	Aqueous solutions:	
Steam	Less than 2.0 cp	200–700
Steam	More than 2.0 cp	100–500§
Steam	Light organics	100–200
Steam	Medium organics	50–100
Steam	Heavy organics	6–60
Steam	Gases	5–50¶
Exchangers		
Hot fluid	Cold fluid	Overall U_D
Water	Water	250–500§
Aqueous solutions	Aqueous solutions	250–500§
Light organics	Light organics	40–75
Medium organics	Medium organics	20–60
Heavy organics	Heavy organics	10–40
Heavy organics	Light organics	30–60
Light organics	Heavy organics	10–40

* *Light organics* are fluids with viscosities of less than 0.5 centipoise and include benzene, toluene, acetone, ethanol, methyl ethyl ketone, gasoline, light kerosene, and naphtha.

† *Medium organics* have viscosities of 0.5 to 1.0 centipoise and include kerosene, straw oil, hot gas oil, hot absorber oil, and some crudes.

‡ *Heavy organics* have viscosities above 1.0 centipoise and include cold gas oil, lube oils, fuel oils, reduced crude oils, tars, and asphalts.

§ Dirt factor 0.001.

|| Pressure drop 20 to 30 psi.

¶ These rates are greatly influenced by the operating pressure.

TABLE 9. TUBE-SHEET LAYOUTS (TUBE COUNTS)
Square Pitch

$\frac{3}{4}$ in. OD tubes on 1-in. square pitch						1 in. OD tubes on $1\frac{1}{4}$ -in. square pitch					
Shell ID, in.	1-P	2-P	4-P	6-P	8-P	Shell ID, in.	1-P	2-P	4-P	6-P	8-P
8	32	26	20	20		8	21	16	14		
10	52	52	40	36		10	32	32	26	24	
12	81	76	68	68	60	12	48	45	40	38	36
$13\frac{1}{4}$	97	90	82	76	70	$13\frac{1}{4}$	61	56	52	48	44
$15\frac{1}{4}$	137	124	116	108	108	$15\frac{1}{4}$	81	76	68	68	64
$17\frac{1}{4}$	177	166	158	150	142	$17\frac{1}{4}$	112	112	96	90	82
$19\frac{1}{4}$	224	220	204	192	188	$19\frac{1}{4}$	138	132	128	122	116
$21\frac{1}{4}$	277	270	246	240	234	$21\frac{1}{4}$	177	166	158	152	148
$23\frac{1}{4}$	341	324	308	302	292	$23\frac{1}{4}$	213	208	192	184	184
25	413	394	376	356	346	25	260	252	238	226	222
27	481	460	432	420	408	27	300	288	278	268	260
29	553	526	480	468	456	29	341	326	300	294	286
31	657	640	600	580	560	31	406	398	380	368	358
33	749	718	688	676	648	33	465	460	432	420	414
35	845	824	780	766	748	35	522	518	488	484	472
37	934	914	886	866	838	37	596	574	562	544	532
39	1049	1024	982	968	948	39	665	644	624	612	600

$1\frac{1}{4}$ in. OD tubes on $1\frac{1}{6}$ -in. square pitch						$1\frac{1}{2}$ in. OD tubes on $1\frac{7}{8}$ -in. square pitch					
10	16	12	10			12	16	16	12	12	
12	30	24	22	16	16	12	16	16	12	12	
$13\frac{1}{4}$	32	30	30	22	22	$13\frac{1}{4}$	22	22	16	16	
$15\frac{1}{4}$	44	40	37	35	31	$15\frac{1}{4}$	29	29	25	24	22
$17\frac{1}{4}$	56	53	51	48	44	$17\frac{1}{4}$	39	39	34	32	29
$19\frac{1}{4}$	78	73	71	64	56	$19\frac{1}{4}$	50	48	45	43	39
$21\frac{1}{4}$	96	90	86	82	78	$21\frac{1}{4}$	62	60	57	54	50
$23\frac{1}{4}$	127	112	106	102	96	$23\frac{1}{4}$	78	74	70	66	62
25	140	135	127	123	115	25	94	90	86	84	78
27	166	160	151	146	140	27	112	108	102	98	94
29	193	188	178	174	166	29	131	127	120	116	112
31	226	220	209	202	193	31	151	146	141	138	131
33	258	252	244	238	226	33	176	170	164	160	151
35	293	287	275	268	258	35	202	196	188	182	176
37	334	322	311	304	293	37	224	220	217	210	202
39	370	362	348	342	336	39	252	246	237	230	224

TABLE 9. TUBE-SHEET LAYOUTS (TUBE COUNTS).—(Continued)
Triangular Pitch

$\frac{3}{4}$ in. OD tubes on $\frac{15}{16}$ -in. triangular pitch						$\frac{3}{4}$ in. OD tubes on 1-in. triangular pitch					
Shell ID, in.	1-P	2-P	4-P	6-P	8-P	Shell ID, in.	1-P	2-P	4-P	6-P	8-P
8	36	32	26	24	18	8	37	30	24	24	
10	62	56	47	42	36	10	61	52	40	36	
12	109	98	86	82	78	12	92	82	76	74	70
$13\frac{1}{4}$	127	114	96	90	86	$13\frac{1}{4}$	109	106	86	82	74
$15\frac{1}{4}$	170	160	140	136	128	$15\frac{1}{4}$	151	138	122	118	110
$17\frac{1}{4}$	239	224	194	188	178	$17\frac{1}{4}$	203	196	178	172	166
$19\frac{1}{4}$	301	282	252	244	234	$19\frac{1}{4}$	262	250	226	216	210
$21\frac{1}{4}$	361	342	314	306	290	$21\frac{1}{4}$	316	302	278	272	260
$23\frac{1}{4}$	442	420	386	378	364	$23\frac{1}{4}$	384	376	352	342	328
25	532	506	468	446	434	25	470	452	422	394	382
27	637	602	550	536	524	27	559	534	488	474	464
29	721	692	640	620	594	29	630	604	556	538	508
31	847	822	766	722	720	31	745	728	678	666	640
33	974	938	878	852	826	33	856	830	774	760	732
35	1102	1068	1004	988	958	35	970	938	882	864	848
37	1240	1200	1144	1104	1072	37	1074	1044	1012	986	870
39	1377	1330	1258	1248	1212	39	1206	1176	1128	1100	1078
1 in. OD tubes on $1\frac{1}{4}$ -in. triangular pitch						$1\frac{1}{4}$ in. OD tubes on $1\frac{9}{16}$ -in. triangular pitch					
8	21	16	16	14		10	20	18	14		
10	32	32	26	24		12	32	30	26	22	20
12	55	52	48	46	44	$13\frac{1}{4}$	38	36	32	28	26
$13\frac{1}{4}$	68	66	58	54	50	$15\frac{1}{4}$	54	51	45	42	38
$15\frac{1}{4}$	91	86	80	74	72	$17\frac{1}{4}$	69	66	62	58	54
$17\frac{1}{4}$	131	118	106	104	94	$19\frac{1}{4}$	95	91	86	78	69
$19\frac{1}{4}$	163	152	140	136	128	$21\frac{1}{4}$	117	112	105	101	95
$21\frac{1}{4}$	199	188	170	164	160	$23\frac{1}{4}$	140	136	130	123	117
$23\frac{1}{4}$	241	232	212	212	202	25	170	164	155	150	140
25	294	282	256	252	242	27	202	196	185	179	170
27	349	334	302	296	286	29	235	228	217	212	202
29	397	376	338	334	316	31	275	270	255	245	235
31	472	454	430	424	400	33	315	305	297	288	275
33	538	522	486	470	454	35	357	348	335	327	315
35	608	592	562	546	532	37	407	390	380	374	357
37	674	664	632	614	598	39	449	436	425	419	407
39	766	736	700	688	672						
$1\frac{1}{2}$ in. OD tubes on $1\frac{7}{8}$ -in. triangular pitch											
12	18	14	14	12	12						
$13\frac{1}{4}$	27	22	18	16	14						
$15\frac{1}{4}$	36	34	32	30	27						
$17\frac{1}{4}$	48	44	42	38	36						
$19\frac{1}{4}$	61	58	55	51	48						
$21\frac{1}{4}$	76	72	70	66	61						
$23\frac{1}{4}$	95	91	86	80	76						
25	115	110	105	98	95						
27	136	131	125	118	115						
29	160	154	147	141	136						
31	184	177	172	165	160						
33	215	206	200	190	184						
35	246	238	230	220	215						
37	275	268	260	252	246						
39	307	299	290	284	275						

TABLE 10. HEAT EXCHANGER AND CONDENSER TUBE DATA

Tube OD, in.	BWG	Wall thick- ness, in.	ID, in.	Flow area per tube, in. ²	Surface per lin ft, ft ²		Weight per lin ft, lb steel
					Outside	Inside	
$\frac{1}{2}$	12	0.109	0.282	0.0625	0.1309	0.0748	0.493
	14	0.083	0.334	0.0876		0.0874	0.403
	16	0.065	0.370	0.1076		0.0969	0.329
	18	0.049	0.402	0.127		0.1052	0.258
	20	0.035	0.430	0.145		0.1125	0.190
$\frac{3}{4}$	10	0.134	0.482	0.182	0.1963	0.1263	0.965
	11	0.120	0.510	0.204		0.1335	0.884
	12	0.109	0.532	0.223		0.1393	0.817
	13	0.095	0.560	0.247		0.1466	0.727
	14	0.083	0.584	0.268		0.1529	0.647
	15	0.072	0.606	0.289		0.1587	0.571
	16	0.065	0.620	0.302		0.1623	0.520
	17	0.058	0.634	0.314		0.1660	0.469
	18	0.049	0.652	0.334		0.1707	0.401
1	8	0.165	0.670	0.355	0.2618	0.1754	1.61
	9	0.148	0.704	0.389		0.1843	1.47
	10	0.134	0.732	0.421		0.1916	1.36
	11	0.120	0.760	0.455		0.1990	1.23
	12	0.109	0.782	0.479		0.2048	1.14
	13	0.095	0.810	0.515		0.2121	1.00
	14	0.083	0.834	0.546		0.2183	0.890
	15	0.072	0.856	0.576		0.2241	0.781
	16	0.065	0.870	0.594		0.2277	0.710
	17	0.058	0.884	0.613		0.2314	0.639
	18	0.049	0.902	0.639		0.2361	0.545
$1\frac{1}{4}$	8	0.165	0.920	0.665	0.3271	0.2409	2.09
	9	0.148	0.954	0.714		0.2498	1.91
	10	0.134	0.982	0.757		0.2572	1.75
	11	0.120	1.01	0.800		0.2644	1.58
	12	0.109	1.03	0.836		0.2701	1.45
	13	0.095	1.06	0.884		0.2775	1.28
	14	0.083	1.08	0.923		0.2839	1.13
	15	0.072	1.11	0.960		0.2896	0.991
	16	0.065	1.12	0.985		0.2932	0.900
	17	0.058	1.13	1.01		0.2969	0.808
	18	0.049	1.15	1.04		0.3015	0.688
$1\frac{1}{2}$	8	0.165	1.17	1.075	0.3925	0.3063	2.57
	9	0.148	1.20	1.14		0.3152	2.34
	10	0.134	1.23	1.19		0.3225	2.14
	11	0.120	1.26	1.25		0.3299	1.98
	12	0.109	1.28	1.29		0.3356	1.77
	13	0.095	1.31	1.35		0.3430	1.56
	14	0.083	1.33	1.40		0.3492	1.37
	15	0.072	1.36	1.44		0.3555	1.20
	16	0.065	1.37	1.47		0.3587	1.09
	17	0.058	1.38	1.50		0.3623	0.978
	18	0.049	1.40	1.54		0.3670	0.831

TABLE 11. DIMENSIONS OF STEEL PIPE (IPS)

Nominal pipe size, IPS, in.	OD, in.	Schedule No.	ID, in.	Flow area per pipe, in. ²	Surface per lin ft, ft. ² /ft.		Weight per lin ft, lb steel
					Outside	Inside	
$\frac{1}{8}$	0.405	40*	0.269	0.058	0.106	0.070	0.25
		80†	0.215	0.036		0.056	0.32
$\frac{1}{4}$	0.540	40*	0.364	0.104	0.141	0.095	0.43
		80†	0.302	0.072		0.079	0.54
$\frac{3}{8}$	0.675	40*	0.493	0.192	0.177	0.129	0.57
		80†	0.423	0.141		0.111	0.74
$\frac{1}{2}$	0.840	40*	0.622	0.304	0.220	0.163	0.85
		80†	0.546	0.235		0.143	1.09
$\frac{3}{4}$	1.05	40*	0.824	0.534	0.275	0.216	1.13
		80†	0.742	0.432		0.194	1.48
1	1.32	40*	1.049	0.864	0.344	0.274	1.68
		80†	0.957	0.718		0.250	2.17
$1\frac{1}{4}$	1.66	40*	1.380	1.50	0.435	0.362	2.28
		80†	1.278	1.28		0.335	3.00
$1\frac{1}{2}$	1.90	40*	1.610	2.04	0.498	0.422	2.72
		80†	1.500	1.76		0.393	3.64
2	2.38	40*	2.067	3.35	0.622	0.542	3.66
		80†	1.939	2.95		0.508	5.03
$2\frac{1}{2}$	2.88	40*	2.469	4.79	0.753	0.647	5.80
		80†	2.323	4.23		0.609	7.67
3	3.50	40*	3.068	7.38	0.917	0.804	7.58
		80†	2.900	6.61		0.760	10.3
4	4.50	40*	4.026	12.7	1.178	1.055	10.8
		80†	3.826	11.5		1.002	15.0
6	6.625	40*	6.065	28.9	1.734	1.590	19.0
		80†	5.761	26.1		1.510	28.6
8	8.625	40*	7.981	50.0	2.258	2.090	28.6
		80†	7.625	45.7		2.000	43.4
10	10.75	40*	10.02	78.8	2.814	2.62	40.5
		60	9.75	74.6		2.55	54.8
12	12.75	30	12.09	115	3.338	3.17	43.8
14	14.0	30	13.25	138	3.665	3.47	54.6
16	16.0	30	15.25	183	4.189	4.00	62.6
18	18.0	20†	17.25	234	4.712	4.52	72.7
20	20.0	20	19.25	291	5.236	5.05	78.6
22	22.0	20†	21.25	355	5.747	5.56	84.0
24	24.0	20	23.25	425	6.283	6.09	94.7

* Commonly known as standard.

† Commonly known as extra heavy.

‡ Approximately.

TABLE 12. FOULING FACTORS*

Temperature of heating medium.....	Up to 240°F		240-400°F†	
Temperature of water.....	125°F or less		Over 125°F	
Water	Water velocity, fps		Water velocity, fps	
	3 ft and less	Over 3 ft	3 ft and less	Over 3 ft
Sea water.....	0.0005	0.0005	0.001	0.001
Brackish water.....	0.002	0.001	0.003	0.002
Cooling tower and artificial spray pond:				
Treated make-up.....	0.001	0.001	0.002	0.002
Untreated.....	0.003	0.003	0.005	0.004
City or well water (such as Great Lakes).....	0.001	0.001	0.002	0.002
Great Lakes.....	0.001	0.001	0.002	0.002
River water:				
Minimum.....	0.002	0.001	0.003	0.022
Mississippi.....	0.003	0.002	0.004	0.003
Delaware, Schuylkill.....	0.003	0.002	0.004	0.003
East River and New York Bay.....	0.003	0.002	0.004	0.003
Chicago sanitary canal.....	0.008	0.006	0.010	0.008
Muddy or silty.....	0.003	0.002	0.004	0.003
Hard (over 15 grains/gal).....	0.003	0.003	0.005	0.005
Engine jacket.....	0.001	0.001	0.001	0.001
Distilled.....	0.0005	0.0005	0.0005	0.0005
Treated boiler feedwater.....	0.001	0.0005	0.001	0.001
Boiler blowdown.....	0.002	0.002	0.002	0.002

† Ratings in the last two columns are based on a temperature of the heating medium of 240 to 400°F. If the heating medium temperature is over 400°F, and the cooling medium is known to scale these ratings should be modified accordingly.

Petroleum Fractions

Oils (industrial):

Fuel oil.....	0.005
Clean recirculating oil.....	0.001
Machinery and transformer oils.....	0.001
Quenching oil.....	0.004
Vegetable oils.....	0.003

Gases, vapors (industrial):

Coke-oven gas, manufactured gas.....	0.01
Diesel-engine exhaust gas.....	0.01
Organic vapors.....	0.0005
Steam (non-oil bearing).....	0.0
Alcohol vapors.....	0.0
Steam, exhaust (oil bearing from reciprocating engines).....	0.001
Refrigerating vapors (condensing from reciprocating compressors).....	0.002
Air.....	0.002

Liquids (industrial):

Organic.....	0.001
Refrigerating liquids, heating, cooling, or evaporating.....	0.001
Brine (cooling).....	0.001
Atmospheric distillation units:	
Residual bottoms, less than 25°API.....	0.005
Distillate bottoms, 25°API or above.....	0.002
Atmospheric distillation units:	
Overhead untreated vapors....	0.0013
Overhead treated vapors.....	0.003
Side-stream cuts.....	0.0013
Vacuum distillation units:	
Overhead vapors to oil:	
From bubble tower (partial condenser).....	0.001
From flash pot (no appreciable reflux).....	0.003

* Standards of Tubular Exchanger Manufacturers Association, 2d ed., New York, 1949.

TABLE 12. FOULING FACTORS.*—(Continued)

Overhead vapors in water-cooled condensers:		Lean oil.....	0.002
From bubble tower (final condenser).....	0.001	Overhead vapors.....	0.001
From flash pot.....	0.04	Gasoline.....	0.0005
Side stream:		Debutanizer, Depropanizer, Depentanizer, and Alkylation Units:	
To oil.....	0.001	Feed.....	0.001
To water.....	0.002	Overhead vapors.....	0.001
Residual bottoms, less than 20°API.....	0.005	Product coolers.....	0.001
Distillate bottoms, over 20°API.....	0.002	Product reboilers.....	0.002
Natural gasoline stabilizer units:		Reactor feed.....	0.002
Feed.....	0.0005	Lube treating units:	
O.H. vapors.....	0.0005	Solvent oil mixed feed.....	0.002
Product coolers and exchangers.....	0.0005	Overhead vapors.....	0.001
Product reboilers.....	0.001	Refined oil.....	0.001
H ₂ S Removal Units:		Refined oil heat exchangers water cooled†.....	0.003
For overhead vapors.....	0.001	Gums and tars:	
Solution exchanger coolers....	0.0016	Oil-cooled and steam generators.....	0.005
Reboiler.....	0.0016	Water-cooled.....	0.003
Cracking units:		Solvent.....	0.001
Gas oil feed:		Deasphalting units:	
Under 500°F.....	0.002	Feed oil.....	0.002
500°F and over.....	0.003	Solvent.....	0.001
Naphtha feed:		Asphalt and resin:	
Under 500°F.....	0.002	Oil-cooled and steam generators.....	0.005
500°F and over.....	0.004	Water-cooled.....	0.003
Separator vapors (vapors from separator, flash pot, and vaporizer).....	0.006	Solvent vapors.....	0.001
Bubble-tower vapors.....	0.002	Refined oil.....	0.001
Residuum.....	0.010	Refined oil water cooled.....	0.003
Absorption units:		Dewaxing units:	
Gas.....	0.002	Lube oil.....	0.001
Fat oil.....	0.002	Solvent.....	0.001
		Oil wax mix heating.....	0.001
		Oil wax mix cooling†.....	0.003

† Precautions must be taken against deposition of wax.

Crude Oil Streams

	0-199°F			200-299°F			300-499°F			500°F and over		
	Velocity, fps											
	Under 2 ft	2-4 ft	4 ft and over	Under 2 ft	2-4 ft.	4 ft and over	Under 2 ft	2-4 ft	4 ft and over	Under 2 ft	2-4 ft	4 ft and over
Dry..	.0.003	0.002	0.002	0.003	0.002	0.002	0.004	0.003	0.002	0.005	0.004	0.003
Salt§	.0.003	0.002	0.002	0.005	0.004	0.004	0.006	0.005	0.004	0.007	0.006	0.005

§ Refers to a wet crude—any crude that has not been dehydrated.

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